Program overview

12-Sep-2023 13:14

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Mechanical Engineering**

Code	Omschrijving	ECTS
Master ME 2023		
ME Core courses (Select at least 5 Courses) for all ME Master		
ME41056	Multibody Dynamics	5
ME41106	Intelligent Vehicles 3ME	5
ME44210	Drive & Energy Systems	3
ME45001	Advanced Heat Transfer	4
ME45042	Advanced Fluid Dynamics	5
ME46000	Nonlinear Mechanics	4
ME46006	Physics for Mechanical Engineers	4
ME46007	Measurement Technology	3
ME46055	Engineering Dynamics	4
ME46085-23	Mechatronic System Design	4
SC42001	Control System Design	5
ME non-technical courses (Select min 3 ECTS, max. 6 ECTS)		
RO47009	Robot & Society	4
TPM013B	Environmental Ethics	5
TPM014B	Ethics of AI	5
TPM015B	Ethics of AI	3
TPM016A	Robots and Society	3
TPM024A	Methods for Risk Analysis and Management	5
TPM403SET	Technology Entrepreneurship and Sustainability	4
TPM404B	Technology Entrepreneurship and Internationalization	5
TPM416A	Turning Technology into Business	6
TPM420A	Ready to Start-up	6
WM0320TU	Ethics and Engineering	3
WM0329TU	Ethics and Engineering	6
WM0349WB	Philosophy of Engineering Science and Design	3
WM0353TU	Climate Ethics	3
WM0375TU	Ethics of Technological Risks	3
WM0801TU	Introduction to Safety Science	3
WM1301TU	Ethics of Transportation	3
WM1302TU	Ethics of Transportation	5
WM1401TU	Ethics of Healthcare Technologies	3
WM1402TU	Ethics of Healthcare Technologies	5
ME-BMD Track A. Biomechanical Design		
ME-BMD Obligatory Courses (27 ECTS)		
BM41040	Neuromechanics & Motor Control	5
ME41006	Musculoskeletal Modeling and Simulation	4
ME41056	Multibody Dynamics	5
ME46007	Measurement Technology	3
RO47006	Human Robot Interaction	5
SC42001	Control System Design	5
ME-BMD Track Courses: Select at least 2		
BM41070	Medical Device Prototyping	6
ME41085	Biomechatronics	4
ME41096	Bio Inspired Design	5
ME46015-23	Precision Mechanism Design	4
ME46115-23	Compliant Mechanisms	4
ME-BMD Electives		
BM41050	Applied Experimental Methods: Medical Instruments	4
BM41055	Anatomy and Physiology	4
BM41131	Tissue Biomechanics	3
BM41155	3D Printing	4
ME41035	Special Topics in Sports Engineering	3
ME41065	System Identification and Parameter Estimation	7
ME41120	Freehand Sketching of Products and Mechanisms	3
ME41125	Introduction to Engineering Research	3
ME46041	Experimental Dynamics	4
ME46050	Advanced Finite Element Methods	4
ME46060	Engineering Optimisation: Concepts and Applications	3
ME46070	Fundamentals of Mechanical Analysis	4
RO47003	Robot Software Practicals	5

RO47004	Machine Perception	5
RO47005	Planning & Decision Making	5
RO47013	Control in Human-Robot Interaction	5
RO47015	Applied Experimental Methods	5
RO47019	Intelligent Control Systems	4
SC42056	Optimization for Systems and Control	3
WI4771TU	Object Oriented Scientific Programming with C++	3
ME-BMD Second Year Project: Select 1		
ME51015	ME-BMD Internship / Research Assignment	15
TUD4040	Joint Interdisciplinary Project	15
ME-BMD Graduation Project Obligatory (45 ECTS)		
ME51010-20	ME-BMD Literature Research	10
ME51035	ME-BMD MSc Thesis	35
ME-EFPT Track B. Energy, Flow and Process Technology		
ME-EFPT Obligatory courses (22 ECTS)		
ME45001	Advanced Heat Transfer	4
ME45042	Advanced Fluid Dynamics	5
ME45160	Advanced Applied Thermodynamics	5
ME45165	Equipment for Heat & Mass Transfer	5
ME46007	Measurement Technology	3
ME-EFPT Track courses: Select at least 2 courses		
ME45030	Turbulence	5
ME45134	Process and Power Plant Design	4
ME45155	Computational Fluid Dynamics for Engineers	5
ME45230	Separation techniques for renewable processes	5
ME-EFPT Elective Courses		
AE4117	Fluid-Structure Interaction	4
AE4140	Gas Dynamics	3
ME45025	Introduction to Multiphase Flow	5
ME45050	Microfluidics: Applied Theory and Lab	3
ME45075	Refrigeration & Heat Pumps Fundamentals	4
ME45111	Buildings as Energy and Indoor Climate Systems	5
ME45170	Turbomachinery	4
ME45190	Chaos in Dynamical Systems	3
ME45201	Electrochemical Energy Storage	3
ME45203	Electrolysers, Fuel Cells and Batteries	4
ME45211-23	Particle-based Modeling of Fluids	5
ME45215	Rheology of Complex Fluids	3
ME45220	Experimental Techniques in Fluid Mechanics	3
ME45225	Multiphysics transport in Energy Materials	3
MT44100	Internal Combustion Engines A	5
SET3070	Thermochemistry of Biomass Conversion	4
WI4014TU	Numerical Analysis	6
WI4019-SP	Non-linear Differential Equations	6
ME-EFPT Second Year Project: Select 1 (15 ECTS)		
ME55015	ME-EFPT Research Assignment	15
TUD4040	Joint Interdisciplinary Project	15
ME-EFPT Graduation Project (45 ECTS)		
ME55010-20	ME-EFPT Literature Research	10
ME55010-20 Toets 1	*Literature Research*	*10*
ME55010-20 Toets 2	*Colloquium Presentation*	*0*
ME55010-20 Toets 3	*Colloquium Attendance*	*0*
ME55035	ME-EFPT MSc Thesis	35
ME-HTE Track C. High-Tech Engineering		
ME-HTE Obligatory Courses (14 ECTS)		
ME46006	Physics for Mechanical Engineers	4
ME46055	Engineering Dynamics	4
ME46085-23	Mechatronic System Design	4
ME46110	Intro lab PME	2
ME-HTE Track Courses I (choose at least 3)		
ME46015-23	Precision Mechanism Design	4
ME46020	Micro- & Nanosystems Design & Fabrication, incl. MEMS Lab.	4
ME46060	Engineering Optimisation: Concepts and Applications	3
ME46070	Fundamentals of Mechanical Analysis	4
ME46300	Optics	4
ME-HTE Track Courses II (choose at least 3)		
ME46010	Intro to Nanoscience and Technology	3
ME46025	Manufacturing for the Micro and Nano Scale	3
ME46035	Stability of Thin-Walled Structures 1	4
ME46041	Experimental Dynamics	4
ME46050	Advanced Finite Element Methods	4
ME46065	Thin Film Materials	3
ME46072	Nonlinear Dynamics	4
ME46095	Multiphysics Modelling using COMSOL	4

ME46115-23	Compliant Mechanisms	4
ME46120	Predictive Modelling	4
ME46125	Micro and Nanofabrication for Cell Biology and Tissue Engineering	3
ME46310	Opto-Mechatronics	4
MS43325	Application of Materials in High Tech Engineering	3

ME-HTE Common Elective Courses

AE4117	Fluid-Structure Interaction	4
AE4896	Space instrumentation	4
AE4ASM516	Material Selection in Mechanical Design	3
AE4ASM526	Spacecraft Thermal Design	3
AE4S12-23	Space systems engineering	4
AP3122	Advanced Optical Imaging	6
AP3391	Geometrical Optics	6
AP3401	Introduction to Charged Particle Optics	6
BM41155	3D Printing	4
CH4011MS	Polymer Science	4
CIE5142	Computational Methods in Non-Linear Solid Mechanic	3
ET4117	Electrical Machines and Drives	4
ET4257	Sensors and Actuators	4
ET4260	Microsystem Integration	4
ET4277	Microelectronics Reliability	4
ET4391	Advanced Microelectronics packaging	3
ME41096	Bio Inspired Design	5
ME45050	Microfluidics: Applied Theory and Lab	3
ME46080	Enriched Finite Element Methods	4
MS43100	Science of Failure	3
MS43210	Advanced Characterisation	4
SC42030	Control for High Resolution Imaging	3
SC42061	Nonlinear Systems Theory	3
SC42065	Adaptive Optics Design Project	3
SC42095	Control Engineering	3
SC42145	Robust Control	3
WI4014TU	Numerical Analysis	6
WI4019-SP	Non-linear Differential Equations	6
WI4260TU	Scientific Programming for Engineers	3

ME-HTE Second Year Project: Select 1 (15 ECTS)

ME56015P	ME-HTE Midterm Review	15
ME56015S	ME-HTE Internship	15
TUD4040	Joint Interdisciplinary Project	15

ME-HTE Graduation Project (45 ECTS)

ME56010-22	ME-HTE Literature Research & Project Proposal	10
ME56035	ME-HTE/OM MSc Thesis	35

ME-MME Track D. Multi-Machine Engineering

ME-MME Obligatory Courses (44 ECTS)

ME44101	Dynamics and Interaction of Material and Equipment	4
ME44106	Structural Design with FEM	4
ME44110	Integrated Design Project for Multi-Machine Systems	5
ME44200	Operations and Maintenance	3
ME44206	Quantitative Methods for Logistics	5
ME44210	Drive & Energy Systems	3
ME44300	Multi-Machine Coordination for Logistics	3
ME44305	System Analysis and Simulation	5
ME46000	Nonlinear Mechanics	4
ME46007	Measurement Technology	3
SC42001	Control System Design	5

ME-MME Elective Courses

ME44115	Discrete Element Method (DEM) simulation	4
ME44125	Reliability and Maintenance of Transport Equipment	3
ME44311	Advanced Operations and Production Management	5
ME44312	Machine Learning for Transport and Multi-Machine Systems	3
MT44001	Mechatronics in MT	5

ME-MME Second year

ME-MME Second Year Project: Select 1 (15 ECTS)

ME54015	ME-MME Research Assignment	15
TUD4040	Joint Interdisciplinary Project	15

ME-MME Graduation Project (45 ECTS)

ME54010-20	ME-MME Literature Research	10
ME54035	ME-MME MSc Thesis	35

ME-OM Track Opto-Mechatronics

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Mechanical Engineering**

Master ME 2023	
Director of Studies	Dr. R. Delfos
Contact for students	onderwijsorganisatie-3me@tudelft.nl
Director of Education	Dr. A. (Arno) Stienen; Dr. R. (René) Delfos (formerly)
Program Coordinator	The MSc track coordinator is the person for questions or problems related to the individual study programme and for monitoring progress. Every student can consult the track coordinator to set up their individual study programme in terms of their specialisation courses, their choice of electives and current ideas about their theme of the master's thesis project. In order to complete the programme in two years, the student should plan to take an average of 30 EC worth of courses per semester (two quarters). At the end of the first and second semester, students are invited to an evaluation meeting to discuss aspects of individual modules and the programme as a whole. Also, at the end of each semester, the student is asked to fill in a questionnaire to enable evaluation of the master's programme.
Prerequisites	Students from an academic BSc in Mechanical Engineering (Werktuigbouwkunde), Aerospace Engineering (Lucht- en Ruimtevaarttechniek) and Marine Technolgy (Maritieme Techniek) are directly admissible to the MSc ME. For academic students with a different background or for HBO-students in Mechanical Engineering, there is a bridging programme of 15-60 ECTS (see studyguide.tudelft.nl - 3ME bridging programmes or doorstroommatrix.nl). The bridging programme is only available in Dutch and therefore not accessible for non-Dutch speaking students.
Introduction 1	The MSc Mechanical Engineering is concerned with all aspects of designing, building, and operating high-tech mechanical engineering equipment and systems. It challenges students to integrate various fields of mathematical and physical sciences and to apply fundamental engineering knowledge in a Mechanical Engineering context. Reasons for undertaking such efforts can be e.g. to better assess or predict the behaviour of ME-systems, to optimise the operation of such a system or to re-design (components of) such a system based on recent scientific developments. The programme prepares well for a career at a university, in internationally operating multinational enterprises or in innovative Dutch small & medium-sized enterprises. The MSc programme ME is a continuation after a BSc programme in Mechanical Engineering. For that reason, the ME programme builds on a solid foundation in the basic fields of mechanical engineering knowledge as laid down in the according BSc programme. The ME programme covers two years of study, each with a study load of 60 ECTS. The total programme involves 120 ECTS and comprises cursory modules, assignments, and a master thesis project.
Exit Qualifications	The graduated Mechanical Engineer in each of the tracks meets, to a sufficient level, the following qualifications: 1. Competent in the scientific discipline Mechanical Engineering. A graduate in Mechanical Engineering is able to: a) apply advanced physics and measurement methods in mechanical systems; b) design, carry out, and evaluate experiments; c) identify, design, and control mechanical systems in an interactive and noisy environment; d) relate scientific knowledge to mechanical systems considering their interaction with the environment. 2. Competent in doing research. A graduate in Mechanical Engineering is able to: a) study a topic by critically selecting relevant scientific literature; b) write a scientific report about their own research; c) analyse mechanical systems at various levels of abstraction; d) generate knowledge within the discipline of Mechanical Engineering. 3. Competent in designing. A graduate in Mechanical Engineering is able to: a) systematically design complex mechanical systems; b) generate innovative contributions to the discipline of Mechanical Engineering. 4. A scientific approach. A graduate in Mechanical Engineering is able to: a) apply paradigms, methods, and tools to (re)design a mechanical system; b) manage their own scientific research independently; c) analyse problems and use modelling, simulation, design, and integration towards solutions. 5. Basic intellectual skills. A graduate in Mechanical Engineering is able to: a) analyse and solve technological problems in a systematic way; b) plan and execute research and design in changing circumstances; c) integrate knowledge in an Research & Development project, considering ambiguity, incompleteness, and limitations; d) identify and acquire lacking expertise; e) critically reflect on their own knowledge, skills, and attitude; f) remain professionally competent; g) take a standpoint with regard to a scientific argument within the research area. 6. Competent in operating and communicating. A graduate in Mechanical Engineering is able to: a) work both independently and in multidisciplinary teams; b) present and report in good English; c) explain and defend outcomes from the research area to academia and industry, to specialists, and to laymen. 7. Considering the temporal and social context. A graduate in Mechanical Engineering is able to: a) evaluate and assess the technological, ethical, and societal impact of their own work; b) act responsibly with regard to sustainability, economy, and social welfare. Furthermore, each of the Mechanical Engineering tracks has some own exit qualifications specific to the track, as defined under the track pages.
Program Structure 1	Before entering the MSc programme, the student should announce for a track and submit a form to the coordinator in the chosen track for approval. The relevant form can be found on the website: https://www.tudelft.nl/en/student/faculties/3me-student-portal/education/related/student-forms/msc-forms. Currently, there are 4 different tracks within the Mechanical Engineering master's programme in which a student can start: 1. Biomechanical Design (ME-BMD); 2. Energy, Flow, and Process Technology (ME-EFPT, formerly ME-EPT); 3. High-Tech Engineering (ME-HTE, formerly ME-PME); 4. Multi-Machine Engineering (ME-MME, formerly ME-TEL). Details of the tracks and their course programmes can be found on their own respective pages in this Studyguide. During the first year, all ME students have a common block of ten fundamental Core Courses in Mechanical Engineering out of which a student selects at least five for their Individual Study Programme (ISP). This block forms the commonality between all ME students. An ME track can appoint a maximum of three out of these as obligatory for the track. In addition, each track has obligatory courses of its own, to form a common block within the track. Furthermore, each track can recommend specific courses fitting best to specific research focus areas within the track. In practice, ME courses have students from their own department, but guests from other departmental tracks as well and often also students from other MSc programmes in 3mE or from other faculties. MSc students in 3mE also take between 3 and 6 ECTS of 'non-technical' (formerly called Social courses). These courses many of them offered by the TPM-faculty are intended to give the student a broader perspective on their education as an engineer, for

	instance through ethics, philosophy, or safety science. The second year of the MSc consists of two elements: 1) Q5 project (15 ECTS): One period of 10 weeks is spent on either an interdisciplinary team project (JIP - Joint Interdisciplinary Project), an individual internship within a company or alike, or an equally individual research assignment. In all cases, there is a staff member of the track department appointed as daily supervisor on the project. The choice for the type of project is made by the student in consultation with the track coordinator to best fit the desired learning path of the student. 2) Master Thesis Project (10+35 ECTS): Each individual student prepares a literature study report and a master's thesis for their graduation project. The thesis work is further evaluated through an oral presentation by the candidate and an oral examination before a master's graduation committee. This committee is composed of at least three scientific staff members, including the thesis supervisor. The graduation committee may also include external examiners from research institutes or from industrial partners. The graduation project is administratively split over 2 courses; the first course is focused on the problem analysis and literature research (resulting in the literature study report; often rounded off by a midterm presentation), the second course is focused on the solution development, chosen methodology, experiments/simulations, results, and conclusions. These parts are graded separately following a rubric.
Program Structure 2	Mechanical Engineering offers an MSc programme of two years. Each course year is divided into two semesters and each semester consists of two periods (quarters). In this Studyguide, these periods will be referred to as p1, p2, p3, and p4. A period includes seven to eight weeks of lectures, followed by two or three examination weeks. For those courses for which written examinations are held, the student will get at least one opportunity per year to do a resit. Resits are generally held in the first period after the regular period a course is offered. Resits for the examinations held in period 4 are scheduled three weeks after the period 4-exams, i.e. mid-July (p5). The study load of a course is expressed in European Credits. This is a result of the European Credit Transfer System (ECTS), which encourages acknowledgement of study results between higher education institutions within the European Union. The study load for one educational year is 60 ECTS. The number of ECTS for a course gives an indication of the weight of (a certain part) of the course. One EC involves approximately 28 hours of study (depending on existing prior knowledge and motivation). These 28 hours include all time spent on the course: lectures, self-study, practicals, examinations, etc.

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Mechanical Engineering**

ME Core courses (Select at least 5 Courses) for all ME Master	
Introduction 1	The MSc ME has a block of 11 Core Courses that are fundamental to Mechanical Engineering. They either build on pre-knowledge from the BSc or form a new basis at a MSc level, applicable in multiple tracks. Select at least five Core Courses. Your track may require three as mandatory and some focus areas within your track even more. Consult your track coordinator for selecting your best fit.

ME41056	Multibody Dynamics	5
Responsible Instructor	J.K. Moore	
Instructor	W. Wolfslag	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Department	3mE Department Biomechanical Engineering	

ME41106	Intelligent Vehicles 3ME	5
Responsible Instructor	Prof.dr. D. Gavrila	
Practical Coordinator	R.M. Ensing	
Contact Hours / Week x/x/x/x	0.6.0.0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	ME41106 Intelligent Vehicles is an introduction course on the (on-board) AI technology of automated driving. Whereas the application setting is that of vehicles, most concepts apply to the broader mobile robotics setting. Students seeking a one-course overview of AI for mobile robotics are thus encouraged to attend, also from outside 3ME (e.g. EEMCS, Civil Engineering and Aerospace Engineering faculties). Introduction Course organization Intelligent Vehicles (IV) domain Motivation for automated driving SAE levels of automation Main components of IV technology Remaining challenges Sensor Overview (camera, radar, LiDAR) 3D Machine vision Perspective camera model Extrinsic and intrinsic camera transformations Stereo vision Object Detection and Classification Detection vs. classification Object proposals Handcrafted features (e.g. HOG) & classification (e.g. linear SVM) End-to-end learning: neural networks, deep learning Performance metrics: confusion matrices, precision vs. recall, ROC curves State Estimation Bayesian Filtering Kalman Filtering Particle Filtering Object Tracking Data Association Track Management Self-Localization & Sensor Fusion Absolute vs. relative localization Ego-motion compensation (e.g. odometry) Extrinsic sensor calibration Environment representations (grids, voxels) Motion Planning Planning as graph-search (the A* algorithm) Trajectory generation and optimization The Road Ahead IV technology currently on the market Future developments The course has a substantial practicum component (about 65% of course work load), where learned concepts are put into practice by means of programming assignments (Python, using Jupyter notebooks).	
Study Goals	At the end of the course students will understand the main methodical components of a intelligent vehicle (mobile robot). They will be able to program basic algorithms and test these on simulated and real-world data. Students will be able to express an educated opinion on the benefits and risks of automated driving, the current developments from driver assistance to driver-less cars, and the forces driving this transformation.	
Education Method	Lectures (2-4 hours per week) Practicum (2 hours per week) Formative feedback: Practica - Pairwise collaboration (alternating random groups) - Pass/Fail (counts for knock-out criteria) - No late submissions or resit (except due to motivated medical conditions) Students can in principle run the software locally on their own computers (laptop/PC). However, we only provide limited support for setting up the necessary software environment (i.e. Anaconda). If the software environment cannot be installed properly (e.g. a few processors like M1 and ARM might not work), the student will need to revert to using the computers in the designated lab rooms.	
Course Relations	This course, ME41106 Intelligent Vehicles *cannot* be selected in combination with RO47004 Machine Perception, due to the content overlap between the courses. This means, for example, that this course ME41106 cannot be selected by students following the MS Robotics program (for which RO47004 is required). This course mainly addresses the on-board AI technology of an intelligent vehicle. Complementary to this course is CIE 5805 Intelligent Vehicles for Safe and Efficient Traffic: Design and Assessment which focuses more on automated vehicles as part of a larger intelligent transportation system (i.e. including smart infrastructure, connected/cooperative driving) and on associated issues at the macro level (e.g. traffic flow efficiency, fuel consumption, behavior adaptation).	
Literature and Study Materials	Slides Hand-outs recommended text book: Artificial Intelligence: A Modern Approach (Russel and Norvig) 3rd edition freely available as PDF Jupyter notebooks	
Prerequisites	Basic linear algebra and probability theory Intermediate Python programming skills, e.g. Numpy, object oriented programming (classes, inheritance), modules, namespaces and scope. Recommended: Pattern Recognition / Machine Learning (RO47002, CSE2510, or equivalent) It is possible to start the course with beginner Python skills if your aim is to take advantage of this course to also improve your programming expertise (a laudable aim, given its importance for both industry and academia). Be aware, however, that this	

	catching up will likely result in an increased study load (e.g. 1-2 EC).
Assessment	Final practicum assignment (50%) - Individual assignment - Late submission: max. one day late for -1.0 grade point - Students having an initial grade of < 5.8 will be offered an opportunity for a remake to obtain a (maximum) grade of 5.8 for this component in the Q3 quarter. Final written exam (50%) - Written exam on campus. Knock-out criteria - At most one of the Pass/Fail practicum assignments is Fail - The minimum final practicum assignment grade is 5 - The minimum final written exam grade is 5 - The minimum final course grade is 6 The final practicum assignment will be evaluated on the computers in the designated lab rooms. If a student chooses to work using own computing equipment, the student needs to test the software on the designated computers before handing-in. Both final practicum assignment and final written exam results remain valid for one year. Each result can separately be transferred to the next academic year, upon request.
Special Information	For more information about this course, see the slides of the Introduction lecture from past academic year at http://intelligent-vehicles.org/teaching/courses/ . TU Delft students might also be able to view the lectures from past academic year on Collegerama.
Department	3mE Department Cognitive Robotics

ME44210	**Drive & Energy Systems**	**3**
Responsible Instructor	Dr.ir. H. Polinder	
Contact Hours / Week x/x/x/x	3/0/0/0 (1st part Q1 2x2 hours, 2nd part Q1 1x2 hours)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	This course is designed for students without a BSc in Electrical Engineering. Proper knowledge of mathematics (differential equations, complex numbers) and mechanics is required.	
Course Contents	- Introduction to drive and energy systems - Electricity and electrical circuits - Magnetism and magnetic circuits - Power electronics - DC machines - AC voltages, currents, powers and phasors - Transformers - Synchronous machines (including PM machines) - Induction machines - Choices in electrical drive and energy systems	
Study Goals	After following this course, students should be able to - Calculate DC and AC (single phase and three-phase) voltages and currents in simple electrical circuits (such as equivalent circuits of electrical machines) with sources, resistive, capacitive and inductive loads using Kirchhoffs laws. - Sketch phasor diagrams with voltage and current phasors. - Calculate active, reactive and apparent power in AC circuits. - Calculate energy stored in capacitors and inductors. - Describe the most important power semiconductors (diodes, IGBTs, MOSFETs) and their switching behaviour. - Describe how power electronic converters use these switches to convert DC from one voltage level to another (choppers or phase legs using IGBTs or MOSFETs), AC into DC (passive rectifiers using diodes or active rectifiers using IGBTs or MOSFETs) or DC into AC (phase legs using IGBTs or MOSFETs). - Derive an expression for the output voltage of a single-phase or three-phase diode bridge rectifier as a function of input voltage in continuous conduction mode (neglecting commutation). - Derive an expression for the output voltage of a DC-DC converter as a function of duty cycle and input voltage in continuous conduction mode. - Calculate induced voltage using Faradays law. - Calculate electromagnetic forces and torques using Lorentz law or the power balance. - Derive voltage equations and equivalent circuits of transformers, DC machines, synchronous machines, induction machines and permanent magnet machines. - Derive performance characteristics based on these equivalent circuits (torque-speed characteristics). - Describe advantages and disadvantages of electrification of drives. - Select suitable machine and drives for different applications.	
Education Method	Lectures	
Books	T. Wildi, "Electrical Machines, Drives and Power Systems"	
Prerequisites	Mathematics, Mechanics	
Assessment	The method of assessment will be: written exam, closed book, partly multiple choice. An equations form will be handed out with the exam. If on campus assessment is not possible (for example, because of measures resulting from COVID-19) the assessment will be as follows: written exam, open book, online, partly multiple choice.	
Permitted Materials during Tests	An equations form will be handed out.	
Department	3mE Department Maritime & Transport Technology	
Contact	h.polinder@tudelft.nl	

ME45001	Advanced Heat Transfer	4
Responsible Instructor	Dr.ir. J.W.R. Peeters	
Instructor	Dr. R. Delfos	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	A BSc-course in Fluid Dynamics and Heat Transfer or in Transport Phenomena. Basic math in 1D (Taylor series & ODE's) and in multiple-dimensions (vector analysis & PDE's) in the BSc-ME in 3ME or equivalent; basic understanding of Fourier series.	
Course Contents	In this course the concepts & mathematics of heat transfer in the engineering context are treated. Elementary understanding of the three modes of heat transfer: conduction, convection and radiation, will be briefly reviewed during the first two lectures. During the remainder of the course, the underlying physics will be emphasized and advanced mathematical formulations will be explained. A large focus in the course will be on the analysis of heat transfer in real-life integrated systems. Subjects in order of appearance: - A refresher on the underlying thermodynamics; energy, enthalpy, specific heats and phase change enthalpy. - A refresher on Conduction, Convection and Radiation. - Integral and differential energy balances in a 1-D and multiple-D continuum; absorption, reaction and dissipation as source terms. - Stationary conduction: cooling fins, multi-dimensional conduction and Laplaces equation; boundary conditions; analytical techniques & numerical techniques; relaxation. - Phase change as a boundary phenomenon; melting and solidification fronts; Jakob number & Stefan condition. - Instationary conduction: Fourier and Biot number; boundary conditions; analytical techniques & numerical techniques; stability criteria. - Forced & Free convection: Nusselt, Stanton, Prandlt & Peclet numbers; Analysis & the physics behind empirical correlations. The role of boundary conditions. - Radiation: radiative exchange between grey bodies, solar radiation, spectral characteristics, surface characteristics.	
Study Goals	More specifically: The student is able to 1. Distinguish between the different modes of heat transfer, and divide real-life systems into subsystems of elementary heat transfer modes in a qualitative and quantitative manner. 2. For all of the below; give the physical interpretation of contributors and terms in balances in words and in sketches. 3. Set up appropriate integral and differential energy balances for one- and multidimensional instationary conduction. 4. Justify and apply simplifications and define the appropriate boundary conditions, including problems containing phase changes, i.e. Stefan conditions. 5. Indicate mathematical solution strategies - both analytical and numerical, and apply those for standard geometries. 6. Distinguish between different modes of convective heat transfer, and distinguish between the different physical mechanisms underlying empirical correlations. Indicate implications when more detailed distributions of convective heat transfer are involved. 7. Estimate the magnitude of radiative heat transfer, distinguish between thermal and short-wave properties and spectral distributions, qualify and quantify the role of surface properties in real-life applications.	
Education Method	- Lectures (2x2 hours per week) including experiments & computer demonstrations. - After each lecture, book assignments, as well as custom assignments are given. Solutions will also be provided. - Weekly instruction hours (2 hours per week), during which students can discuss heat transfer problems with the instructors and/or during which instructors show how to solve an assignment. - Total number of contact hours: 6 hours per week.	
Computer Use	Computers are used for demonstrations of the lecture material during the course on the basis of TU Delft software. Students are invited to solve assignment problems using Matlab/Python/Maple - in some cases using given code, in others to start from scratch.	
Literature and Study Materials	Book: as a lead "Basic Heat and Mass Transfer" or just called "Mills" is used: A.F. Mills and C.F.M. Coimbra, 3rd edition, Temporal Publishing, ISBN 978-0-9963053-0-3 (available through MechEng study association 'Gezelschap Leeghwater'. The 2nd edition as by Pearson is out of print but still widely available; it is identical, but cheaper, and of lousy bookbinding quality. Derived versions like "Basis Heat Transfer", or "Heat Transfer" by Anthony Mills are nearly identical. Alternative books on heat transfer from your Bachelor's course (including Cengel & Incropera) can be used as well, but you might want to check with the teacher; just bring it to the first Office Hour if it's not standard. Lecture slides, short videos, Matlab scipts & other study materials required will be provided via Brightspace.	
Assessment	The assessment consists of a hand-written closed-book exam, which you will have to complete in a pre-determined timeslot of 3 hours. You are allowed to bring the equation sheet posted on Brightspace; you are also allowed to make your own notes on the equation sheet itself.	
Permitted Materials during Tests	Only a non-programmable non-graphical calculator like Casio fx-82 MS is accepted. One (1) equation sheet of 2 sides A4 can be taken and/or is provided.	
Department	3mE Department Process & Energy	

ME45042	Advanced Fluid Dynamics	5
Responsible Instructor	Dr. A.J.L.L. Buchner	
Instructor	Dr. D.S.W. Tam	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Expected prior knowledge	Undergraduate level course in thermo-fluids (WB2542) Knowledge of vector analysis, multivariable and differential calculus is essential for this course (WBMT1050, WBMT2048)	
Course Contents	This course surveys the principal concepts and methods of fluid mechanics. Topics include control volume analysis for mass and momentum, the derivation of Navier-Stokes equations from mass and momentum conservation for incompressible flows, boundary conditions at fluid/fluid and fluid/solid interfaces, dimensional analysis, interfacial tension, inviscid flows, circulation and vorticity, lift and drag, boundary layers, lubrication theory, classical instabilities and waves. Classical solutions of the Navier -Stokes equations are derived and Concepts are illustrated through practical examples from engineering, geophysics and biological fluid dynamics.	
Study Goals	At the end of this class, the student is able to formulate and to use the fundamental governing equations of incompressible, laminar fluid mechanics in classical flow regimes, and to apply these to a new problem in flow technology. More specifically, the student must know how to: 1. formulate the conservation equations for mass and momentum 2. derive the governing equations of motion for an incompressible flow. (Starting from the conservation equations for mass and momentum, the constitutive equation for a Newtonian fluid, and the appropriate boundary conditions at a fluid/solid and fluid/fluid interface) 3. compute the stress tensor and the aero/hydrodynamic forces on a mechanical system 4. derive the vorticity equation and the consequence on the generation and transport of vorticity in 2D (Kelvin circulation theorem) 5. perform a dimensional or scaling analysis to identify the dominant terms in the governing equations. Also, use the non-dimensional numbers introduced in class to determine the flow regime (Reynolds, Strouhal, Froude, Bond, Capillary, Weber) 6. simplify the Navier-Stokes equations for the case of an inviscid flow (the Euler equations), a boundary-layer, a unidirectional flow, a creeping (Stokes) flow and in the lubrication limit. 7. apply appropriately the different forms of the Bernoulli equation The student must be able to derive classical solutions to the Navier-Stokes equations discussed in class. The student must be able to determine whether these classical solutions can be applied to a new problem and use them to compute the flow fields. These solutions include: 8. For ideal flows: a source, a sink, an irrotational vortex, a dipole, corner flows and the superposition of any combination of these using complex potentials (e.g. the 2D irrotational flow around a cylinder and the interpretation of Lift (Joukowski theorem) and Drag (d'Alembert Paradox)) 9. 3D axisymmetric irrotational flow around a sphere for steady and unsteady flows (added mass) 10. For viscous flows: Unidirectional laminar flows, Couette and Poiseuille flows, flow in a lubrication layer 11. Blasius solution for the boundary layer over a flat plate 12. Waves and instabilities in fluid mechanics, gravity waves, Kelvin-Helmholtz instability When facing a new problem in fluid dynamics, the student should be able to: 13. Write all the relevant boundary conditions 14. Use the appropriate formulation of mass and momentum transport (at the control volume level or at the differential level i.e. Navier Stokes equations) 15. Simplify the governing equations using a dimensional/scaling analysis 16. Solve the governing equations for simple flows, when analytically possible 17. Use well-known solutions derived in class to model flows appropriately for the new problem. 18. Model the effect of aero/hydrodynamic forces on the dynamics of a simple mechanical system	
Education Method	Lectures: 3 hours/week Recitations: 1 hour/week Office Hours: 1 hour/week	
Books	Fluid Mechanics Sixth Edition Kundu, Cohen and Dowling	
Assessment	Midterm Written Exam (35%) Final Written Exam (65%)	
Department	3mE Department Process & Energy	
Contact	email: a.j.buchner@tudelft.nl	

ME46000	Nonlinear Mechanics	4

Responsible Instructor Dr. C. Ayas

Instructor Dr. A.M. Aragon

Instructor Dr. F. Alijani

Contact Hours / Week x/x/x/x 0/4/0/0

Education Period 2

Start Education 2

Exam Period 2
3

Course Language English

Course Contents

Introduction

Nonlinear mechanics (NM) builds upon the engineering mechanics (statics and dynamics) courses as embedded within the BSc curriculum. This MSc course will complete the education in engineering mechanics for a large group of students. Thus, this course should prepare them as good as possible for common problems faced by mechanical engineers without a specialisation in engineering mechanics. On the other hand, a group of students will continue their training in dynamics and/or statics. For this reason, this course serves as a first step into the more advanced engineering mechanics topics.

Given the broad background of the students participating in this course, abstract formulations will be avoided. Moreover, many topics will be presented as introductions, creating awareness among the students.

Detailed content:

Introduction to nonlinearities in statics and dynamics
Principle of virtual work
Principle of minimum potential
Green-Lagrange strain tensor
2nd Piola-Kirchhoff stress tensor and equilibrium equations
Alternative stress and strain tensors,
Geometrical nonlinearity (finite rotations, finite deformations, geometric stiffness effects)

Introduction to nonlinear material models
Nonlinear elastic models
Simple plasticity models (elastic-ideally plastic, isotropic and kinematic hardening)
Residual stresses
Application to simple truss and beam problems
3D plasticity models work (yield surface, flow rule)

Introduction to nonlinear dynamics
Studying simple discrete nonlinear systems using the phase plane
Quantitative analysis of weakly nonlinear single-degree-of freedom systems using general perturbation theory
Periodic behaviour of simple nonlinear oscillators, limit cycles and softening/hardening responses
Basics of bifurcation theory and stability for simple nonlinear dynamic systems

Introduction to the finite element method in elastostatics (strong, weak or variational, and Galerkin forms of the boundary value problem with linearized kinematics)
The finite element method for large deformation (total Lagrangian framework)
Computational tools needed to solve the discrete system of equations

Study Goals

Learning goals:

Regarding nonlinear mechanics in general (25%), students:
are aware of the range of nonlinearities that are encountered in practical problems, including both static as well as dynamic settings,
are capable of detecting when the nonlinear analysis might be required for practical problems,
have an awareness of the role of nonlinearities in both static as well as dynamic systems,
are capable of formulating and solving simple nonlinear engineering mechanics problems.
can distinguish when geometrical nonlinearity may be relevant (finite rotations, finite deformations, geometric stiffness effects),
can make analytical estimates for nonlinear problems, e.g., based on minimum of potential energy.

Regarding physical nonlinearity (25%), students:
have an awareness of different nonlinear material models,
are familiar with and know when to use nonlinear elastic models,
are able to describe the essential features of a nonlinear elastic model,
are aware of simple plasticity models (elastic-ideally plastic, isotropic and kinematic hardening) to solve simple problems,
are aware and can describe the practical relevance of yielding and the generation of residual stresses
are able to describe how 3D plasticity models work (yield surface, flow rule),

Regarding nonlinear dynamics (25%), students:
can perform a qualitative analysis of simple discrete nonlinear systems using the phase plane,
can carry out quantitative analysis of weakly nonlinear single-degree-of-freedom systems using general perturbation theory,
can discuss the periodic behaviour of simple nonlinear oscillators, limit cycles and softening/hardening responses,
are aware of the basics of bifurcation theory and stability for simple nonlinear dynamic systems.

Finally, regarding computational nonlinear mechanics and the finite element method (25%), students:
can perform relatively simple nonlinear simulations,
are aware of computational tools needed to solve nonlinear equations, including topics such as rate equations, physical stiffness, geometric stiffness, iterative solutions procedures, incremental procedures, incremental iterative procedures, path following, dynamic and static transient simulations, convergence, stability points,
can state the strong form of a boundary value problem of an elliptic equation, its variational form, and derive the discrete equations using linearized kinematics
understand the procedure to derive the discrete equations in finite deformation setting using a total Lagrangian formulation (students understand the linearization process)
can state the strong form of a boundary value problem of an elliptic equation, its variational form, and derive the discrete equations using linearized kinematics
understand the procedure to derive the discrete equations in finite deformation setting using a total Lagrangian formulation (students understand the linearization process)

Education Method	Contact hours: Lectures (4 hours per week) Weekly tutor session (1 hour per week) Study materials: lecture slides Suggested literature: Nonlinear Finite Element Analysis of Solids and Structures, 2nd Edition; Rene De Borst, Mike A. Crisfield, Joris J. C. Remmers, Clemens V. Verhoosel; ISBN: 978-0-470-66644-9; 540 pages August 2012 Nonlinear Finite Elements for Continua and Structures, 2nd Edition; Ted Belytschko, Wing Kam Liu, Brian Moran, Khalil Elkhodary; ISBN: 978-1-118-63270-3; 830 pages December 2013, ©2013 Nonlinear Solid Mechanics: A Continuum Approach for Engineering Gerhard A. Holzapfel; ISBN: 978-0-471-82319-3; 470 pages February 2000 Applied Mechanics of Solids. Allan F. Bower, CRC Press, ISBN:978-1-4398-0247-2 (available freely from www.solidmechanics.org) Lecture Notes on Nonlinear Vibrations, Richard H. Rand, freely available at http://audiophile.tam.cornell.edu/randdocs/nlvibe52.pdf. Nayfeh, A. H., & Mook, D. T. (2008). Nonlinear oscillations. John Wiley & Sons.
Assessment	Written (closed-book) exam.
Department	3mE Department Precision & Microsystems Engineering

ME46006	Physics for Mechanical Engineers	4
Responsible Instructor	Dr. G.J. Verbiest	
Instructor	Prof.dr. P.G. Steeneken	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Students are expected to have Physics knowledge at the level of the final exam of the physics exam at Dutch highschools (VWO) and to have Mathematics knowledge at the level of the BSc education in Mechanical Engineering (e.g. WBMT2048). If they have not, they can still do the course but it will require more self-study from the book of Tipler.	
Summary	The course ME46006 Physics for Mechanical Engineers, aims to provide mechanical engineering students with a broad background in physics. The course offers basic understanding and skills in the field of physics, that will provide a background that can be useful in many engineering situations. It can serve as a starting point for more advanced physics studies and can also help mechanical engineers to interact with electrical engineers and physicists in their future career. Since the course aims for a broad coverage of the field of physics, the course material includes many different subjects at a basic to intermediate level. In contrast to other courses, this course favours to cover a relatively large number of different topics in order to provide a broad background, instead of covering a small number of topics in-depth. This means that the amount of material that needs to be studied is larger than for other courses. A bachelor degree in mechanical, electrical, aerospace, industrial engineering or applied physics is considered prerequisite.	
Course Contents	- Waves - Electrostatics and electricity - Magnetism - Electrodynamics - Optics - Interference and diffraction - Outlook into quantum mechanics	
Study Goals	At the end of the course, students should have knowledge, and should be able to solve problems on the following topics: Waves (Chapters 15&16 of Tipler): - The wave equation - Propagation, reflection and transmission of waves - The Doppler effect Electrostatics (Chapters 21-24): - Coulomb's law and electric forces - Electric fields and Gauss' law - Voltage and capacitors Electricity (Chapters 25 & 29): - Ohm's law - DC circuits Magnetism (Chapters 27-28, 30): - Gauss' law for magnetism - Lorentz', Ampere's, Lenz' and Faraday's law - Electromagnetic motors and inductors - Maxwell's equations Optics (Chapter 31-34): - Propagation and properties of light - Light-matter interaction - Imaging systems with light - Interference and diffraction - Limits of optical systems and quantum mechanics	
Education Method	Learning methods include power point presentations, class demonstrations, and recommended homework. Lecture notes will be made available on the Brightspace. Recommended homework will be on the concepts discussed in the class and solutions will be made available on the Brightspace. The course ends with a written exam.	
Literature and Study Materials	The course will closely follow selected chapters of the book of Tipler. In addition lecture slides will be provided on Brightspace.	
Books	Physics for Engineers and Scientists, 6th edition by Tipler and Mosca.	
Assessment	Written exam.	
Department	3mE Department Precision & Microsystems Engineering	

ME46007	Measurement Technology	3
Responsible Instructor	Dr.ir. G.E. Elsinga	
Instructor	Dr.ir. M.J. Tummers	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Summary	This master level course covers fundamental aspects of measurements useful for a mechanical engineer. It covers measurement of various physical quantities, signal conditioning, error analysis and statistical analysis of the data. A bachelor degree in mechanical, electrical, aerospace, industrial engineering or applied physics is considered prerequisite.	
Course Contents	topics: - General introduction to measurements - Measurement system behavior - Hypothesis testing - Uncertainty analysis - Sensor probes: variety of experimental techniques - Signal conditioning - Data acquisition and processing - Imaging and image processing - 3D techniques	
Study Goals	The general aim of this course is to provide students with a background in the theory of engineering measurements. At the end of the course, the students should be able to (1) analyze measurement systems and devices using the laws of mechanics and general physics, (2) evaluate the accuracy, sensitivity, spatial and temporal resolution of a measurement, (3) analyze and interpret datasets that contain errors, (4) design a measurement scheme and signal conditioning in order to enhance the significance of the results.	
Education Method	The main learning methods include lectures, take-home assignments and a textbook. The lectures cover the different topics of the course. (In case of unforeseen circumstances or measures resulting from COVID-19, the lectures will be replaced by pre-recorded videos of the lectures and/or live online-sessions. Such changes will be communicated on Brightspace.) Lecture slides will be made available on Brightspace. The recommended textbook is given below, however, you are encouraged to refer other books as well. Additional reading material, if necessary, will be made available on Brightspace. Assignments will be provided to practice the relevant problems and the concepts discussed in the class. They will be made available on Brightspace. The solutions to the assignments will be discussed in the class. The course ends with a written exam.	
Literature and Study Materials	The prescribed books will be followed during the course. Lecture slides will be made available on Brightspace. Additional study materials will be provided on Brightspace if necessary.	
Books	Theory and design for mechanical measurements, 6th edition by Richard S. Figliola & Donald E. Beasley	
Assessment	The examination will be as follows: Closed book written exam. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be: Remote (online) open book with several fraud prevention measures.	
Department	3mE Department Process & Energy	

ME46055	Engineering Dynamics	4
Responsible Instructor	Dr. F. Alijani	
Instructor	R.A. Norte	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	Multibody Dynamics A(ME41050), Multibody Dynamics B (ME41050), Nonlinear mechanics(ME46000), Nonlinear dynamics (ME46072).	
Course Contents	The dynamic behavior of structures (and systems in general) plays an essential role in engineering mechanics and in particular in the design of controllers. In this master course, we will discuss how the equations describing the dynamical behavior of a structure and of a mechatronical system can be set up. Fundamental concepts in dynamics such as equilibrium, stability, linearization and vibration modes are discussed. If time permits, also an introduction to numerical integration techniques is proposed. The course will discuss the following topics: --Review of the virtual work principle and Lagrange equations --Linearization around an equilibrium position: vibrations --Free vibration modes and modal superposition --Forced harmonic response of nondamped and damped structures -- Introduction to time integration techniques	
Study Goals	The student is able to select different ways of setting up the dynamic equations of mechanical systems, to perform an analysis of the system in terms of linear stability and vibration modes and to properly use mode superposition techniques for computing transient and harmonic responses. More specifically, the student must be able to : 1. Explain the dynamic principle of virtual work and Lagrange equations, and discuss their relation to the basic Newton laws. 2. Describe the concept of kinematic constraints and identify a proper set of degrees of freedom to describe a dynamic system. 3. Use Lagrange equations via employing the minimum set of degrees of freedom to find the governing equations of dynamic systems, and construct these equations for systems with kinematic constraints. 4. Use Hamiltons principle to find the governing equations of motion of dynamics systems. 5. Find equilibrium positions and construct the linearized equations of motion by using different contributions of the kinetic and potential energies. 6. Analyze the linear stability of dynamic systems according to their state space formulation. 7. Explain and use the concept of free vibration modes and frequencies for multi degree of freedom systems. 8. Apply the orthogonality properties of modes to describe the forced response of damped and undamped systems.	
Education Method	Lecture	
Literature and Study Materials	Course material: Lecture notes (available through brightspace) References from literature: 1. M. Géradin and D. Rixen. Mechanical Vibrations. Theory and Application to Structural Dynamics. Wiley & Sons, 2nd edition, 1997. 2. J. Ginsberg. Engineering Dynamics. Cambridge University Press, 2008. 3. D.J. Inman, Engineering Vibration. PrenticeHall, 1996. 4. L. Meirovitch. Principles and Techniques of Vibrations. PrenticeHall, 1997.	
Assessment	written exam + an assignment or a quiz	
Remarks	In case an assignment is given- The assignment will involve the use of a Matlab-like software 2. Old course code: WB1418-07	
Department	3mE Department Precision & Microsystems Engineering	

ME46085-23	Mechatronic System Design	4
Responsible Instructor	Dr. S.H. Hossein Nia Kani	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	The following are strictly required to be able to follow the course: - Basics of Multi-body Dynamics, including the ability to form equations of motion. - Basics of Electromagnetism, particularly Maxwell's equations. - Fundamentals of signals and systems (Laplace transforms, transfer function, bode plot, Nyquist stability). It is expected that students are able to use MATLAB or are interested in spending time to learn it while passing the course. This is required to complete the assignments.	
Course Contents	Course Contents Mechatronic system design deals with the design of controlled motion systems by the integration of functional elements from a multitude of disciplines. It starts with thinking how the required function can be realised by the combination of different subsystems according to a Systems Engineering approach. It should be noted that the control principles used in this course place a strong emphasis on frequency domain methods by linearising the system at its working point with the use of Bode- and Nyquist plots. The main reason for this emphasis is the strong focus in other control related courses on (non-linear) time domain related methods while linearised frequency domain related methods are still dominantly applied in the industry. A mechatronic engineer should be able to work with both methods and use them where appropriate. Some supporting disciplines, like power-electronics and electromechanics, are not part of the BSc program of mechanical engineers. For this reason this course introduces these disciplines in connection with mechanical dynamics and PID-motion control principles to realise an optimally designed motion system. The target application for the lectures are motion systems that combine high speed movements with extreme precision. The course covers the following three main subjects: 1: Dynamics of motion systems in the time and frequency domain, including analytical frequency transfer functions that are represented in Bode and Nyquist plots. 2: Electromechanical actuators, mainly based on the electromagnetic Lorentz principle. Reluctance force and piezoelectric actuators will be shortly presented to complete the overview. 3: Motion control in the frequency domain with PID and model-based feedforward control-principles that effectively deal with the mechanical dynamic anomalies (resonances and eigenmodes) of the plant. The other relevant discipline, electronics and position measurement systems is dealt with in another course: ME46005, Physics and measurement. The most important educational element that will be addressed is the necessary knowledge of the physical phenomena that act on motion systems, to be able to critically judge results obtained with simulation software. The lectures challenge the capability of students to match simulation models with reality, to translate a real system into a sufficiently simplified dynamic model and use the derived dynamic properties to design a suitable, practically realiseable controller. This course increases the understanding what a position control system does in reality in terms of virtual mechanical properties like stiffness and damping that are added to the mechanical plant by a closed loop feedback controller. It is shown how a motion system can be analysed and modelled top-down with approximating (scalar and linearised) calculations by hand, giving a sufficient feel of the problem to make valuable concept design decisions in an early stage. With this method students learn to work more efficiently by starting their design with a quick and dirty global analysis to prove feasibility or direct further detailed modelling in specific problem areas.	
Study Goals	Can analyse and derive improvements to the dynamic behaviour of an actuator-driven mechanical structure with maximum 6th order plant dynamics (incl actuator and amplifier) by means of Bode and Nyquist plots. Can select and calculate a single-axis functional electromagnetic actuator for a given specification, working according to the Lorentz and reluctance force generation principle. Can select a proper amplifier to drive the above mentioned electromagnetic actuator. Can understand the basic concept of piezoelectric actuators and their application in mechatronic systems Can identify and apply feedforward and feedback P-, PD- or PID-motion controller settings for a given plant, consisting of a dynamically realistic power amplifier, electromagnetic actuator and mechanical structure with an ideal sensor, to achieve a stable system targeting a specified maximum bandwidth or disturbance rejection.	
Education Method	Lectures are given in 12*2 lecture hours, with presentations on the theory and practice of active-controlled motion systems. In addition, students will work on assignments to implement the theory learned in the course.	
Literature and Study Materials	Only the book is allowed at the examination. Therefore, no printed copy of the first edition, no computers, e-readers, smartphones, or other items are permitted. However, small notes written on the pages of the book are allowed.	
Books	The Design of High Performance Mechatronics - 3rd Revised Edition by R. Munnig Schmidt, G. Schitter, A. Rankers and Jan Van Eijk	
Assessment	The Final Assignment will account for 50% of the final grade. The remaining 50% of the score will be based on an open-book written examination. The examination will consist of questions covering the above-mentioned study goals.	
Contact	s.h.hosseinniakani@tudelft.nl	

SC42001	Control System Design	5
Responsible Instructor	Dr.ir. A.J.J. van den Boom	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	State-space description of single-input, single-output linear dynamic systems, interconnections, block diagrams. Linearization, equilibria, stability, Lyapunov functions and the Lyapunov equation Dynamic response, relation to modes, the matrix exponential. Realization of transfer function models by state space descriptions, coordinate changes, canonical forms. Controllability, stabilizability, uncontrollable modes and pole-placement by state-feedback. Application of LQ regulator. Observability, detectability, unobservable modes, state-estimation observer design Output feedback synthesis and separation principle. Reference signal modeling, integral action for zero steady-state error; Analysis in robust stability and robust performance. Basics of model predictive control (MPC). Different model-structures. Prediction models in state-space setting. Standard predictive control scheme. Relation standard form with GPC, LQPC and other predictive control schemes. Solution of the standard predictive control problem. Stability of MPC.	
Study Goals	By taking this course, the student - will be able to master the introduced theoretical concepts in systems theory and feedback control design - will be able to practically apply these concepts to design projects and tasks - and will be able to relate the learned concepts and techniques to other more specialized ones, to potentially integrate them by taking adjacent courses. - will be able to translate a predictive control problem into a standards setting and solve the predictive control problem. More specifically, the student will be able to: - Translate differential equation models into state-space and transfer function descriptions - Rationalize differences between state-space and transfer function approaches - Linearize a system, determine its equilibrium points, analyze directly its local stability, leverage Lyapunov theory to study general stability properties - Describe the effect of eigenvalue/pole locations to the dynamic system response in time/frequency domain. Contrast step and impulse responses. Analyze transients and steady-state - Investigate model controllability. Formulate and apply the procedure of pole-placement by state-feedback, as well as LQ optimal state-feedback control - Derive observability properties. Formulate and apply the procedure of state estimation and build converging observers - Formulate the separation principle and employ it for the design of output feedback - Build reference models and achieve zero steady-state error using integral control. - set up a predictive control problem. - Solve the standard predictive control problem; - Master the main analytical details in stability proofs of predictive control schemes;	
Education Method	Lectures 4/0/0/0	
Literature and Study Materials	Textbook (its use is strongly recommended): K.J. Astrom, R.M. Murray, Feedback Systems: An Introduction for Scientists and Engineers; second edition; Princeton University Press, Princeton and Oxford, 2009 Available online for download: http://www.cds.caltech.edu/~murray/books/AM08/pdf/fbs-public_24Jul2020.pdf The link for downloading the lecture notes for model predictive control will be available in the first quarter.	
Assessment	Written examination. Due to COVID-19 the exam may be an online exam. Disclaimer: information may change depending on the developments around the coronavirus.	
Department	3mE Department Delft Center for Systems and Control	

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Mechanical Engineering**

ME non-technical courses (Select min 3 ECTS, max. 6 ECTS)	
Introduction 1	Engineering is more than just the technical courses of your track: an engineer works in society and this is represented in your Individual Study Programme, (Form 2 to be filled out during Q1). The way you make this connection is up to you; there is a wide variety of courses to choose from; some courses will fit your programme better than others. You can also consult your track coordinator about this. At least one course needs to be selected. No more than 6 ECTS count within the 120 EC degree programme.

RO47009	Robot & Society	4
Responsible Instructor	O. Kudina	
Contact Hours / Week x/x/x/x	0.0.2.0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	

Course Contents

Introduction

Robots and Society takes a critical look at the design, development, production, and implementation of robots and AI used in society. The class will build upon traditional robot ethics and ethics of technology concepts and approaches applied specifically to current technical developments in the fields of robotics and AI. The course will provide students with the skills of critical reflection, ethical sensitivity, and deliberation about the ethical, societal, design, and governance issues at stake concerning these technologies. Thus, the course should prepare students for common problems they will face in their current studies and prepare them for their future careers in the technical or humanities disciplines. The course serves as a first step in: uncovering societal issues related to robotics and AI, understanding how and why they are points of concern, and applying conceptual tools to mitigate and/or overcome such issues.
Given the broad background of the students who will participate in this course, abstract formulations will be avoided. Moreover, many of the topics will be presented as introductions, creating awareness among the students.

Detailed content:

Introduction to ethics of technology
Technology as a socio-technical system
Practice-based approach to values
Technological mediation of morality
Technology as a Social Experiment
Value Change

Introduction to robot ethics
Ethics of Human-Robot Interaction (HRI) (e.g. social script, anthropomorphism, gender, inclusivity, etc.)
Ethics of Robot Applications (e.g. healthcare robots, Digital Voice Assistants, military drones, etc.)
Ethics of Robot Design and Development (e.g. Design for Values, Value hierarchy, Ethics as an accompaniment model, etc.)

Introduction to ethics of AI
AI as a Black box
Bias in training data
Principle of explainability
Power and politics of AI
Ethics of AI in public sectors (e.g. predictive policing, facial recognition, Covid19 tracking apps, etc.)

Study Goals

Learning goals:

After completing the course, the students:
are aware of the range of views on the relationship between robots and AI and society, from technology being neutral to technology being value laden to the technology as a socio-technical system,
are capable of detecting when certain assumptions are appealed to in a debate about the ethics of technology, and robots and AI in society specifically,
have awareness for the role of these concepts and approaches to thinking in the current debate surrounding robotics and AI,
are capable of formulating a position on the ethical issues at stake in current applications of robotics and AI,
develop evaluative judgments on the quality of their own work and of the performance of others (including peers) by implementing and providing feedback.

Education Method

Contact hours:
Lectures (online) and tutorials (on campus), weekly quizzes based on the readings, an exam, and a final paper

Weekly quizzes:
Weekly quizzes will assess the students' ability to understand the reading and lecture material from one week to the next and will prepare them for the exam.

Tutorials:
Students are expected to attend two tutorials over the duration of the course (on campus), one in Part A and one in Part B of the course
Students are expected to write, prior to the tutorial, a short reflective paragraph discussing the assigned journalistic engagement to be discussed during the tutorial

Exam (on campus, in a computer room):
Throughout the exam (on campus, in a computer room, open book), the students will be asked to identify, define and illustrate concepts used in the course, to make arguments using ethical principles, theories or frameworks they will learn in the course, and to defend or critique claims made about robotic and AI-based technologies discussed in the course.

Final paper
Students will be invited to form groups of 3-4 to reflect in a written form on a technology from the course. In the paper, students are asked to identify a specific ethical issue raised by that technology, to evaluate it ethically and to make recommendations on how to develop or apply the technology in a responsible way. The assignment will proceed in the following stages throughout the course: students will first be asked to submit a paper proposal, then a draft paper, after which each student will also peer review one other paper on a similar topic. The final paper will be submitted in Week 3.10.

Grade calculation:
Mandatory attendance in tutorials and short formative assignments (quizzes and reflective tutorial posts),
Exam (multiple choice and/or short open questions),
Final paper.
Grades will be calculated as follows: exam (60%), final paper (40%).

* The course design might undergo slight changes in view of unforeseen events and circumstances, e.g. the measures resulting from COVID-19.

	Study materials: Readings (available on Brightspace). Become available during the course. Lecture slides, available on Brightspace after the lecture. Quizzes, available before the lectures . Suggested literature: Asaro, P. M. (2006). What should we want from a robot ethic. International Review of Information Ethics, 6(12), 9-16. Van de Poel, I. (2013). Why new technologies should be conceived as social experiments. Ethics, Policy & Environment, 16(3), 352-355. February 2000 Johnson, D. G. (2015). Technology with no human responsibility?. Journal of Business Ethics, 127(4), 707-715. Kudina, O., & Verbeek, P. P. (2019). Ethics from within: Google Glass, the Collingridge dilemma, and the mediated value of privacy. Science, Technology, & Human Values, 44(2), 291-314. van Wynsberghe, A., & Robbins, S. (2019). Critiquing the reasons for making artificial moral agents. Science and engineering ethics, 25(3), 719-735.
Assessment	Exam (60%), final paper (40%) * The course design might undergo slight changes in view of unforeseen events and circumstances, e.g. the measures resulting from COVID-19.
Tags	Artificial intelligence Ethics Philosophy
Department	3mE Department Cognitive Robotics

TPM013B	**Environmental Ethics**	**5**
Module Manager	A.R. Gammon	
Responsible for assignments	A.R. Gammon	
Co-responsible for assignments	Dr. J.B. van Grunsven	
Contact Hours / Week x/x/x/x	0/0/0/2 2 tutorial sessions in week 4.4 & 4.6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	

TPM014B	Ethics of AI	5
Module Manager	Dr. J.M. Duran	
Co-responsible for assignments	Dr. J.B. van Grunsven	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course aims to learn to argue about moral, social, and epistemic values that students involved with AI (computer scientists, engineers) will encounter in their professional practice. The course focuses on contemporary pressing topics such as: 1. Introduction to theories of normative ethics: utilitarianism, deontology, virtue ethics. The distinction between descriptive and normative claim, epistemology and ethics, values, norms, and virtues. 2. Ethical and epistemological issues in AI: identification of six different sources of ethical and epistemological concerns in AI. 3. Bias in AI: Automated-decision making systems are polluted with different forms of bias that can be found at a different levels. Here we address the different forms of bias and what can be done about it. 4. Explanatory AI and the right to explanation: The GDPR grants the so-called "right to an explanation." But what does this means, and how far have we got into explanatory AI? In this unit, we explore the scope and limits of current approaches to explainable AI and how this affects users' rights 5. Privacy: there are different ways of studying and understanding privacy. In addition, this unit discusses the value of privacy in AI in comparison with other values, such as explainability, accuracy, bias, and fairness. 6. Trustworthy AI: What does it mean to trust an AI system? is there more than one way to trustworthy AI? can we design trust? In this unit, we will address questions such as these. We will also present and discuss the two major approaches to trustworthy AI: transparency and computational reliabilism. Lights and shadows of both. Students taking the course will develop the following academic skills: reading philosophical texts, critical analysis, oral presentation and discussion, and essay writing.	
Study Goals	The learning objectives of the course are: 1. Students will recognize and dissect current epistemic, normative, and societal issues emerging in the context of the design and use of AI. 2. Students can identify strengths and weaknesses in epistemological and normative claims about AI. 3. Students are able to critically assess ethical, societal, and epistemological arguments related to AI 4. Students are able to recognise modes of reasoning in ethics.	
Education Method	The course will have lectures and tutorials. Students will be involved in individual as well as group activities. The methodology used is philosophical analysis, reflection, and discussion.	
Assessment	Assignments in tutorials (ca. 20% of the final grade), a final exam (ca. 40% of the final grade), and a final paper (ca. 40% of the final grade). In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be a timed take-home exam (open book).	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Tags	Artificial intelligence Ethics Philosophy Technology	
Category	MSc level	

TPM015B	Ethics of AI	3
Module Manager	Dr. J.M. Duran	
Co-responsible for assignments	Dr. J.B. van Grunsven	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course aims to learn to argue about moral, social, and epistemic values that students involved with AI (computer scientists, engineers) will encounter in their professional practice. The course focuses on contemporary pressing topics such as: 1. Introduction to theories of normative ethics: utilitarianism, deontology, virtue ethics. The distinction between descriptive and normative claim, epistemology and ethics, values, norms, and virtues. 2. Ethical and epistemological issues in AI: identification of six different sources of ethical and epistemological concerns in AI. 3. Bias in AI: Automated-decision making systems are polluted with different forms of bias that can be found at a different levels. Here we address the different forms of bias and what can be done about it. 4. Explanatory AI and the right to explanation: The GDPR grants the so-called "right to an explanation." But what does this means, and how far have we got into explanatory AI? In this unit, we explore the scope and limits of current approaches to explainable AI and how this affects users' rights 5. Privacy: there are different ways of studying and understanding privacy. In addition, this unit discusses the value of privacy in AI in comparison with other values, such as explainability, accuracy, bias, and fairness. 6. Trustworthy AI: What does it mean to trust an AI system? is there more than one way to trustworthy AI? can we design trust? In this unit, we will address questions such as these. We will also present and discuss the two major approaches to trustworthy AI: transparency and computational reliabilism. Lights and shadows of both. Students taking the course will develop the following academic skills: reading philosophical texts, critical analysis, and oral presentation and discussion.	
Study Goals	The learning objectives of the course are: 1. Students will recognize and dissect current epistemic, normative, and societal issues emerging in the context of the design and use of AI. 2. Students can identify strengths and weaknesses in epistemological and normative claims about AI. 3. Students are able to critically assess ethical, societal, and epistemological arguments related to AI 4. Students are able to recognise modes of reasoning in ethics.	
Education Method	The course will have lectures and tutorials. Students will be involved in individual as well as group activities. The methodology used is philosophical analysis, reflection, and discussion.	
Assessment	Assignments in tutorials (ca. 35% of the final grade), and a final exam (ca. 65% of the final grade). In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be a timed take-home exam (open book).	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Tags	Artificial intelligence Ethics Philosophy Technology	
Category	MSc level	

TPM016A	Robots and Society	3
Module Manager	O. Kudina	
Instructor	T.N. Coggins	
Instructor	O. Kudina	
Instructor	M.J. Ley	
Co-responsible for assignments	Dr. J.B. van Grunsven	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	

Course Contents

Introduction

Robots and Society takes a critical look at the design, development, production, and implementation of robots and AI used in society. The class will build upon traditional robot ethics and ethics of technology concepts and approaches applied specifically to current technical developments in the fields of robotics and AI. The course will provide students with the skills of critical reflection, ethical sensitivity, and deliberation about the ethical, societal, design, and governance issues at stake concerning these technologies. Thus, the course should prepare students for common problems they will face in their current studies and prepare them for their future careers in the technical or humanities disciplines. The course serves as a first step in: uncovering societal issues related to robotics and AI, understanding how and why they are points of concern, and applying conceptual tools to mitigate and/or overcome such issues.
Given the broad background of the students who will participate in this course, abstract formulations will be avoided. Moreover, many of the topics will be presented as introductions, creating awareness among the students.

Detailed content:

Introduction to ethics of technology
- Technology as a socio-technical system
- Practice-based approach to values
- Technological mediation of morality
- Technology as a Social Experiment
- Value Change

Introduction to robot ethics
- Ethics of Human-Robot Interaction (HRI) (e.g. social script, anthropomorphism, gender, inclusivity, etc.)
- Ethics of Robot Applications (e.g. healthcare robots, Digital Voice Assistants, military drones, etc.)
- Ethics of Robot Design and Development (e.g. Design for Values, Value hierarchy, Ethics as an accompaniment model, etc.)

Introduction to ethics of AI
- AI as a Black box
- Bias in training data
- Data and transparency
- Power and politics of AI
- Ethics of AI in public sectors (e.g. predictive policing, facial recognition, Covid19 tracking apps, etc.)

Study Goals

Learning goals:

After completing the course, the students:
- are aware of the range of views on the relationship between robots and AI and society, from technology being neutral to technology being value laden to the technology as a socio-technical system,
- are capable of detecting when certain assumptions are appealed to in a debate about the ethics of technology, and robots and AI in society specifically,
- have awareness for the role of these concepts and approaches to thinking in the current debate surrounding robotics and AI,
- are capable of formulating a position on the ethical issues at stake in current applications of robotics and AI.

Education Method

Lectures (weekly, on campus, seven sessions) and tutorials (on campus, three sessions), weekly quizzes based on the readings, an exam (on-campus, computer room, open book).

* The course design might undergo slight changes in view of unforeseen events and circumstances, e.g. the measures resulting from COVID-19.

Literature and Study Materials

Study materials:
- Readings (available on Brightspace). Become available during the course.
- Lecture slides, available on Brightspace after the lecture.
- Quizzes, available on Brightspace.

Suggested literature:
- Nyholm, S. (2020). Humans and robots: Ethics, agency, and anthropomorphism. Rowman & Littlefield Publishers.
- Asaro, P. M. (2006). What should we want from a robot ethic. International Review of Information Ethics, 6(12), 9-16.
- Hagendorff, T. (2020). The ethics of AI Ethics: An evaluation of guidelines. Minds & Machines 30, 99120
- Van de Poel, I. (2013). Why new technologies should be conceived as social experiments. Ethics, Policy & Environment, 16(3), 352-355.
- Kudina, O., & Verbeek, P. P. (2019). Ethics from within: Google Glass, the Collingridge dilemma, and the mediated value of privacy. Science, Technology, & Human Values, 44(2), 291-314.
- van Wynsberghe, A., & Robbins, S. (2019). Critiquing the reasons for making artificial moral agents. Science and engineering ethics, 25(3), 719-735.

Assessment

To pass the course, the students need to complete the following:
- Mandatory attendance in tutorials (on campus, three times over the course)
- Short formative assignments (quizzes, assignments during the tutorials),
- Exam (on campus, computer room, open book, multiple choice and/or short open questions).

Grades will be calculated as follows: exam (100%).

Weekly quizzes:
- Weekly online quizzes will assess the students' ability to understand the reading and lecture material from one week to the next

	and will prepare them for the exam. Tutorials: During each tutorial, students will complete a reflective exercise (e.g. a debate) and produce an analytic piece (e.g. a presentation) on a specific case of implementing robotic systems in society. Exam (on-campus, computer room, open book): Throughout the exam, the students will be asked to identify, define and illustrate concepts used in the course, to make arguments using ethical principles, theories or frameworks they will learn about in the course, and to defend or critique claims made about robotic and AI-based technologies discussed in the course. * The course design might undergo slight changes in view of unforeseen events and circumstances, e.g. the measures resulting from COVID-19.
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/
Elective	Yes
Tags	Artificial intelligence Ethics Philosophy
Category	MSc level

TPM024A	**Methods for Risk Analysis and Management**	**5**
Module Manager	Dr. M. Yang	
Instructor	Dr. M. Yang	
Responsible for assignments	Prof.dr.ir. G.L.L.M.E. Reniers	
Co-responsible for assignments	Prof.dr.ir. P.H.A.J.M. van Gelder	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course consists of three blocks following a risk analytical approach: (A) introduction, (B) analysis, (C) assessment, treatment and follow-up. In block A Students will get familiar with the concepts and definition used in safety science and the basic steps of a risk analysis. In block B the course will focus on identification of hazards and qualitative/semi quantitative risk analysis. A selection of methods is explored to get acquainted with its application and limitations. For instance: Fault tree analysis, Event tree analysis, Bowtie diagram, Reliability Block Diagram, Markov Chain Analysis, Bayesian networks, Decision making under uncertainty. In block C the assessment of the outcomes of a risk analysis are put central. This includes acceptance, stakeholder perspectives, decision making and monitoring.	
Study Goals	After taking this course the student will have knowledge about: * the basic concepts of safety science and risk analysis * risk acceptance and decision making about risks After following this course, the student will be able to: * apply structured methods for risk identification, risk assessment, risk evaluation, and risk treatment.	
Education Method	The following methods may be used: Lectures, exercises and workgroup assignments. Students are expected to form groups to work on the projects applying risk-based methods in design or operation of socio-technical systems.	
Literature and Study Materials	The following textbook will be partially used: Engineering Risk Management (Meyer and Reniers, 2016; De Gruyter Publishing) Other required and additional material will be distributed via Brightspace	
Assessment	The primary assessment item of the course is a group project with an expected group size of 5-6 (depends on the class size). Assessment will be carried out on basis of groups at different stages of the project period. It will consist of three assignments, one presentation, and one final report. In total, there are 5 summative assessments. The grade distribution is: assignment 1 (proposal) - 20%, assignment 2 (methodology) -20%, assignment 3 (case study and results) - 20%, assignment 4 (oral presentation) - 20%, and assignment 5 (final report) - 20%. The minimum grade for each assessment item is 5,0. Feedback on the graded assignments can be reviewed by students upon appointment with the teach staff of this course within a week after the release of grades. The students are allowed one re-sit per assignment. It is not allowed to request a re-sit for an assignment for which students have received a pass (6,0 or higher). The re-sit format needs to be discussed with the teaching staff in line with examination regulations. In case the student is granted an extra re-sit by the Board of Examiners, this re-sit must be taken within the current study year.	
Elective	Yes	
Category	MSc level	

TPM403SET	Technology Entrepreneurship and Sustainability	4
Module Manager	H. Khodaei	
Instructor	F. Delgado Medina	
Instructor	H. Khodaei	
Instructor	Dr.ing. V.E. Scholten	
Responsible for assignments	H. Khodaei	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	No specific knowledge is required prior to entering the course.	
Course Contents	This course is aimed at high-tech entrepreneurs who have great ideas which they like to implement in established organizations, through intrapreneurship or corporate venturing, or use to start and grow themselves new companies. The course is specifically focused on the business opportunities related to new energy technologies. New, emerging, sometimes radically different energy technologies provide significant opportunities for alternative products and services, new business models and new ways of working for more efficient approaches to resource exploitation and energy production and consumption. Yet to harness these opportunities, it often requires to abandon existing paradigms and search in landscapes more distant from current ones. Entrepreneurial activity is a key driver here and the need to develop the skills to develop and act on innovative business opportunities is increasingly vital. By using evidence-based content, methods, and models this course guides you in the analyses of new venture opportunities by assessing and investigating the commercial potential of implementing new technologies either from a social challenge or technology perspective. Opportunity recognition, product-service innovation, sustainable business modelling, and new venture growth dynamics in the context of networks of established firms and the wider stakeholders are core elements of this course. This course is the third course in the SET Economics & Society profile.	
Study Goals	Our goal is to demystify the new venture process, and to help you build the skills to identify and act on innovative opportunities now, and in the future. Upon finishing the course you will be able to: 1. understand how entrepreneurial ideas in the energy sector are shaped and what influences the development of them into viable sustainable business opportunities in the context of both start-ups and corporate venturing; 2. understand the role of the innovation process, how it depends on the technology and institutional context and be able to apply the generic pattern of development and diffusion for radically new high-tech products or systems; 3. understand the principles of market segmentation, and apply different niche-strategies to commercialize radical new high-tech products; 4. understand and apply various methods for developing sustainable/ circular business models for the creation of successful new ventures or innovate existing business models and transform them into desired ones; 5. Understand different circular economy barriers and strategies, and apply different circular strategies to make business models more circular; 6. understand the growth stages of new ventures and the role of network-based facilitators, the networks of established firms in the energy sector and the role of stakeholder management.	
Education Method	The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learnings from each learning block.	
Literature and Study Materials	Trott, P., Hartmann, D., Scholten, V., van der Duin, P., & Ortt, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA Journal articles: Will be posted on Brightspace-site. Slides presented in class: Will be posted on Brightspace-site.	
Assessment	The assessment is composed of several elements: 1. Active participation during the lectures and learning block assignments. You will be assessed based on the active participation, completing assigned readings, and on time assignment submission of each learning blocks. This will account for the 30% of the final grade. 2. Group report and final presentation. Proposed group assignment. This consists of two parts. The first part is about analysing the business model of a start-up, identify how the business model works and reflect on it by the literature and come with suggestion to improve or adapt it to new technological opportunities. Then the second part is to identify an established firm in the energy sector that is related to the field of the start-up and analyse for this established firm what the corporate venturing activities are that they have to support new innovations either through internal venturing or by acquiring external ventures. Then analyse how the start-up could contribute to the large firm in terms of making strategic or operational transitions into new energy systems and vice versa what the benefits are for the start-up to become a venture within the established firm. Students will present and submit a final report and will be judged on their ability to understand both theoretical and practical aspects of bringing a technology to the market, and to analyse energy-related business challenges and opportunities in the field of entrepreneurship and innovation. The group grade will be adjusted for individual participation. This will account for the 40% of the final grade. 3. Individual: Theory quizzes (30%) All marks will be expressed on a scale of 1-10.	
Enrolment / Application	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Targetgroup	Master students from all master programs can apply.	

TPM404B	Technology Entrepreneurship and Internationalization	5
Module Manager	H. Khodaei	
Instructor	F. Delgado Medina	
Instructor	H. Khodaei	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course offers theoretical fundamentals of international entrepreneurship. You create a case study of an existing start-up (born global) analysing its entrepreneurial journey and how it fits the broad technology development process. This course helps you prepare for a wide array of career paths, from starting or joining an international start-up to academic research on the topics of International entrepreneurship. Throughout the course you will develop the skills to analyse the emergence of innovative business opportunities and how these are exploited. Using proven content, methods, and models for new venture opportunity assessment and analysis, you will learn how to analyze the commercial potential of implementing new technologies either by analyzing social challenges or analyzing the value of technology for various user applications. Innovation, opportunity recognition, strategic decision making, business modelling, internationalization marketing and growth stages of high-tech start-ups are core elements of the course.	
Study Goals	Upon finishing the course you will be able to: LO 1 Analyse and apply the innovation process within startup firms and compare it to established firms. LO 2 Describe and apply opportunity identification process based on opportunity analysis canvas. LO 3 Identify and analysis international opportunities across different markets. LO 4 Describe and apply the born-global start-up growth process. LO 5 Demonstrate understanding of the entrepreneurial ecosystem, the main stakeholders and their role in the start-up process. LO 6 Explain and apply methods for developing business models for high-tech start-up and analyse how they can change over time. LO 7 Describe and analysis the barriers to internationalization and evaluate the niche strategies. LO 8 Analyse an existing born-global start-up and present your findings about the dynamics of its growth.	
Education Method	The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learning from each learning block. A final presentation and final report based on a startup-case analyses provided by a team of students will close the course. In case of unforeseen circumstances or measures resulting from COVID-19, the presentations will be facilitated in an online mode (for example through zoom or Skype). Book: Trott, P., Hartmann, D., Scholten, V., van der Duin, P., & Ortt, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA Journal articles: Will be posted in Brightspace Slides presented in class: Will be posted in Brightspace	
Assessment	The assessment is as follows: - Team: Case analyses (30%) and a final report and presentation (40%) - Individual: theory quizzes (30%) All marks will be expressed on a scale of 1-10. In case of unforeseen circumstances or measures resulting from COVID-19, the presentations and individual assessment will be facilitated in an online mode (for example through zoom or Skype)	
Elective	Yes	

TPM416A	Turning Technology into Business	6
Module Manager	Dr. L. Hartmann	
Responsible for assignments	Dr. L. Hartmann	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Turning Technology into Business (TTiB) aims to equip participants with a strong conceptual foundation to actively understand the dynamic process of technology-based entrepreneurship. Participants learn how business strategies are best formulated and how (through entrepreneurship) technology can create value. TTiB consists of seven lectures, introducing the theoretical backgrounds of technological, market and business analyses. A unique aspect of TTiB is that existing technologies (developed and patented by the TU Delft) are used as case subjects. The patent project is the focal point of the curriculum. Each group of 4-5 students will be assigned an original patent, and is expected to evaluate the commercial potential of this technology. This includes choosing promising applications (products) for the technology, and making recommendations for the most suitable business model to commercialize the technology. The patent project should provide a coherent and structured answer to the central question: which strategy is most likely to generate business from this patent? In addition to equipping participants with a thorough knowledge of the course subject, the section Technology, Strategy & Entrepreneurship (TSE) hopes that this course will be the starting point for a variety of university spin-offs in which the students will participate. To accommodate this, TSE offers the course TPM420A- Ready to Startup!	
Study Goals	This course aims to equip students with a strong conceptual foundation to an active understanding of two domains: (1) The dynamic process of technological innovation through concepts such as technology life-cycles, dominant design, disruptive technologies, Schumpeterian competition and the diffusion of innovations (2) How business strategies are formulated and, through entrepreneurship, technology can create value. This multi-faceted process of technology commercialization process is addressed in terms of assessing technology position, discovering market opportunities, competitive analysis, appropriability and the various modes of entrepreneurship.	
Education Method	7 interactive lectures, participant-centered case studies, homework assignments, classroom assignments, individual group coaching. Final Presentations in February.	
Literature and Study Materials	Will be provided during the course	
Assessment	Each group writes a report on their findings, analysis and recommendations for the technology (patent) they used as a case for this course. On the Final Day, the group has to give a 10 minute oral presentation. The report makes up 80% of the final grade. This team grade will be modified according to the individual student's class contributions and his/her performance within the group (peer review).	
Enrolment / Application	MAXIMUM CAPACITY is limited to 75 participants. Register on BrightSpace from October 1. Because of the huge popularity of this course, there is a selection procedure which consists of pre-course assignments. One of these assignments is a letter of motivation: why do you want to participate in this course? Based on the motivation and the other pre-course assignments, the final selection of 75 participants will be made.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Targetgroup	Advanced Master students, PDEng and PhD students, and staff researchers from all faculties within the TU Delft.	
Category	MSc level	

TPM420A	Ready to Start-up	6
Module Manager	Ir. B.R. Joseph	
Module Manager	Dr.ing. V.E. Scholten	
Responsible for assignments	Dr.ing. V.E. Scholten	
Co-responsible for assignments	Dr. L. Hartmann	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Summary	This course is offered twice in a year; First starting and ending in Q2 and second starting mid Q3 until end Q4 Ready to Startup (RtS) is an elective master course that has been the crown jewel of entrepreneurial education at TU Delft for over two decades. Bringing together business experts and students eager to start companies, it has one simple goal: turn engineers into successful founders.	
Course Contents	RtS focuses predominantly on Business Strategy: gaining market traction, prototyping, growing the team, devising a financial plan and fundraising strategy, protecting IP, and foundational activities (setting up a company, sharing ownership, etc.) The course has the prerequisite that teams should be formed and have business ideas that are analysed and validated. It is meant for students who plan to start a company. After RtS, teams must be at the level required for YES!Delft incubation and Delft Enterprises investments.	
Study Goals	At the end of the course, you will be able to - Reflect on your current business model and identify further necessary validation activities. Analyse market entry strategies and consider alternative models. - Learn how to negotiate, attract first users, and seek endorsement of stakeholders (such as investors and advisors). - Plan your core operational activities such as company foundation, IP protection, product development strategy and team development - Devise financial plans and appropriate fundraising strategies - Convey a convincing narrative of your business and be able to provide constructive feedback to other startup plans. - Demonstrate effective entrepreneurial behaviour by making swift decisions based on newly gathered information and adjusting your model accordingly.	
Education Method	The course is spread out over 8 sessions (online in case of persistent Covid-19 restrictions for on campus education) in the afternoon (13h30-17h30) where the core elements of a startup plan are a presented. You work on assignments that are aimed at specific elements of the core startup plan, and you will by apply them to your startup project and present your findings during the course. In this way you make evident progress with your startup. The course combines state of the art business methods, translated to the needs of starting entrepreneurs, with consults from practictioners. Topics are discussed using case studies. You will present your progress, provide peer-feedback as well as receive advice from mentor experts. The course ends with a jury pitch and an executive summary.	
Literature and Study Materials	Handouts and literature are available trough in Brightspace.	
Prerequisites	Enthusiasm, entrepreneurial spirit and a good business idea.	
Assessment	The course has 3 grading components that are Covid-19 proof. - Individual: Active online participation and constructive peer review feedback on other business plans (30%) - Team: class-based assignments (20%) - Team: final startup plan report (40%), supported by a jury pitch (10%)	
Enrolment / Application	Application process you need to register in advance to the course. Teams of students with clear ideas about a business they want to start supported with evidence that it addresses a need, will have priority to enter. All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Elective	Yes	
Targetgroup	Master students from all master programs can apply.	
Category	MSc level	

WM0320TU	Ethics and Engineering	3
Module Manager	Dr. F. Santoni De Sio	
Instructor	M. Sand	
Co-responsible for assignments	M. Sand	
Contact Hours / Week x/x/x/x	4/0/4/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	This course is identical to the initial part of the course WM0329TU. You will explore the ethical and social aspects and problems related to technology and to your future work as professional or manager in the design, development, management or control of technology. You will be introduced to and make exercises with a range of relevant aspects and concepts, including professional codes, philosophical ethics, individual and collective responsibilities, value pluralism and relativism, responsibility within organisations, responsible conduct of companies and the role of law. You will analyse legal, political and organisational backgrounds to existing and emerging ethical and social problems of technology, and you will explore possibilities for resolving, diminishing or preventing these problems.	
Study Goals	After having completed the course you: can better recognise and analyse ethical and social aspects and problems inherent in technology and in the work of professionals and managers active in the design, development, management and control of technology. have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds. are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to technology and the work of professionals and managers. are better prepared to perform your future work as a professional or manager in the design, development, production and control of technology in an ethical and socially responsible way.	
Education Method	A series of 6 lectures and interactive work sessions with partial assessment, concluded with a written test.	
Literature and Study Materials	Chapters from the Reader Ethics and Engineering, available at Nextprint and as PDF files on Brightspace; articles available as PDF on Brightspace; Powerpoint slides, materials used in the working groups, and lecture notes.	
Assessment	Written exam (60%), presentation and active participation during the working group sessions (40%). In order to pass the course, a minimal grade of 5.5 should be obtained *both* in the exam *and* in the working group, and the average grade should be higher than 5.75. It is possible to re-take the exam without re-taking the working groups, in this case the working group grade will be kept. If you re-take the exam, it is also possible to re-attend the working groups to improve the final grade.	
Enrolment / Application	Enrolment via Brightspace is required for this course. This is needed in order to plan the workgroups. Please enroll as soon as possible and check Brightspace for more information on the enrolment on the enrolment procedure and timeline. The Brightspace page will be activated a couple of weeks before the start of the course.	
Remarks	The course is run twice each year in the first and third quarter. The course is identical to the initial part of the course wm0329tu (6 ects).	
Elective	Yes	
Category	MSc niveau	

WM0329TU	Ethics and Engineering	6
Module Manager	Dr. F. Santoni De Sio	
Instructor	M. Sand	
Co-responsible for assignments	M. Sand	
Contact Hours / Week x/x/x/x	4/0/0/0 or 0/0/4/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	The first part of this course completely coincides with the WM320TU course. The WM329TU course is the WM320TU course plus the writing of an essay. Please check the WM320TU Brightspace for all relevant information about lectures, work sessions, exam, and the essay. Please note that the essay should be written in the same quarter in which you will attend the WM0320TU course. You will explore the ethical and social aspects and problems related to technology and to your future work as professional or manager in the design, development, management or control of technology. You will be introduced to and make exercises with a range of relevant aspects and concepts, including professional codes, philosophical ethics, individual and collective responsibilities, value pluralism and relativism, responsibility within organisations, responsible conduct of companies and the role of law. You will analyse legal, political and organisational backgrounds to existing and emerging ethical and social problems of technology, and you will explore possibilities for resolving, diminishing or preventing these problems.	
Study Goals	After having completed the course you: can better recognise and analyse ethical and social aspects and problems inherent in technology and in the work of professionals and managers active in the design, development, management and control of technology. have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds. are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to technology and the work of professionals and managers. are able to write an opinion article on ethical or societal aspects of technology, according to academic standards of writing are better prepared to perform your future work as a professional or manager in the design, development, production and control of technology in an ethical and socially responsible way.	
Education Method	A series of 6 lectures and interactive work sessions (including role playing sessions) concluded with a written test.	
Literature and Study Materials	Chapters from the Reader Ethics and Engineering, available at Nextprint and as PDF files on Brightspace; articles available as PDF on Brightspace; Powerpoint slides, materials used in the working groups, and lecture notes.	
Assessment	50%: grade for the participation in the WM320TU course, which consists of a written exam (60%), presentation, and active participation during the working group sessions (40%). 50%: grade of the essay	
Enrolment / Application	Enrolment via Brightspace is required for this course. This is needed in order to plan the number of workgroups. Please enrol as soon as possible and in any case before the start of the course via Brightspace. All relevant information about the enrolment and timeline for the essay will be published on the WM0320TU Brightspace page.	
Remarks	The course is run twice each year, starting in quarters 1 and 3. The first part of this course (concluded with the written test) coincides with the course WM0320TU.	
Elective	Yes	
Category	MSc niveau	

WM0349WB	Philosophy of Engineering Science and Design	3
Module Manager	Dr. J.M. Duran	
Instructor	Dr. J.M. Duran	
Responsible for assignments	Dr. J.M. Duran	
Co-responsible for assignments	Ir. S.J. Zwart	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Course contents: (1) The goals of science; the character and scope of scientific claims. (2) The goals of engineering design; the nature of technical artefacts; the value-neutrality of technology. (3) The scientific method and the validation of scientific claims. (4) Methods of engineering design; the character and scope of design claims; the decision-making aspect of design. (5) The development of science; the objectivity of science; the notion of scientific progress. (6) The development of technology; social determinism and technical determinism. (7) The place and role of values in science and in engineering design.	
Study Goals	This course aims first of all to support students in developing a reflective and critical attitude with regard to empirical research underlying engineering science and engineering design at an academic level. Additionally, and in support of this primary goal, it aims to make students acquainted with views on the nature and goals of science and engineering design and how in these activities facts and values both have their role to play and how they interact.	
Education Method	The course is taught in the form of seven weekly sessions of a plenary lecture of two hours, and five seminars of two hours during which students articulate and discuss their answers to a number of questions in smaller groups. Each seminar will be prepared and chaired by a team of about three students. For this task general and specific instructions, recommendations and suggestions will be available.	
Assessment	Assessment is through (1) a final individual exam, (2) participation in the seminars and (3) performance in preparing and chairing a seminar session. The exam result forms 60 percent of the final grade, assessment of general seminar participation 20 percent and assessment of the preparing and chairing role also 20 percent. In the expectation that no COVID restrictions will be in place while the course is being offered, the exam will be a closed-book on-campus exam, which may be either hand-written or digital (Möbius). Which of the two will be announced at the start of the course.	
Tags	Artificial intelligence Ethics Philosophy Technology	
Category	MSc level	

WM0353TU	Climate Ethics	3
Responsible Instructor	A. Melnyk	
Module Manager	Prof.dr.ir. B. Taebi	
Instructor	A. Melnyk	
Responsible for assignments	A. Melnyk	
Co-responsible for assignments	Prof.dr.ir. B. Taebi	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Climate change is considered to be one of the most urgent problems the world is currently facing. Two types of solutions are proposed namely mitigation and adaptation. Both climate change mitigation and adaptation are heavily relying on the progress of technology. Addressing the climate problem requires, however, more than just developing and applying a certain technology. It demands considerations of whose interests and whose rights are at stake. It also requires reflection on how and "under which condition" technology can change and shape the world for the better. In this course we will focus on the following ethical aspects of climate change: - Past emissions and responsibility to deal with climate change - Implications of global warming on human safety and security - The distribution of burdens and benefits, emission rights and international justice - Future generations and intergenerational justice - New developments after the Paris Agreement The course will be offered in English and it is primarily for MSc students. Enrollment on Bright Space is sufficient.	
Study Goals	- Creating awareness about climate change induced by humans - Insights into ethical aspects of climate change and into implications for law and policy-making - Being able to understand the ethical and social conditions under which technological solutions can be successful - Being able to understand and reflect on the international negotiations and agreements for dealing with climate change	
Education Method	Lectures and discussion sessions. Students are expected to actively participate in the class, which is why attendance is mandatory.	
Literature and Study Materials	We will study different papers and book chapters on the ethics of climate change. The required reading material along with instruction on how to acquire this material will be announced on BB.	
Assessment	Class evaluation (1/5) and written exam (4/5). More instruction regarding the exam will be provided in class.	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/	
Elective	Yes	
Category	MSc level	

WM0375TU	Ethics of Technological Risks	3
Module Manager	Prof.dr. S. Roeser	
Instructor	K. Maheshwari	
Instructor	Prof.dr. S. Roeser	
Responsible for assignments	Prof.dr. S. Roeser	
Co-responsible for assignments	Prof.dr.mr.ir. N. Doorn	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course investigates ethical aspects of technological risks. Debates about technological risks related to, for example, energy technologies, robotics and biotechnology frequently culminate in stalemates. This is due to the complexities and intricacies inherent to such debates as they involve scientific information and uncertainties, as well as ethical and emotional considerations. Conventional, quantitative approaches focus primarily on statistical information to risk. They do not explicitly incorporate important ethical considerations such as justice, fairness and autonomy. Emotions such as compassion, care and feelings of responsibility can draw attention to such ethical aspects of risky technologies. Taking emotions and ethical considerations seriously can lead to more fruitful deliberations between different stakeholders in which relevant concerns are taken seriously and are explicitly reflected upon. This course will study how approaches to ethical aspects of risk can lead to more morally responsible decision making and design of technological innovations.	
Study Goals	This course will provide students with an understanding of approaches to risk ethics and students will get experience with discussing and evaluating ethical aspects of risk	
Education Method	Self-study, lectures and tutorials (in smaller groups, ca 25-35 students, depending on the total number of students). All students will hold one group presentation (ca 5 students) at one of the lectures. Attending lectures and tutorials and giving a presentation is obligatory.	
Assessment	On campus written exam (individual assignment), this will count for 100% of the grade for this course. Students need to have a pass (5.75 or higher) in order to pass the course. There is a resit of the written exam in the 4th quarter (date to be announced). In case of unforeseen circumstances, eg related to COVID-19 measures, the on campus exam may be replaced with an online (eg take home) exam. [There is also a 5 ECTS version of this course, i.e. WM0376TU; the only difference is that for the 5 ECTS version students also write an essay in addition to the exam.]	
Special Information	Contact person for additional questions: S.Roeser@tudelft.nl	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Tags	Ethics Philosophy	
Category	MSc level	

WM0801TU	Introduction to Safety Science	3
Module Manager	Dr. A.P. Afghari	
Instructor	Dr. A.P. Afghari	
Instructor	Prof.dr.ir. P.H.A.J.M. van Gelder	
Co-responsible for assignments	Dr. E. Papadimitriou	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	Students wishing to graduate with a final project in the area of safety supervised by the Safety Science and Security Group should have followed this course or one of the other introductory courses run by the group (WM0808TU, WM0821TU or WM0822TU)	
Expected prior knowledge	Technical academic BSc and common (scientific) sense	
Summary	This course focuses on the development of a safety-oriented pattern of thinking and a holistic approach towards systems safety and security. The tools that will be learned in this course will be helpful in recognizing, understanding, and analyzing hazards; and assessing risk in contemporary complex systems. Areas included are (1) Hazard analysis and safety assessment, (2) Human reliability assessment, and (3) Safety and security management.	
Course Contents	Bayesian inference Consequence assessment Decision making under uncertainty Fault & event trees Human factors in safety science Introduction to system safety engineering Reliability block diagram Risk ranking Safety performance measurement System safety assessment	
Study Goals	At the end of this course, students will be able to: - Explain the principles of safety, security and risk management - Discuss the role of human factors in systems safety - Explain the principles of systems decompositioning and causal analysis of undesired events using fault/event trees and Bayesian networks - Analyse data with quantitative methods for safety assessment and risk estimation - Apply quantitative methods on systems for risk-informed decision making - Perform security risk analysis using game theoretical and Bayesian analysis techniques	
Education Method	Plenary lectures (mandatory) Exercises (carried out during lectures, and individually or in small groups of 4-5 students)	
Literature and Study Materials	Clifton A. Ericson, Hazard Analysis Techniques for System Safety, John Wiley and Sons; ISBN: 978-0471720195. Harold E. Ronald and Brian Moriarty. (1990). System Safety Engineering and Management, John Wiley and Sons, 2nd Edition.	
Prerequisites	Technical background preferred but certainly not mandatory	
Assessment	This course has only one summative assessment in the form of a final exam. There is no partial assessment. In addition, several formative assessments are conducted during the lectures including quizzes and group activities. Formative assessments are not graded. The final exam is individual, computer-based, closed-book and on-campus. This means that students need to be present on campus and sit behind a computer to take the exam. They are NOT allowed to access the teaching materials of the course during the exam, for instance via a USB stick on the desktop computer or via printed material. They are NOT allowed to use the internet during the exam. The minimum grade of the final exam in order to pass the course needs to be 5.75 or higher. The final grade is determined by the grade of the final exam (the final exam counts 100% for the final grade). The final exam is scheduled on Monday 22 January 2024. Students can re-sit/retake the final exam too. The re-sit exam is scheduled on Monday, 11 March 2024. To view the exam after publishing the grades, students must make an appointment with the main teacher of the course and meet (in person) with the teacher at a defined location.	
Enrolment / Application	Enrol through Brightspace. Minor students are automatically enrolled into the course.	
Remarks	This course can be expanded to a System Safety/Reliability project	
Elective	Yes	
Category	MSc level	

WM1301TU	Ethics of Transportation	3
Module Manager	Dr. F. Santoni De Sio	
Co-responsible for assignments	Dr. J.B. van Grunsven	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	In this course you will explore some ethical and societal aspects and problems related to the design, development, management and use of transportation systems, with a particular focus on their digitalisation and robotisation. You will be introduced to a range of relevant aspects and concepts, including Responsible Innovation, moral dilemmas with self-driving cars, philosophical theories of freedom and justice, moral and legal responsibility, privacy, and the future of human employment. You will analyse ethical, legal and social problems of transportation in relation to case studies, and you will explore possibilities for resolving, diminishing or preventing these problems.	
Study Goals	After having completed the course you: can better recognise and analyse ethical and social aspects and problems inherent in (automated) transportation and in the work of professionals and managers active in the design, development, management and control of (automated) transportation systems can see these problems within the broader background of moral, legal and political philosophy have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to (automated) transportation and the work of professionals and managers in this domain. are better prepared to perform your future work as a professional or manager in the design, development, production and control of (transportation) technology in an ethical and socially responsible way	
Education Method	A series of 6 lectures and some working group sessions, focused on the discussion and application of the theories presented in the course	
Assessment	Written exam and assessment participation in working groups	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Category	MSc level	

WM1302TU	Ethics of Transportation	5
Module Manager	Dr. F. Santoni De Sio	
Co-responsible for assignments	Dr. J.B. van Grunsven	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course is identical to the WM1301 course, with the addition of a final essay. In this course you will explore some ethical and societal aspects and problems related to the design, development, management and use of transportation systems, with a particular focus on their digitalisation and robotisation. You will be introduced to a range of relevant aspects and concepts, including Responsible Innovation, moral dilemmas with self-driving cars, philosophical theories of freedom and justice, moral and legal responsibility, privacy, and the future of human employment. You will analyse ethical, legal and social problems of transportation in relation to case studies, and you will explore possibilities for resolving, diminishing or preventing these problems.	
Study Goals	After having completed the course you: can better recognise and analyse ethical and social aspects and problems inherent in (automated) transportation and in the work of professionals and managers active in the design, development, management and control of (automated) transportation systems can see these problems within the broader background of moral and political philosophy have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to (automated) transportation and the work of professionals and managers in this domain. are better prepared to perform your future work as a professional or manager in the design, development, production and control of (transportation) technology in an ethical and socially responsible way can write a short academic essay of critical analysis of an ethical or societal issue related to transportation technology	
Education Method	A series of 6 lectures, active participation in some working groups, and the writing of an essay on ethical, social or legal aspects of technology to be written under the supervision of an instructor from the section Ethics of Technology at TBM	
Assessment	Written exam, assessment participation in working groups and the writing of an essay the ethical, social or legal aspect of technology to be written under the supervision of the course manager or an instructor from the section Ethics of Technology at TBM. In order to pass the course both the exam and the essay should be graded with at least 5.5 (and the average should be sufficient, that is higher than 5.75)	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Category	MSc level	

WM1401TU	Ethics of Healthcare Technologies	3
Module Manager	Dr. S.M. Copeland	
Co-responsible for assignments	Prof.dr. S. Roeser	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introduction Ethics and Healthcare Technologies introduces students to the range of ethical issues that tend to arise during the development, design and application of devices used in healthcare or for medical purposes. The class will gain insight into the different aspects of healthcare ethics, such as the distinction between medical research and practice, the ethics of using human or animal research participants, and the ethics of technology, as well as value-sensitive design and responsible innovation approaches within the context of healthcare technologies. The course will provide students with skills in critical reflection, sensitivity to ethical concerns, and the ability to take up considerations of value to society as well as governance in the development and applications of technology. Assignments and cases examined in class will address contemporary developments in these technologies, the reasons behind new regulations that pertain, as well as giving a historical perspective on the ethical principles that underlie current approaches. Thus, the course should prepare students for common problems they will face in their current studies and prepare them for addressing issues that will arise in their future careers. Topics include: Introduction to ethics in medicine and healthcare: The research-practice and normative-descriptive distinctions, Bioethics, and the ethics of clinical research (historical and present), Medical devices and some relevant regulations & codes. Applying ethical frameworks e.g., utilitarianism, principalism, care ethics; Scientific integrity and the social value of research. Introduction to ethics of technology: Technology as a socio-technical system, Innovation in healthcare, Value-sensitive design, responsible innovation. Specific issues in developing healthcare technologies, such as: wearable and other health tracking devices; transhumanism and augmentation; human and animal research ethics; genetic modification; 3D tissue/organ printing; datamining, imagery and related privacy issues; the role of AI, machine learning and robotics.	
Study Goals	Regarding healthcare ethics & ethics of innovation in healthcare, students can: identify and describe various canonical forms of debate and ethical approaches within healthcare and biomedical ethics, compare diverging viewpoints in the field and assess the strength of an argument made in the field, evaluate different ethical issues as they arise at different points along the development and application trajectory. Regarding ethics and technology, students can: identify and describe normative aspects of technology; analyse and evaluate the ethical and social aspects of a case in contemporary healthcare technology development and application, and make recommendations for value sensitive design.	
Education Method	Contact hours: Plenary sessions that include blended learning activities, 2x Tutorials, Online discussion forums. Study materials: Reader (online) - becomes available during the course; Lecture slides, available on Brightspace after the lecture; Online materials to prepare for in-class activities, including self-directed research.	
Assessment	Grade calculation WM1401 (3ECs): Mandatory attendance in tutorials and short formative assignments, Exam (multiple choice and/or short open questions).	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Category	MSc level	

WM1402TU	Ethics of Healthcare Technologies	5
Module Manager	Dr. S.M. Copeland	
Co-responsible for assignments	Prof.dr. S. Roeser	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introduction Ethics and Healthcare Technologies introduces students to the range of ethical issues that tend to arise during the development, design and application of devices used in healthcare or for medical purposes. The class will gain insight into the different aspects of healthcare ethics, such as the distinction between medical research and practice, the ethics of using human or animal research participants, and the ethics of technology, as well as value-sensitive design and responsible innovation approaches within the context of healthcare technologies. The course will provide students with skills in critical reflection, sensitivity to ethical concerns, and the ability to take up considerations of value to society as well as governance in the development and applications of technology. Assignments and cases examined in class will address contemporary developments in these technologies, the reasons behind new regulations that pertain, as well as giving a historical perspective on the ethical principles that underlie current approaches. Thus, the course should prepare students for common problems they will face in their current studies and prepare them for addressing issues that will arise in their future careers. Topics include: Introduction to ethics in medicine and healthcare: The research-practice and normative-descriptive distinctions, Bioethics, and the ethics of clinical research (historical and present), Medical devices and some relevant regulations & codes. Applying ethical frameworks e.g., utilitarianism, principalism, care ethics; Scientific integrity and the social value of research. Introduction to ethics of technology: Technology as a socio-technical system, Innovation in healthcare, Value-sensitive design, responsible innovation. Specific issues in developing healthcare technologies, such as: wearable and other health tracking devices; transhumanism and augmentation; human and animal research ethics; genetic modification; 3D tissue/organ printing; datamining, imagery and related privacy issues; the role of AI, machine learning and robotics.	
Study Goals	Regarding healthcare ethics & ethics of innovation in healthcare, students can: identify and describe various canonical forms of debate and ethical approaches within healthcare and biomedical ethics, compare diverging viewpoints in the field and assess the strength of an argument made in the field, evaluate different ethical issues as they arise at different points along the development and application trajectory. Regarding ethics and technology, students can: identify and describe normative aspects of technology; analyse and evaluate the ethical and social aspects of a case in contemporary healthcare technology development and application, and make recommendations for value sensitive design. Regarding skills, students will have: developed skills in finding and interpreting relevant policy, regulations and codes of ethics for the field; formulated, debated and defended a substantiated position on the ethical issues at stake in current applications of technology in health care.	
Education Method	Contact hours: Plenary sessions that include blended learning activities, 2x Tutorials, Online discussion forums. Study materials: Reader (online) - becomes available during the course; Lecture slides, available on Brightspace after the lecture; Online materials to prepare for in-class activities, including self-directed research.	
Assessment	Grade calculation WM1402 (5ECs): Mandatory attendance in tutorials and short formative assignments, Exam (multiple choice and/or short open questions), Final paper or project (group).	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Category	MSc level	

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Mechanical Engineering**

ME-BMD Track A. Biomechanical Design	
Program Coordinator	Ir. Bob van Vliet E Coordinator-BMD-3mE@tudelft.nl
Introduction 1	Biomechanical Design (BMD) is one of six tracks within MSc Mechanical Engineering. It is rooted in the Department of Biomechanical Engineering (BMechE), where students receive advanced education in the design and engineering of robotic devices, mechatronic design, control engineering, and biological principles. Biomechanical systems are technical systems designed to interact with biological systems, or designed following the principles of biological systems. Examples are telemanipulation systems, such as those used in surgical robots, in space and in the off-shore industry, where a human controls a master robot whilst a slave robot mimics these actions. Other examples of biologically inspired design are endoscopes with the flexibility and steerability of an octopus tentacle, humanoid robots walking as humans do and intelligently collaborating robots using local interactive information exchange, as humans do.
Exit Qualifications	The exit qualifications for the master Mechanical Engineering can be found in the common section of this studyguide. For the track BMD, the specific exit qualifications are: 1. Competent in the scientific discipline Mechanical Engineering Biomechanical Design. A graduate in Mechanical Engineering Biomechanical Design is able to: a) apply advanced physics and measurement methods in mechanical and biological systems; b) optimise the interaction between biological and mechanical systems and to think of innovative technical solutions; c) analyse the motions of linked rigid body systems in two and three dimensions including systems with various kinematic constraints; d) use multibody dynamics models, appreciate the limitations, and draw sensible conclusions about the modelled system; e) reproduce important concepts of human perception, cognition, and action. 2. Competent in doing research. A graduate in Mechanical Engineering Biomechanical Design is able to: a) explain the interaction between humans and machines, ranging from manual to supervisory control; b) apply existing techniques to measure and model human behaviour when interacting with machines; c) explain the advantages and disadvantages of automation and the effects of automation on humans; d) explain how human skills develop and how feedback influences skill acquisition. 3. Competent in designing. A graduate in Mechanical Engineering Biomechanical Design is able to: a) design autonomous operating systems, capable of human-like actions and interaction; b) apply unconventional biological approaches in engineered systems. 4. A scientific approach. A graduate in Mechanical Engineering is able to: a) carry out human-subject research in an ethical manner; b) apply methods for conducting research or design projects, in particular related to human behaviour, interaction between humans and their technical environment, and intelligent machines; c) solve multidisciplinary problems between engineering and human factors science. 5. Considering the temporal and social context. A graduate in Mechanical Engineering Biomechanical Design is able to a) explain the historic development of the discipline, its technological and scientific boundaries, and the necessity of life-long learning to maintain the desired level.

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Mechanical Engineering**

ME-BMD Obligatory Courses (27 ECTS)

BM41040	Neuromechanics & Motor Control	5
Responsible Instructor	Dr.ir. W. Mugge	
Instructor	Prof.dr. F.C.T. van der Helm	
Instructor	Prof.dr.ir. A.C. Schouten	
Instructor	Prof.dr. H.E.J. Veeger	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Relevant TU Delft BSc Courses: -Wiskunde 2 Lineaire Algebra WBMT1051 -Wiskunde 3 Differentiaal Vergelijkingen WBMT2048 T2 -Wiskunde 4 Kansrekening & Statistiek and Numerieke Wiskunde WBMT2049 T1&T2 -Rigid Body Dynamics WB2630 -Signaalanalyse WB2235 -Systeem- en Regeltechniek WB3240	
Course Contents	Humans have an extraordinary flexibility in generating movements. With little attention most of us can perform complex tasks such as walking or riding a bicycle. The central nervous system integrates available sensory information and select the proper motor command for the muscles. The skeletal system enables the transfer of forces required for body movement while the muscles produce the forces to accelerate it. Stability is maintained through both passive (intrinsic muscle properties) and active (reflexes) contributions to the visco-elasticity of the joints. In this course we investigate how humans generate movements, stabilize posture, and integrate sensory feedback into control actions, while acknowledging the kinematics and the mechanical properties of the human motor system. This course addresses fundamental aspects and clinical applications. During the course the role of subsystems (e.g. muscles, sensory systems) and their interactions as well as uncertainty in sensory signals are discussed with a focus on reaching movements and posture maintenance. During the course it will be shown how this knowledge can help in understanding impaired motor function after neurological disorders, like Stroke and Parkinsons disease.	
Study Goals	After this course the student must be able to: - Describe the motion of bony segments of the human skeleton system with respect to each other using kinematic analysis methods. - Calculate muscle forces from known joint motion using different optimization techniques (forward dynamics and inverse optimization). - Describe the physiology of the human motor and sensory systems as well as the central nervous system with respect to motor control. - Build and use neuromusculoskeletal models to simulate movements using motion equations. - Explain the neuromuscular system from a control engineering perspective, including the interaction between intrinsic and reflexive feedback. - Describe how the central nervous system deals with uncertainty, integrates sensory feedback and adapts to new situations. - Is able to deduce how impairments of different processes involved in motor control due to neuromuscular disorders affect motor control and adaptation. - Is able to critically judge the value of theories about postural and movement control posed in scientific literature.	
Education Method	Lectures and practicals (4 hours per week) and homework assignments	
Assessment	Written/Digital exam, computer tests (digital midterm) and assignments	
Tags	Lab Research Matlab Mechanics Mechatronics Modelling Optimalisation Signals and Systems Technology	
Department	3mE Department Biomechanical Engineering	
Contact	Preferably by email to the Responsible Instructor	

ME41006	Musculoskeletal Modeling and Simulation	4
Responsible Instructor	A. Seth	
Instructor	Ir. J.D.C. Groenhuis	
Contact Hours / Week x/x/x/x	0/0/0/3	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Human and animal movement is often smooth, powerful, and graceful. Animal movement is generated through a cascade of nervous, muscular, and skeletal system dynamics. Even the simple tasks of walking and running require the coordination of hundreds of muscles that pull on bones that move our limbs and generate forces on the environment so we can move. When these systems are disrupted through disease and injury, they result in movement disorders that can be debilitating. In this course you will learn to construct and simulate musculoskeletal models that generate movement. You will learn how to apply models to improve measurements of movement and to predict behavior, thus extracting more insights about human (and animal) movement than from observations alone. We will start with the constituent components of bones, joints, muscles, and tendons and their dynamics, as the building blocks of musculoskeletal systems. You will learn inverse techniques to extract unmeasured quantities, like joint kinematics, work, and power from observations and to use forward dynamic simulations to predict behaviors and answer what if? type questions. You will apply these models and techniques to answer a research question of your own formulation	
Study Goals	The goal of this course is to teach you how to use state-of-the-art computational tools to model and simulate the human musculoskeletal system. Learning will be guided by lectures, reading articles, practical labs and a final project. Upon successfully completing this course, you will be able to: 1. Model Develop and analyze mathematical models of the human musculoskeletal system using OpenSim and MATLAB. 2. Simulate Create inverse and forward dynamic simulations of human movement. Perform inverse dynamic simulations and optimizations using experimental data to compute quantities that cannot be easily measured (e.g., muscle forces). Perform forward dynamic simulations to reproduce observed movements and predict new movements. 3. Analyze Use musculoskeletal simulation to answer questions about human movement (e.g., how does a muscular impairment affect muscle coordination?), and validate results against experimental data. 4. Critique Read journal papers in the field; establish the main contributions and limitations of the work. Summarize your work using a thoughtful structure and relate your work to the broader field. You will also review the work of your peers and provide feedback on clarity, correctness and impact. 5. Formulate a research question and write a corresponding proposal.	
Education Method	Flipped Classroom method. Lecture slides, reading materials, and assignment descriptions provided through Brightspace. Class time is used to review reading materials and answer questions on concepts and practical labs. The bulk of the work is performing challenge based labs and a final project of your choice.	
Assessment	60% Three practical computer lab assignments (20% each) 40% Final project (proposal, presentation, abstract, reproducibility, reviews) Pass 6/10 grade total	
Department	3mE Department Biomechanical Engineering	

ME41056	Multibody Dynamics	5
Responsible Instructor	J.K. Moore	
Instructor	W. Wolfslag	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Department	3mE Department Biomechanical Engineering	

ME46007	Measurement Technology	3
Responsible Instructor	Dr.ir. G.E. Elsinga	
Instructor	Dr.ir. M.J. Tummers	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Summary	This master level course covers fundamental aspects of measurements useful for a mechanical engineer. It covers measurement of various physical quantities, signal conditioning, error analysis and statistical analysis of the data. A bachelor degree in mechanical, electrical, aerospace, industrial engineering or applied physics is considered prerequisite.	
Course Contents	topics: - General introduction to measurements - Measurement system behavior - Hypothesis testing - Uncertainty analysis - Sensor probes: variety of experimental techniques - Signal conditioning - Data acquisition and processing - Imaging and image processing - 3D techniques	
Study Goals	The general aim of this course is to provide students with a background in the theory of engineering measurements. At the end of the course, the students should be able to (1) analyze measurement systems and devices using the laws of mechanics and general physics, (2) evaluate the accuracy, sensitivity, spatial and temporal resolution of a measurement, (3) analyze and interpret datasets that contain errors, (4) design a measurement scheme and signal conditioning in order to enhance the significance of the results.	
Education Method	The main learning methods include lectures, take-home assignments and a textbook. The lectures cover the different topics of the course. (In case of unforeseen circumstances or measures resulting from COVID-19, the lectures will be replaced by pre-recorded videos of the lectures and/or live online-sessions. Such changes will be communicated on Brightspace.) Lecture slides will be made available on Brightspace. The recommended textbook is given below, however, you are encouraged to refer other books as well. Additional reading material, if necessary, will be made available on Brightspace. Assignments will be provided to practice the relevant problems and the concepts discussed in the class. They will be made available on Brightspace. The solutions to the assignments will be discussed in the class. The course ends with a written exam.	
Literature and Study Materials	The prescribed books will be followed during the course. Lecture slides will be made available on Brightspace. Additional study materials will be provided on Brightspace if necessary.	
Books	Theory and design for mechanical measurements, 6th edition by Richard S. Figliola & Donald E. Beasley	
Assessment	The examination will be as follows: Closed book written exam. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be: Remote (online) open book with several fraud prevention measures.	
Department	3mE Department Process & Energy	

RO47006	Human Robot Interaction	5
Responsible Instructor	Prof.dr.ir. J.C.F. de Winter	
Instructor	Dr.ir. D.A. Abbink	
Contact Hours / Week x/x/x/x	0.6.0.0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The course on Human-Robot Interaction aims to equip students with the knowledge necessary to design robots that can function safely, comfortably, and effectively in environments inhabited by humans. The course offers both theoretical and empirical insights into the physical and cognitive aspects of human-robot interaction. Students will learn to model the fundamental principles of this interaction, critically examine existing theories and models, and assess various design options by reflecting on and experiencing experimental research.	
Course Contents Continuation	The course will cover key principles of human information-processing, human error, the interaction between humans and automation, human-machine interfaces, and operator state assessment (which includes factors like visual attention, workload, and situation awareness). The lectures will delve into the performance of human operators in roles ranging from manual control (direct control) and supervisory control of automated systems, to shared control of human-robot systems. Additionally, students will learn essential elements of experimental design for human-subject studies. The course finds its main applications in areas such as intelligent vehicles and industrial robotics. To supplement the core curriculum, guest lectures will address specialized topics like haptic interfaces, motion comfort in automated vehicles, eye movements in supervisory control, and collaborative human-robot interactions.	
Study Goals	Upon completing this course, students will be able to: - Explain the mechanisms of human information processing, which includes sensation, perception, decision-making, and action execution. - Explain the key theories and models related to human error, learning, and individual differences. - Describe and model direct human interactions with robots and other machines. - Articulate how humans supervise and interact with robots at higher levels of automation. - Critically examine concepts such as trust, overreliance, and behavioural adaptation in human-robot interaction. - Describe and reflect on the design of interfaces that facilitate effective human-robot cooperation. This includes classical interface design principles, but also more modern haptic interfaces as well as linguistic interfaces such as ChatGPT. - Adapt robotic functionality to human needs, either by teaching the robot or through operator state assessment. - Process experimental data and bio-signals to infer human visual attention, workload, and situation awareness. - Design and critically evaluate experimental studies involving human subjects for the assessment of robotic systems. - Provide constructive feedback to peers.	
Education Method	The course is taught through lectures, totalling four hours per week. Some of the lecture hours will be practical or interactive in nature. In addition to the regular lectures, there will also be guest lectures. Participation in sessions related to practical exercises is obligatory. Students will also participate in an experiment.	
Literature and Study Materials	The course materials will be made available on Brightspace. This includes: (1) the slides, (2) notes accompanying the lectures, and (3) scientific articles and book chapters. Example literature: - Parasuraman, R. (1997). Humans and automation: Use, misuse, disuse, abuse. Human Factors, 39, 230253. https://doi.org/10.1518/001872097778543886 - De Winter, J. C. F., Petermeijer, S. M., & Abbink, D. A. (in press). Shared control versus traded control in driving: A debate around automation pitfalls. Ergonomics. https://doi.org/10.1080/00140139.2022.2153175 - De Winter, J. C. F., & Dodou, D. (2017). Human subject research for engineers: A practical guide. SpringerBriefs in Applied Sciences and Technology. https://doi.org/10.1007/978-3-319-56964-2 - Mulder, M., Pool, D. M., Abbink, D. A., Boer, E. R., Zaal, P. M. T., Drop, F. M., Van Paassen, M. M. (2017). Manual control cybernetics: State-of-the-art and current trends. IEEE Transactions on Human-Machine Systems, 48, 468485. https://doi.org/10.1109/THMS.2017.2761342	
Assessment	Exam (45% of the final grade): This will be an on-campus, open-book digital examination. Additionally, the course incorporates two homework assignments: Assignment 1 is a group task related to manual control, while Assignment 2 is an individual task related to supervisory control of robotic systems and human-state assessment. Additionally, students will participate in a human-subject experiment.	
Department	3mE Department Cognitive Robotics	

SC42001	Control System Design	5
Responsible Instructor	Dr.ir. A.J.J. van den Boom	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	State-space description of single-input, single-output linear dynamic systems, interconnections, block diagrams. Linearization, equilibria, stability, Lyapunov functions and the Lyapunov equation Dynamic response, relation to modes, the matrix exponential. Realization of transfer function models by state space descriptions, coordinate changes, canonical forms. Controllability, stabilizability, uncontrollable modes and pole-placement by state-feedback. Application of LQ regulator. Observability, detectability, unobservable modes, state-estimation observer design Output feedback synthesis and separation principle. Reference signal modeling, integral action for zero steady-state error; Analysis in robust stability and robust performance. Basics of model predictive control (MPC). Different model-structures. Prediction models in state-space setting. Standard predictive control scheme. Relation standard form with GPC, LQPC and other predictive control schemes. Solution of the standard predictive control problem. Stability of MPC.	
Study Goals	By taking this course, the student - will be able to master the introduced theoretical concepts in systems theory and feedback control design - will be able to practically apply these concepts to design projects and tasks - and will be able to relate the learned concepts and techniques to other more specialized ones, to potentially integrate them by taking adjacent courses. - will be able to translate a predictive control problem into a standards setting and solve the predictive control problem. More specifically, the student will be able to: - Translate differential equation models into state-space and transfer function descriptions - Rationalize differences between state-space and transfer function approaches - Linearize a system, determine its equilibrium points, analyze directly its local stability, leverage Lyapunov theory to study general stability properties - Describe the effect of eigenvalue/pole locations to the dynamic system response in time/frequency domain. Contrast step and impulse responses. Analyze transients and steady-state - Investigate model controllability. Formulate and apply the procedure of pole-placement by state-feedback, as well as LQ optimal state-feedback control - Derive observability properties. Formulate and apply the procedure of state estimation and build converging observers - Formulate the separation principle and employ it for the design of output feedback - Build reference models and achieve zero steady-state error using integral control. - set up a predictive control problem. - Solve the standard predictive control problem; - Master the main analytical details in stability proofs of predictive control schemes;	
Education Method	Lectures 4/0/0/0	
Literature and Study Materials	Textbook (its use is strongly recommended): K.J. Astrom, R.M. Murray, Feedback Systems: An Introduction for Scientists and Engineers; second edition; Princeton University Press, Princeton and Oxford, 2009 Available online for download: http://www.cds.caltech.edu/~murray/books/AM08/pdf/fbs-public_24Jul2020.pdf The link for downloading the lecture notes for model predictive control will be available in the first quarter.	
Assessment	Written examination. Due to COVID-19 the exam may be an online exam. Disclaimer: information may change depending on the developments around the coronavirus.	
Department	3mE Department Delft Center for Systems and Control	

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Mechanical Engineering**

ME-BMD Track Courses: Select at least 2

BM41070	Medical Device Prototyping	6
Responsible Instructor	Dr.ir. T. Horeman	
Instructor	Dr. B.J. van Straten	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Previous experience in design is a preferred. courses that can help you to prepare are: -BioInspired technology ME41095 -Introduction to Biomedical Engineering BME-MI-087 -Fundamentals of biomechanical Engineering BM41140	
Course Contents	In the course Medical Device Design students develop and produce a sustainable solution to a problem in the medical field in collaboration with their supervisor, clinicians, technical support and sometimes even manufacturing companies. The course offers students to work on a design assignment that encompasses the complete design cycle from problem analysis, development to actual production of the prototype with the support of professional instrument makers. Students will select a design assignment from a number of options proposed by healthcare workers/clinicians (e.g. surgeons, rehabilitation doctors) or companies. The students work in groups of 3 to 4 students under close supervision of an instructor on the assignment, to end up with a working prototype. The course is finalised with a public presentation for clinicians and companies and a report. In bi-weekly meetings of the instructors with all participating groups students are expected to present and discuss the progress in their project in an informal setting.	
Study Goals	The student must be able to: 1. Employ a design task with a multidisciplinary team to solve a real technical problem in a medical environment: translate the clinical problem as presented in the assignment into a practical, technical solution, i.e. do a problem analysis, specify design requirements, come up with a conceptual design, create a cardboard model; obtain feedback on the clinical feasibility of the concept from the medical assignor to further detail the design. bring "green" elements in their design related to reuse, repair, refurbish and recycle to minimise the impact on the environment during production, use and disposal (live cycle) 2. Realize the fabrication of a prototype medical device in collaboration with instrument makers and production companies: create detailed drawings, CAD model; formulate a production plan. 3. Evaluate the performance of the new prototype: test the technical functionality of the device and/or the clinical applicability of the device in a medical setting; reflect on previously made design choices based on the performance of the prototype. 4. Present the design to a multidisciplinary audience of technicians, clinicians and (medical) companies.	
Education Method	Design project	
Assessment	Prototype, report and final presentation	
Remarks	-Students are expected to write a motivation letter before the selection process -Summative feedback is given by a team of experts to guide you towards your end goal.	
Department	3mE Department Biomechanical Engineering	

ME41085	Biomechatronics	4
Responsible Instructor	Dr.ir. G. Smit	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	Biomechatronics is a contraction of biomechanics and mechatronics. In this course the central focus is on function and coordination of the human motion apparatus, and on the design of assistive devices for the support of the function of the motion apparatus. Examples are assistive devices like an orthosis, a prosthesis or Functional Electrical Stimulation of muscles. The goal is to provide some function to patients with functional deficiencies.	
Study Goals	After successfully completing this course, you are be able to: 1. make a Problem Analysis of a given assignment in the field of the human motion apparatus and its interaction with an assistive device a. identify the underlying cause [pathology] of the problem as presented in the assignment b. describe and explain the possible treatment options for the pathology of 1a c. translate the result of 1a into technical Design Objectives without reference to any solutions d. derive qualitative and quantitative Design Specifications from the Design Objectives, and categorize and prioritize these 2. optimize the assistive device application given in the assignment in energetic and control aspects a. select and apply appropriate Design Methodology and Design Methods b. generate a variety [typically at least three] of Conceptual Designs c. judiciously select the most appropriate Conceptual Design d. demonstrate the plausibility or feasibility of the Conceptual Design, with special emphasis to the patient benefits	
Education Method	Lectures (2 hours per week)	
Literature and Study Materials	Course material: A reader is available through brightspace References from literature: D.B. Popovic and T. Sinkjaer Control of Movement for the Physically Disabled Springer (2000) ISBN-13: 978-1852332792 D.H. Plettenburg Upper Extremity Prosthetics. Current status & evaluation VSSD (2006) ISBN-13: 978-9071301759	
Prerequisites	Assumed prior knowledge when entering the course, a basic understanding of: Statics Dynamics Control Theory Engineering Design	
Assessment	Assignment: Report made in a group (~2 persons)	
Remarks	Old course code: WB2432	
Department	3mE Department Biomechanical Engineering	

ME41096	Bio Inspired Design	5
Responsible Instructor	Dr.ir. P. Breedveld	
Instructor	B. van Vliet	
Contact Hours / Week x/x/x/x	4.4.0.0	
Education Period	1 2	
Start Education	1	
Course Language	English	
Expected prior knowledge	Mechanics and basic design experience in hand sketching and CAD software, like SolidWorks.	
Course Contents	The course Bio-Inspired Design gives an overview of non-conventional mechanical approaches in nature and shows how this knowledge can lead to more creativity in mechanical design and to better (simpler, smaller, more robust) solutions than with conventional technology. The course discusses a large number of biological organisms with smart constructions and unusual internal mechanisms and presents a number of technical examples and designs of bio-inspired (surgical) instruments, devices, robots, vehicles and machines. Examples of topics: Life and its creation, engineering in biology, strength at low weight, stiffness with soft structures, robustness and redundancy, storing energy in compliant springs, energetically efficient muscle configurations, clamping with hands, claws, suction, glue, dry - and wet adhesion, biological vibration systems, biological walking, swimming, crawling and flying, locomotion of micro- and single-celled-organisms, biological sensors, soft robots, walking robots, flying robots, snake-like surgical instruments and devices. To teach students how to create smart and truly innovative designs, students are trained with the ACCREx creative design method that was developed at the 3mE department Biomechanical Engineering. Alternating intuitive brainstorming with logical, scientific abstracting and categorizing, ACRREx structures and guides a design process towards fundamentally new design solutions. Structure of the course: 1. Engineering & biology - Life and its creation - Finding biological information - Creating creativity 2. Bioconstruction (how are creatures constructed) - Mechanical stiffness & motion - Hydrostatic stiffness & motion 3. Biopropulsion (how do creatures move) - Flying at macro- & microscales - Floating, rolling & adhesive devices - Sliding, crawling & snake-like devices - Walking, compliant devices & compliant actuation 3. Bioclamping (how do creatures grasp) - Friction & biological grasping - Biological hands & prosthetic hands - Biological adhesion	
Study Goals	The student must be able to: 1. describe methods for creative design 2. identify mechanical working principles and phenomena of biological creatures - explain their construction, motion, and/or processing mechanisms - formalize the essence of these mechanisms in models - derive non-conventional design principles from these models 3. implement these design principles in innovative mechanical devices - summarize the transition process from the biological to the mechanical domain - present their design in drawings or preferably in working models	
Education Method	Interactive lectures Interludia with driven scientists in a talkshow-like setting Student presentations about the design process Reflective Q&A sessions	
Assessment	Students are subdivided into multidisciplinary student groups. Each student group receives a design assignment for which a novel, bio-inspired solution has to be found. During the design process, the student groups are coached by the teachers and have to give a number of presentations about their progress to the other student groups. The examination consists of three parts: 1. Each student group hands in a scientific paper covering a problem analysis, the biological background, the design process, the final solution and an evaluation. 2. Each student group gives a final presentation to the teachers and the other student groups. 3. Each student group fabricates a demonstration model that shows the essence of the final solution. The final mark of each individual student is based on a weighted average between these group results and his/her functioning within the student group.	
Department	3mE Department Biomechanical Engineering	

ME46015-23	Precision Mechanism Design	4
Responsible Instructor	Prof.dr.ir. J.L. Herder	
Instructor	F.G.J. Broeren	
Contact Hours / Week x/x/x/x	0/0/4/2	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course focuses on cultivating a new design mindset among students and equipping them with advanced skills in mechanisms theories. The goal is to enable students to effectively analyse complex real-life problems and extract their essential elements. Through this course, students will develop the expertise needed to construct the next generation of precision machines. In this course, students apply advanced mechanics concepts to a real world design assignment, provided by industry. Students perform conceptual, detailed and critical design steps, all within a systems engineering context. In this process, students develop the capacity to identify problematic areas, generate diverse design alternatives, and make informed design decisions. Key principles explored throughout the course include designing lightweight and rigid mechanisms, achieving precise kinematic constraints, creating low-hysteresis mechanisms, and designing precision manipulators.	
Study Goals	After the course, the student understands the principles of conceptual design of precision mechanisms. They are able to recognize and prioritize problem areas, generate design alternatives and make adequate design decisions. Theoretical part (Q3): - Analyse the degrees of freedom and constraints of a mechanism using Grübler counting and Screw Theory - Design flexure mechanisms freedom and constraint topologies - Design and analyse precision mechanisms with respect to stiffness - Analyse the effects of hysteresis and microslip - Design and analyse kinematic couplings between machine components - Design precision mechanisms with respect to dynamics and energy management Assignment part (Q4): - Identify functional requirements for a real-life (precision) machine design - Divide a real-life (precision) machine design problem into sub-assignments - Coordinate sub-assignment to constitute a coherent final solution to a real-life problem - Generate various machine design concepts - Define design specifications that can be assessed - Evaluate machine design concepts and justify the conclusions - Determine critical dimensions of a machine design based on estimates and calculations - Estimate the performance of a detailed machine design	
Education Method	The course consists of two parts: in Q3, the focus is on the theoretical skills, while Q4 is mainly devoted to the design project. In Q3, there are two hours of lecture and two workshop hours per week. In the lectures, theoretical concepts are presented. The workshops serve to increase familiarity with and intuition for these concepts through hands-on exercises. In Q4, students work together in large design teams to perform a design assignment, presented by a Dutch High-Tech company. In the lectures, students receive the necessary information on the system design process. This information is provided just-in-time to make it directly applicable in the project. Throughout the quarter, intermediate reports are written to finalize the conceptual, detailed and critical design phases. These reports are subjected to peer-feedback. A bundling of these reports will form the end-product of the design project.	
Assessment	An individual written exam on the theoretical part concludes Q3. Design reports written in groups conclude Q4. The final grade is the average of the two.	

ME46115-23	Compliant Mechanisms	4
Responsible Instructor	D. Farhadi Machekposhti	
Instructor	F.G.J. Broeren	
Instructor	Prof.dr.ir. J.L. Herder	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Prior knowledge or skills required in this course are: Statics (TU Delft B.Sc. course example: WB1630); Strength of Material (TU Delft B.Sc. course example: WB1631); Linear Algebra (TU Delft B.Sc. course example: WBMT1051); Rigid-Body Dynamics (TU Delft B.Sc. course example: WB2630)	
Course Contents	While engineering usually relies on complexity to achieve more functionality and better performance, compliant mechanisms can outperform classical solutions by the less is more principle. Compliant mechanisms transmit and transform force and motion by undergoing elastic deformation of their flexible segments, as opposed to rigid body parts and joints. While it may be easier to fabricate compliant mechanisms, it is challenging to understand their underlying systems as they have a monolithic structure and their design is complex because kinematics and kinetics cannot be treated separately. This course will enable students to synthesize, design, model, build and test a compliant mechanism and experience a blend of creativity, theory and hands-on. Based on a design assignment, students apply the knowledge from the lectures (learning objectives) to their project, create a 2D/3D design, produce this by laser cutting or additive manufacturing, and generate different theoretical models (Pseudo Rigid Body model/Beam theory and Finite Element model) that will be compared with experimental data of their design. The course will conclude with a mini-symposium where students will present their work in the format of poster, presentation and demo.	
Study Goals	After successful completion of the course, students will be able to : Describe the different classes of existing compliant mechanisms and rigid-body mechanisms, and the similarities and differences between them. Apply kinematic analysis and synthesis in rigid-body mechanism design. Apply pseudo-rigid-body modelling, beam theory, and FEM modeling to a compliant mechanism. Determine functional requirements, strategies, and specs in the design of compliant mechanisms for a given application. Design a compliant mechanism for a given application using kinematic synthesis, modeling methods, and experiments. Communicate the design assignment in the form of a report, presentation, poster, and pitch based on the given format.	
Education Method	Students will master the subject by applying theory and methods to their final design assignment. The course consists of lectures, invited lectures, in-class assignments, a design project in small groups of students, presentations, discussions, and a final report. In the case of the COVID-19 type of situations, the form of the course may be slightly changed in that the lectures may be partly online. As much as possible the hands-on nature of the course will be maintained.	
Books	Compliant Mechanisms 1st Edition, by Larry L. Howell (Author)	
Assessment	The grade is based on a final group design project (70% of the final grade) and a written exam (30% of the final grade). The final group design project consists of a design paper and presentation with a given format that addresses the learning objectives of the course. Timeline: Written exam: end of Q1. Final report submission: end of Q2.	
Remarks	For this course, you need to be interested in design and have a background in mechanical engineering, civil engineering, or aerospace engineering from your BSc. Knowing kinematics and FEM (in the format of software packages) is an advantage. You will learn most of the tools needed to do your design project during the course.	

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Mechanical Engineering**

ME-BMD Electives

BM41050	Applied Experimental Methods: Medical Instruments	4
Responsible Instructor	Prof.dr. J.J. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	In this course, students carry out experimental research in instrument-tissue interactions within the context of application of minimally invasive techniques such as needle or catheter placement. Key in this course is that students familiarize themselves with making an experimental design, setting up an experiment and investigate the response to your experimental manipulations. The results should allow you to critically evaluate the chosen experiment parameters and data analysis methods.	
Study Goals	The student must be able to: Select and apply appropriate research methods. Create an experimental platform. Use suitable methods to analyse the data. Present the results in a concise report.	
Education Method	Students work in groups of two. They acquire basic knowledge through self-study and hands-on experience in experimental work.	
Assessment	Written report (at most 4 pages, two-column scientific paper format).	
Department	3mE Department Biomechanical Engineering	

BM41055	Anatomy and Physiology	4
Responsible Instructor	Dr.ir. F.J.H. Gijsen	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Course Contents	In this course, we will focus on the following topics: 1. Introduction to human physiology (human body, homeostasis, mass transport) 2. Cell physiology (plasma membrane, cytoplasm, nucleus, cell growth and reproduction) 3. Musculoskeletal system (bones, joints, musles) 4. Nervous system (nerve cells, neurophysiology, central nerve system, peripheral nervous system and reflex activity, autonomic nervous system) 5. Endocrine system (hormones and endocrine organs) 6. Cardiovascular system: blood (blood cells, blood flow, blood pressure, vascular compliance, clotting, Newtonian flow). 7. Cardiovascular system: heart (anatomy, coronary circulation, cardiac muscle fibres, cardiac output). 8. Respiratory system (respiration, transport and exchange of oxygen and carbon dioxide, control mechanisms of respiration)	
Study Goals	The student is able to describe the anatomy and the function of several physiological systems. The student must be able to: Describe the anatomy and the function of the musculoskeletal, nervous, endocrine, cardiovascular, immune, respiratory, digestive, urinary and reproductive systems Describe the control mechanisms involved in the different physiological systems. For detailed description of learning goals, see first slides of the lectures on Brightspace	
Education Method	1 time per week 2 lectures of 45 minutes Self study before every lesson; cases and examples during (interactive) lectures Course material: Elaine N. Marieb and Katja Hoehn Human Anatomy & Physiology, Global Edition, 10th and llth edition, Pearson	
Assessment	You will take two digital exams for this course, in Q1 and Q2. These will cover the material of the respective quarters. Both exams account for 45% of your grade. The (digital) resit is in the Q3 exam period and will cover all of the material; this will replace the grade of both exams. In addition, you will take weekly online quizzes which account for 10% of your final grade.	
Department	3mE Department Biomechanical Engineering	

BM41131	Tissue Biomechanics	3
Responsible Instructor	A.C. Akyildiz	
Instructor	Dr.ir. M.P. Peirlinck	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course involves lectures and interactive group discussions. In the lectures, the students will be introduced to the biomechanics of various tissue types in the human body, including bone, cartilage, tendon, ligament, muscle, and vessel tissues. The students will be guided through the topics of these tissue types focusing on structure-function relationship, material behavior (experimental examples), combined with modeling approaches and clinical application. The fundamental knowledge on continuum mechanics, required for a good understanding in tissue biomechanics, will be provided as well. Additionally, experts will be invited to talk about their current scientific research on tissue biomechanics, representing the rapidly developing field of tissue biomechanics. During the group discussions, the students will be expected to read and take a deeper dive into selected tissue-specific research articles and discuss the topics background, approach, methods, results, and limitations with their peers. Active participation in these group discussions is obligatory in order to take the final examination.	
Study Goals	After successful completion of this course, the student should be able to: 1. Explain the biomechanical constituents, hierarchical structure and mechanical function of biological tissues, and how they relate to each other, 2. Discuss the basic concepts of continuum mechanics that are relevant for tissue biomechanics, 3. Examine the mechanical behavior of biological tissues using continuum mechanics framework, 4. Evaluate recent scientific research on tissue biomechanical modeling and experimentation, and its applicability in the clinic.	
Education Method	Lectures, group discussions	
Literature and Study Materials	Book / online material recommended to complement lectures: An introduction to biomechanics: solids and fluids, analysis and design (Jay D Humphrey & Sherry L O'Rourke) Nonlinear solid mechanics: a continuum approach for engineering (Gerhard Holzapfel) Continuummechanics.org	
Assessment	Group assignments Written exams	
Department	3mE Department Biomechanical Engineering	

BM41155	3D Printing	4
Responsible Instructor	Dr. J. Zhou	
Responsible Instructor	Prof.dr. A.A. Zadpoor	
Instructor	Ing. M.A. Leeflang	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	in Track I (Musculoskeletal Biomechanics)and Track II (Medical Devices and Bioelectronics) of MSc Biomedical Engineering BME) and in the specialization of Bio-Inspired Technology (ME-BMD-BITE)	
Expected prior knowledge	BSc in mechanical engineering, materials engineering, industrial design engineering and aerospace engineering with a basic knowledge of materials (e.g. WB2330)	
Parts	- Overview of AM technologies, their principles and applications - Materials selection (polymers, ceramics, metals and composites) - Powder-bed fusion processes - Powdered materials and powder fabrication - Powder characteristics in relation to AM - Structural integration during melting and sintering - Extrusion-based systems - Design for functionality and 3D printability Software issues and AM limitations - Other powder-based manufacturing technologies as alternatives to AM - Post-processing of AM products - Mechanical testing of AM products - Case studies AM for engineering and medical applications - Assignment and presentations	
Course Contents	Additive manufacturing (AM), also known as 3D printing the process of joining materials to make objects from 3D model data, usually layer upon layer, as opposed to conventional subtractive manufacturing technologies, is set to revolutionize manufacturing across a wide range of industries. The course 3D printing - AM technologies is designed to equip students with the fundamental knowledge of AM and basic hands-on experience in AM. The topics include various AM processes, their principles, capabilities and limitations, materials used for AM processes, powdered material fabrication and characterization, CAD designs for AM, alternative powder-based manufacturing technologies, current and future AM applications, as well as the mechanical evaluation of AM parts. The primary focus is placed on powder-based AM for metallic parts. In addition to a series of lectures, project groups, each composed of 3-5 students, will create their own designs of the objects made of polymer PLA, formulate requirements in terms of geometry, dimensions, tolerances, mass, surface finish, functionality and mechanical properties of these objects, redesign them by making use of SolidWorks, print them, test and evaluate 3D printed products against the requirements, and document the projects in the form of reports and presentations. The new design is expected to reflect the distinctive features and freedom that AM has to offer (thin walls, lattice structures, form-free channels, integrated parts/functions, and customization), but within the capacity of the printing machines (bounding box size, minimum feature size or wall thickness). The report and presentation are expected to contain: reason for the selection of the object, weakness of the existing design, new design features, actual testing results and recommendations.	
Study Goals	Students will be familiar with the concept of AM, various AM processes, AM applications, materials for AM, product design for AM, post-processing techniques and evaluation methods. In addition, students will gain basic hands-on experience in product development via AM. Once successfully completing the course, the students will: - Have a fundamental knowledge of 3D printing/AM technologies for a wide range of applications - Be able to explain the benefits, advantages, challenges and limitations of individual AM technologies as well as alternative powder-based manufacturing technologies - Have an insight into selective laser sintering and melting (SLS/SLM) processes regarding process control in relation to the geometry and mechanical properties of the products - Become familiar with the common (metallic) materials for 3D printing and their characterization techniques - Understand printable geometry design rules and the common methods of digital file creation - Become familiar with the common techniques for post-processing - Become familiar with the common methods for the mechanical evaluation of 3D printed materials and functional parts.	
Education Method	Lectures and student projects Attending lectures and performing assignment are both compulsory	
Computer Use	- Object design with Software (SolidWorks Student Engineering Kit)	
Course Relations	Biomaterials, Tissue biomechanics and other engineering courses in MSc biomedical engineering, materials science and engineering, mechanical engineering, marine technology, aerospace engineering, design for interaction, integrated design and strategic design	
Literature and Study Materials	- Additive Manufacturing Technologies, Ian Gibson, David Rosen, Brent Stucker and Mahyar Khorasani, 3rd Edition, 2021, Springer (chapters: 1, 2, 3, 5, 6, 7, 10, 15, 16, 18, 19 and 21) - Lecture materials	
Books	- Additive Manufacturing Technologies, Ian Gibson, David Rosen, Brent Stucker and Mahyar Khorasani, 3rd Edition, 2021, Springer (chapters: 1, 2, 3, 5, 6, 7, 10, 15, 16, 18, 19 and 21)	
Reader	- Lecture materials	
Assessment	Written exam	
Department	3mE Department Biomechanical Engineering	

ME41035	Special Topics in Sports Engineering	3
Responsible Instructor	J.K. Moore	
Contact Hours / Week x/x/x/x	x/x/x/4	
Education Period	4	
Start Education	4	
Exam Period	4 Different, to be announced	
Course Language	English	
Course Contents	*WARNING* This course is an intensive, full time 2 week summer course that will occur during the months of June, July, or August. If in June, it will conflict with your other courses so plan accordingly for the work load. Check the course website for up-to-date information: https://moorepants.github.io/me41035/ Special Topics in Sports Engineering is an inter-university course for Master students in Mechanical Engineering, Movement Sciences, Sport Sciences, and other related MSc programs. The course is organized as a two-week intensive course and comprises lectures, demonstrations, practicals, hands-on research, and a final field test. The course will be taught by staff from the six participating universities. Due to the ongoing coronavirus situation, this year's course will again be online. The course is organized around a basic theme relevant for sports engineering and this year's theme will be "Maximizing cycling performance". During the course students will work out what aspects determine cycling performance, and collect data (through experiments or literature research) that are needed to develop and feed a simulation program for the estimation of the optimal bicyclerider combination and the maximal performance humanly possible. The course's final activity will be a test ride to quantify the differences between actual performance and predicted performance. In this course, students will have to answer the question: Given a particular bike, what will be your own predicted time over a given distance and track and how well does this match reality? The prediction should be based on a power-based simulation model of cycling and the relevant bicycle and rider dependent parameters, which have to be collected experimentally. Answering this question will require insight in relevant parameters, but also collecting these parameters, for each individual student with his or her individual bike. The bike in question can be chosen freely and might therefore be a top-end racing bike as well as your grandmother's shopping bike.	
Study Goals	After following this course, students should understand the complexity of maximizing sports performance and the importance of the inclusion of materialathlete interaction. More specifically, students should be: - Familiar with the Power Equation concept and be able to apply this to cycling; have knowledge of methodological aspects of sports research, in particular error propagation, manmachine interaction (closed loop complexity), measurement techniques, internal and external validity. - have insight in the organizational and psychological complexities of sports innovation. - Able to measure key parameters needed for power equations, related to their own field and have experience in the measurement of key parameters in adjacent fields; - able to provide a cycling performance simulation program with the parameters necessary to evaluate performance on a realistic level; - able to collect and present to fellow group members, data on parameters for such a simulation program. - present research findings through an individual portfolio, and a group presentation/poster/brief oral.	
Education Method	lectures, workshops, practicals, presentations, team work	
Assessment	A pass/fail grade will be given based on participation in all practicals, presentations, writing reports, and a peer review.	
Department	3mE Department Biomechanical Engineering	

ME41065	System Identification and Parameter Estimation	7
Responsible Instructor	M.L. van de Ruit	
Instructor	Prof.dr.ir. A.C. Schouten	
Instructor	Dr.ir. W. Mugge	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic control engineering and experience with Matlab	
Course Contents	System identification deals with the estimation of physical properties of (dynamic) systems from measured data (inputs and outputs), which is essential for the understanding of system behavior. System identification is particularly useful for biological systems where physical entities are generally not known a priori. The course provides mathematical a background of system identification in both time domain and frequency domain with a strong focus on practical application. Major topics to be studied are: spectral densities, nonparametric black-box models, discrete time domain models (parametric grey-box), closed loop systems (utilizing feedback), translation of transfer functions into physical parameters, optimization routines and design of perturbation signals. During the course examples from both technical and physiological systems (i.e. estimate the behavior of a driver or the dynamics of a human joint) will be discussed.	
Study Goals	At the end of the course the student is able to. LO1: Signals a: explain the importance of input signal design and signal acquisition properties for the identification of system dynamics. b: select optimal input signals to identify unknown system dynamics and test them using MATLAB and Simulink. LO2: Systems a: illustrate how system dynamics can be accurately estimated using non-parametric (non)linear time-invariant models in time- and frequency domain. b: estimate nonparametric models of unknown systems from measured input and output signals using MATLAB and Simulink. LO3: Models a: explain what model structure is best fits a non-parametric estimate and which corresponding optimization algorithm to use to find the model parameters. b: parameterize physical models from nonparametric estimates employing optimization techniques using MATLAB and Simulink.	
Education Method	Self study, lectures (2x2 hours per week) including (home) work assignments	
Prerequisites	recommended prior knowledge - Probability theory and statistics (e.g. WBMT2049-T1) - Signal analysis (e.g. WB2235) - Systems and control engineering (e.g. WB3240)	
Assessment	Final closed book exam (60%), digital midterm (40%) tests and assignments (pass/fail)	
Tags	Analysis Mathematics Matlab Mechanics Modelling Numeric Methods Programming Signals Signals and Systems Small groups Technology	
Department	3mE Department Biomechanical Engineering	
Contact	By email SIPE-3ME@tudelft.nl	

ME41120	Freehand Sketching of Products and Mechanisms	3
Responsible Instructor	Ir. J.W. Hoftijzer	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Freehand sketching is a communicative and effective method for engineers and designers to convey ideas and share concepts with the various partakers in a design or development project. Although the definitions of a drawing and sketch overlap, in this class a sketch is defined as a 2-d (orthographic) or 3-d (mostly 2-point perspective) representation of an object or concept. The course will help you learn to observe, to make choices in what to sketch, to understand special relationships better, to depict them manually, and to improve your motor skills (draw with vigour).	
Study Goals	The main objectives of the course concern four elements of learning how to draw: 1.'Construction (set up) and proportions' Student is able to methodologically set up (construct) your design (idea/concept) in 2d and 3d, considering accuracy, proportions and clarity. 2.'Technique and skills' Student demonstrates skills and vigour in his/her sketches (concerning pace, accuracy, efficacy), using various sketching tools (fineliner, marker..). 3.'Tonal value and spaciousness' Student is able to apply the methodologically suitable tonal value and colour in a sketch, aiming for clarity of the sketch and of the object/design. 4.'Composition' Student is able to produce a clear and attractive sketch (-sheet) composition (size, viewpoints, relationships, distribution, partition, focus, sequence)	
Education Method	Classes include lectures, demonstrations, student drawing exercises Students who actively participate in class and do all homework assignments will be well prepared for all assessments in the course.	
Literature and Study Materials	Students need to buy markers and marker-proof paper. For details see Brightspace. For some extra information on perspective drawing, please take a look at www.delftdesigndrawing.com. In case interested, recommended books to consider: Eisen/Steur: 'Sketching/ Sketching the Basics' Robertson: 'How to draw'	
Prerequisites	No expected prior knowledge, ability or skill to draw or sketch.	
Assessment	If not otherwise announced, all assignments are due at the start of the class. Grade assessment of each assignment is based on the following criteria: Meet all requirements, e.g. three drawings on A3 paper with correct views Composition of each drawing, taking into consideration white space along the edges of the paper and labeling area for title, name and date. Quality of line weight and differentiation between construction lines and edges of object. Expression of markings, e.g. do the lines or shading show timidity or confidence?	
Enrolment / Application	Class size is limited to a total of 20 students. Interested students should write an application email to FreeHandSketching-3ME@tudelft.nl, including -a motivation for taking this course -commitment to attend all lectures and practicals.	
Remarks		
Department	3mE Department Biomechanical Engineering	
Contact	FreeHandSketching-3ME@tudelft.nl/ j.w.hoftijzer@tudelft.nl	

ME41125	Introduction to Engineering Research	3
Responsible Instructor	Dr.ing. L. Marchal Crespo	
Instructor	L. Hoogendam	
Instructor	Drs. A.E. Kam	
Contact Hours / Week x/x/x/x	0.0.0.4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Students must have have basic programming skills in order to perform simulations or process data, for example in Matlab, Python, or R. The ability to derive the equations of motion and to simulate and design a controller for a mechanical system is helpful for this course.	
Course Contents	1) Literature review 2) Academic Writing 3) Dealing with uncertainty in experimental data 4) Generating reliable results 5) Preparing data and code for FAIR use 6) Reviewing 7) Dealing with feedback 8) Oral presentation	
Study Goals	Learning objectives By the end of this course you should be able to: 1) review the literature on a chosen topic 2) write a formal academic paper following the standard structure elements of an academic paper about an engineering topic 3) analyze the validity of data and statistical analyses 4) generate reliable simulations and code 5) assemble your data and your code for FAIR use 6) assess peers work using criteria provided by the instructor 7) revise an academic text, using the lecturers and/or peer feedback 8) present your own research to a mixed audience	
Education Method	Lectures, active learning activities, material provided on Brightspace, assignments.	
Assessment	Assessment is based on assignments and the final presentation, as follows: 1) 25%: your first draft academic paper 2) 25%: quality of the feedback provided to others (in two rounds) 3) 25%: rebuttal letter and final version of your paper. 4) 25%: your final presentation.	
Enrolment / Application	The course will only be offered if a minimum of 20 participants is reached.	
Department	3mE Department Biomechanical Engineering	

ME46041	Experimental Dynamics	4

Responsible Instructor
D. de Klerk

Responsible Instructor
Dr. M.V. van der Seijs

Contact Hours / Week x/x/x/x
0/0/2/2

Education Period
3
4

Start Education
3

Exam Period
Different, to be announced

Course Language
English

Expected prior knowledge
Engineering Dynamics (ME46055)
Linear algebra
Different equations

Course Contents
This course focuses on learning to understand and solve problems in the field of experimental dynamics. Motto: In theory, theory and practice are the same... In practice they are not. This course concentrates on pointing where those differences originate from. It'll be valuable for any who performs and analyses measurements in practice and tries to match his / her simulation to the experiment.

This course specifically goes away from becoming "book-smart": instead it challenges you to formulate and motivate answers from various (incomplete) sources of data, using an understanding of mechanical vibrations principles and linear algebra. As such, it will trigger the curious mind to solve problems as they really occur in fields of engineering, such as the automotive and high-tech industry, where vibrations play an important role in product development.

The course builds on fundamentals of mechanical vibrations as taught in the course Engineering Dynamics (ME46055), which is considered a prerequisite.

Other than the specific course content, you will be challenged on:
* Comprehending the concept of degrees-of-freedom: what do they really mean, how does this relate to matrix equations and aspects such as matrix rank and decomposition.
* Being resourceful and independent, as to be expected from a academically trained engineer in your professional career.
* Making reports that are compelling, focusing on argumentation and appearance of plots such that they best convey the message.

Experimental Dynamics Theory
- Modelling of dynamic system in various domains: from the numerical world (FE, mass/damping/stiffness) to the experimental world described by FRF and everything in between.
- A practical look at Frequency (Fourier) Analysis: Discrete Fourier Transformation, FFT, Leakage, Aliasing and Windowing. Know it or you'll mess up your experiments.
- The power of Transfer and Frequency Response Functions (FRF); why are they so useful?
- Harmonic excitation (with frequency stepping), hammer impact excitation, stochastic excitation.
- Experimental Modal Analysis: Do's and don't, pitfalls & challenges in practice.
- Experimental Dynamic Substructuring. An alternative FEM formulation which can also use experimental data.
- Virtual Point Transformation. A method to transform your experimental model in a test-based super-element model compatible with Substructuring.
- Transfer Path Analysis, a useful way to identify source excitation and system sensitivities.

Experimental assignments
You will be working on two group assignments, meant to illustrate concepts as discussed in the lectures and to deepen insight. Teams of four students each, carry out multiple assignments involving the analysis of vehicle and helicopter vibration measurements, and an assignment involving Substructuring, Virtual Point Transformation and Transfer Path Analysis.

Study Goals
In general the student is able to understand and analyze dynamic measurements, being aware of possible pitfalls.

More specifically, the student must be able to:
1. describe the effects of Quantization, Leakage, Aliasing and Windowing in measurements and measurement equipment.
2. explain the principle of extracting modal parameters (resonance frequency, mode shape, damping ratio) from system response both in the time domain and in the frequency domain
3. discuss relative merits of different excitation techniques (shaker with frequency sweep, impact hammer, shaker with random excitation)
4. analyse operational measurements with FFT and waterfall diagrams.
5. Explain the different concepts of Transfer Path Analysis.
6. Carry out an (experimental) Dynamic Substructuring analysis along with a Component (based) TPA analysis

Education Method
Weekly classes followed by 2 group assignments.

Computer Use
MATLAB: the two assignments can be worked out in MATLAB. Some prior experience in the basic syntax of MATLAB is warranted.

Literature and Study Materials
Lecture notes and your own search on the web

Prerequisites
Engineering Dynamics (ME46055)

Assessment
First group assignment: analysis of vehicle and helicopter measurements. You will conduct actual vibration measurements on a vehicle. Afterwards, you will analyze your own measurement data with your group and write a report. You will use MATLAB to process and analyze the data. There is also an additional measurement of a helicopter added to the assignment. The recording is of a helicopter pass-by and you are asked to figure out the speed and height of the helicopter flying but also how many blades the helicopter has and what height the microphone was placed on the runway.

Second group assignment: Experimental Substructuring and Transfer Path Analysis. You will simulate the dynamic behavior of an assembly consisting of two substructures. Measurements on these substructures will be simulated using a toolbox for MATLAB.

Department
3mE Department Precision & Microsystems Engineering

ME46050	Advanced Finite Element Methods	4

Responsible Instructor	Dr. A.M. Aragon
Contact Hours / Week x/x/x/x	0/0/4/0
Education Period	3
Start Education	3
Exam Period	3 4
Course Language	English
Required for	Enriched Finite Element Methods (ME46080)
Expected prior knowledge	Basic knowledge on finite element analysis and programming (any language though Python is recommended).
Summary	The finite element method (FEM) is undoubtedly the established procedure for solving problems in solid mechanics. This courses gives a thorough introduction to the finite element method. For a simple 1D bar in elastostatics equilibrium, we derive the variational formulation and work out the resulting finite element discrete system of equations. This is then generalized to higher dimensions and other problems. We also look at the p-version of the finite element method (p-FEM). For these methodologies, we look at a priori and a posteriori error estimates that help us understand the convergence properties.
Course Contents	The outline for the course is: Overview of the standard finite element method (FEM) The problem of linear elastostatics in one dimension - The strong (classical) form - The weak (variational) form - The Galerkin formulation - Orthogonality of Galerkin error - The finite element discrete equations - Iso-parametric mapping The elastostatics problem in higher dimensions - Strong form - Weak form - Discrete formulation Heat conduction The p-version of the finite element method (p-FEM) phierarchical elements in 1-D pFEM in higher dimensions - Integration and geometrical mapping - Prescribing essential (Dirichlet) boundary conditions h, p, and hp convergence Problem classification based on regularity A priori error estimates A posteriori error estimate for h-FEM Computational aspects Finite element implementation The assembly process and numerical integration Checking the implementation (high-order patch tests and construction of manufactured solutions)
Study Goals	The study goals for this course are: - Derive the finite element discrete equations, starting from the equilibrium equation (or any other elliptic partial differential equation); - Identify the limitations of the finite element method and understand the means to overcome them; - Evaluate the performance of different methodologies on academic problems; - Extend the software package provided to simulate complex problems.
Education Method	Lectures and projects that involve programming.
Assessment	The course will be assessed by one project (to be performed individually) that will be handed in at the end the quarter. The project will involve coding and using the resulting code to solve the problems.
Department	3mE Department Precision & Microsystems Engineering
Contact	Phone: +31 15 278 22 67 Email: a.m.aragon@tudelft.nl

ME46060	Engineering Optimisation: Concepts and Applications	3
Responsible Instructor	Dr.ir. M. Langelaar	
Instructor	Dr.ir. L.F.P. Noel	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge	Basic knowledge of mechanical engineering and mathematics. MATLAB is used for exercises and the final project, students without MATLAB experience must gain this through self-study. Familiarity with Finite Element modelling is helpful.	
Course Contents	Formulation of optimization problems Typical characteristics of optimization problems Minimization without constraints Constrained minimization Direct and gradient-based optimization algorithms Approximation concepts Sensitivity analysis Topology optimization	
Study Goals	After successfully completing this course, you will be able to formulate a proper optimization problem in order to solve a given design problem, to select a suitable approach for solving this problem numerically, and apply such a numerical optimization procedure to the formulated problem. Furthermore, you will be able to interpret results of completed optimization procedures. More specifically, you will be able to: 1. formulate an optimization model for various design problems 2. identify optimization model properties such as monotonicity, (non-)convexity and (non-) linearity 3. identify optimization problem properties such as constraint dominance, constraint activity, well boundedness and convexity 4. apply Monotonicity Analysis to optimization problems using the First Monotonicity Principle 5. perform the conversion of constrained problems into unconstrained problems using penalty or barrier methods 6. compute and interpret the Karush-Kuhn-Tucker optimality conditions for constrained optimization problems 7. describe the complications associated with the use of computational models in optimization 8. illustrate the use of compact modeling and response surface techniques for dealing with computationally expensive and noisy optimization models 9. perform design sensitivity analysis using variational, discrete, semi-analytical and finite difference methods 10. identify a suitable optimization algorithm given a certain optimization problem 11. perform design optimization using the optimization routines implemented in the Matlab Optimization Toolbox 12. derive a linearized approximate problem for a given constrained optimization problem, and solve the original problem using a sequence of linear approximations 13. describe the basic concepts used in structural topology optimization.	
Education Method	Lectures (2x2 hours per week), exercises, online tests. Exercises and online tests count for an optional bonus of maximum 0.8 points on the final grade. There are no possibilities to redo/retake these, and the bonus only applies to reports handed in at the first report deadline. Further details on the bonus arrangement and calculation are given through Brightspace.	
Literature and Study Materials	References from literature: R.T. Haftka and Z. Gürdal: Elements of Structural Optimization, and various articles made available on Brightspace.	
Books	Course material: P.Y. Papalambros et al. Principles of Optimal Design: Modelling and Computation. 3rd edition.	
Assessment	The course is concluded through an optimization project and report. This can be done individually or in pairs. Deadlines for handing in the final report will be announced at the start of the course. Brightspace/MATLAB exercises and tests (optional) during education period. Also these can be done individually or in pairs. To pass the course, the final grade must be a 5.75 or higher. The final grade is determined by adding the obtained bonus to the report grade, followed by rounding to the nearest half/full mark (.5/.0). The maximum bonus score is 0.8. The bonus only applies to reports with grade >= 5.5 and submitted before the first report deadline.	
Remarks	Former course code: WB1440	
Percentage of Design	80%	
Design Content	The course is focusing on design optimization.	
Department	3mE Department Precision & Microsystems Engineering	
Contact	Dr.ir. Matthijs Langelaar Email: m.langelaar@tudelft.nl	

ME46070	Fundamentals of Mechanical Analysis	4
Responsible Instructor	Dr. C. Ayas	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	In this course, the students will acquire the basic know-how to formulate the equations describing the mechanical behaviour of continuous media. Mathematical tools, namely tensor algebra and tensor calculus will be described. Subsequently, kinematics in a geometrically non-linear setting will be discussed in detail along with Eulerian and Lagrangian descriptions. Non-linear stress measures will be used to formulate conservation of mass, linear and angular momentum. The course contents are the foundation of mechanical analysis.	
Study Goals	Students must be able to apply the basics of tensor algebra and calculus in direct and index notation. describe and formulate the deformation of a continuum both for small and finite deformations. analyse time derivatives of motion of continuous media. analyse the relation between different deformation, strain, and strain rate tensors. analyse the relations between different stress tensors analyse the balance of mass, linear and angular momentum both in Lagrangian and Eulerian settings.	
Education Method	Lectures (four hours per week) and homework exercises	
Literature and Study Materials	Slides Gerhard A. Holzapfel, "Nonlinear Solid Mechanics: a Continuum Approach for Engineering", Wiley, 2000 References from literature: R. Aris, "Vectors, Tensors and the Basic Equations of Fluid Mechanics", Dover, 1962. Fung, Y.C., "Foundations of Solid Mechanics", Prentice-Hall, 1965. M.E. Gurtin, "An Introduction to Continuum Mechanics, Mathematics in Science and Engineering", vol. 158, Academic Press, New York, 1982. R.W. Ogden, "Nonlinear elastic deformations", Ellis Horwood Ltd., 1984	
Prerequisites	Basic knowledge of engineering mechanics (statics, mechanics of materials, dynamics, and continuum mechanics courses from BSc) and calculus and linear algebra (Mathematics course of the BSc) is required.	
Assessment	The examination will be as follows: written exam on campus. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be: an open-book online exam with fraud prevention measures or proctoring.	
Permitted Materials during Tests	No electronic devices!	
Department	3mE Department Precision & Microsystems Engineering	

RO47003	Robot Software Practicals	5
Responsible Instructor	J.F.P. Kooij	
Instructor	G.A. van der Hoorn	
Contact Hours / Week x/x/x/x	6.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic programming experience (e.g. python or matlab) at the BSc level, familiarity with the use of functions, variables, and control structures (for-loops, if-statements, etc.).	
Course Contents	This course teaches students the basics for creating and collaborating on academic robotics software. It aims to give students sufficient knowledge and experience such that they can proceed learning more about robotics programming by themselves as needed. The course does this by providing an overview of key technologies, and letting students develop basic skills often used in real world robotic software development. The main objective is to obtain hands-on experience - beginners level - with Linux, Git & Gitlab, C++, ROS, and integrate with common external libraries such as OpenCV and PCL. This course does not teach new robotics theory (e.g. perception, planning, control, etc.) which is covered in other courses. The allotted 5 ECTS means that students are expected to invest approximately 11 hours/week outside of contact hours. Be aware that active participation in all the lab and peer-review assignments is required, hence it is not recommended to do this course if you cannot commit to a regular time investment over the whole education period.	
Course Contents Continuation	Linux: Although Windows is the most prevalent operating system on desktop PCs, Linux became very popular for embedded systems such as robots. Developing for embedded Linux systems is most easily done in Linux itself, and this course aims to familiarize students with the use of Linux on the desktop and terminal. It includes the following topics: OS architecture, File system, Shell, Scripting. Git/gitlab: All exercises will be submitted using Git, a widely used version control system to manage and collaborate on coding projects. During the course, students will practice basic tasks with git, such as solving code conflicts, and merging the work of lab partners. Using an issue tracker and giving constructive feedback to peers is also an important aspect that will be practiced. C++: C++ is one of the most relevant programming languages for robotics. Being an object oriented language brings great advantages compared to its predecessor C. This practical gives students practical knowledge on C++ programming. The course encompasses the following topics: Introduction to C++, common compilation tools, basic programming in C++, classes and objects. ROS: The Robot Operating System (ROS) is a flexible framework for writing robot software. It is a collection of tools, libraries, and conventions that aim to simplify the task of creating complex and robust robot behaviour across a wide variety of robotic platforms. During the course, students will learn to use existing ROS tools, create their own software components in C++, integrate them into an application, and test their solutions using a physics simulator. others: You will also get some practice using external libraries in your robotics project, such as OpenCV or PCL. OpenCV is a computer vision library designed for computational efficiency and with a strong focus on real-time applications. The Point Cloud Library (PCL) is an open-source project for point cloud processing. Note that this course is *not* about understanding the methods or theory on which these libraries are based, but rather on integrating the provided functionality in novel software components.	
Study Goals	--------------------------------------- Knowledge, insight, judgment and skills --------------------------------------- After completing this course, students will be able to 1. use the command line on a Linux system, perform basic file operations and job management tasks, and automate tasks with shell scripts; 2. use Git to commit and integrate incremental code changes together with a lab partner, and resolve any code integration conflicts; 3. use a Gitlab server to exchange code changes, and use its issue tracker to provide and receive feedback on each other's code; 4. solve programming tasks in C++, compile the code using Make, debug compile-time, link-time and run-time problems; 5. create new C++ ROS packages, use common ROS development tools, and utilize the ROS API in their C++ programs; 6. investigate and integrate external libraries into their own ROS components, such as OpenCV or PCL; 7. add code comments and basic documentation to their code, find and interpret documentation of external libraries; ------------------------------------- Transferable and interpersonal skills ------------------------------------- 8. develop evaluative judgments on the quality of your own work and of the performance of others (including peers) by implementing and providing feedback; 9. collaborate and coordinate with lab partner on coding tasks.	
Education Method	Lectures, which will introduce new course content (2 hours per week). Interactive Sessions will provide class wide feedback on past lab assignments, tutorials, quiz exercises, Q&A (2 hours per week) Practicum session, during which there will be TAs around to help you understand the lab assignments (2 hours per week) Robotics Software Practicals focuses on improving programming and collaborative software development skills. The main method to develop these skills is by gaining hands-on experience through lab assignments, and receiving feedback on the submitted work. Lectures will introduce the various topics, provide context and tips, and will discuss common pitfalls and solutions. Still, the nature of the assignments means that students are expected to invest sufficient time for self-study and practice outside the fixed lab and lecture contact hours. Lab assignment There are 4 obligatory lab assignments, each covering two weeks, namely 1. Linux, Git (Formative) 2. C++ programming (Formative) 3. ROS (Formative) 4. Final Assignment (Summative, demonstrates capacity to use Linux, Git, C++, and ROS). A manual will be provided for each lab assignment before the start of the first practicum session for the assignment. The manual will contain detailed instructions, guidelines, optional exercises, and an assignment that must be submitted to an online repository for your lab group.	
Literature and Study Materials	1. A Gentle Introduction to ROS, Jason M. OKane. http://www.cse.sc.edu/~jokane/agitr/ 2. Provided lab manuals, and lecture slides that will be made available on Brightspace	

Practical Guide	For the practicum assignments, bring a reasonably recent laptop with a x86-64 CPU (i.e. a normal" Intel or AMD CPU; but NOT the new MacBook Air with M2 chip, NOT a Chromebook, etc. These use different CPU architecture which means that the software we provide will not run! A MacBook with an Intel x86 CPU is fine). A laptop with an Nvidia GPU would make the robot simulations run even smoother. Please note the following lab rules: The first three lab assignments are made in groups of two, as you will learn to collaborate on code with your lab partner using gitlab. You can find a lab partner using the special Brightspace forum, but of course also by talking to others after the first lecture, or by looking for a Brightspace lab group with a single enrolled person. To enrol in the practicum, and therefore participate in the course, you and your partner *must* enrol on Brightspace in a lab group before the lab enrol deadline (this deadline will be announced in first lecture). If you plan to drop the course, inform the lab coordinator in time before the start of the next lab assignment, but while you are still enrolled, you should finish the ongoing lab assignment with your lab partner. The scheduled practicum sessions are intended for you to obtain help on the assignments from the TAs. Outside the practicum sessions, you can post questions on the TA forum on Brightspace. To submit your work, you and your partner must push your contributions to a provided gitlab repository. Note that using git and the gitlab webserver for code collaboration is part of the learning objectives of this course. You are responsible for testing the code that you hand-in via gitlab, so make sure that you have tested your work thoroughly, and that the correct version has been uploaded to your gitlab repository. For the first three lab assignments, there will be a peer-review task after the lab deadline has passed. You will learn during the course how to provide feedback, and how to use the gitlab issue tracker to manage issues and comments. The lectures will afterwards discuss positively peer-reviewed lab solutions with all students for additional feedback by a teacher. Active participation in each lab assignment is required to participate in the next assignment. Active participation entails that you submit your solution before the given deadline, and that you complete the required peer-review tasks. You may only use the gitlab code repository that the lab coordinator will create for your group. You are *NOT* allowed to put your coding homework on github.com, gitlab.com, bitbucket.com, google drive, dropbox, etc. (neither with public nor private settings!) TU Delft ethical guidelines apply to reports, code, etc. If your name is on it, you claim it is your work, do *not* copy parts of other student groups code. You are *not* allowed to share your code, solutions with others apart from your lab partner and TAs. Generally, do *not* show your groups work to other students. Plagiarism will be reported to exam committee.
Assessment	# Assessment overview Formative: First three lab assignments, peer-review tasks, discussion on common mistakes and solutions of past hand-in assignments during the lecture by teacher, multiple-choice practice questions. Summative: The final lab assignment, and digital exam. # Grade calculation You final course grade (minimum is a 6 to pass) determined by Final lab assignment (grade, weight 25%, minimum a 5.5 to pass), Digital exam (grade, weight 75%, minimum a 5.5 to pass) Any penalties (see Deadlines below) A resit will be available for the exam, but not the final lab assignment. # Lab assignments You must enroll for the obligatory lab assignments before the announced enrollment deadline (given during first lecture), so that the lab coordinator can fix the groups and setup everybodys code repositories in time. If you do not enroll, you cannot make the lab assignments, or make the exam. In the first three assignments you must collaborate with a lab partner. Your group will receive and give formative feedback through student peer-review (what goes well, what to improve). The fourth and final lab assignment is made individually, and you receive a grade (weights 25% for final grade), based on a rubric for the quality of the solution, code, documentation, and use of Git. This assignments requires you to apply all knowledge from the previous lab assignments to solve a well-defined robotics programming task. Your (groups) lab work is evaluated on the following three Lab Criteria (LC): - LC1: solution quality (does the solution solve the given task?) - LC2: code quality (are there any bugs or problems with the code?), and - LC3: documentation quality (is there a clear readme, does the code contain sensible comments?). # Deadlines, submission requirements, and penalties While most of the lab assignments are formative, your active participation is required and there are strict submission and peer-review requirements: - LC4: Each partner must fulfil each labs minimum submission requirements regarding use of Git, and Gitlab, file layout, and build instructions, before the lab deadline. Each assignments unique submission requirements will be listed in the corresponding lab manual. - LC5: You must participate in the peer-review tasks for the first three lab assignments, and complete these before the given peer-review deadlines. Failing to comply with these criteria will result in penalties on your final *course* grade: - After lab submission deadline: -1 grade point per day for not completing the assignments minimum submission requirements, with a maximum of -2 per assignment. Not enrolling in time before the lab automatically results in -2. - After peer review deadline: immediate -0.5 grade point for not fully completing the peer-review task (only applies to those enrolled for the lab). NB: - These criteria apply to both lab partners individually. If you comply with the criteria but your partner does not, only your partner will receive the penalty. - Penalties apply to your final course grade, so if you do not submit for one lab (-2 penalty) you could only complete the course with an 8 at best. - There are NO redos or extra assignment opportunities to remove a given penalty. You will have to redo the lab next year and comply with the deadlines then for a clean slate. # Exam There will be an individually-made digital exam, for which you will receive a grade (weights 75% for final grade, a minimum of 5.5 to pass). Questions are based on content discussed in the lectures, and on practical situations that you have encountered during the lab assignments. The exam will also require you to reflect on and refer to your own coding solutions for the final lab assignment, therefore to participate in the exam you *must* have participated in the final lab assignment. Practice questions will be provided before the exam so you can assess your own understanding first.
Department	3mE Department Cognitive Robotics

RO47004	Machine Perception	5

Responsible Instructor Prof.dr. D. Gavrila

Practical Coordinator R.M. Ensing

Contact Hours / Week x/x/x/x 0.6.0.0

Education Period 2

Start Education 2

Exam Period 2
3

Course Language English

Course Contents

This course provides an overview of machine perception techniques in robotics, with a focus on intelligent vehicles. Interested students outside 3ME (e.g. EEMCS, Civil Engineering and Aerospace Engineering faculties) with the proper background (see Prerequisites below) are encouraged to attend.

Introduction
 Machine perception in robotics
 Course organization
Sensor Overview (camera, radar, LiDAR, tactile)
3D Machine Vision
 Perspective camera model
 Extrinsic and intrinsic camera transformations
 Stereo vision
Radar Sensing and Processing
 Principles of FMCW Radar
 Distance, velocity and angle measurements
 Use in object detection
Lidar Sensing and Processing
 Point cloud representations
 Use in object detection
Tactile Sensing and Processing
 Piezoresistive, capacitive, piezoelectric and optical sensing
 Contact and slippage
 Shape extraction and elasticity
Visual Object Detection and Classification
 Detection vs. classification
 Object proposals
 Handcrafted features (e.g. HOG) & classification (e.g. linear SVM)
 End-to-end learning: neural networks, deep learning
 Performance metrics: confusion matrices, precision vs. recall, ROC curves
State Estimation
 Bayesian Filtering
 Kalman Filtering
 Particle Filtering
Object Tracking
 Data Association
 Track Management
Self-Localization & Sensor Fusion
 Absolute vs. relative localization
 Ego-motion compensation (e.g. odometry, ICP algorithm)
 Extrinsic sensor calibration
 Environment representations (grids, voxels)

The course has a substantial practicum component (about 65% of course work load), where learned concepts are put into practice by means of programming assignments (Python, using Jupyter notebooks).

Study Goals

After completing this course, students will be able to:
 explain the role of Machine Perception (MP) in Robotics, and describe possible applications
 explain the measurement principles of the relevant sensors
 explain the principles of well-established methods for low- to high-level sensor processing
 analyze a MP problem, consider available sensor and computational resources, and select the appropriate MP methods to apply
 write Python code in relevant frameworks to visualize data and implement MP methods
 perform MP experiments, evaluate the results, and draw sound conclusions

Education Method

 Lectures (2-4 hours per week)
 Practicum (2 hours per week)

Formative feedback: Practica
- Pairwise collaboration (alternating random groups)
- Pass/Fail (counts for knock-out criteria)
- No late submissions or resit (except due to motivated medical conditions)

Students can in principle run the software locally on their own computers (laptop/PC). However, we only provide limited support for setting up the necessary software environment (i.e. Anaconda). If the software environment cannot be installed properly (e.g. a few processors like M1 and ARM might not work), the student will need to revert to using the computers in the designated lab rooms.

Course Relations

This course, RO47004 Machine Perception *cannot* be selected in combination with ME41106 Intelligent Vehicles, due to the content overlap between the courses.

Literature and Study Materials

 Slides
 Hand-outs
 Jupyter notebooks

Prerequisites

 Basic linear algebra and probability theory
 Intermediate Python programming skills, e.g. Numpy, object oriented programming (classes, inheritance), modules, namespaces and scope.
 Recommended: Pattern Recognition / Machine Learning (RO47002, CSE2510, or equivalent)

It is possible to start the course with beginner Python skills if your aim is to take advantage of this course to also improve your programming expertise (a laudable aim, given its importance for both industry and academia). Be aware, however, that this catching up will likely result in an increased study load (e.g. 1-2 EC).

Assessment

Final practicum assignment (50%)

	- Individual assignment - Late submission: max. one day late for -1.0 grade point - Students having an initial grade of < 5.8 will be offered an opportunity for a remake to obtain a (maximum) grade of 5.8 for this component in the Q3 quarter. Final written exam (50%) - Written exam on campus. Knock-out criteria - At most one of the Pass/Fail practicum assignments is Fail - The minimum final practicum assignment grade is 5 - The minimum final written exam grade is 5 - The minimum final course grade is 6 The final practicum assignment will be evaluated on the computers in the designated lab rooms. If a student chooses to work using own computing equipment, the student needs to test the software on the designated computers before handing-in. Both final practicum assignment and final written exam results remain valid for one year. Each result can separately be transferred to the next academic year, upon request.
Special Information	For more information about this course, see the slides of the Introduction lecture from past academic year at http://intelligent-vehicles.org/teaching/courses/ . TU Delft students might also be able to view the lectures from past academic year on Collegerama.
Department	3mE Department Cognitive Robotics

RO47005	Planning & Decision Making	5
Responsible Instructor	Dr. J. Alonso Mora	
Contact Hours / Week x/x/x/x	0.6.0.0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course provides an overview of motion planning and decision-making techniques and their practical application in robotics. Planning and Decision-making are critical components of autonomy in robotic systems. These components are responsible for making decisions that range from path planning and motion planning to coverage and task planning to taking actions that help robots understand the world around them better. This course studies underlying algorithmic techniques used for planning and decision-making in robotics and examines case studies in ground and aerial robots, mobile manipulation platforms and multi-robot systems. The students will learn the algorithms and implement them in a series of programming-based projects. In particular, we will first cover the fundamentals of planning (workspace, configuration space, representation of obstacles and robot models). We will then cover a broad set of methods, which include reactive methods and feedback control; discrete, combinatorial and probabilistic planning; planning under differential constraints; planning under uncertainty; learning in planning; and planning for multi-robot systems. Finally, we will briefly touch upon task assignment and vehicle routing. Course structure: Introduction Fundamentals: 3D rotations, configuration space, algorithm properties Differential constraints: recap of modeling & control of manipulators, ground robots, aerial robots Graph-search fundamentals (depth/breadth first, Dijkstra, A*, heuristics) Combinatorial algorithms (cell decomposition, shortest-path roadmaps, motion primitives) Sampling-based algorithms (PRM, RRT, RRT*) Reactive algorithms: collision avoidance, potential fields, velocity obstacles Kinodynamic planning Trajectory optimization (fundamentals, global, local, Model Predictive Control) Planning under uncertainty (Markov Decision Processes, learning) Multi-robot motion planning (joint configuration, optimization-based, coverage) Task assignment (Hungarian, auction, linear program) Intro to vehicle routing	
Study Goals	After completing this course, students will be able to: explain the role of Motion Planning and Decision-Making (PDM) in Robotics, and describe possible applications identify different classes of robotic systems, the associated mathematical models for their kinematics, dynamics and motion and the relevant spaces for PDM, such as the configuration and state spaces describe and compare algorithms for motion planning and decision-making of mobile robots and multi-robot systems, including: reactive algorithms, optimization-based algorithms, combinatorial algorithms, sampling-based algorithms and learning-based methods analyze a PDM problem, consider constraints, objectives and select the appropriate PDM methods to apply design and implement motion planners of moderate complexity to solve motion-planning tasks and navigate mobile robots perform PDM experiments, evaluate the results, and draw sound conclusions present the findings of a PDM experiment clearly in a report using graphs and scientific English understand PDM experiments descriptions found in scientific literature, and reproduce results	
Education Method	Lecture hours/week: 4 Q&A sessions/week: 2 Practical assignments and project (home) hours/week: 6 Self-study hours/week: 5	
Books	Planning Algorithms, S. LaValle http://planning.cs.uiuc.edu/ And additional papers/references provided by the lecturer.	
Prerequisites	Linear Algebra Programming in Python Rigid body dynamics (recommended) Dynamics & Control (recommended) Probability Theory (recommended)	

Assessment	Practicum assignments (10%)
	 Project (40%) Collaboration in groups of 3 students
	   - No resit (repeat the following year)
	 Written Exam (50%) individual
	
	 Knock-out criteria
	   - Minimum grade for the project is a 5.0
	   - Minimum grade for the exam is a 5.0	

Department	3mE Department Cognitive Robotics	

RO47013	Control in Human-Robot Interaction	5
Responsible Instructor	Dr. M. Wiertlewski	
Instructor	L. Peternel	
Contact Hours / Week x/x/x/x	0.0.6.0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	

Course Contents

Robots are increasingly pervasive in our lives and need to safely interact with humans in unstructured environments. This course covers the design principles and control strategies to ensure a safe and intuitive interaction between humans and these collaborative robots. A particular attention is put to the fundamentals needed to build and control robotic systems that physically interact with humans.

The course is split between interactive theory lectures and hands-on practical assignments using haptic interfaces. During the practical assignments, the theory is applied: first during a guided scenario, then with an open-ended mini-project. The first two assignments will teach you how to program real-time haptic rendering and to implement a teleoperation loop. The mini-projet will cover how to implement haptic guidance and validate the approach with human subjects.

After the practical assignments, additional lectures expand the perspective beyond the fundamental theory to industrial and consumer oriented applications. The course ends on an overview of the current state-of-the-art and remaining open challenges of the human-robot interaction field.

THEORY SUBJECTS

Fundamentals of physical interaction
- Motivation and relevance
- Existing examples

Impedance control
- Motivation behind modulating impedance of a robot
- Impedance and admittance control principles
- Hybrid force/impedance controller design
- Manipulability and its impact on interaction
- Implemented examples

Teleoperation
- Applications: when and why?
- Types: unilateral and bilateral
- Scaling position commands and force feedback

Tele-impedance
- Actively adapt robot impedance to achieve the desired interaction
- Impedance-command interfaces
- Implemented examples

Force feedback with virtual environments
- Providing human with physical virtual sensations
- Haptic rendering of virtual impedances
- Contact and Friction

Control of haptic interfaces
- Study of the stability with the Virtual Wall problem
- Effect of sampling on stability
- Transparency and Z-width to evaluate performance of haptic devices

Surface haptics (touch panel applications)
- Vibrotactile
- Ultrasonic
- Electroadhesion

Evaluation of human robot interaction
- Measuring performance with Fitts and Steering Law
- Learning effects with haptic guidance

Ergonomics in human-robot collaboration
- Basic human biomechanics
- Ergonomic metrics (manipulability, joint torque, muscle fatigue)
- Implemented examples

Haptic guidance
- Virtual fixtures
- Dynamic guidance
- Shared control

Mechatronics of HRI devices
- Sensors, actuators and linkages
- Designing the mechanical impedance
- Trade-off between performance and workspace
- Serial elastic actuators and soft robotics

Future outlook
- State-of-the-art research
- Emerging technologies
- Conclusion

PRACTICAL ASSIGNMENTS

	Assignment 1: Haptic rendering (individual) - Implement virtual environment - Implement impedance/admittance control - Observe and analyze stability issues Assignment 2: Teleoperation (individual) - Design a hybrid force/impedance controller - Derive and visualize manipulability ellipsoid - Implement teleoperation and tele-impedance - Create a real-time UDP communication Assignment 3: mini-project (individual or by group of 2) Design and build a scenario to test the effect of haptic guidance on operators performance Analyze the results and evaluate the performance of haptic guidance Present the results to peers (and give feedback)
Study Goals	After this course, students will be able to: - explain impedance-based robot control theory and its advantages - apply the theory to different robots and tasks - analyze the implementation issues and control performance - create haptic feedback schemes to enhance human performance
Education Method	CONTACT HOURS - Lectures (2 hours per week on average) - Practical assignments (2 hours per week on average) FEEDBACK - Practical assignments (40%) (no resit) + individual reports - Exam (60%) individual - Homework individual self progress check Knock-out criteria: - Average of practical assignment grades needs to be a 5.5 or above - Minimum grade for the exam is a 5.5 - Average of both assignments has to be 6.0
Literature and Study Materials	- Slides (released shortly before lectures) - Scientific papers - Lecture recordings
Prerequisites	- Human-Robot Interaction (Q2 course) - Dynamics and Control (Q1 course) - Programming (Python)
Assessment	Summative assessment: - Exam - Reports on practical assignments and implementation analysis - Presentation of the final project Formative assessment: - Homework assignments Exam: The exam will be a computer exam in ANS on campus. In the case of COVID-19 restrictions, the exam will be held online (still in ANS). Grade calculation: -If and only if the exam grade is 5.5 or higher and the average assignment grade is 5.5 or higher, the grade will consist of 60% of the exam grade and 40% of the average score on the assignments. The final grade must be 6.0 or higher to pass the course. Assignment submission lateness criteria: <1 hour: -1 point <1 day: -2 points <2 days: -4 points <3 days: -6 points >3 days: no points given
Department	3mE Department Cognitive Robotics

RO47015	Applied Experimental Methods	5
Responsible Instructor	Dr. Y. Vardar	
Instructor	Dr.ir. D.A. Abbink	
Contact Hours / Week x/x/x/x	0.0.0.x	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Study Goals	After completing this course, students will be able to: independently design, execute and analyse a human factors experiment formulate a clear hypothesis in the field of human factors that can be tested experimentally, in a realistic amount of time with a realistic amount of means set up a methodologically sound experiment to test this hypothesis o select, adapt or build an experimental setup o gather reliable experimental data on test subjects o choose appropriate metrics o apply correct statistical techniques o choose the most relevant results, and construct the most clear figures to show them critically reflect on your own experimental findings communicate clearly through presentations and writing	
Education Method	Students will perform a mini-graduation project in six weeks, in small groups (2-3 students). They will independently design, perform, analyze and present a human factors experiment, guided by four supervisors.	
Literature and Study Materials	Scientific papers	
Assessment	The final grade is composed of the following sub grades 10% Presentation 1: Hypothesis 10% Presentation 2: Methods 10% Presentation 3: Analyzing Results (two main results figures, explain) 20% First Draft for Paper 50% Final Paper In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment can take place in a similar way.	
Elective	Yes	
Tags	Broad Design Group Dynamics/Project Organisation Group work Integrated Intensive Involved Lab Research Project planning / management Small groups	
Department	3mE Department Cognitive Robotics	

RO47019	Intelligent Control Systems	4
Responsible Instructor	Dr. C. Della Santina	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Fundamentals of Control theory Fundamentals of Coding in Python and MatLab	
Course Contents	Theory and application of intelligent control system i.e., the combination of machine learning and control theory. Topics: * Introduction to nonlinear control and ICS * Learning nonlinear dynamical systems (direct and inverse) * Sum of Basis functions * Artificial neural networks, deep learning, and other advanced neural architecture in the context of ICS * Gaussian processes * Reinforcement learning * Imitation learning * Adaptive control * Iterative learning control * Examples of applications	
Study Goals	At the end of this course, you will be able to: LO1 Explain the fundamentals of 'intelligent control' techniques - namely iterative control, Reinforcement learning, Model learning for control, etc. LO2 Implement 'intelligent control' techniques in Python. LO3 Reason on the expected performance of a given intelligent controller when applied to a given control problem. LO4 Propose an original, intelligent control scheme that can provide motor intelligence to a nonlinear and uncertain system (e.g., robot within a partially known environment).	
Education Method	Lectures and one assignment.	
Course Relations	Course title until 20/21 "Knowledge Based Control Systems" Course code until 21/22 "SC42050"	
Literature and Study Materials	Lecture notes: R. Babuska. Knowledge-Based Control Systems. Slides and other course material (video lectures, software, demos, notes) can be downloaded from BrightSpace.	
Assessment	The examination will be as follows: Practical assignment and written exam, open book. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be: Remote open book with several fraud prevention measures. In accordance with the master program regulations: * The students will have the opportunity for a resit if they fail the written examination. * The groups that are graded between 5 and 5.75 in an assignment, will have two weeks to integrate the assessors feedback. If this is done in a satisfactory way, they can be rewarded with a mark that is not higher than 6/10. The exact instruction for the addition will be announced on Brightspace.	
Tags	Artificial intelligence Complex Mathematics Intensive Lineair Algebra Mathematics Modelling Programming Project Proofs	
Department	3mE Department Cognitive Robotics	
Contact	Cosimo Della Santina, C.DellaSantina@tudelft.nl	

SC42056	Optimization for Systems and Control	3
Responsible Instructor	Prof.dr.ir. B.H.K. De Schutter	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	basic experience with Matlab	
Course Contents	Essentially, almost all engineering problems are optimization problems. If a civil engineer designs a bridge, then one of the main objectives is to obtain the cheapest design or the design that can be implemented most rapidly, where of course several specifications and constraints such as size, strength, safety, etc. have to be taken into account. When developing a new type of engine, we look for the most economical design, the cheapest design, or the design with the highest performance. A process engineer wants a production unit to deliver a final product of maximal quality, with minimal expenditure of energy or with maximal output flow. When composing a portfolio, a financial engineer tries to maximize the expected profits, subject to the given risk constraints. So we encounter optimization problems in almost every engineering field. How can we solve such an optimization problem? That is the topic that will be addressed in this course. We will in particular discuss several classes of optimization problems and next focus on how to select most efficient numerical algorithms to solve a given optimization problem. The examples and case studies of this course are primarily oriented towards systems and control. More specifically, the following topics will be treated: * Linear Programming * Quadratic Programming * Nonlinear Optimization * Constraints in Nonlinear Optimization * Convex Optimization * Global Optimization * Matlab Optimization Toolbox * Multi-Objective Optimization * Integer Optimization	
Study Goals	After this course the students should be able to select the most efficient and best suited optimization algorithm for a given optimization problem. They should have insight into the basic operation of some of the most important and relevant optimization algorithms. They should be able to reduce the complexity of the problem using simplifications and/or reformulations so as to augment the efficiency of the solution approach. Finally, they should be able to solve the resulting optimization problem in practice.	
Education Method	lectures + 2 assignments Depending on the corona measures, the lectures will be taught on-campus or online. Attending the lectures is not mandatory, provided the recorded lectures and/or lecture materials are the processed via self-study.	
Assessment	exam (individual, counts for 75% of the final marks) + 2 assignments (group assignments, assessed through written reports, count for 25% of the final marks) Important: Partial marks for the exam or the assignments do not carry over from one academic year to the next. Depending on the of measures resulting from COVID-19, the prescribed exam will either be an open-book online exam, or if the opportunity arises for examinations on campus, a closed-book written exam (fully closed-book, no calculators nor formulas sheets allowed).	
Permitted Materials during Tests	none	
Department	3mE Department Delft Center for Systems and Control	
Contact	questions are preferably asked via the Discussion forum on Brightspace or (for questions on the material of the lectures) during the online or on-campus lectures	

WI4771TU	Object Oriented Scientific Programming with C++	3
Responsible Instructor	Dr. M. Möller	
Contact Hours / Week x/x/x/x	0/2/0/0 & 0/4/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic programming knowledge in at least one programming language (e.g. Matlab, C/C++, Fortran)	
Course Contents	Introduction to C++ programming including new concepts from the C++11/14/17/20 standard. Object-oriented scientific programming.	
Study Goals	The student will be able to apply modern software design patterns and state-of-the-art programming techniques to implement (numerical) algorithms in C++ (using the latest standards C++11/14/17/20). The student will build up practical experience in turning a textbook (numerical) algorithm into an efficient and maintainable C++ code. The student will be able to develop larger software projects in teams and test the implementation systematically.	
Education Method	Hands-on lectures and practical sessions in Q2. Problem-oriented teaching approach starting from a concrete (mathematical) problem (taken from introductory numerical analysis course) and building up knowledge in C++/scientific programming up to the level required to solve the problem. Further self-study is required to build up in-depth practical experience. Implementation of final project (in groups of 1-3 students) starting mid Q2 and with submission deadline end Q2 / beginning Q3.	
Literature and Study Materials	Complete course material (slides and source code of demo applications) will be provided. Though not mandatory, the book "A Tour of C++" by Bjarne Stroustrup, C++ In-Depth Series, Addison-Wesley, ISBN-13 978-0-321-95831-0 gives a good overview of the covered material.	
Prerequisites	Basic programming knowledge in at least one programming language (e.g. Matlab, C/C++, Fortran)	
Assessment	The final grade of the course consists of the following components: - Homework programming assignments (10 in total) for each session during Q2 (1/3 of final grade). - Group programming project (group work) starting mid Q2 with submission deadline end Q2 (2/3 of final grade). Final grade calculation: (1/3 * homework assignments) + (2/3 * group programming project) but it is only valid when all homework programming assignments have been completed with sufficient results. Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Homework programming assignments: submit new set of homework - Group programming project: submit improved version Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	K. Cools	

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Mechanical Engineering**

ME-BMD Second Year Project: Select 1

ME51015	ME-BMD Internship / Research Assignment	15
Responsible Instructor	B. van Vliet	
Contact Hours / Week x/x/x/x	0/0/x/0	
Education Period	None (Self Study)	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The Internship consists of three months (1 quarter, full-time) of research or development work outside TU Delft to get working experience in another environment than TU Delft. This could either be a company, a medical center, research institute, or another university within the country or abroad.	
Study Goals	The primary goal of the Internship is to gain experience in working as an engineer in a professional environment. After an Internship, you should be able to: - reflect critically on your work, and on your personal and professional development. - reflect on how well the experienced working environment and work fits with your skills, interests, values, and character.	
Education Method	In your Internship, you undertake a project task defined in consultation with the host organization and your TUD supervisor (must be a professor, associate professor, or assistant professor). If you plan to do an Internship abroad, you must also consult the International Office (internationaloffice-3me@tudelft.nl). Internships are usually arranged via one of the staff members of the BioMechanical Engineering department (BMechE). The ME -BMD master track coordinator must give approval before the Internship is started, via the Internship Application Form, which can be found on the following website: https://www.tudelft.nl/en/student/faculties/3me-student-portal/education/related/student-forms/internship-forms/. In the Internship Application, in addition to the project definition, you also formulate a number of personal learning goals for your professional development. ===CHECK THE ME-BMD BRIGHTSPACE PAGE FOR MORE AND UP-TO-DATE INFORMATION===	
Assessment	The internship is assessed pass/fail. You are required to write a brief report about your work and your experience during the Internship. Some students write this in the form of a scientific paper. It will contain: 1) Internship report - About the contents 2) Internship reflection - About your learning goals	
Department	3mE Department Biomechanical Engineering	

TUD4040	Joint Interdisciplinary Project	15
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Project Coordinator	Ir. B.J.E. de Bruin	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Summary	JIP consists of three sets of activities: 1. The project which takes place at a company. Students are responsible for the project work (collaboration, integration of different perspectives, the content quality of their work). They plan their activities via a scrum method, are challenged to realise field trips, consult with experts and realise the necessary research work. The team keeps a Scrum log, run a blog for interaction with the outside world and every team member keeps a personal development log. 2. Meeting Professionals lectures and workshops about specialised topics, by academic staff or senior professionals from the companies involved. 3. Plenary meetings and design reviews for all the teams, company coaches and academic staff. At the meetings the students present their intermediate resp. final outcomes and plans, and are provided with feedback.	
Course Contents	The aim of the Joint Interdisciplinary Project is to prepare students to contribute to solving impactful technological challenges. The projects not only demand good engineering working knowledge but also experience with interdisciplinary and systems theory, and both knowledge and mindsets of innovation and entrepreneurial behavior. The project brief is provided by renowned companies like Airbus, Arcadis, etc. Teams of interdisciplinary student teams guided by a company coach and offered academic and industry expertise, are invited to realise an innovative problem solution to a complex problem and contributing to the sustainable development goals.	
Study Goals	1. Cognitive abilities attributable to interdisciplinary learning Demonstrate the ability to engage in perspective-taking; Develop structural knowledge pertaining to the problem; Integrate knowledge and modes of thinking drawn from two or more disciplines; Produce an interdisciplinary understanding of complex problem or intellectual question. 2. Scientific and intellectual development Capable to analyze scientific and societal consequences (economic, social, cultural, environmental) of the innovation; 3. Research and design capabilities Demonstrate engineering skills: technical skills, interpreting results, creativity, usability for company/institute; Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; 4. Collaboration and communication in an interdisciplinary team Demonstrate behavioral competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to respect different disciplines and adapt to different cultures); Show ability to write a technical report: structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (30 pages); Present work performed in a structured way through an oral presentation to their peers and customer. 5. Self-adjustment and reflection capabilities Plan and control the project efficiently considering resources and methodology; Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career. Cognitive abilities attributable to interdisciplinary learning; The ability to integrate (scientific and practical technological)knowledge from different disciplines to solve complex problems Scientific and intellectual development The capacity to evaluate the ethical, scientific and societal consequences of the proposed innovation Research and design capabilities The ability to create reasonable and relevant research or design , according to the academic standards of the involved disciplines Collaboration and communication in an interdisciplinary team Demonstrate behavioural competences and skills relevant for teamwork and effective communication with different stakeholders Self-adjustment and reflection capabilities To carry out regular reflections on professional and personal development and being able to improve upon those reflections Understand contemporary and societal issues in their work.	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Assessment	The assessment criteria pertain to the process, product and presentation and are assessed at an individual and team level, during 3 review sessions at three levels of accomplishment (each review assesses at a different level): Interdisciplinary work Scientific reasoning and ethical mindset Innovation process Presentation and communication Academic staff, supported by the input from company coaches, grade the students during the Problem Statement, the Midterm and the Final Review. The final group mark for JIP is differentiated per person and is based on: The team presentations - 30% The Problem definition, progress report and final report 30% The individual contribution to the team 20% The final individual reflection report 5% The Blog 5%	
Remarks	Project Outcomes (product, model, system) are presented at: 1. Problem Statement Review 2. Midterm meeting 3. Final Review 4. Symposium presentation	

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Mechanical Engineering**

ME-BMD Graduation Project Obligatory (45 ECTS)	
Introduction 1	The 2nd Master's year has a flexible structure, in which the Literature Research (10 ECTS) and the Master's thesis (35 ECTS) are compulsory for everyone. In a nominal situation this will start in the 2nd quarter of an academic year. In the first quarter, students can choose between: 1. 15 ECTS Joint Interdisciplinary Project; 2. 15 ECTS Internship or Research Assignment.

ME51010-20	ME-BMD Literature Research	10
Responsible Instructor	B. van Vliet	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	None (Self Study)	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	This course combines three elements. First and foremost (T1), the literature study itself. This is a preparatory study for your thesis project (ME51035). In practice, ME51010-20 and ME51035 together form a single overall project, conducted under the same supervisor. In addition (T2), you will give a colloquium presentation on the results of this literature study (usually combined with a description the approach you will take in your thesis project). Third (T3), you are required to attend at least 10 colloquium sessions (organized by the BMechE department).	
Study Goals	1. Knowledge, insight, and academic skills By the end of this course, you should be able to: - Search the literature systematically; - Formulate a research question; - Select representative scientific sources from several (for example, economic, ethical- environmental-, social-, health) perspectives relevant to the assignment; - Draw conclusions related to the research problem and give recommendations towards new research opportunities; - Write a literature review in academic English. 2. Transferable skills & interpersonal skills By the end of this course, you should be able to: - Demonstrate a scientific attitude; - Manage your own learning process (including time management).	
Education Method	Self study ===CHECK THE ME-BMD BRIGHTSPACE PAGE FOR MORE AND UP-TO-DATE INFORMATION===	
Assessment	Three tests: T1 Literature Report (grade) T2 Colloquium Presentation (pass/fail) T3 Colloquium Attendance (pass/fail)	
Department	3mE Department Biomechanical Engineering	

ME51035	ME-BMD MSc Thesis	35
Responsible Instructor	B. van Vliet	
Contact Hours / Week x/x/x/x	-/x/x/x	
Education Period	None (Self Study)	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	You will set up and carry out your own research project.	
Study Goals	The overall objective of this assignment is to demonstrate a sufficient academic level in the field of Biomechanical Design.	
Education Method	Self study with regular supervision from staff members and professor. ===CHECK THE ME-BMD BRIGHTSPACE PAGE FOR MORE AND UP-TO-DATE INFORMATION===	
Assessment	Thesis report, oral exam, and defence.	
Department	3mE Department Biomechanical Engineering	

ME-EFPT Track B. Energy, Flow and Process Technology

Program Coordinator	Dr. B.P. (Brian) Tighe T +31 15 278 1103 E B.P.Tighe@tudelft.nl
Introduction 1	There is widespread agreement that human activities, and specifically industrial processes, are sustainable if they promote development which meets the needs of the present without compromising the ability of future generations to meet their own demands. That, of course, poses formidable and urgent technical challenges. This Master's programme offers the motivated student the chance to play a fundamental role in this primal human enterprise. The knowledge acquired by specializing in EFTP gives students the technological understanding and skills (theoretical, numerical and experimental) they need to help develop the energy and process technologies of the next generation. Moreover, they become aware of the importance of sustainable development to our society and acquire the tools necessary to promote it. This track opens up opportunities to pursue a career either in industry, where design and development play a major role, or in academia, where science dominates. Most EFP graduates find process and energy-related jobs. The MSc programme is divided in parts, which are summarized in the "Curriculum" table.
Exit Qualifications	The exit qualifications for the master Mechanical Engineering can be found in the common section of this studyguide. For the track EFPT, the specific exit qualifications are: 1. Competent in the scientific discipline Mechanical Engineering Energy, Flow & Process Technology. A graduate in Mechanical Engineering Energy, Flow & Process Technology is able to: a) apply knowledge of fluid dynamics, applied thermodynamics, heat- and mass transfer to the disciplines covering the domain of energy technology, fluid flow, and process technology. 2. Competent in doing research. A graduate in Mechanical Engineering Energy, Flow & Process Technology is able to: a) apply knowledge of fluid dynamics, applied thermodynamics, heat- and mass transfer at an advanced level in theoretical and/or experimental research to solve problems and/or to generate new knowledge. 3. Competent in designing. A graduate in Mechanical Engineering Energy, Flow & Process Technology is able to: a) analyse, design, and evaluate systems in the area of energy technology, fluid flow, and process technology in particular for details of equipment for heat- and mass transfer; b) analyse technical designs related to equipment. 4. Considering the temporal and social context. A graduate in Mechanical Engineering Energy, Flow & Process Technology is able to: a) describe and implement sustainable development based on thermodynamic fundamentals.

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Mechanical Engineering**

ME-EFPT Obligatory courses (22 ECTS)

ME45001	Advanced Heat Transfer	4
Responsible Instructor	Dr.ir. J.W.R. Peeters	
Instructor	Dr. R. Delfos	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	A BSc-course in Fluid Dynamics and Heat Transfer or in Transport Phenomena. Basic math in 1D (Taylor series & ODE's) and in multiple-dimensions (vector analysis & PDE's) in the BSc-ME in 3ME or equivalent; basic understanding of Fourier series.	
Course Contents	In this course the concepts & mathematics of heat transfer in the engineering context are treated. Elementary understanding of the three modes of heat transfer: conduction, convection and radiation, will be briefly reviewed during the first two lectures. During the remainder of the course, the underlying physics will be emphasized and advanced mathematical formulations will be explained. A large focus in the course will be on the analysis of heat transfer in real-life integrated systems. Subjects in order of appearance: - A refresher on the underlying thermodynamics; energy, enthalpy, specific heats and phase change enthalpy. - A refresher on Conduction, Convection and Radiation. - Integral and differential energy balances in a 1-D and multiple-D continuum; absorption, reaction and dissipation as source terms. - Stationary conduction: cooling fins, multi-dimensional conduction and Laplaces equation; boundary conditions; analytical techniques & numerical techniques; relaxation. - Phase change as a boundary phenomenon; melting and solidification fronts; Jakob number & Stefan condition. - Instationary conduction: Fourier and Biot number; boundary conditions; analytical techniques & numerical techniques; stability criteria. - Forced & Free convection: Nusselt, Stanton, Prandlt & Peclet numbers; Analysis & the physics behind empirical correlations. The role of boundary conditions. - Radiation: radiative exchange between grey bodies, solar radiation, spectral characteristics, surface characteristics.	
Study Goals	More specifically: The student is able to 1. Distinguish between the different modes of heat transfer, and divide real-life systems into subsystems of elementary heat transfer modes in a qualitative and quantitative manner. 2. For all of the below; give the physical interpretation of contributors and terms in balances in words and in sketches. 3. Set up appropriate integral and differential energy balances for one- and multidimensional instationary conduction. 4. Justify and apply simplifications and define the appropriate boundary conditions, including problems containing phase changes, i.e. Stefan conditions. 5. Indicate mathematical solution strategies - both analytical and numerical, and apply those for standard geometries. 6. Distinguish between different modes of convective heat transfer, and distinguish between the different physical mechanisms underlying empirical correlations. Indicate implications when more detailed distributions of convective heat transfer are involved. 7. Estimate the magnitude of radiative heat transfer, distinguish between thermal and short-wave properties and spectral distributions, qualify and quantify the role of surface properties in real-life applications.	
Education Method	- Lectures (2x2 hours per week) including experiments & computer demonstrations. - After each lecture, book assignments, as well as custom assignments are given. Solutions will also be provided. - Weekly instruction hours (2 hours per week), during which students can discuss heat transfer problems with the instructors and/or during which instructors show how to solve an assignment. - Total number of contact hours: 6 hours per week.	
Computer Use	Computers are used for demonstrations of the lecture material during the course on the basis of TU Delft software. Students are invited to solve assignment problems using Matlab/Python/Maple - in some cases using given code, in others to start from scratch.	
Literature and Study Materials	Book: as a lead "Basic Heat and Mass Transfer" or just called "Mills" is used: A.F. Mills and C.F.M. Coimbra, 3rd edition, Temporal Publishing, ISBN 978-0-9963053-0-3 (available through MechEng study association 'Gezelschap Leeghwater'. The 2nd edition as by Pearson is out of print but still widely available; it is identical, but cheaper, and of lousy bookbinding quality. Derived versions like "Basis Heat Transfer", or "Heat Transfer" by Anthony Mills are nearly identical. Alternative books on heat transfer from your Bachelor's course (including Cengel & Incropera) can be used as well, but you might want to check with the teacher; just bring it to the first Office Hour if it's not standard. Lecture slides, short videos, Matlab scipts & other study materials required will be provided via Brightspace.	
Assessment	The assessment consists of a hand-written closed-book exam, which you will have to complete in a pre-determined timeslot of 3 hours. You are allowed to bring the equation sheet posted on Brightspace; you are also allowed to make your own notes on the equation sheet itself.	
Permitted Materials during Tests	Only a non-programmable non-graphical calculator like Casio fx-82 MS is accepted. One (1) equation sheet of 2 sides A4 can be taken and/or is provided.	
Department	3mE Department Process & Energy	

ME45042	Advanced Fluid Dynamics	5
Responsible Instructor	Dr. A.J.L.L. Buchner	
Instructor	Dr. D.S.W. Tam	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Expected prior knowledge	Undergraduate level course in thermo-fluids (WB2542) Knowledge of vector analysis, multivariable and differential calculus is essential for this course (WBMT1050, WBMT2048)	
Course Contents	This course surveys the principal concepts and methods of fluid mechanics. Topics include control volume analysis for mass and momentum, the derivation of Navier-Stokes equations from mass and momentum conservation for incompressible flows, boundary conditions at fluid/fluid and fluid/solid interfaces, dimensional analysis, interfacial tension, inviscid flows, circulation and vorticity, lift and drag, boundary layers, lubrication theory, classical instabilities and waves. Classical solutions of the Navier -Stokes equations are derived and Concepts are illustrated through practical examples from engineering, geophysics and biological fluid dynamics.	
Study Goals	At the end of this class, the student is able to formulate and to use the fundamental governing equations of incompressible, laminar fluid mechanics in classical flow regimes, and to apply these to a new problem in flow technology. More specifically, the student must know how to: 1. formulate the conservation equations for mass and momentum 2. derive the governing equations of motion for an incompressible flow. (Starting from the conservation equations for mass and momentum, the constitutive equation for a Newtonian fluid, and the appropriate boundary conditions at a fluid/solid and fluid/fluid interface) 3. compute the stress tensor and the aero/hydrodynamic forces on a mechanical system 4. derive the vorticity equation and the consequence on the generation and transport of vorticity in 2D (Kelvin circulation theorem) 5. perform a dimensional or scaling analysis to identify the dominant terms in the governing equations. Also, use the non-dimensional numbers introduced in class to determine the flow regime (Reynolds, Strouhal, Froude, Bond, Capillary, Weber) 6. simplify the Navier-Stokes equations for the case of an inviscid flow (the Euler equations), a boundary-layer, a unidirectional flow, a creeping (Stokes) flow and in the lubrication limit. 7. apply appropriately the different forms of the Bernoulli equation The student must be able to derive classical solutions to the Navier-Stokes equations discussed in class. The student must be able to determine whether these classical solutions can be applied to a new problem and use them to compute the flow fields. These solutions include: 8. For ideal flows: a source, a sink, an irrotational vortex, a dipole, corner flows and the superposition of any combination of these using complex potentials (e.g. the 2D irrotational flow around a cylinder and the interpretation of Lift (Joukowski theorem) and Drag (d'Alembert Paradox)) 9. 3D axisymmetric irrotational flow around a sphere for steady and unsteady flows (added mass) 10. For viscous flows: Unidirectional laminar flows, Couette and Poiseuille flows, flow in a lubrication layer 11. Blasius solution for the boundary layer over a flat plate 12. Waves and instabilities in fluid mechanics, gravity waves, Kelvin-Helmholtz instability When facing a new problem in fluid dynamics, the student should be able to: 13. Write all the relevant boundary conditions 14. Use the appropriate formulation of mass and momentum transport (at the control volume level or at the differential level i.e. Navier Stokes equations) 15. Simplify the governing equations using a dimensional/scaling analysis 16. Solve the governing equations for simple flows, when analytically possible 17. Use well-known solutions derived in class to model flows appropriately for the new problem. 18. Model the effect of aero/hydrodynamic forces on the dynamics of a simple mechanical system	
Education Method	Lectures: 3 hours/week Recitations: 1 hour/week Office Hours: 1 hour/week	
Books	Fluid Mechanics Sixth Edition Kundu, Cohen and Dowling	
Assessment	Midterm Written Exam (35%) Final Written Exam (65%)	
Department	3mE Department Process & Energy	
Contact	email: a.j.buchner@tudelft.nl	

ME45160	Advanced Applied Thermodynamics	5
Responsible Instructor	Dr. B.P. Tighe	
Instructor	Dr. O. Moultos	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	EFPT track	
Expected prior knowledge	Thermodynamics at the level of a BSc in Mechanical Engineering	
Course Contents	- Review of fundamentals of thermodynamics, including thermodynamic potentials, Maxwell relations, and the 1st and 2nd Law - Equilibrium in one-component systems: equations of state, chemical potential, fugacity, Clausius-Clapeyron, etc. - Mixtures: ideal gas mixtures, partial molar quantities, Gibbs-Duhem, vapor- liquid equilibrium, activity coefficient models, and osmosis - Model equations of state - First introduction to statistical mechanics	
Study Goals	After this course, you will be able to calculate phase and material properties in thermodynamic equilibrium for multi-phase and/or multi-component mixtures under ideal and non-ideal process conditions. You will be able to use equations of state and activity coefficient models, and to calibrate and validate those models using thermodynamic data.	
Education Method	Lectures, optional homework, review session	
Computer Use	homework assignments require scientific computing tools such as python, Matlab (or comparable) and basic programming methods such as for loops	
Books	Holyst Poniewierski	
Assessment	Written final exam; open book	
Department	3mE Department Process & Energy	

ME45165	Equipment for Heat & Mass Transfer	5
Responsible Instructor	Prof.dr. K. Hooman	
Instructor	Dr. H.B. Eral	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	Advanced reaction & separation systems Process plant design	
Expected prior knowledge	ME4500 (Advanced heat transfer)	
Summary	The course covers basic principles and methods required for the design and selection of heat and mass transfer equipment (for instance heat exchangers and thermal separation processes). This allows for the integration of heat & mass transfer principles to design process equipment.	
Course Contents	Rating and sizing of heat and mass transfer equipment such as heat exchangers, distillation, absorption, tripping, crystallization, adsorption, extraction and membrane separations will be covered. In particular: classification of various separation and heat transfer processes, the separation efficiency in single and multi-stage processes (and the influence of the process conditions and contacting modes on the separation). design procedures of different separation processes based on limitations of the thermodynamic equilibria. quantification of the influence of process condition on the design procedures of different separation processes based on limitations of the thermodynamic equilibria. characteristics of equipment for various separation processes the role of mass transfer in the sizing and design of the separation equipment. methods for the design and dimensioning of separation equipment. the use of computer tools for the design of heat exchangers and separation processes	
Study Goals	At the end of the course, the student will be able to rate and size heat and mass transfer equipment. In particular the student: 1. Can calculate overall heat transfer coefficient for heat exchangers; 2. Can select heat transfer equipment for given application; 3. Can apply effectiveness and Number of Transfer Units method to design heat exchangers; 4. Can apply pinch technology and heat integration for heat exchanger network. 5. Can describe and formulate thermodynamics equations and models for the selection and design of thermal separation processes, 6. Comprehends the role of mass transfer in the dimensioning of equipment for separation processes 7. Can determine the separation efficiency for single and multi-stage separations and can quantify the influence of the process conditions and the contacting modes on the separation. 8. Can use computer based process simulation tools to analyze and design separation equipment	
Education Method	Lectures, assignments, self study, team work, tutorials	
Computer Use	Simulation of performance of distillation trays Standard engineering packages including ASPEN	
Literature and Study Materials	Course material: 1: H.B.Eral, A.De Haan, B.Schuur, "Industrial Separation Processes" Second Edition, De Guyter 2020 ISBN-13: 978-3110654738 2: Henley, E.J., Seader, J.D., and Roper, D.K. Separation Process Principles, third edition (or 4th Edition), International student version, J. Willey & Sons, 2011. 3: VDI Heat Atlas (2010) ISBN 9783540778776, Springer. 4: Course sheets / relevant papers 5: R K Sinnott, Gavin P Towler5: Chemical engineering design, Coulson & Richardson's Chemical Engineering Series; Butterworth-Heinemann, an imprint of Elsevier: Kidlington, Oxford, United Kingdom, 2020. 6: Kays, W. M.; London, A. L. Compact Heat Exchangers, 3rd ed.; McGraw-Hill: New York, 1984. 7: Kroger Detlev G. Air-Cooled Heat Exchangers and Cooling Towers; Penwell Corp: Tulsa, Okl., 2004. Further reading Incropera, Dewitt, Fundamentals of heat and mass transfer, 1985, John Wiley and Son Geoffrey F. Hewitt, "Heat Exchanger Design Handbook", ISBN Online: 978-1-56700-422-9, Imperial College, England, UK	
Assessment	Formative: In-class exercises/quizzes Summative: Group project report & presentation (75%) Individual exam (25%)	
Permitted Materials during Tests	Course books and course handouts	
Percentage of Design	80	
Design Content	Thermohydraulic design of heat exchangers; heat exchanger network, dimensioning pieces of equipment used in the abovementioned processes	
Department	3mE Department Process & Energy	

ME46007	Measurement Technology	3
Responsible Instructor	Dr.ir. G.E. Elsinga	
Instructor	Dr.ir. M.J. Tummers	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Summary	This master level course covers fundamental aspects of measurements useful for a mechanical engineer. It covers measurement of various physical quantities, signal conditioning, error analysis and statistical analysis of the data. A bachelor degree in mechanical, electrical, aerospace, industrial engineering or applied physics is considered prerequisite.	
Course Contents	topics: - General introduction to measurements - Measurement system behavior - Hypothesis testing - Uncertainty analysis - Sensor probes: variety of experimental techniques - Signal conditioning - Data acquisition and processing - Imaging and image processing - 3D techniques	
Study Goals	The general aim of this course is to provide students with a background in the theory of engineering measurements. At the end of the course, the students should be able to (1) analyze measurement systems and devices using the laws of mechanics and general physics, (2) evaluate the accuracy, sensitivity, spatial and temporal resolution of a measurement, (3) analyze and interpret datasets that contain errors, (4) design a measurement scheme and signal conditioning in order to enhance the significance of the results.	
Education Method	The main learning methods include lectures, take-home assignments and a textbook. The lectures cover the different topics of the course. (In case of unforeseen circumstances or measures resulting from COVID-19, the lectures will be replaced by pre-recorded videos of the lectures and/or live online-sessions. Such changes will be communicated on Brightspace.) Lecture slides will be made available on Brightspace. The recommended textbook is given below, however, you are encouraged to refer other books as well. Additional reading material, if necessary, will be made available on Brightspace. Assignments will be provided to practice the relevant problems and the concepts discussed in the class. They will be made available on Brightspace. The solutions to the assignments will be discussed in the class. The course ends with a written exam.	
Literature and Study Materials	The prescribed books will be followed during the course. Lecture slides will be made available on Brightspace. Additional study materials will be provided on Brightspace if necessary.	
Books	Theory and design for mechanical measurements, 6th edition by Richard S. Figliola & Donald E. Beasley	
Assessment	The examination will be as follows: Closed book written exam. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be: Remote (online) open book with several fraud prevention measures.	
Department	3mE Department Process & Energy	

ME-EFPT Track courses: Select at least 2 courses

ME45030	Turbulence	5
Responsible Instructor	Prof.dr.ir. J. Westerweel	
Instructor	Dr.ir. G.E. Elsinga	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Advanced Fluid Dynamics (or equivalent)	
Course Contents	The course provides an introduction to the physical aspects of turbulent flows.	
Study Goals	(1) basic equations that describe turbulence; (2) transition to turbulence; (3) turbulence closure models; (4) energy cascade; (5) turbulent dispersion; (6) selected topics in turbulence	
Education Method	lectures and instructions	
Literature and Study Materials	Turbulence (2016) by Nieuwstadt, Boersma & Westerweel	
Assessment	Written exam	
Tags	Fluid Mechanics Mechanics Numeric Methods Physics	
Department	3mE Department Process & Energy	

ME45134	Process and Power Plant Design	4
Responsible Instructor	Prof.dr.ir. W. de Jong	
Instructor	Dr. L.L. Cutz IJchajchal	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Design of equipment for mass and heat transfer Thermodynamics of mixtures and phases separation	
Course Contents	This course provides the fundamentals for conceptually designing processes of chemical and energy conversion plants, where basic unit operations such as reactions, energy conversion, heat and mass transfer, separations, etc. are designed, modeled and optimized. The conceptual process design involves the design of the flow sheet, creating the topology of the process by the optimizing the sequence of reactors, separation units and energy conversion systems to achieve the required product specification in a sustainable and economically feasible way. In the course the students learn how to select, sequence and model the different unit operations as well as the selection and rough sizing of the required equipment. In addition the students will learn how to integrate the different units into a process flow sheet, and how to model, optimize and rank of the different design alternatives. During the course the students get a hands-on experience with modern process design tools (a.o. Aspen Plus), which will be used in a larger assignment in which a conceptual design of a sustainable, and (if possible) economically feasible process will be setup and simulated. The course includes the following elements: Intro; context and scope of conceptual process design; Modelling the thermodynamics of mixtures Modelling of key unit operations Hierarchical approach (Douglas method) to conceptual process design o reaction and separation systems o homogeneous systems o heterogeneous systems Simulation and optimization Superstructures Costing and project evaluation Flow-sheet design using Aspen Presentation (pre-recorded) of the case studies	
Study Goals	The student can conceptually design and optimize chemical process plants. More specifically, the student must be able to: 1. describe the context and scope of a conceptual design of a chemical plant, 2. model and design the most important unit operations of chemical/energy conversion process plants, 3. make a conceptual process design using the hierarchical Douglas approach for process design 4. select, sequence and model the different unit operations in the flowsheet of a chemical process plant and determine the main dimensions of the required reaction/separation equipment, 5. apply process flowsheeting tools for the design and simulation of process flowsheets for a chemical plant including the recycle streams 6. Use the process simulation tools to evaluate, optimize and rank the different design alternatives for a process.	
Education Method	Lectures (plenary and working), assignments, pre-recorded presentations and oral examination	
Literature and Study Materials	Lecture slides Book (selected chapters): Robin Smith Chemical Process Design and John Wiley & Sons, U.S.A. (either first or second edition)	
Assessment	The examination will be as follows: Oral exam [50%, must be 6 or higher] + two sets of (report + recorded presentation) [10+40%]. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be: exactly the same. The assessment is Covid-19 independent.	

ME45155	Computational Fluid Dynamics for Engineers	5
Responsible Instructor	Dr.ir. M.J.B.M. Pourquie	
Instructor	Dr. P. Simões Costa	
Instructor	Prof.dr. R. Pecnik	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	WB2542 (thermofluids), ME45042 or ME45043 (advanced fluid dynamics), WBMT2049 T2 and T3 or similar (basic numerical analysis), some elementary programming skill (python, matlab or any other)	
Course Contents	This is a course on computational fluid dynamics related to process and energy systems. Some elementary knowledge of numerical methods (discretisation of a derivative, numerical solution of ordinary differential equations) is assumed. See expected prior knowledge. In general the course will cover: (1) general conservation equations for mass, heat and momentum transport (Navier-Stokes equations) (2) numerical solution of multi-dimensional, stationary and time-dependent problems as encountered in low-speed CFD applications; (2.1) numerical schemes for temporal discretization; (2.2) numerical schemes for spatial discretization; (2.3) imposition boundary conditions (3) turbulence modeling (4) wave equation (5) numerical methods for high-speed (compressible) flows (6) high performance computing	
Study Goals	After learning the content of the course the student will have the following capabilities: (1) Describe the two most popular methods in commercial CFD, finite differences and finite volumes (2) solve simple demonstrative problems in fluid flow and heat transfer by programming them in python or matlab, using finite differences and finite volumes (3) recognize the effects of numerical methods on the solution, such as numerical diffusion and numerical dispersion and to explain how to make these effects smaller (4) recognize numerical instability, to list several ways to avoid it and to analyze stability of simple methods analytically (5) solve fluid flow and heat transfer problems with the commercial CFD package Fluent or the open source CFD package OpenFOAM, which includes the following: set up a calculation, generate a simple grid, set appropriate boundary conditions, choose suitable discretisation, interpret and validate the results. Alternatively, in some cases, instead of using a commercial package, students can decide to write and validate their own code which they will use for calculating a fluid mechanics problem.	
Education Method	Lectures, practical exercises (2x2 hours per week)	
Literature and Study Materials	Course material: (1) Sheets/handouts (2) J.H. Ferziger and M. Peric, Computational methods for Fluid Dynamics, Springer Verlag. (3) P. Moin, Fundamentals of Engineering Numerical Analysis, Cambridge University Press, 2001. References from literature: (1) C. Hirsch, Numerical computation of internal and external flows, Volume I Fundamentals of numerical discretization, Volume II Computational methods for inviscid and viscous flows, Chicester, Wiley & Sons, 1988, 1990 (2) C.A.J. Fletcher, Computational techniques for Fluid Dynamics, Volume I Fundamental and general techniques, Volume II Specific techniques for different flow categories, Berlin, Springer, 2-nd ed. 1991. (3) H.K. Versteegh, W. Malalasekara, An introduction of computational fluid dynamics. The finite volume method. Second edition. Pearson Education.	
Assessment	Written exam + assignment with report; written exam may be online type depending on circumstances	
Department	3mE Department Process & Energy	

ME45230	Separation techniques for renewable processes	5
Responsible Instructor	H. Bazyar	
Instructor	M. Ramdin	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basics of - Thermodynamics - Heat and Mass transfer	
Course Contents	- Classical reaction and separation techniques (basics): We will briefly mention the available methods of separation in gas-liquid and liquid-liquid systems such as distillation, extraction, absorption/desorption, membranes, etc. This is a quick repetition of some parts of the course Equipment for heat and mass transfer. You are not required to follow this course. But it will be beneficial. Problem definition with these state-of-the-art techniques as well as water and climate crisis. - Renewable process and design of gas-gas separation/purification: based on membranes for gas separation (polymeric, inorganic, etc.), Electrochemical hydrogen separation, direct air capture and post-combustion CO2 capture techniques based on adsorption, absorption, etc. -Renewable process and design of liquid-liquid separation/purification: Electrochemical conversion of CO2 (electrolysis), extraction of chemicals (fuels) from CO2 electrolysis processes, bipolar membrane based electrodialysis for ammonia production, as well as CO2 capture, Capacitive deionization (CDI) for separation of ions from liquids. - Renewable process and design of gas-liquid separation/purification: porous membrane contactors for removal of gases from liquids, i.e., Ammonia/CO2 removal from product/waste streams, controlled dissolution of gases in liquids (oxygenation). - Process design and techno-economics for renewable reaction and separation systems: mass and energy balance, sizing, and costing of the separation units, accounting for CO2 emissions, economic evaluation (net present value, levelized costs, payback time).	
Study Goals	By the end of this course, you should be able to: LO1: Explain separation processes for gas-gas, gas-liquid, and liquid-liquid systems, using absorption, membrane separation and electrochemical processes LO2: Assess the techno-economics and environmental impacts of these separation techniques using economic analysis and mass/energy balance LO3: Employ lab-based set-ups for production of renewable fuels using these separation techniques and techno-economic analysis	
Education Method	Lectures, assignments (practical or theoretical), final exam Course materials: lecture slides, published articles, books, videos Books: 1- Membrane technology and applications (by Richard W. Baker) 2- Membrane separation processes, Theories, problems, and solutions (by A. Fuzi Ismail, T. Matsuura) 3- Plant design and economics of chemical engineers (by M. Peters and K. Timmerhaus) Other required books will be introduced during the course.	
Literature and Study Materials	Some parts of these book will be also used: - Separation process essentials by Alan M. Lane - Multidisciplinary advances in efficient separation processes by Chernyshova, Ponnurangam and Liu	
Assessment	A combination of a written exam (offline or online) and design projects. Minimum final grade is 6.0. The written exam is closed book (with a handwritten formula sheet) and accounts for 50% of the final grade. The design projects account for the other 50% of the final grade. You will work in groups for the design projects which are performed in laboratory under the supervision of an experienced person. At the end of the design projects, each group should hand in a report and present their results.	
Department	3mE Department Process & Energy	

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Mechanical Engineering**

ME-EFPT Elective Courses

AE4117	Fluid-Structure Interaction	4
Responsible Instructor	Dr.ir. A.H. van Zuijlen	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Fluid-Structure interaction is a topic that covers many important and complex problems in engineering where the interaction between the mechanical behaviour of a solid structure is significantly influenced by surrounding fluids (water, air, etc) and where, in turn, the aero/hydro-dynamic forces are modified by the deformation of the structure. Although it was pioneered by aeronautics engineers to study the static and dynamic deformation of wings under aerodynamic forces (aeroelasticity), fluid-structure interaction analysis involves also the description of interaction phenomenon in constructions (e.g. wind induced vibrations), vibro-acoutics, blood flow in elastic arteries or ink flow in an actuated printer head. In the past, many semi-analytical approaches were developed to describe fluid-structure interaction. Today, complex problems interaction problems are investigated using engineering codes that couple structural models to fluid models. In this course, we will recall the basics of fluid and solid mechanics and discuss some important numerical issues appearing when coupling fluid-structure models. In particular we will shortly introduce the Finite Volume Discretization of the fluid, discuss the expression of the fluid equation on moving meshes (Arbitrary Eulerian Lagragian formulation) and discuss time integration issues of the coupled problem. We will go in more details in discussion the issues of time-integration of the coupled problem. Also the issue on sharing forces/displacements across the interface between the fluid and the structural mesh will be handled in more details.	
Study Goals	In this course you will learn the modeling philosophies and computational techniques required to set up and simulate fluid-structure interaction (FSI) problems. The course focuses on the algorithmic aspects and provides the necessary background to choose (and motivate) the appropriate methodologies to simulate an FSI problem using a partitioned solution strategy in a robust and efficient manner. After the course you will be able to: 1. Derive the Navier-Stokes equations to describe fluid flow, 2. Explain the concepts of Eulerian, Lagrangian and Arbitrary Lagrangian-Eulerian reference frames and relate them to the governing equations, 3. Derive, apply and interpret the (Discrete) Geometric Conservation Law, 4. Formulate the FSI problem in terms of dynamic and kinematic interface (coupling) conditions, 5. Explain the different approaches to impose the coupling conditions when dealing with spatial and temporal discretization, 6. Give an overview of mesh deformation methods and their properties, 7. Explain basic concepts of time integration methods (implicit vs. explicit schemes and multi-step vs. multi-stage schemes), 8. Apply time integration on a coupled problem using a partitioned approach, 9. Motivate the choice for a specific (mesh deformation/interpolation, partitioning) method for a given application,	
Education Method	Lectures (2 hours per week), group assignments	
Computer Use	Use Matlab for assignment and illustration.	
Literature and Study Materials	Course material: Lecture notes (available through Brightspace) References from literature: Structural acoustics and vibration; mechanical models, variational formulations and discretization, R. Ohayon, C. Soize, Academic Press, 1998, ISBN 0-12-524945-4 A modern course in aeroelasticity, Earl H. Dowell, Kluwer Academic Pub.,1995, isbn 0-7923-2788-8 Fluid-Structure interaction: applied numerical methods, H. Morand. R. Ohayon, Wiley ed., 1995, isbn0-471-94459-9	
Assessment	Written report & oral examination	

AE4140	Gas Dynamics	3
Responsible Instructor	Dr.ir. F.F.J. Schrijer	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2 none	
Course Language	English	
Expected prior knowledge	AE2130-III Compressible aerodynamics (chapter 7 - 12 of Fundamentals of Aerodynamics by J.D. Anderson) WI3150TU Partial differential equations Knowledge of computer programming (Python, Matlab or similar) in order to successfully complete the tasks	
Course Contents	- How to apply the basic laws of mechanics and thermodynamics to describe compressible flows. - How can we properly describe a compressible flow field using the Euler equation + jump relations. - Understand the concept of characteristics and invariants in the context of linear and non-linear flows. - How to apply characteristics for non-linear flows for the computation of isentropic unsteady flows. - Understand the relation between shock waves and characteristics. - Application of Hugoniot and Poisson curves to solve a Riemann problem. - Definition of characteristics for 2D steady flows, and similarity with 1D unsteady flows (time-like and space-like). - How can use the method of characteristics and method of waves to compute a 2D compressible flow field. - Investigate the effect of viscosity and heat transfer in a 1D flow (Fanno and Rayleigh flow).	
Study Goals	At the end of this course, the student will be able to: - Understand aerodynamic concepts and apply aerodynamic theory for compressible flows. - Apply the fundamental equations of fluid mechanics and thermodynamics to describe compressible flows; derive the governing equations for compressible flow and discuss the terms. - Derive the jump relations for the Euler equations and describe their relation to shock waves. - Discuss the role of entropy in combination with the jump relation for a correct description of a flow field. - Explain the concept of characteristics and invariants for 1D unsteady and 2D steady flows and how to use them for flow field computations (MOC). - Understand the role of characteristics in shock wave formation, elaborate on the theory of simple waves. - Derive the equations governing 1D flows through channels and nozzles in presence of viscosity and heat transfer. Explain the physical phenomena and processes that occur.	
Education Method	Lectures and self study	
Literature and Study Materials	Reader (can be dowloaded from Brightspace) Further recommended literature M.J. Zucrow, J.D.Hoffman, Gasdynamics.- vol. 1, 1976 ISBN 047198440X. John D. Anderson jr., Modern Compressible Flow with Historical Perspective 2nd edition MacGraw-Hill 1990; H.W. Liepman, A. Roshko, Elements of Gasdynamics, Dover Publications, 2001	
Prerequisites	Compressible aerodynamics Fundamentals of Aerodynamics by J.D. Anderson ch 7 - 10	
Assessment	Take home tasks + oral examination	

ME45025	Introduction to Multiphase Flow	5
Responsible Instructor	Dr.ir. W.P. Breugem	
Instructor	Prof.dr.ir. R.A.W.M. Henkes	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	A multiphase flow is the simultaneous flow of two or more immiscible phases of matter like gasses, liquids and/or solids. Such flows are important in many environmental and industrial processes and equipment. Examples are gas and oil transport, dredging, steel making, refineries and chemical plants, sediment transport in rivers, cloud formation, etc. This course gives a broad introduction to the fundamental fluid mechanics of such flows. Topics include: classification of multiphase flows into dispersed and separated multiphase flows, relevant dimensionless numbers, stability of gas/liquid two-phase flows, spatial/statistical averaging of the Navier-Stokes equations and continuum modelling, flow pattern maps and mechanistic modelling of multiphase pipe flows, physics of particles (droplets, bubbles, solids), dynamics of particle-laden flow, effects of turbulence, experimental techniques and computational methods for multiphase flows. Examples will be given of the use of multiphase flow theory in industrial applications.	
Study Goals	At the end of this course the student is able to: 1.a classify multiphase flows into separated and dispersed multiphase flows. 1.b formulate appropriate interfacial conditions for a fluid/fluid and a fluid/solid interface. 1.c formulate relevant dimensionless numbers for multiphase flows. 1.d explain the concept of dynamic similarity of multiphase flows. 1.e reflect on scaling choices when designing a laboratory experiment or setting up a numerical simulation. 2.a apply a linear stability analysis to two-phase flows such as for instance a liquid jet in a gas (Rayleigh-Plateau instability) and two horizontal fluid layers in relative motion with each other (Kelvin-Helmholtz instability). 2.b give physical mechanisms for the transition of stratified to non-stratified (bubble or slug) multiphase flows in both (tilted) horizontal and vertical pipes. 3.a apply averaging methods (such as statistical, spatial and temporal averaging methods) to the Navier-Stokes equations for describing the macroscopic behavior of multiphase flows. 3.b formulate appropriate closure models needed to solve the averaged Navier-Stokes equations. 4.a classify gas/liquid pipe flows into stratified, slug and bubble flows, dependent on the superficial gas and liquid velocities. 4.b apply simplified 1D (cross-section averaged) models to each of the flow regimes. 4.c formulate appropriate closure models in order to compute the streamwise pressure drop and liquid holdup in two-phase (gas/liquid) pipe flow. 5.a classify dispersed multiphase flows into flows with solid particles, gas bubbles and/or liquid droplets. 5.b describe similarities and differences between the dynamics of solid particles, gas bubbles and liquid droplets. 5.c apply the Newton-Euler equations of motion to an analysis of the dynamics of solid particles. 5.d describe the settling of a solid sphere under gravity in a stagnant fluid in terms of the Archimedes number and the fluid/solid mass density ratio. 5.e describe possible interactions between solid particles and turbulence of the carrier phase. 6.a explain changes in the shape of a rising gas bubble in a stagnant fluid dependent on the Reynolds number, Eotvos number and Haberman-Morton number. 6.b estimate the terminal rise velocity of a bubble in a stagnant liquid from the equation of motion and an appropriate model for the drag. 6.c give physical mechanisms for the breakup of bubbles/droplets in turbulence. 7. describe the dynamics of some particle-laden flows such as sediment transport through a pipe and/or sediment transport in rivers. 8. describe the merits and shortcomings of different experimental methods for multiphase flows. 9. describe the merits and shortcomings of different numerical methods for simulating multiphase flows. 10.a give examples of multiphase flow in industrial processes. 10.b describe the types of models used to design multiphase industrial processes.	
Education Method	Oral lecture 0/0/2/2; Plenary discussion of homework assignments 0/0/2/2	
Assessment	Written exam. The student can earn bonus points by making homework assignments. The final grade is the sum of the grade for the written exam and the bonus points from the homework.	
Department	3mE Department Process & Energy	

ME45050	Microfluidics: Applied Theory and Lab	3
Responsible Instructor	Dr. D.S.W. Tam	
Instructor	H. Bazyar	
Instructor	M.K. Ghatkesar	
Instructor	Prof.dr. U. Staufer	
Contact Hours / Week x/x/x/x	HC 0.0.2.0 + PT 0.0.0.4	
Education Period	3 4	
Start Education	3	
Exam Period	3 4 5	
Course Language	English	
Expected prior knowledge	Bachelor degree level knowledge in fluid mechanics (WB2542). The measurement techniques course (ME46007) is recommended but not mandatory.	
Course Contents	In this master level course, you will be introduced to the important theoretical concepts of fluid flow (low Reynolds number hydrodynamics, transport phenomena on the micron scale, electrical double layer, electroosmosis and electrophoresis) relevant to the general design and fabrication of microfluidic devices and circuits (e.g., channels, valves and flow controllers) . You will apply these general concepts in a group project, which includes the design, the fabrication, and the testing of a microfluidic device (e.g., droplet generator, liquid steering in channels with valves/pumps, flow rate measurement, visualization of laminar and parabolic flow, diffusional mixing and effect of flow rate, microchip gel electrophoresis for dye separation).	
Study Goals	After completion of the course, you will be able to: 1. Apply the theoretical concepts of fluid flow to the design of a microfluidic device. 2. Be able to design, fabricate and motivate the chosen fabrication technique for a simple microfluidic device. 3. Set up and conduct experiments with microfluidic devices with pumps, flow controllers, digital imaging and data acquisition. 4. Analyze the acquired experimental data. 5. Produce well-structured technical figures, reports and presentations to communicate methods, results, and conclusions in a scientific manner.	
Education Method	Q3 corresponds to 1 EC of work (indicative). It will include the lectures on theoretical concepts necessary to the design of the microfluidic device. You will work in a group of 4 students on the design aspect of the project. Q4 corresponds to 2 ECs of work (indicative). It will include lab sessions on microfabrication. You will work with the same group as in Q4 on the fabrication and the testing of your microfluidic device.	
Books	Lecture material will include material from: 1. Henrik Bruus, Theoretical Microfluidics, Oxford University Press, 2007 2. Brian J. Kirby, Micro- and Nanoscale Fluid Mechanics: Transport in Microfluidic Devices, Cambridge University Press, 2010	
Assessment	In Q3: Graded design group report (35%) In Q4: Graded final group report (50%) and Final presentation (15%)	
Elective	Yes	
Tags	Design Fluid Mechanics Project Transport phenomena	
Contact	d.s.w.tam@tudelft.nl	

ME45075	Refrigeration & Heat Pumps Fundamentals	4
Responsible Instructor	M. Ramdin	
Instructor	Prof.dr.ir. T.J.H. Vlugt	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic knowledge on the first and second law of thermodynamics (a short introduction to this will also be provided during the lectures). Students who are missing this we explain the most important concepts again)	
Summary		
Course Contents	The course consists of the following content: -Introduction of thermodynamic cycles for heating and cooling. The working field of refrigeration and heat pumps; -Overview of the most important refrigeration / heat pumps systems: Mechanically driven heat systems, thermally driven systems, solid state systems and gas cycles. Comparison of these systems; -Components of refrigeration and heat pump cycles (compressors, valves, condensers, evaporators, turbines etc.); -analysis of cooling and heating cycles using the 1st and second law of thermodynamics, exergy analysis; -gas liquification; -techno-economic analysis of refrigeration and heat pump cycles; -Working fluids. Refrigerants for mechanical vapour compression; refrigerating machines: limits of application, evaluation of ODP and GWP; -Gas cycle refrigerating machines. Gas-phase cycles; -Magnetic refrigeration / heat pumping;	
Study Goals	By the end of this course, the students are able to: explain, reproduce and apply thermodynamic concepts in relation to refrigeration machines and heat pumps, considering their economic and environmental impact. recognize and apply the working principles, specific characteristics and application range of the most relevant methods to generate cooling and heating. explain and apply the concepts of gas liquefaction perform an energy and exergy analysis of refrigeration machines and heat pumps. evaluate the impact of the alternative concepts for refrigerating machines and heat pumps. sizing of heat pump and refrigeration equipment. perform techno-economic analysis of heat pumps and refrigeration systems.	
Education Method	Lectures	
Literature and Study Materials	Course sheets / relevant papers / additional material provided during the lectures and on Brightspace -Moran&Shapiro, Fundamentals of Engineering Thermodynamics, 9th edition -Dincer, Refrigeration systems and applications (available online via the TU Delft library) -Kiss/Infante Ferreira, Heat pumps in chemical process industry (available online via TU Delft library)	
Assessment	written exam	
Department	3mE Department Process & Energy	

ME45111	Buildings as Energy and Indoor Climate Systems	5
Responsible Instructor	Dr. L.C.M. Itard	
Course Coordinator	Dr. L.C.M. Itard	
Module Manager	Dr. L.C.M. Itard	
Instructor	A. Rasooli	
Instructor	Dr.ir. W.H. van der Spoel	
Co-responsible for assignments	A.I. Joseph Thaddeus	
Contact Hours / Week x/x/x/x	4.0.0.0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	This course is suitable to students who want to acquire specific knowledge on buildings' energy systems in relation to NZEB buildings, Smart buildings and their inclusion in smart energy grids, Urban energy or modelling of facade components. The course is suitable for students with a bachelor in mechanical or civil engineering, physics or students from architecture, TPM, ID with a strong background in maths.	
Course Contents	This course is an introduction to the fundamentals of buildings as energy and indoor climate control systems. It entails four main aspects: thermal behaviour of buildings, energy conversion &distribution systems for buildings, thermal comfort & indoor air quality and finally humid air properties & air handling. The course is based on a system modelling & simulation engineering approach and is useful for those interested in the design and simulation of smart low energy buildings with a good thermal indoor climate. Thermal behaviour of buildings: This part is the core of the course. You will study the dynamic thermal behaviour of buildings and the way it can be simulated by heat balances in order to predict indoor temperatures, needed heating and cooling capacities and energy consumption. During the course a dynamic hourly simulation, accounting for heat storage in buildings thermal mass, is set up for a modern office room, based on a set of linearized partial differential equations. Different numerical solving methods are studied. The use of models for indoor climate control and to achieve low energy buildings is demonstrated . Energy conversion &distribution systems for buildings: Energy conversion and related distribution systems used in practice are reviewed. Pro and contras are given. The integration of building physics, indoor climate and energy conversion systems is studied. The design process of buildings energy systems (Heating, Ventilation and Air Conditioning (HVAC)) is clarified. Thermal comfort & indoor air quality: Energy conversion systems and the building itself must ensure thermal comfort and a healthy indoor air quality. The thermal comfort of human beings is studied and simulated by mathematical models based on heat and mass balances. The relationship between thermal comfort, building and HVAC is clarified. The main parameters of indoor air quality are briefly treated. Humid air properties & air handling: Ventilation flows are of main importance for both energy use and indoor air quality. The thermodynamic properties of humid air are therefore studied and applied to understand the working of air handling units.	
Study Goals	The student is able to apply the fundamentals of indoor climate and energy control in buildings in order to design sustainable systems including relevant parts of the building envelope, the indoor climate equipment (HVAC) and the related energy conversion system. More specifically, the student will be able to: 1. Derive the mathematical equations describing the thermal performance of the various components 2. Combine these equations in order to simulate the whole system consisting of the weather, building, occupants, HVAC equipment and energy conversion system 3. Use dynamic simulations to answer questions about the design, such as capacities, energy use, comfort. 4. Determine the change of air conditions by the various components of an air handing installation 5. Translate the fuzzy term "comfort" into design requirements that can be checked afterwards by measurable variables and to describe the limitations of this technical approach 6. Calculate the heating and cooling capacity of a confined space by means of simple hand calculation and by dynamic simulation, taking into account the building mass. 7. Present the pro and cons of indoor climate systems used in practice, as well as of sustainable alternatives 9. Make optimal designs of simple systems based on simulation Education method Lectures, videos, exercises, assignments and excursion. Depending on TUDelft regulations around COVID-19 the lectures, exercises and assignment may be given online and use parts of Edx MOOCs in the PCP series Buildings as Sustainable Energy systems. Excursion will take place only if allowed by regulations. The course is built around a case study that runs like a common thread between the different course items. The case study is an self-paced individual assignment divided into 4 parts, that will be discussed at regular times. It addresses the simulation of the dynamic thermal behaviour of a building in order to define the cooling and heating capacities, indoor temperatures, thermal comfort and energy consumption. Finally you will optimize the building towards a zero energy building.	
Education Method	Lectures, videos, exercises, assignments and excursion. Depending on TUDelft regulations around COVID-19 the lectures, exercises and assignment may be given online and use parts of Edx MOOCs in the PCP series Buildings as Sustainable Energy systems. Excursion will take place only if allowed by regulations. The course is built around a case study that runs like a common thread between the different course items. The case study is an self-paced individual assignment divided into 4 parts, that will be discussed at regular times. It addresses the simulation of the dynamic thermal behaviour of a building in order to define the cooling and heating capacities, indoor temperatures, thermal comfort and energy consumption. Finally you will optimize the building towards a zero energy building. Course materials -Reader -All materials (slides, lectures, exercises, additional readings) of the MOOC Dynamic Energy Modelling of Buildings: Thermal Simulation and parts of the MOOC program Buildings as Sustainable Energy systems	
Assessment	Written exam (50% of grade) + Midterm quizz (50% of grade). The midterm quiz relates closely to the case study. An average grade of 6 (rounded to the nearest 0.5) AND a minimal grade of 5 for the written exam are required to pass.	

ME45170	Turbomachinery	4
Responsible Instructor	Prof.dr. R. Pecnik	
Responsible Instructor	Prof.dr.ir. S.A. Klein	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Thermodynamics, Fluid dynamcis	
Course Contents	The course consists of the following parts: - Introduction and recap of the required thermodynamics. This class will give an overall overview of the relevance of turbomachinery, the general basics and the history. Furthermore the basic thermodynamics will be repeated (1 class) - Turbomachinery: basic theory and the application to the design of turbomachinery (Conservation equations and velocity triangles, turbines, compressors and pumps, 3D flow, radial machinery) (5 classes) - Application of compressor, pump and turbine maps. Limitations in the operation (stall, surge, NPSH etc) (1 class) - Gas turbine: components (turbine, compressor, combustor), gas turbine cycles, performance calculation,	
Study Goals	The student is able to describe the fundamental aspects of the working principle of turbomachinery. More specifically, the student must be able to: 1. reproduce and apply thermodynamic principles of turbomachinery 2. describe the basic fluid dynamics of turbomachinery and the way work is transferred between the working fluid and the turbomachinery, including the impact on performance and limitations of operations 3. apply compressor, pump and turbine curves in practical applications 4. describe different gas turbine cycles 5. describe the influence of different losses on the performance (power and efficiency) of gas turbines 6. synthesize different gas turbine cycles with respect to their application. 7. describe the principles of combustion as applied in gas turbines with respect to functionality and emissions 8. describe the part load characteristics of gas turbines and its relation to the functional characteristics of gas turbine components	
Education Method	Lectures and homework assignments	
Assessment	Closed book exam (60%) and home work assignments (40%). All homework assignments must be submitted. The grade for the homework assignments will only be included in the final grade if the exam grade is above 5.5/10	
Tags	Energy Energy & Industry Fluid Mechanics Thermodynamics	
Department	3mE Department Process & Energy	

ME45190	Chaos in Dynamical Systems	3
Responsible Instructor	Prof.dr.ir. W. van de Water	
Instructor	Prof.dr.ir. J. Westerweel	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	Working knowledge of ordinary differential equations, and how to solve them practically on the computer.	
Course Contents	Chaos is the seemingly erratic behavior of simple deterministic, but nonlinear dynamical systems. The systems that we are used to in Mechanics and Physics are almost always linear, as they explore tiny excursions from equilibrium. However, once systems become nonlinear, we are in for surprises. Their state can change dramatically upon variation of a parameter (bifurcations), or they may become chaotic altogether. We will first discuss the route to chaos, where we will already encounter the scaling concepts that return in the description of synchronization, chaotic fluid flow and turbulence. The route to chaos is often universal, and scaling concepts demonstrate why. Systems with no friction are called "Hamiltonian", the route to chaos in Hamilton systems follows a beautiful geometry cartoon, with striking pictures and little mathematics. We will discuss the basic concepts of chaotic behavior, such as the continued sensitivity to variation of initial conditions and the fractal structure of phase space. Then we will look at chaos and synchronization in nonlinear coupled oscillators, again from the scaling perspective. In the meantime we will have practiced bifurcation using "path following" software.	
Study Goals	With this course you will be able to recognize universal routes to chaos, understand synchronization of dynamical systems, and the application of chaos to fluid flow and turbulence.	
Education Method	8 Lectures of 2 hours. There are 4 blocks, corresponding to the 4 assignments. A block has 2 lecture hours, and 2 "studio-classroom" hours, where we will work on the assignments.	
Computer Use	We will use Matlab, but for each assignment there is a template that gets you started.	
Literature and Study Materials	Lecture notes are available	
Assessment	There will be 4 assignments where you play with chaotic dynamical systems using Matlab. The subjects of these assignments cover the course subjects. The last assignment even involves a simulation of a turbulent flow.	
Department	3mE Department Process & Energy	
Contact	w.vandewater@tudelft.nl	

ME45201	Electrochemical Energy Storage	3
Responsible Instructor	J.W. Haverkort	
Instructor	Dr. R. Kortlever	
Contact Hours / Week x/x/x/x	0/0/4/0 (2 lecture hours and 2 seminar hours)	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic thermodynamics. Basic chemistry and knowledge of kinetics is not required, but will be helpfull.	
Course Contents	One of the main challenges that mankind faces in the 21st century is the energy transition from fossil fuels and feedstocks to renewable sources. Most renewable energy sources, such as solar and wind energy, are intermittent. Therefore, large scale energy storage is required to successfully increase the share of these renewable energy sources. One way of achieving this is by storing energy using electrochemical systems and devices. This course introduces the principals of electrochemical energy storage and conversion to students new to field. Examples of topics discussed during this course are the thermodynamics and kinetics of electrochemical systems, transport phenomena, the electrochemical double layer, electrocatalysis and an introduction to electrochemical methods and measurements. We will discuss these topics in the light of electrochemical applications such as batteries, electrolysers and fuel cells.	
Study Goals	After successful completion of this course, the student is able to independently: 1. Apply electrochemical nomenclature to describe the electrochemical cell and electrochemical measurements; 2. Explain the concept of the electrochemical double layer and mass transport in the electrolyte; 3. Apply basic principles of electrochemical thermodynamics, such as the Nernst equation, overpotentials and (standard) equilibrium potentials; 4. Apply basic principles of electrochemical kinetics, such as the Tafel equation and the Buttler-Volmer model; 5. Explain the principles that determine the electrocatalytic performance of materials; 6. Interpret scientific literature reports on electrochemical energy storage and conversion systems and devices.	
Education Method	Combination of two (online, dependent on the situation) lecture hours per week and self-study. Homework assignments will be given after every lecture and will be discussed in class.	
Course Relations	A suitable follow-up course may be ME45203: Electrolyzers, Fuel Cells and Batteries.	
Assessment	In case there is an opportunity for examinations on campus, the method of assessment will be a written exam. If, due to measures resulting from COVID-19, on campus examination is not possible the prescribed assessment will be as follows: oral exam and presentation on a literature report.	
Department	3mE Department Process & Energy	

ME45203	Electrolysers, Fuel Cells and Batteries	4
Responsible Instructor	J.W. Haverkort	
Instructor	Dr. R. Kortlever	
Contact Hours / Week x/x/x/x	0.0.0.4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Familiarity with transport phenomena, basic fluid dynamics, and elementary differential equations. A working knowledge of electrochemistry and in particular the Butler-Volmer equation is also assumed. Ideal preparation include the courses ME45201, CH3513, or CH3222. A previous course on fuel cells, batteries, or similar, or additional self-study may also be sufficient.	
Course Contents	Electrochemistry plays a pivotal role in the transition towards a sustainable energy future. Renewable electrical energy can be stored in batteries or used to synthesize chemicals or fuels in electrochemical reactors, like an electrolyzer producing hydrogen. This course will introduce to a student, the knowledge and skills required to understand, model and optimize such electrochemical devices. Content overview: Porous electrode theory, multiphase flow, and transport models will be used to derive analytical current-voltage relations of various electrochemical devices including fuel cells, batteries, redox flow batteries, and electrolysers. Some of the topics that will be discussed in this course include: - The Nernst-Planck equations for ionic transport in binary electrolytes. - Porous electrode theory using the concept of an electrochemical Thiele modulus. - The flooded agglomerate and relative permeability models for gas diffusion electrodes. - The single particle and Newman's P2D battery models. - Tobias model for gas-liquid electrolysers.	
Study Goals	After successful completion of this course, the student is able to independently: 1. explain and apply porous electrode theory. 2. derive 0D, 1D, and 2D mathematical models to describe the current-voltage relationship of various electrochemical cells including flow, multiple phases, and porous electrodes. 3. optimize an electrochemical cell design with respect to, for example, energy efficiency.	
Education Method	Two lecture hours + two exercise hours per week.	
Course Relations	This course is a natural sequel to ME45201, CH3513, and CH3222 for students interested in analytical modelling of electrochemical systems.	
Books	Electrolyzers, Fuel Cells and Batteries : Analytical Modelling. (J.W. Haverkort)	
Assessment	Written exam.	

ME45211-23	Particle-based Modeling of Fluids	5
Responsible Instructor	Dr. O. Moultos	
Instructor	Dr. R.M. Hartkamp	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge	Basic thermodynamics. Basic programming skills are beneficial.	
Course Contents	This course provides a theoretical and hands-on introduction to particle-based modeling with an emphasis on molecular simulation. We intend to give an overview of computational tools used for studying thermodynamic, structural, and transport properties of fluids and granular matter. Concepts that will be treated include the essential statistical mechanics, inter- and intramolecular interactions, and Monte Carlo and molecular dynamics simulation methods. This course aims to provide future engineers with the physical intuition and computational tools needed to address a wide range of industrial problems. Various energy-related and environmental applications will be discussed. The lectures will address the following topics: - Application areas of particle-based modeling in engineering - Interactions between particles - Monte Carlo and molecular dynamics - Computation of thermodynamic, transport, and structural properties of fluids - Visualization and analysis of simulation results	
Study Goals	1. Develop basic particle-based simulation programs 2. Analyze which simulation method is appropriate for a given application. 3. Calculate structural and thermodynamic quantities and transport coefficients from simulation output. 4. Convert microscopic information of a system (e.g., atomic positions and velocities) into macroscopic observables (e.g., pressure, energy, diffusion, viscosity). 5. Describe the most common molecular simulation methods and understand their theoretical basis.	
Education Method	Lectures	
Assessment	Assignments	

ME45215	Rheology of Complex Fluids	3
Responsible Instructor	Dr. B.P. Tighe	
Instructor	Dr. E.B.A. Hinderink	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Familiarity with the mechanical concepts of stress, strain, and strain rate is assumed. Familiarity with the mathematical concept of tensors is also assumed. A prior course in continuum solid or fluid mechanics is therefore useful. Familiarity with python, Matlab, Mathematica, or similar is assumed and will be required to complete the homework.	
Course Contents	This course will introduce the student to rheological methods for characterizing and modeling complex fluids, i.e. systems whose deformation and flow differs from simple Newtonian fluids and Hookean solids. Examples of complex fluids include suspensions, pastes, slurries, foams, emulsions, glasses, granular media, polymer melts, gels, and fibrous networks. The student will learn to identify and model characteristic features of complex fluid rheology, including shear thinning/thickening, strain stiffening, and yielding. The student will also acquire knowledge of rheometric tools used to acquire rheological data, and how to analyze rheometric data to develop a rheological model. Course content will be illustrated with applications drawn from mechanical engineering, chemical engineering, civil engineering, and (bio)physics.	
Study Goals	After completing this course, the student will be able to describe the elastic/viscous/plastic response of complex fluids and viscoelastic solids. More specifically, after this course you should be able to: 1. Identify signature rheological features of non-Newtonian phenomena such as yielding, shear thinning, shear thickening, dilatancy, strain stiffening, shear banding, thixotropy, and gelation/jamming. 2. Analyze rheometric data from standard geometries (e.g. Couette, parallel plates, cone-plate) and standard flow modes (including steady shear flow, oscillatory shear, stress relaxation, flow start-up, and creep flow). 3. Implement standard models of constitutive relations for complex fluids, including the following: Maxwell, Kelvin-Voigt, standard linear solid, Herschel-Bulkley, power law fluid, Cross, and Carreau. 4. Discuss relevant techniques and analysis to quantify the non-linear flow behavior of complex fluids when subjected to large deformations during processing.	
Education Method	- traditional lecture format introducing theory and fundamentals - in-class problem sessions - guest lectures by TU Delft staff members. Lecturers will highlight the applications perspective and describe their use of rheometry/rheology in their own research.	
Computer Use	The homework will include elements of data analysis that can be performed in, e.g., python, Matlab, or Mathematica. Familiarity with at least one such software package is assumed.	
Books	To be announced	
Assessment	- homework assignments: theory and fundamentals, data analysis, model development - final report	
Department	3mE Department Process & Energy	

ME45220	Experimental Techniques in Fluid Mechanics	3
Responsible Instructor	Dr. A. Laskari	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Bachelor degree level knowledge in fluid mechanics (WB2542) and measurement techniques (ME46007) is essential for this course.	
Course Contents	In this elective master level course, the students learn to use a variety of experimental techniques in different flow setups, through a combination of lectures and hands-on experiments.	
Study Goals	After completion of the course, you will be able to: - Carry out different experimental measurements in fluid flows. This will include planning/implementation of good experimental practices/safety and experimentation with different scientific equipment. - Analyse the acquired experimental data. This will include proper appraisal, classification, and processing of the datasets. - Compare different experimental techniques. - Evaluate the uncertainty, validity and significance of the analysed data. - Produce well-structured technical figures, reports and presentations to communicate methods, results, and conclusions in a scientific manner.	
Education Method	Lectures: During the first few weeks the students follow a series of lectures relevant to the experimental methods, planning, data analysis. Experiments (mandatory): Subsequently, students follow supervised laboratory sessions (in groups of 4-5) during which they perform experiments in different flow setups, employing the techniques discussed in the lectures.	
Literature and Study Materials	The prescribed book will be followed during the lectures and lecture slides will be made available on Brightspace. Apart from those students are encouraged to refer to other books and journal papers with relevant content and techniques.	
Books	Springer Handbook of Experimental Fluid Mechanics (2007) Cameron Tropea, Alexander Yarin, John F. Foss	
Assessment	One graded group report (with individually graded components) contributing 80% to the final grade. Presentation of findings in a final presentation session (20% of the final grade).	
Elective	Yes	
Tags	Fluid Mechanics	
Department	3mE Department Process & Energy	
Contact	a.laskari@tudelft.nl	

ME45225	Multiphysics transport in Energy Materials	3
Responsible Instructor	L. Botto	
Instructor	Dr. R.M. Hartkamp	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Elements of fluid dynamics, heat/mass transfer and thermodynamics. Familiarity with modelling via PDEs.	
Course Contents	This is a course in modelling of transport phenomena in energy storage materials and devices. It is an excellent introduction to multiphysics problems for students interested in learning how one can simulate or model the coupling of heat, mass and electricity in the devices that drive the energy transition. Example of such devices are batteries, supercapacitors, hydrogen/ CO2 storage devices, or materials for heat storage. Many courses treat transport phenomena as separate subjects, while this course unifies different transport phenomena exploiting analogies between different transport fields, with an applied emphasis on energy materials. The course is particularly suitable for student wishing to later specialize in numerical simulation of conservation laws (e.g. courses on CFD), as an introduction on the transport theories used in the modelling of electrochemical systems and heat transfer systems, or for students wishing to later specialize in experimental, theoretical or simulation-based fluid dynamics. Different from other transport phenomena courses, this course covers the modelling of diffusive and convective transport of generic species in materials that are not homogeneous, but are made of different phases in the form of compacted dry powders, solid foams, inks, slurries etc. For example, the course covers how to estimate the effective electrical or heat conductivity of materials used as battery electrodes, a key issue when modelling batteries. Among other topics, the courses will give a practical introduction to Comsol Multiphysics, probably one of the most widespread software for the simulation of general multiphysics problems. Topics include: MATHEMATICAL MODELLING: Mathematical analogies between conservation laws for heat, mass, electrons, ions and electric fields; Modes of microscopic transport: diffusion, convection, reaction/adsorption and their microscale modelling; From micro to macro: upscaling microscopic fluxes to the macroscale; the science behind engineering correlations; Extract general constitute laws for multiphysics fields: the most important ideas; Averaging of linear and non-linear equations; Bounds on transport properties for statistically homogeneous, periodic or percolating fields; effective conductivities of composites MICRO/NANOPHYSICS: Coupling of mass, momentum or heat transport with electric fields: electrokinetic transport in bulk and at interfaces; Upscaling adsorption in porous media: application to hydrogen and CO2 adsorption; How to analyse images, e.g. Scanning Electron Microscope images, to estimate transport properties in mesoporous materials NUMERICAL SIMULATIONS: How to use commercial codes (e.g. Comsol or Fluent) to compute experimentally testable transport properties; basic approaches to the discretizations of coupled multiphysics problems APPLICATIONS: use of the theory in practical case studies: battery electrodes, high-surface-area capacitors, hydrogen storage media, membranes for CO2 sequestration, multicomponent/multiphase heat storage materials	
Study Goals	After completion of the course, you will be able to: - Read confidently scientific or technical literature related to multiphysics transport in energy materials - Analyse experimental or simulation data for multiscale mesoporous materials to extract key transport properties such as, for example, multiphase heat/mass/electron conductivities - Develop computational models of engineering transport in energy materials and equipment, and evaluate the accuracy of meso- and macro-scale transport correlations used in these models - Learn how to be the go-to-person in multi-physics modelling in an industrial or scientific organisation, by learning the unique jargon and method of analysis of this field - Save time in your research by learning how to develop back-of-the-envelop calculations	
Education Method	The course is based on lectures on theory and practice, and in-class discussion of practical case studies. Lectures will be complemented by a few assignments analysing particular technologies. There is no single textbook: the study material is taken from book chapters, research knowledge, or scientific articles and will be given by the instructor. Some books will be suggested by the instructor for the students interested in learning certain topics more in depth.	
Assessment	Assessment is based on a written exam, a presentation on a topic agreed with the instructor, and a short report on a lab-based experience.	
Department	3mE Department Process & Energy	
Contact	For information about the course, feel free to contact the course instructor Dr. Lorenzo Botto at l.botto@tudelft.nl (the instructor will be happy with sharing lecture notes from the past year with students interested in evaluating whether or not they should take the course)	

MT44100	Internal Combustion Engines A	5
Responsible Instructor	Prof.dr.ir. B.J. Boersma	
Instructor	R.H. van Till	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	PERFORMANCE Introduction: Engine types and construction, historical overview and present applications. Performance parameters: fuel economy, power density and emissions, basic principles and main paramaters of modern diesel engines. Sizing of main dimensions and database of engines. Thermodynamic analysis of cylinder process, including explanation of generalised polytropic process. Performance analysis of diesel engines: important trends made clear by using a realistic Seiliger process and reasonable estimation of mechanical and heat losses. TURBOCHARGING Introduction: mechanical charging versus exhaust driven turbocharger, constant pressure versus pulse system, turbocharging explained in P-V and T-S diagram. Gas exchange: mechanisms of air supply and air swallow capacity of 2- and 4-stroke engines, mechanisms of gas disposal, i.e. blow down, exhaust and scavenging. Principles of interaction between the turbocharger and the engine: power balance of turbocharger and flow balance between engine and turbine, energy in the exhaust gas, off-design performance of a turbocharged diesel engine. Enhanced turbocharging: waste gate, Variable Turbine Geometry (VTG), sequential turbocharging, two-stage turbocharging, including recent modelling developments. Modelling: classification of diesel engine simulation models, physical balances, blockdiagram of engine + turbocharger. NEW IN 2017: Introduction to GASENGINES (LNG as a fuel)	
Study Goals	The student must be able to: 1. Recognise the technical and economical importance of the diesel engine relative to other energy generating installations. 2. Explain the complexity and interdependency of the main performance parameters of a diesel engine and be able to calculate these parameters in several problem settings. 3. Formulate the limitations when determining the main dimensions of a diesel engine and be able to calculate these dimensions in several problem settings. 4. Analyse the thermodynamic processes in the diesel engine, both in the cylinder and in the turbocharger and be able to calculate the thermodynamical process data and resulting engine performance. 5. Discuss the importance of main performance parameters as well as trends and limitations from the perspective of the user 6. Explain the principles of a turbocharger as an example of direct waste heat usage to increase the power density of the engine. 7. Explain the gas exchange mechanisms and be able to contrast the differences between the 2- and 4-stroke diesel engine. 8. Explain the complex interaction between engine and turbocharger at part load and be able to calculate part load performance with a simplified method. 9. Discuss the importance, trends and limitations of turbocharging from the perspective of the user. 10. Identify the relevant characteristics of LNG fuelled engines (including Dual Fuel engines) 11. Apply the knowledge gained during the course in a practical laboratory test.	
Education Method	The student prepares the lecture by reading chapters and preparing (preferably challenging) questions. During the lectures these questions are discussed as well as questions raised by the tutor. The lecturing is focussed on subjects that are agreed to need further explanation.	
Assessment	Written exam, open book. In case of a remote/online exam, the exam will contain several fraud revention measures, like Oath of integrity, Check handwritten explanations/ answers on similarity, Oral check after exam, Several time slots for different questions, Shorter timeslot, Different versions of one question, Shuffled questions, Parametrize questions, Plagiarism check.	
Remarks	Previous code: WB4408-A	
Department	3mE Department Maritime & Transport Technology	

SET3070	Thermochemistry of Biomass Conversion	4
Responsible Instructor	Prof.dr.ir. W. de Jong	
Co-responsible for assignments	Dr. L.L. Cutz IJchajchal	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	To start this course you need to have a good knowledge of BSc. level thermodynamics, chemistry, kinetics, basic reactor modelling (batch, plug flow, continuously well stirred reactor), basic process engineering (mass, energy, momentum balances), and physical transport phenomena. In case you think you miss some of this pre-existing knowledge, please perform selfstudy of the first six chapters of the book 'Biomass as a Sustainable Energy Source for the Future', Wiley (2015), eds. De Jong and Van Ommen	
Summary	Biomass is available in abundance on a global level; it is naturally stored solar energy in a wide diversity of chemical structures. Among the renewable energy sources, biomass can be stored most easily, of course depending once again on the resources characteristics. Due to the heterogeneous nature of biomass also many different conversion technologies have been developed in the context of energy conversion chains with combustion being the oldest performed by mankind. This course deals with the background and design and implementation of different technologies to convert biomass thermochemically.	
Course Contents	In this course you will learn about the following aspects of biomass conversion based on thermo-chemical conversion technologies. - Biomass Characterisation (biomass types, properties, analyses) - Pretreatment (harvesting, size reduction, drying, torrefaction) - Thermochemical conversion (combustion, gasification and pyrolysis; reactor types) - Transportation fuels / biorefinery (syngas conversion, economics) - Process design (structured design approach)	
Study Goals	During this course, the students will obtain knowledge of and insight into energy conversion technologies based on biomass thermo-chemical conversion processes for a wide variety of applications, like supply of power, heat and fuels. They will be offered practical problems to solve and projects to perform in this area. These main learning objectives can be divided into several smaller learning objectives: 1) The student has knowledge on the typical biomass composition, related properties (like e.g. heating value, ash melting behaviour) and can critically identify and select fuel characterization techniques. 2) The student is able to identify and perform basic capacity calculations concerning thermo-chemical biomass conversion systems for small and larger scale (combined) heat and power generation as well as systems for transportation fuel production. 3) The student is able to set up reaction equations (stoichiometry) for a wide range of energy supply related fuel conversion processes. 4) The student can set up and perform heat and mass balance calculations regarding single phase and simple multi-phase biomass thermo-chemical conversion reactors and extended (downstream) systems for power, heat and fuels generation. 5) The student knows how to perform basic ideal gas phase chemical reaction equilibrium calculations to calculate product compositions and extents of reactions given initial reaction mixture composition, pressure and temperature. 6) The student can understand and describe reaction kinetic expressions and also apply them in order to solve (design) problems related to the conversion of fuels and the formation of emission components. 7) The student demonstrates good knowledge of mass and heat transfer mechanisms and phenomena applied to the chemical conversion of both liquid and solid biomass (derived) fuels. 8) The student can derive equations for idealised model reactors (well-stirred reactor and plug flow reactor concepts) and apply these simplified reactor concepts to solve engineering problems related to the thermo-chemical conversion of biomass fuels, in particular combustion, gasification and pyrolysis. 9) The student is able to perform calculations concerning the conversion of small particles; important in this respect is the identification of the regime of conversion. Related dimensionless numbers such as the Thiele Modulus and the Biot number are key indicators in this respect. 10) The student is able to understand the basics of thermo-chemical conversion processes and to translate this into designing and evaluating such (sometimes hybrid) systems, including their economic evaluation based on characteristic parameters like NPV, POT, and ROI. 11) The student demonstrates good communication skills, both oral and written, concerning presentation of (applied) scientific results.	
Education Method	This course consists of six 2-hour lectures, also using blended learning methods. Each lecture is followed by a 2-hour work session or excursion(s)/lab visit(s) in the same week -depending possibly on an actual pandemic situation- focused on working out minor assignments including literature surveys, problem solving and work on the major project. In the week of the exams you will present the major assignment project results.	
Course Relations	This course is a core course in the Sustainable Energy Biomass Energy profile.	
Literature and Study Materials	Book: "Biomass, a sustainable energy source for the future", de Jong and van Ommen, Wiley., 1st edition, Wiley (2015). Lecture slides	
Assessment	The final grade of the course consists of the following components: * the solution to the (up to three) minor assignments (team work) (40%) * a larger final assignment (team work); (50%) * evaluation of the final assignment's presentation; (10%) Final grade calculation: (0.4 * minor group assignments) + (0.5 * larger group assignemtn) + (0.1 * presentation) Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: * minor assignments: submit 1 new minor assignment * lagers assignment: resubmit report * Presentation: resit (new presentation) Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr. L.L. Cutz IJchajchal	

WI4014TU	Numerical Analysis	6
Responsible Instructor	Dr. N.V. Budko	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2	
Course Language	English	
Course Contents	Classification of partial differential equations and boundary and initial conditions. Maximum principle, uniqueness and non-negativity of the solution, elements of variational calculus. Finite- Difference, Finite-Volume and Finite-Element discretization methods. Solution methods for time-dependent and nonlinear problems. Direct and iterative linear solvers.	
Study Goals	Students will learn: the principles of finite-difference, finite-volume, and finite-element discretizations; the accuracy, consistency, and convergence of numerical solutions; the basic ideas of numerical linear algebra. They will be able to implement a simple PDE solver and to work with existing sophisticated numerical software.	
Education Method	Lectures (possibly, online), homework assignments, programming assignments, written report.	
Literature and Study Materials	[1] J.van Kan, A.Segal, and F.Vermolen, Numerical methods in scientific computing. VSSD, Delft, 2005, improved 2008, ISBN-13 978-90-71301-50-6 [2] Python programming: - Python Help: https://www.python.org/about/help/ - Numpy for Matlab users: https://docs.scipy.org/doc/numpy-dev/user/numpy-for-matlab-users.html [3] FENICS Project: https://fenicsproject.org/ - Installation: https://fenicsproject.org/download/ - Tutorial: https://fenicsproject.org/tutorial/	
Prerequisites	Calculus of functions of several variables, vector calculus, introductory course on numerical analysis (numerical integration, error analysis), introductory course on partial differential equations, some experience with Python or Matlab.	
Assessment	Theoretical and computational assignments, graded reports. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Tags	Mathematics Modelling Numeric Methods	
Co-Instructor	Prof.dr.ir. M.B. van Gijzen	

WI4019-SP	Non-linear Differential Equations	6
Responsible Instructor	Dr.ir. W.T. van Horssen	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Introductory course on Differential Equations e.g. Boyce and DiPrima Chapters 2-7, 9.	
Course Contents	Existence and uniqueness of solutions of initial value problems, Gronwall's lemma, Autonomous systems, Critical points, Periodic solutions, Stability theory, Linear systems and Floquet theory, Perturbation theory and asymptotic methods, Poincare-Lindstedt method, Averaging method, Multiple time-scales method, Elementary bifurcations.	
Study Goals	After following this course the student will be able to apply fundamental and advanced, analytical techniques and methods to study properties of solutions of nonlinear ordinary differential equations. The mathematical methods and techniques are mentioned under "course contents". After studying the lecture notes and after making the exercises successfully the student can start to work on research problems in the field of nonlinear differential equations.	
Education Method	Lectures and a MAXIMA practical work. In total 14 lectures of 2 hours each will be given. The MAXIMA practical work is about 4 hours work. Following and preparing the lectures and the MAXIMA practical work takes about 60 to 70 hours. Making the 30 take-home exercises takes about 60 hours. Preparing the 45 minutes oral exam takes about 30 hours.	
Literature and Study Materials	F. Verhulst, Nonlinear Differential Equations and Dynamical Systems, Springer-Verlag 2nd edition, ISBN 3-540-60934-2.	
Assessment	The final grade of the course consists of the following components: - Take home exam (50%) - oral exam (50%) Final grade calculation: (0.5 * take home exam) + (0.5 * oral exam) The take-home exam consists of 30 exercises. The oral exams will be held at the end of the 2nd quarter, and the student is only allowed to take part in this oral exam when at least 70% of the take-home exercises are made correctly and when the MAXIMA practical work is followed successfully. In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5 take home exam: resubmit deliverable oral exam: oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Mechanical Engineering**

ME-EFPT Second Year Project: Select 1 (15 ECTS)

ME55015	ME-EFPT Research Assignment	15
Responsible Instructor	Dr. B.P. Tighe	
Instructor	Dr. R. Delfos	
Instructor	Dr.ir. M.J. Tummers	
Contact Hours / Week x/x/x/x	x/x/0/0	
Education Period	None (Self Study)	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Students gain experience with the "doing" of EFPT-related engineering in a real-world context by completing a research project at an organization external to the Process & Energy Department, for example a company, government lab, or another department at TU Delft. Students must be supervised by one staff member from the external organization, and one staff member from the Process & Energy department. Course credit cannot be earned for experience obtained prior to enrolling in the EFPT MSc track. Students may not do the Research Assignment at an external organization where they have prior work experience. In exceptional circumstances an exemption may be made for large organizations with multiple divisions; in this case the student must obtain permission from the Master Coordinator in advance.	
Study Goals	After completing this course, the student will be able to: - Carry out engineering-related tasks in a professional rather than classroom environment - Define and implement a research plan aligned with the external organizations goals - Apply contents and skills developed in the first-year courses to advance the research plan - Document progress and outcomes in a manner useful to other members of the organization	
Education Method	hands-on	
Assessment	Students deliver a final report documenting their research assignment. The supervisors may require a presentation at their discretion.	
Department	3mE Department Process & Energy	

TUD4040	Joint Interdisciplinary Project	15
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Project Coordinator	Ir. B.J.E. de Bruin	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Summary	JIP consists of three sets of activities: 1. The project which takes place at a company. Students are responsible for the project work (collaboration, integration of different perspectives, the content quality of their work). They plan their activities via a scrum method, are challenged to realise field trips, consult with experts and realise the necessary research work. The team keeps a Scrum log, run a blog for interaction with the outside world and every team member keeps a personal development log. 2. Meeting Professionals lectures and workshops about specialised topics, by academic staff or senior professionals from the companies involved. 3. Plenary meetings and design reviews for all the teams, company coaches and academic staff. At the meetings the students present their intermediate resp. final outcomes and plans, and are provided with feedback.	
Course Contents	The aim of the Joint Interdisciplinary Project is to prepare students to contribute to solving impactful technological challenges. The projects not only demand good engineering working knowledge but also experience with interdisciplinary and systems theory, and both knowledge and mindsets of innovation and entrepreneurial behavior. The project brief is provided by renowned companies like Airbus, Arcadis, etc. Teams of interdisciplinary student teams guided by a company coach and offered academic and industry expertise, are invited to realise an innovative problem solution to a complex problem and contributing to the sustainable development goals.	
Study Goals	1. Cognitive abilities attributable to interdisciplinary learning Demonstrate the ability to engage in perspective-taking; Develop structural knowledge pertaining to the problem; Integrate knowledge and modes of thinking drawn from two or more disciplines; Produce an interdisciplinary understanding of complex problem or intellectual question. 2. Scientific and intellectual development Capable to analyze scientific and societal consequences (economic, social, cultural, environmental) of the innovation; 3. Research and design capabilities Demonstrate engineering skills: technical skills, interpreting results, creativity, usability for company/institute; Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; 4. Collaboration and communication in an interdisciplinary team Demonstrate behavioral competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to respect different disciplines and adapt to different cultures); Show ability to write a technical report: structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (30 pages); Present work performed in a structured way through an oral presentation to their peers and customer. 5. Self-adjustment and reflection capabilities Plan and control the project efficiently considering resources and methodology; Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career. Cognitive abilities attributable to interdisciplinary learning; The ability to integrate (scientific and practical technological)knowledge from different disciplines to solve complex problems Scientific and intellectual development The capacity to evaluate the ethical, scientific and societal consequences of the proposed innovation Research and design capabilities The ability to create reasonable and relevant research or design , according to the academic standards of the involved disciplines Collaboration and communication in an interdisciplinary team Demonstrate behavioural competences and skills relevant for teamwork and effective communication with different stakeholders Self-adjustment and reflection capabilities To carry out regular reflections on professional and personal development and being able to improve upon those reflections Understand contemporary and societal issues in their work.	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Assessment	The assessment criteria pertain to the process, product and presentation and are assessed at an individual and team level, during 3 review sessions at three levels of accomplishment (each review assesses at a different level): Interdisciplinary work Scientific reasoning and ethical mindset Innovation process Presentation and communication Academic staff, supported by the input from company coaches, grade the students during the Problem Statement, the Midterm and the Final Review. The final group mark for JIP is differentiated per person and is based on: The team presentations - 30% The Problem definition, progress report and final report 30% The individual contribution to the team 20% The final individual reflection report 5% The Blog 5%	
Remarks	Project Outcomes (product, model, system) are presented at: 1. Problem Statement Review 2. Midterm meeting 3. Final Review 4. Symposium presentation	

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Mechanical Engineering**

ME-EFPT Graduation Project (45 ECTS)	
Introduction 1	The 2nd Master's year has a flexible structure, in which the Literature Research (10 ECTS) and the Master's thesis (35 ECTS) are compulsory for everyone. In a nominal situation this will start in the 2nd quarter of an academic year. In the first quarter, students can choose between: 1. 15 ECTS Joint Interdisciplinary Project; 2. 15 ECTS Internship or Research Assignment.

ME55010-20	ME-EFPT Literature Research	10
Responsible Instructor	Dr. B.P. Tighe	
Instructor	Dr. R. Delfos	
Instructor	Dr.ir. M.J. Tummers	
Contact Hours / Week x/x/x/x	x/x/x/x (colloquium attendance)	
Education Period	None (Self Study)	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	The EFPT Literature Study is integrated with the Thesis and must be conducted simultaneously with ME5035. The Literature Study positions the student to carry out the research project in their Thesis. This includes performing a search of the relevant scientific literature, formulating a well-posed research question, and developing a research plan to answer the question. Students will also study how other researchers approach problems related to EFPT by attending 8 department colloquia.	
Study Goals	By the end of this course, the student should be able to: Search the literature systematically; Formulate a research question; Select representative scientific sources from several (for example, economic, ethical, environmental, medical, social, health) perspectives relevant to the assignment; Draw conclusions related to the research problem and give recommendations towards new research opportunities; Write a literature review in academic English. Demonstrate a scientific attitude; Manage your own learning process (including time management).	
Education Method	Self study	
Assessment	T1 Literature Research (written report) T2 Presentation (pass/fail) T3 Colloquium Attendance (pass/fail) The report should serve as the (nearly) complete first chapter of the students thesis. The presentation will summarize the report and will be assessed by a jury of staff members from the Process & Energy Department.	
Department	3mE Department Process & Energy	

ME55010-20 Toets 1	*Literature Research*	*10*

ME55010-20 Toets 2	*Colloquium Presentation*	*0*

ME55010-20 Toets 3	*Colloquium Attendance*	*0*

ME55035	ME-EFPT MSc Thesis	35
Responsible Instructor	Dr.ir. M.J. Tummers	
Responsible Instructor	Dr. B.P. Tighe	
Instructor	Dr. R. Delfos	
Contact Hours / Week x/x/x/x	0/x/x/x	
Education Period	None (Self Study)	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Students perform an independent research project that integrates knowledge from coursework, internship, etc.. Students develop detailed expertise in a topic relevant to energy, flow, and/or process technology, under the supervision of a staff member from the P&E department. Co-supervisors are possible under certain conditions, see the Master's Coordinator.	
Study Goals	After completing this course, the student will be able to: - reproduce relevant theory at the level of MSc textbooks - [research project] develop new theory through the use of standard mathematical, numerical or experimental methods - [design project] develop a new design, or design method, through the use of standard design methods and analyses - Generate results that can function as a basis for a publication or design - describe the outcome of the thesis in an report with academically appropriate structure, referencing and clarity, and give a presentation on the basis of the contents of the thesis. - thesis in an requirements in terms of structure, referencing and clarity - demonstrate responsibility for the proper progress and completion of the project - demonstrate relevant practical skills, e.g. (experimental, computer, and/or data acquisition. Is aware of safety and operates accordingly - demonstrate critical attitude towards own results and responsiveness to feedback	
Education Method	Varies according to the project.	
Assessment	Oral exam, thesis report and defense	
Department	3mE Department Process & Energy	

ME-HTE Track C. High-Tech Engineering

Contact for students	E. Matroos T +31 15 27 86581 E e.matroos@tudelft.nl
Program Coordinator	Dr.ir. J.F.L. (Hans) Goosen T +31 15 278 6500 E J.F.L.Goosen@tudelft.nl
EC Program	

Introduction 1

The ME MSc Track in High-Tech Engineering (HTE) educates engineers in the technological knowledge and skills they need to design a new generation of both the products and the required equipment that will enable even greater achievements. Starting from the fundamentals of physics and mechanics, students gain the insights and understanding they will need to push beyond the current limits. The programme includes analysis, design and implementation of solutions, using analytical models, computational methods and experimental work to reach new performance and understanding. With this focus on the ultimate in mechanical engineering the program confronts students with the daunting conceptual and design challenges of developing (and utilising) tools for precision mechanical engineering. Although the emphasis is on high-tech equipment and instrumentation, the same knowledge and methodology applies to energy systems, medical equipment, automotive and aerospace design and many other fields of mechanical engineering, enabling these future engineers to address the needs of our modern society.

The organising department Precision and Microsystems Engineering (PME) has close links with companies involved in consumer goods, IC manufacturing, industrial equipment, the automotive industry, aerospace and biomechanics. Many of our research projects are undertaken in collaboration with industry and directed towards future needs.

Education
The HTE programme consists of several "building blocks" which for the most part can be shaped according to the student's own interest, as long as it is in focus with the research themes. The scope for freedom of choice gradually increases in the course of the programme. The HTE track offers five focus areas. The programme starts with compulsory courses which are common to all. Once the focus area has been chosen, "research area-specific" components are introduced, which include star electives and common electives. The last part of the programme offers the greatest opportunity for students to pursue studies in their area of interest and encompasses a design/literature/research assignment, sometimes an industrial traineeship and a master's thesis. Both can also be performed within the subject of another HTE research area. This can be done in the Netherlands or abroad.

The HTE track encourages students to take a part of their study abroad, through courses or individual assignments. Students have the opportunity to participate in an Erasmus exchange programme at European institutions. In the past, students have also visited Japan, Brazil, China, South Africa, Dubai, the USA, Australia, Switzerland and Sweden amongst others.

Master's thesis
The master's thesis will be undertaken at the University and sometimes in industry. Most projects are done in close collaboration with industry or with regular feedback from the industry, or related to ongoing research programmes. This work includes aspects of modelling/simulation, design, realisation and experimental verification in varying ratios depending on the area of interest and subject.

Introduction 2

Focus Areas

Next to the HTE obligatory courses students choose a research focus in which they want to deepen their knowledge. Focus Areas within the High-Tech Engineering track are:

Mechatronic System Design (MSD)
aims at designing integrated systems of mechanisms, sensors, actuators and control to perform complex tasks while interacting in a multi-physical environment, typically at high speed and high accuracy, at various length scales. Recent trends include distributed motion, as in compliant mechanisms, as well as distributed actuation and sensing, and control techniques based on fractional order calculus and reset strategies.

Engineering Dynamics (ED)
studies the time-dependent linear and non-linear motion of mechanical structures to engineer dynamical systems. Material properties, thermodynamic interactions and physical actuation forces are studied for enhanced performance of high-speed devices, using mathematical and experimental methods to elucidate and control their complex motions. Explore the ultimate limits of high-frequency nano-electromechanical systems of atomic-scale dimensions.

Micro and Nano Engineering (MNE)
bridges the gap between the ultimate small and the macro world. Students learn to develop and optimise production and assembly processes and technologies which make use of phenomena at the nanometre level. The primary focus within the Micro and Nano Engineering group is on the production and assembly of precise and small parts and products of micrometer and nanometre scale.

Engineering Mechanics (EM)
deals with physics of mechanics and its experimental, mathematical and numerical tools, design procedures and innovative designs. It covers the foundations of mechanical engineering: the theoretical and experimental analysis of the statics and dynamics of structures and mechanical systems. Basic themes covered are Solid Mechanics, Dynamics, Computational Mechanics, Structural design and Optimisation.

Opto-Mechatronics (OM)
covers the fundamentals of optics in theory and practice as well as understanding and design of high-end optical systems and digital mirror devices. This expertise is combined with mechatronic system design treating dynamics and motion control, adaptive optics and design principles for precision positioning and thermo-mechanical stability. The focus area OM within the track HTE is currently identical to the new Track OM. This gives students more flexibility in choosing electives and liaising with a larger student population.

Introduction 3

Career Prospects

The track prepares students to fulfil key positions in companies such as NXP, ADM, ASML, VDL, Airbus, Allseas, Siemens, SKF, BMW, MI Partners and Hittech and research institutes such as TNO and NLR. Many of these organisations are keen to hire our students to be future leaders or to join multidisciplinary teams working on mechanical innovations. Research Oriented students with interest in teaching general basic aspects of Mechanical Engineering have also excellent perspectives for an academic career. Other opportunities include research at institutes or in the private sector and positions at engineering consultancy.

Exit Qualifications

The exit qualifications for the master Mechanical Engineering can be found in the common section of this studyguide.

For the track HTE, the specific exit qualifications are:
1. Competent in the scientific discipline Mechanical Engineering High-Tech Engineering. A graduate in Mechanical Engineering High-Tech Engineering is able to:
a) apply standard linear and non-linear methods, in theory and experiment;

	b) develop computational techniques for linear and non-linear analysis; c) evaluate concepts of mechatronic systems, including tribology, bearing concepts, and vibration isolation; d) synthesise special and temporal discretisation techniques, multi domain integration and interaction, and linear and non-linear solution techniques; e) describe and analyse microsystems and their working principles; f) apply numerical modelling techniques for linear and non-linear problems, simulation of multi-physics problems, and numerical optimisation techniques. 2. Competent in designing. A graduate in Mechanical Engineering High-Tech Engineering is able to: a) describe microfabrication techniques and use this in design of microsystems; b) apply scaling laws and characterisation techniques; c) design high precision mechatronic systems and compliant mechanisms.
Program Structure 1	First year: ME obligatory core courses: 11 ECTS ME recommended social course: 3 (choose at least 3 max 6 ECTS) ME recommended courses: 5 ECTS (choose at least 5 ECTS) HTE core courses I, Introlab: 2 ECTS) HTE core courses II: 19_27 (choose ast least 5 out of 7) HTE (recommended) ectives: 12_20 ECTS Second year: ME56010 Literature Assignment and Project Proposal: 10 ECTS ME56015 Internship or Midterm review: 15 ECTS ME56035 MSc master project: 35 ECTS (in most cases combined with ME56010 & ME56015 midterm) For more information please check the curriculum tabs within the study guide and/ or send an email to e.matroos@tudelft.nl

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Mechanical Engineering**

ME-HTE Obligatory Courses (14 ECTS)

ME46006	Physics for Mechanical Engineers	4
Responsible Instructor	Dr. G.J. Verbiest	
Instructor	Prof.dr. P.G. Steeneken	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Students are expected to have Physics knowledge at the level of the final exam of the physics exam at Dutch highschools (VWO) and to have Mathematics knowledge at the level of the BSc education in Mechanical Engineering (e.g. WBMT2048). If they have not, they can still do the course but it will require more self-study from the book of Tipler.	
Summary	The course ME46006 Physics for Mechanical Engineers, aims to provide mechanical engineering students with a broad background in physics. The course offers basic understanding and skills in the field of physics, that will provide a background that can be useful in many engineering situations. It can serve as a starting point for more advanced physics studies and can also help mechanical engineers to interact with electrical engineers and physicists in their future career. Since the course aims for a broad coverage of the field of physics, the course material includes many different subjects at a basic to intermediate level. In contrast to other courses, this course favours to cover a relatively large number of different topics in order to provide a broad background, instead of covering a small number of topics in-depth. This means that the amount of material that needs to be studied is larger than for other courses. A bachelor degree in mechanical, electrical, aerospace, industrial engineering or applied physics is considered prerequisite.	
Course Contents	- Waves - Electrostatics and electricity - Magnetism - Electrodynamics - Optics - Interference and diffraction - Outlook into quantum mechanics	
Study Goals	At the end of the course, students should have knowledge, and should be able to solve problems on the following topics: Waves (Chapters 15&16 of Tipler): - The wave equation - Propagation, reflection and transmission of waves - The Doppler effect Electrostatics (Chapters 21-24): - Coulomb's law and electric forces - Electric fields and Gauss' law - Voltage and capacitors Electricity (Chapters 25 & 29): - Ohm's law - DC circuits Magnetism (Chapters 27-28, 30): - Gauss' law for magnetism - Lorentz', Ampere's, Lenz' and Faraday's law - Electromagnetic motors and inductors - Maxwell's equations Optics (Chapter 31-34): - Propagation and properties of light - Light-matter interaction - Imaging systems with light - Interference and diffraction - Limits of optical systems and quantum mechanics	
Education Method	Learning methods include power point presentations, class demonstrations, and recommended homework. Lecture notes will be made available on the Brightspace. Recommended homework will be on the concepts discussed in the class and solutions will be made available on the Brightspace. The course ends with a written exam.	
Literature and Study Materials	The course will closely follow selected chapters of the book of Tipler. In addition lecture slides will be provided on Brightspace.	
Books	Physics for Engineers and Scientists, 6th edition by Tipler and Mosca.	
Assessment	Written exam.	
Department	3mE Department Precision & Microsystems Engineering	

ME46055	Engineering Dynamics	4
Responsible Instructor	Dr. F. Alijani	
Instructor	R.A. Norte	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	Multibody Dynamics A(ME41050), Multibody Dynamics B (ME41050), Nonlinear mechanics(ME46000), Nonlinear dynamics (ME46072).	
Course Contents	The dynamic behavior of structures (and systems in general) plays an essential role in engineering mechanics and in particular in the design of controllers. In this master course, we will discuss how the equations describing the dynamical behavior of a structure and of a mechatronical system can be set up. Fundamental concepts in dynamics such as equilibrium, stability, linearization and vibration modes are discussed. If time permits, also an introduction to numerical integration techniques is proposed. The course will discuss the following topics: --Review of the virtual work principle and Lagrange equations --Linearization around an equilibrium position: vibrations --Free vibration modes and modal superposition --Forced harmonic response of nondamped and damped structures -- Introduction to time integration techniques	
Study Goals	The student is able to select different ways of setting up the dynamic equations of mechanical systems, to perform an analysis of the system in terms of linear stability and vibration modes and to properly use mode superposition techniques for computing transient and harmonic responses. More specifically, the student must be able to : 1. Explain the dynamic principle of virtual work and Lagrange equations, and discuss their relation to the basic Newton laws. 2. Describe the concept of kinematic constraints and identify a proper set of degrees of freedom to describe a dynamic system. 3. Use Lagrange equations via employing the minimum set of degrees of freedom to find the governing equations of dynamic systems, and construct these equations for systems with kinematic constraints. 4. Use Hamiltons principle to find the governing equations of motion of dynamics systems. 5. Find equilibrium positions and construct the linearized equations of motion by using different contributions of the kinetic and potential energies. 6. Analyze the linear stability of dynamic systems according to their state space formulation. 7. Explain and use the concept of free vibration modes and frequencies for multi degree of freedom systems. 8. Apply the orthogonality properties of modes to describe the forced response of damped and undamped systems.	
Education Method	Lecture	
Literature and Study Materials	Course material: Lecture notes (available through brightspace) References from literature: 1. M. Géradin and D. Rixen. Mechanical Vibrations. Theory and Application to Structural Dynamics. Wiley & Sons, 2nd edition, 1997. 2. J. Ginsberg. Engineering Dynamics. Cambridge University Press, 2008. 3. D.J. Inman, Engineering Vibration. PrenticeHall, 1996. 4. L. Meirovitch. Principles and Techniques of Vibrations. PrenticeHall, 1997.	
Assessment	written exam + an assignment or a quiz	
Remarks	In case an assignment is given- The assignment will involve the use of a Matlab-like software 2. Old course code: WB1418-07	
Department	3mE Department Precision & Microsystems Engineering	

ME46085-23	Mechatronic System Design	4
Responsible Instructor	Dr. S.H. Hossein Nia Kani	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	The following are strictly required to be able to follow the course: - Basics of Multi-body Dynamics, including the ability to form equations of motion. - Basics of Electromagnetism, particularly Maxwell's equations. - Fundamentals of signals and systems (Laplace transforms, transfer function, bode plot, Nyquist stability). It is expected that students are able to use MATLAB or are interested in spending time to learn it while passing the course. This is required to complete the assignments.	
Course Contents	Course Contents Mechatronic system design deals with the design of controlled motion systems by the integration of functional elements from a multitude of disciplines. It starts with thinking how the required function can be realised by the combination of different subsystems according to a Systems Engineering approach. It should be noted that the control principles used in this course place a strong emphasis on frequency domain methods by linearising the system at its working point with the use of Bode- and Nyquist plots. The main reason for this emphasis is the strong focus in other control related courses on (non-linear) time domain related methods while linearised frequency domain related methods are still dominantly applied in the industry. A mechatronic engineer should be able to work with both methods and use them where appropriate. Some supporting disciplines, like power-electronics and electromechanics, are not part of the BSc program of mechanical engineers. For this reason this course introduces these disciplines in connection with mechanical dynamics and PID-motion control principles to realise an optimally designed motion system. The target application for the lectures are motion systems that combine high speed movements with extreme precision. The course covers the following three main subjects: 1: Dynamics of motion systems in the time and frequency domain, including analytical frequency transfer functions that are represented in Bode and Nyquist plots. 2: Electromechanical actuators, mainly based on the electromagnetic Lorentz principle. Reluctance force and piezoelectric actuators will be shortly presented to complete the overview. 3: Motion control in the frequency domain with PID and model-based feedforward control-principles that effectively deal with the mechanical dynamic anomalies (resonances and eigenmodes) of the plant. The other relevant discipline, electronics and position measurement systems is dealt with in another course: ME46005, Physics and measurement. The most important educational element that will be addressed is the necessary knowledge of the physical phenomena that act on motion systems, to be able to critically judge results obtained with simulation software. The lectures challenge the capability of students to match simulation models with reality, to translate a real system into a sufficiently simplified dynamic model and use the derived dynamic properties to design a suitable, practically realiseable controller. This course increases the understanding what a position control system does in reality in terms of virtual mechanical properties like stiffness and damping that are added to the mechanical plant by a closed loop feedback controller. It is shown how a motion system can be analysed and modelled top-down with approximating (scalar and linearised) calculations by hand, giving a sufficient feel of the problem to make valuable concept design decisions in an early stage. With this method students learn to work more efficiently by starting their design with a quick and dirty global analysis to prove feasibility or direct further detailed modelling in specific problem areas.	
Study Goals	Can analyse and derive improvements to the dynamic behaviour of an actuator-driven mechanical structure with maximum 6th order plant dynamics (incl actuator and amplifier) by means of Bode and Nyquist plots. Can select and calculate a single-axis functional electromagnetic actuator for a given specification, working according to the Lorentz and reluctance force generation principle. Can select a proper amplifier to drive the above mentioned electromagnetic actuator. Can understand the basic concept of piezoelectric actuators and their application in mechatronic systems Can identify and apply feedforward and feedback P-, PD- or PID-motion controller settings for a given plant, consisting of a dynamically realistic power amplifier, electromagnetic actuator and mechanical structure with an ideal sensor, to achieve a stable system targeting a specified maximum bandwidth or disturbance rejection.	
Education Method	Lectures are given in 12*2 lecture hours, with presentations on the theory and practice of active-controlled motion systems. In addition, students will work on assignments to implement the theory learned in the course.	
Literature and Study Materials	Only the book is allowed at the examination. Therefore, no printed copy of the first edition, no computers, e-readers, smartphones, or other items are permitted. However, small notes written on the pages of the book are allowed.	
Books	The Design of High Performance Mechatronics - 3rd Revised Edition by R. Munnig Schmidt, G. Schitter, A. Rankers and Jan Van Eijk	
Assessment	The Final Assignment will account for 50% of the final grade. The remaining 50% of the score will be based on an open-book written examination. The examination will consist of questions covering the above-mentioned study goals.	
Contact	s.h.hosseinniakani@tudelft.nl	

ME46110	Intro lab PME	2
Responsible Instructor	Dr.ir. J.F.L. Goosen	
Course Coordinator	E. Matroos	
Contact Hours / Week x/x/x/x	12/12/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Required for	Mandatory for all students of ME-HTE. This course is only open to ME-HTE students.	
Course Contents	This course consists of 3 parts: part a) Introduction week Planned in the first week of the study year: from Monday till Friday (5 days) part b) Introduction to LaTeX/ Academic Writing ..* LaTeX is a high-quality typesetting system; it includes features designed for the production of technical and scientific documentation. LaTeX is the de facto standard for the communication and publication of scientific documents. part c) Complete 2 practicals (of one or two morning or afternoon sessions, sometimes with a plenary session as introduction of the subject) the organized practicals will most likely include: 1. Labview (Q1) 2. PLC (Q1 and/or Q2) 3. Functional Analysis (Q2) 4. Computational Methods (Q2) 5. Python (Q3) 6. Surface Roughness (Q3) 8. Modal analysis of microsystems (Q3) 9. Linear Motor (Q4) Furthermore the department will organize some "comeback meetings" to: - evaluation of past quarter/semester (Q2) - explanation of the second year/ how to find a Thesis Project including explanation of our rules and regulations (Q3) - a Master Thesis Market (Q4)	
Study Goals	Learning a number of basic skills relevant for the ME-HTE Master and become acquainted with the lab infrastructure of the organizing PME research department.	
Education Method	Presentations, lectures, instructions, assignments.	
Literature and Study Materials	- handouts of powerpoint presentations. - instruction manuals - selected documentation on the infrastucture used. - exercise manuals	
Assessment	Most practical sessions are assessed by a report and/or a portfolio of generated software source code and/or a demonstration of the workings (in the case of programming activities). Final grade: Satisfied (Voldaan) / Not Satisfied You will receive a final grade if you have: a) Participated in the Introweek b) Completed the Latex / Academic writing workshop c) Finished a least 2 practicals (including the report)	
Enrolment / Application	Make sure to enroll in BrightSpace to receive all relevant information concerning offered practicals, dates etc. Only ME-HTE students will be accepted. For more information contact the Daily Coordinator of the Master ME-HTE, Eveline Matroos via: e.matroos@tudelft.nl	
Remarks	Information about introlab practical sessions will be shared with our new ME-HTE students via the daily coordinator, Eveline Matroos. Announcements concerning the registration, dates for practical sessions en information concerning the content of these different sessions is placed on the BrightSpace Page of ME46110 Introlab. Information concerning the introduction week of ME_HTE is shared with registered students in july/mid augustus via mail. Also this information will be placed on the student information BS page of Mechanical Engineering - High Tech Engineering track. For urgent questions please send an e-mail to e.matroos@tudelft.nl	
Department	3mE Department Precision & Microsystems Engineering	

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Mechanical Engineering**

ME-HTE Track Courses I (choose at least 3)

ME46015-23	Precision Mechanism Design	4
Responsible Instructor	Prof.dr.ir. J.L. Herder	
Instructor	F.G.J. Broeren	
Contact Hours / Week x/x/x/x	0/0/4/2	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course focuses on cultivating a new design mindset among students and equipping them with advanced skills in mechanisms theories. The goal is to enable students to effectively analyse complex real-life problems and extract their essential elements. Through this course, students will develop the expertise needed to construct the next generation of precision machines. In this course, students apply advanced mechanics concepts to a real world design assignment, provided by industry. Students perform conceptual, detailed and critical design steps, all within a systems engineering context. In this process, students develop the capacity to identify problematic areas, generate diverse design alternatives, and make informed design decisions. Key principles explored throughout the course include designing lightweight and rigid mechanisms, achieving precise kinematic constraints, creating low-hysteresis mechanisms, and designing precision manipulators.	
Study Goals	After the course, the student understands the principles of conceptual design of precision mechanisms. They are able to recognize and prioritize problem areas, generate design alternatives and make adequate design decisions. Theoretical part (Q3): - Analyse the degrees of freedom and constraints of a mechanism using Grübler counting and Screw Theory - Design flexure mechanisms freedom and constraint topologies - Design and analyse precision mechanisms with respect to stiffness - Analyse the effects of hysteresis and microslip - Design and analyse kinematic couplings between machine components - Design precision mechanisms with respect to dynamics and energy management Assignment part (Q4): - Identify functional requirements for a real-life (precision) machine design - Divide a real-life (precision) machine design problem into sub-assignments - Coordinate sub-assignment to constitute a coherent final solution to a real-life problem - Generate various machine design concepts - Define design specifications that can be assessed - Evaluate machine design concepts and justify the conclusions - Determine critical dimensions of a machine design based on estimates and calculations - Estimate the performance of a detailed machine design	
Education Method	The course consists of two parts: in Q3, the focus is on the theoretical skills, while Q4 is mainly devoted to the design project. In Q3, there are two hours of lecture and two workshop hours per week. In the lectures, theoretical concepts are presented. The workshops serve to increase familiarity with and intuition for these concepts through hands-on exercises. In Q4, students work together in large design teams to perform a design assignment, presented by a Dutch High-Tech company. In the lectures, students receive the necessary information on the system design process. This information is provided just-in-time to make it directly applicable in the project. Throughout the quarter, intermediate reports are written to finalize the conceptual, detailed and critical design phases. These reports are subjected to peer-feedback. A bundling of these reports will form the end-product of the design project.	
Assessment	An individual written exam on the theoretical part concludes Q3. Design reports written in groups conclude Q4. The final grade is the average of the two.	

ME46020	Micro- & Nanosystems Design & Fabrication, incl. MEMS Lab.	4
Responsible Instructor	M.K. Ghatkesar	
Instructor	Dr.ir. W.J. Westerveld	
Instructor	Dr.ir. J.F.L. Goosen	
Contact Hours / Week x/x/x/x	0/0/5/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Basics of statics (e.g. Statica WB1630-20), basics of dynamics (e.g. Introductory dynamica WB1135) and basics of electronic circuits. The expected prior knowledge is not mandatory but helps.	
Course Contents	Micro and Nanosystems are devices which perform a task by a combination of actuation, sensing, and signal processing. They are either monolithically built or conceived as a hybrid and then encapsulated in a package to control the interaction with the environment. In this course we will look at the following questions: What are the key process steps and units in building micro and nanosystems and how are they linked into a proper technology? What are the design processes that are needed to build a micro- or nano-electromechanical system? How is a M/NEMS characterized?	
Study Goals	Learn about basic micro and nano fabrication techniques and link them together to form a technology. Design a sensing or actuating device using a generic technology platform Know how to practically use and characterise MEMS. At the end of course, student should be capable to do the following with MEMS: Discuss microfabrication techniques to produce micro/nanosystems Design various types of actuators and sensors Analytically evaluate a device Take measurements, plot, analyze and infer their behaviour Design a mask layout to fabricate the designed device Present their approach of design to their colleagues and convince As the course involves lot of teamwork, students are also encouraged to develop team spirit, organization of activities, good communication and distribution of tasks.	
Education Method	Lectures, Lab experiments, Design of a device according to the specs, Tour of a micro fabrication cleanroom, Design contest, In the practical part (MEMS Lab) of the course, students will characterize various existing MEMS devices.	
Literature and Study Materials	Copies of the lecture slides will be made available	
Books	The course will make use of the following book: Foundations of MEMS by Chang Liu Publishers: Pearson Prentice Hall Students are advised to acquire this book, there will be no separate reader.	
Assessment	1) Two written reports on the experimental work (MEMS laboratory) (20% + 20%). 2) Design assignment presentation (10%). 2) A written report on the design assignment (50%).	
Special Information	The MEMS laboratory will take place every week. Due to space limitations, we divide students into two groups. While one group does experiments, the other group works on the design assignment.	
Design Content	The final product of the design will be a MEMS designed by the student team. The report should describe the design rationale, analytical/simulated performance analysis and a mask design to implement the same.	
Department	3mE Department Precision & Microsystems Engineering	

ME46060	Engineering Optimisation: Concepts and Applications	3
Responsible Instructor	Dr.ir. M. Langelaar	
Instructor	Dr.ir. L.F.P. Noel	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge	Basic knowledge of mechanical engineering and mathematics. MATLAB is used for exercises and the final project, students without MATLAB experience must gain this through self-study. Familiarity with Finite Element modelling is helpful.	
Course Contents	Formulation of optimization problems Typical characteristics of optimization problems Minimization without constraints Constrained minimization Direct and gradient-based optimization algorithms Approximation concepts Sensitivity analysis Topology optimization	
Study Goals	After successfully completing this course, you will be able to formulate a proper optimization problem in order to solve a given design problem, to select a suitable approach for solving this problem numerically, and apply such a numerical optimization procedure to the formulated problem. Furthermore, you will be able to interpret results of completed optimization procedures. More specifically, you will be able to: 1. formulate an optimization model for various design problems 2. identify optimization model properties such as monotonicity, (non-)convexity and (non-) linearity 3. identify optimization problem properties such as constraint dominance, constraint activity, well boundedness and convexity 4. apply Monotonicity Analysis to optimization problems using the First Monotonicity Principle 5. perform the conversion of constrained problems into unconstrained problems using penalty or barrier methods 6. compute and interpret the Karush-Kuhn-Tucker optimality conditions for constrained optimization problems 7. describe the complications associated with the use of computational models in optimization 8. illustrate the use of compact modeling and response surface techniques for dealing with computationally expensive and noisy optimization models 9. perform design sensitivity analysis using variational, discrete, semi-analytical and finite difference methods 10. identify a suitable optimization algorithm given a certain optimization problem 11. perform design optimization using the optimization routines implemented in the Matlab Optimization Toolbox 12. derive a linearized approximate problem for a given constrained optimization problem, and solve the original problem using a sequence of linear approximations 13. describe the basic concepts used in structural topology optimization.	
Education Method	Lectures (2x2 hours per week), exercises, online tests. Exercises and online tests count for an optional bonus of maximum 0.8 points on the final grade. There are no possibilities to redo/retake these, and the bonus only applies to reports handed in at the first report deadline. Further details on the bonus arrangement and calculation are given through Brightspace.	
Literature and Study Materials	References from literature: R.T. Haftka and Z. Gürdal: Elements of Structural Optimization, and various articles made available on Brightspace.	
Books	Course material: P.Y. Papalambros et al. Principles of Optimal Design: Modelling and Computation. 3rd edition.	
Assessment	The course is concluded through an optimization project and report. This can be done individually or in pairs. Deadlines for handing in the final report will be announced at the start of the course. Brightspace/MATLAB exercises and tests (optional) during education period. Also these can be done individually or in pairs. To pass the course, the final grade must be a 5.75 or higher. The final grade is determined by adding the obtained bonus to the report grade, followed by rounding to the nearest half/full mark (.5/.0). The maximum bonus score is 0.8. The bonus only applies to reports with grade >= 5.5 and submitted before the first report deadline.	
Remarks	Former course code: WB1440	
Percentage of Design	80%	
Design Content	The course is focusing on design optimization.	
Department	3mE Department Precision & Microsystems Engineering	
Contact	Dr.ir. Matthijs Langelaar Email: m.langelaar@tudelft.nl	

ME46070	Fundamentals of Mechanical Analysis	4
Responsible Instructor	Dr. C. Ayas	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	In this course, the students will acquire the basic know-how to formulate the equations describing the mechanical behaviour of continuous media. Mathematical tools, namely tensor algebra and tensor calculus will be described. Subsequently, kinematics in a geometrically non-linear setting will be discussed in detail along with Eulerian and Lagrangian descriptions. Non-linear stress measures will be used to formulate conservation of mass, linear and angular momentum. The course contents are the foundation of mechanical analysis.	
Study Goals	Students must be able to apply the basics of tensor algebra and calculus in direct and index notation. describe and formulate the deformation of a continuum both for small and finite deformations. analyse time derivatives of motion of continuous media. analyse the relation between different deformation, strain, and strain rate tensors. analyse the relations between different stress tensors analyse the balance of mass, linear and angular momentum both in Lagrangian and Eulerian settings.	
Education Method	Lectures (four hours per week) and homework exercises	
Literature and Study Materials	Slides Gerhard A. Holzapfel, "Nonlinear Solid Mechanics: a Continuum Approach for Engineering", Wiley, 2000 References from literature: R. Aris, "Vectors, Tensors and the Basic Equations of Fluid Mechanics", Dover, 1962. Fung, Y.C., "Foundations of Solid Mechanics", Prentice-Hall, 1965. M.E. Gurtin, "An Introduction to Continuum Mechanics, Mathematics in Science and Engineering", vol. 158, Academic Press, New York, 1982. R.W. Ogden, "Nonlinear elastic deformations", Ellis Horwood Ltd., 1984	
Prerequisites	Basic knowledge of engineering mechanics (statics, mechanics of materials, dynamics, and continuum mechanics courses from BSc) and calculus and linear algebra (Mathematics course of the BSc) is required.	
Assessment	The examination will be as follows: written exam on campus. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be: an open-book online exam with fraud prevention measures or proctoring.	
Permitted Materials during Tests	No electronic devices!	
Department	3mE Department Precision & Microsystems Engineering	

ME46300	Optics	4
Responsible Instructor	Dr.ir. S.F. Pereira	
Instructor	Dr. N. Bhattacharya	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introduction to optics: light propagation, the vectorial nature of light, interference, diffraction, optical spectra, geometrical optics	
Study Goals	At the completion of this course the student will be able to - have a good understanding of the basics of optics (i.e. making new connections between learned concepts and applying what has been learned to new situations) - explain the basic laws of polarization, birefringence, interference and diffraction - apply the theory to solve simple problems - put the theory in practice by performing experiments.	
Education Method	Lectures, exercise sections and practicals	
Books	Textbook that will be used in this course: Introduction to Modern Optics, G.T. Fowles, second edition, ISBN 100486659577 plus lecture notes and slides (available on Brightspace)	
Assessment	Individual written exam (90% of the grade). If it is required that the exam will be online, we will have an oral check after the exam and similarity check. The grade of the practicals (10% of the final grade) will be the average of the written group reports.	
Department	3mE Department Precision & Microsystems Engineering	

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Mechanical Engineering**

ME-HTE Track Courses II (choose at least 3)

ME46010	Intro to Nanoscience and Technology	3
Responsible Instructor	Dr. S. Caneva	
Instructor	Prof.dr. U. Staufer	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	Micro and Nano Engineering within the PME track	
Expected prior knowledge	ME46005 or equivalent course on classic physics (mechanics, electrodynamics, waves)	
Summary	Introduction into concepts, methods, instruments, and processes used in nanotechnology	
Course Contents	Nanoscience emerged from the analysis of basic physical, chemical and biological phenomena at the atomic to sub-micrometer scale range. By having investigated and explained fundamental questions, it broke ground for what is now often cited as being one of the most important areas for future technology developments. Based on nanoscientific concepts, new materials, processes, and devices are expected to emerge within the next few years. Nanoscience has developed its own professional jargon with expressions from its parent disciplines, which have to be know if one wants to communicate within the nanoscience community. This course establishes this basic knowledge and introduces the major instruments and methods used in nanoscience and -technology. It thus lays the base for participating in the above mentioned developments. The following chapters will be treated: The basics: - Photon: the quantization of light - de Broglie: Electrons behave like a wave - Uncertainty Principle - Wave-function - Particle-in-a-Box; quantum numbers - quantum mechanical tunnelling - Pauli principle - the structure of atom and molecules; spectroscopy Seeing and sensing at the nanoscale: - The family of scanning probe microscopes "SXM" and - The electron microscopes TEM and SEM, - Nanopore and nanogap sensing; Working material at the nanoscale: - Surface modifications by means of SXM -> highly controlled, low throughput - Bottom up synthesis by chemical means -> high throughput, challenging assembly Carbon, the amazingly diverse building block: - bottom up: the C-atom, hydro-carbons, C60, carbon nanotubes - top down: diamond, graphite, graphene Applications, e.g.: - Nanoparticles as artificial dyes, labels - Sensors (mass, pressure, bio)	
Study Goals	By the end of the course, you should be able to: LO1: Explain the basic concepts and expressions used to describe nanoscale structures and processes; LO2: Describe the measurements tools developed to probe features and properties of nanoscale materials; LO3: Discuss the top-down and bottom-up techniques used for the preparation of nanomaterials.	
Education Method	Lectures will be given on-site. Mandatory student participation: For each lecture, a team of max. 3 students will be assigned to present a 5 min summary of the previous lecture. Optional problem sets will be given and discussed on a weekly basis.	
Books	The book closest to the lecture is: Chin Wee Shon, Sow Chorng Haur and Andrew TS Wee, Science at the Nanoscale - An Introductory Textbook Pan Stanford Publishing, Singapore, 2010 ISBN: 13 978 981 4241 03 8 ISBN: 10 981 421 03 2 [Concise book with exercises and indications for further readings. Some topics are more detailed and other less than what I can do during the lecture. Some topics are not treated.] Alternative with more emphasis on Quantum Mechanics: Introduction to Nanoscience S.M. Lindsay Oxford University Press, Oxford New York, 2010 ISBN 9780199544219 The part on Quantum Mechanics can be found in any university level Physics textbook under the rubric 'Introduction into Modern Physics' or similar. Examples are Alonso-Finn Vol. 3 'Quantum and Statistical Physics' or Chapters 34 - 36 of Tipler-Mosca 'Physics for Scientists and Engineers' E. Meyer, H.J. Hug and R. Bennewitz, Scanning Probe Microscopy - the lab on a tip, Springer Verlag, Berlin, Heidelberg, New York, 2004. ISBN: 3 540 43180 2. [Covers the most important aspects of nanotools, some topics of the course are not treated] E. L. Wolf, Nanophysics and Nanotechnology, Wiley-VCH, Weinheim, 2004 ISBN 3 527 40407 4. [Gives a good introduction to some aspects of nanotechnology, however does not cover the full course and not the full depth.]	
Reader	There is no special reader, however, students will be able to download/watch: - presentation slides; - videos shown in the lectures; - supplementary texts on specific topics.	

Assessment	The examination will be as follows: Written exam on campus. Details will be published on Brightspace and discussed in class. If less than 10 candidates register for the exam, an oral exam will be given instead. Details will be published on Brightspace and discussed in class.
Exam Hours	In case of an oral exam, a Doodle poll will be circulated where students will have to register for a specific time-slot (20 minutes) in addition to the regular registration on OSIRIS.
Department	3mE Department Precision & Microsystems Engineering
Contact	e-mail: S.Caneva@tudelft.nl (preferred) Office: 34.G-1-420 phone: +31 15 27 81673

ME46025	**Manufacturing for the Micro and Nano Scale**	**3**
Responsible Instructor	Dr. A. Hunt	
Instructor	Dr.ir. J.F.L. Goosen	
Instructor	Dr.ir. M. Tichem	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Students are expected to have a basic knowledge on manufacturing technologies, e.g. gained through BSc education. Knowledge on micro/nano engineering is an advantage.	
Course Contents	This course introduces students to micro and nanomanufacturing, for the creation of new generations of micro/nano-enabled materials, devices and products. The focus is particularly on those techniques and applications that are complementary to semiconductor lithography. The students will learn about the challenges for, as well as principles and techniques that allow the establishment of a controlled and high-throughput manufacturing process. Since many innovative manufacturing approaches and resulting applications are still in the research phase, students will be introduced to selected state-of-art research in the field. The course will link manufacturing (research) to a variety of existing and emerging applications. Topics addressed in the course are: 1. Overview of micro and nanomanufacturing. 2. Selected manufacturing techniques: 2a. Multi-scale additive manufacturing 2b. Two-photon polymerization; 2c. Inkjet Printing 2d. Nano Imprint Lithography; 2e. Micro-laser cutting; 2f. Subtractive micromanufacturing; 2g. Micro-assembly. 3. Volume scale-up and integration of manufacturing processes for micro/nano-enabled materials and devices. Design for micromanufacture and assembly.	
Study Goals	In this course you will need to learn to: (1) Explain the fundamentals, pros and cons of the key manufacturing methods at micro- and nanoscale, that other than the semiconductor lithography and enable the manufacturing and the assembly of state-of-the-art materials, devices and products. (2) Evaluate the manufacturing needs for realising micro- and nanoscale devices, using individual and combined processes of the above and state-of-the-art manufacturing methods. (3) Analyse feasible process combinations for manufacturing micro- and nanoscale device designs.	
Education Method	Lectures; student team project.	
Literature and Study Materials	Handouts with lecture slides, selected papers and other (background) reading material. Provided on-line during course period.	
Assessment	Course assessment consists of (1) a test that bases on the lecture materials; and (2) a course project that bases on a manufacturing problem, as explained below. The final grade is a combination of the test and project result. (1) Passing the course requires a positive result in a test. This test bases on the manufacturing methods, process integration and design-for-fabrication topics that are covered in the lectures. (2) Passing the course requires a completing a course project for a positive grade. The project goal is analysing a specific manufacturing problem, it is conducted in small groups of 3-5 people, and it results in a graded report.	
Remarks	Participation in lectures and attendance during contact hours is expected, that is part of the teaching method. Students who cannot attend the lectures should not choose this course. Lectures are not recorded. The teaching period will be split in two phases: during the first phase lectures will be provided, during the second phase students work on the project, and (obligatory) feedback moments are being planned. Next to the responsible teacher, guest speakers from within and outside the Department are involved in the course.	
Elective	Yes	
Tags	Analysis Challenging Design English proficiency Industry Mechanics Process Project Small groups Technology	
Department	3mE Department Precision & Microsystems Engineering	

ME46035	Stability of Thin-Walled Structures 1	4
Responsible Instructor	Prof.dr.ir. A. van Keulen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	- Stability of elastic structures - Buckling and post-buckling behavior - Variational statements - Discrete and continuous models - The effect of geometric nonlinearity - Geometric stiffness - The effect of imperfections - Design sensitivities - Computational procedures - Relevance for practise	
Study Goals	Students understand the effect of geometric nonlinearities on the stability of elastic structures. Students are able to analyze the elastic stability of a structure using both analytical as well as computational techniques	
Education Method	Lectures in which students make their own notes. During the lectures small assignments will be provided.	
Assessment	Oral + Report	
Remarks	Old course code: WB1405A	
Department	3mE Department Precision & Microsystems Engineering	

ME46041	Experimental Dynamics	4
Responsible Instructor	D. de Klerk	
Responsible Instructor	Dr. M.V. van der Seijs	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge	Engineering Dynamics (ME46055) Linear algebra Different equations	
Course Contents	This course focuses on learning to understand and solve problems in the field of experimental dynamics. Motto: In theory, theory and practice are the same... In practice they are not. This course concentrates on pointing where those differences originate from. It'll be valuable for any who performs and analyses measurements in practice and tries to match his / her simulation to the experiment. This course specifically goes away from becoming "book-smart": instead it challenges you to formulate and motivate answers from various (incomplete) sources of data, using an understanding of mechanical vibrations principles and linear algebra. As such, it will trigger the curious mind to solve problems as they really occur in fields of engineering, such as the automotive and high-tech industry, where vibrations play an important role in product development. The course builds on fundamentals of mechanical vibrations as taught in the course Engineering Dynamics (ME46055), which is considered a prerequisite. Other than the specific course content, you will be challenged on: * Comprehending the concept of degrees-of-freedom: what do they really mean, how does this relate to matrix equations and aspects such as matrix rank and decomposition. * Being resourceful and independent, as to be expected from a academically trained engineer in your professional career. * Making reports that are compelling, focusing on argumentation and appearance of plots such that they best convey the message. Experimental Dynamics Theory - Modelling of dynamic system in various domains: from the numerical world (FE, mass/damping/stiffness) to the experimental world described by FRF and everything in between. - A practical look at Frequency (Fourier) Analysis: Discrete Fourier Transformation, FFT, Leakage, Aliasing and Windowing. Know it or you'll mess up your experiments. - The power of Transfer and Frequency Response Functions (FRF); why are they so useful? - Harmonic excitation (with frequency stepping), hammer impact excitation, stochastic excitation. - Experimental Modal Analysis: Do's and don't, pitfalls & challenges in practice. - Experimental Dynamic Substructuring. An alternative FEM formulation which can also use experimental data. - Virtual Point Transformation. A method to transform your experimental model in a test-based super-element model compatible with Substructuring. - Transfer Path Analysis, a useful way to identify source excitation and system sensitivities. Experimental assignments You will be working on two group assignments, meant to illustrate concepts as discussed in the lectures and to deepen insight. Teams of four students each, carry out multiple assignments involving the analysis of vehicle and helicopter vibration measurements, and an assignment involving Substructuring, Virtual Point Transformation and Transfer Path Analysis.	
Study Goals	In general the student is able to understand and analyze dynamic measurements, being aware of possible pitfalls. More specifically, the student must be able to: 1. describe the effects of Quantization, Leakage, Aliasing and Windowing in measurements and measurement equipment. 2. explain the principle of extracting modal parameters (resonance frequency, mode shape, damping ratio) from system response both in the time domain and in the frequency domain 3. discuss relative merits of different excitation techniques (shaker with frequency sweep, impact hammer, shaker with random excitation) 4. analyse operational measurements with FFT and waterfall diagrams. 5. Explain the different concepts of Transfer Path Analysis. 6. Carry out an (experimental) Dynamic Substructuring analysis along with a Component (based) TPA analysis	
Education Method	Weekly classes followed by 2 group assignments.	
Computer Use	MATLAB: the two assignments can be worked out in MATLAB. Some prior experience in the basic syntax of MATLAB is warranted.	
Literature and Study Materials	Lecture notes and your own search on the web	
Prerequisites	Engineering Dynamics (ME46055)	
Assessment	First group assignment: analysis of vehicle and helicopter measurements. You will conduct actual vibration measurements on a vehicle. Afterwards, you will analyze your own measurement data with your group and write a report. You will use MATLAB to process and analyze the data. There is also an additional measurement of a helicopter added to the assignment. The recording is of a helicopter pass-by and you are asked to figure out the speed and height of the helicopter flying but also how many blades the helicopter has and what height the microphone was placed on the runway. Second group assignment: Experimental Substructuring and Transfer Path Analysis. You will simulate the dynamic behavior of an assembly consisting of two substructures. Measurements on these substructures will be simulated using a toolbox for MATLAB.	
Department	3mE Department Precision & Microsystems Engineering	

ME46050	Advanced Finite Element Methods	4
Responsible Instructor	Dr. A.M. Aragon	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	Enriched Finite Element Methods (ME46080)	
Expected prior knowledge	Basic knowledge on finite element analysis and programming (any language though Python is recommended).	
Summary	The finite element method (FEM) is undoubtedly the established procedure for solving problems in solid mechanics. This courses gives a thorough introduction to the finite element method. For a simple 1D bar in elastostatics equilibrium, we derive the variational formulation and work out the resulting finite element discrete system of equations. This is then generalized to higher dimensions and other problems. We also look at the p-version of the finite element method (p-FEM). For these methodologies, we look at a priori and a posteriori error estimates that help us understand the convergence properties.	
Course Contents	The outline for the course is: Overview of the standard finite element method (FEM) The problem of linear elastostatics in one dimension - The strong (classical) form - The weak (variational) form - The Galerkin formulation - Orthogonality of Galerkin error - The finite element discrete equations - Iso-parametric mapping The elastostatics problem in higher dimensions - Strong form - Weak form - Discrete formulation Heat conduction The p-version of the finite element method (p-FEM) phierarchical elements in 1-D pFEM in higher dimensions - Integration and geometrical mapping - Prescribing essential (Dirichlet) boundary conditions h, p, and hp convergence Problem classification based on regularity A priori error estimates A posteriori error estimate for h-FEM Computational aspects Finite element implementation The assembly process and numerical integration Checking the implementation (high-order patch tests and construction of manufactured solutions)	
Study Goals	The study goals for this course are: - Derive the finite element discrete equations, starting from the equilibrium equation (or any other elliptic partial differential equation); - Identify the limitations of the finite element method and understand the means to overcome them; - Evaluate the performance of different methodologies on academic problems; - Extend the software package provided to simulate complex problems.	
Education Method	Lectures and projects that involve programming.	
Assessment	The course will be assessed by one project (to be performed individually) that will be handed in at the end the quarter. The project will involve coding and using the resulting code to solve the problems.	
Department	3mE Department Precision & Microsystems Engineering	
Contact	Phone: +31 15 278 22 67 Email: a.m.aragon@tudelft.nl	

ME46065	Thin Film Materials	3
Responsible Instructor	Dr. J.G. Buijnsters	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	This is a mixed lectures and project-based course. The course consists of at least 8 plenary lectures and a group project: - The lectures will cover the basics of thin films: deposition techniques, vacuum technology, film growth, microstructure and application domains. The lectures have the purpose to provide the necessary background and technological information to fuel the group projects. - The project consists of carrying out in a group (typically ~3 students) a literature research on a specific thin-film topic, which can be chosen from a shortlist of topics or is self-chosen. The students will present their joint group research in oral and written form. Plenary progress presentations will be given to the class. By the end of the course, each group will hand in a final paper and will give a final presentation to conclude their work.	
Course Contents	The course consists of 7-8 plenary lectures and a student project. The topics of the lectures are 1. Introduction, liquid deposition techniques 2. Physical vapor deposition (evaporation & sputtering) 3. Chemical vapor deposition 4. Atomic layer deposition 5. Thin-film diamond 6. Nucleation & Growth modes The project consists in carrying out in a group of few students a literature research on a specific thin film topic, chosen out of the following: 1. hard coatings 2. self-lubricating coatings 3. adhesive layers 4. thin films in MEMS/NEMS 5. thin-film based batteries 6. thin-film photovoltaics 7. coatings for implants The students can also propose a topic of their own interest. The students will present their progress in several plenary presentations. The class will challenge them with critical questions. The students will also prepare a review paper on their topic that will be peer reviewed by a different group of students. At the end of the course a final presentation will be given by all groups.	
Study Goals	At the end of the course, the student is able to: 1. describe the various liquid and vacuum deposition technologies 2. discuss the pros and cons of each of these techniques 3. identify deposition-microstructure-properties relationships for a given deposition method 4. explain the main mechanisms of thin film growth 5. find, extract and order relevant information from open literature 6. explore new trends in thin films and their applications 7. professionally present information in written and oral form 8. critically assess the information found 9. critically assess the information presented by the other groups 10. cooperate with groupmates such that everybody is responsible for the content.	
Education Method	Lectures and assignments (literature study in group, presentations, discussions, paper, and peer review). Preference will be given to on-campus lectures and on-campus presentations.	
Literature and Study Materials	Course material will be shared by the teacher and posted on Brightspace.	
Assessment	The grade will be composed as follows: 1. Final paper 50% 2. Final presentation 30% 3. Mid-term presentations 20% A facultative, individual multiple-choice quiz is held towards the end of the lecture series. In addition a peer evaluation form will be used to assess whether all team members have equally contributed to the work.	
Department	3mE Department Precision & Microsystems Engineering	

ME46072	Nonlinear Dynamics	4
Responsible Instructor	Dr. F. Alijani	
Instructor	Dr. G.J. Verbiest	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Knowledge of linear dynamics, basics of nonlinear mechanics and differential equations. The essential prerequisite is single-variable calculus, including curve-sketching, Taylor series, and separable differential equations. In a few places, multivariable calculus (partial derivatives, Jacobian matrix, divergence theorem) and linear algebra (eigenvalues and eigenvectors) are used. Fourier analysis is not assumed, and is developed where needed. Introductory physics is used throughout. Other scientific prerequisites would depend on the applications considered, but in all cases, a first course should be adequate preparation.	
Course Contents	Every mechanical and physical system is nonlinear. A thorough understanding of the effects of nonlinearities on the dynamics of these systems is therefore often essential, especially when amplitude of speeds are large. Nonlinear dynamics focuses on the measurement and mathematical description of the motion in nonlinear systems. It thus enables the design of new nonlinear functionalities and maximization of operating ranges while avoiding failure. Nonlinearities cause a wide variety of interesting new physical and dynamic phenomena that cannot be seen in linear systems. Examples include resonances with multiple coexisting solutions and chaotic dynamics. The nonlinear differential equations that govern nonlinear systems have implications that reach far beyond the realm of mechanics, including physics, meteorology, fluid dynamics, chemistry and biology. This course focuses on teaching the fundamental principles of nonlinear dynamics, to bring students to a level where they master the essential concepts of nonlinear dynamics and can apply them to solve practical cases. The course will discuss the following topics: Introduction to nonlinear dynamics together with a brief review of linear vibrations Local geometric theory, stability of equilibrium points and phase space diagrams Limit cycles and self-sustained oscillations Static and dynamic bifurcations Analytic techniques for nonlinear dynamics and perturbation methods Nonlinear resonances: Primary, sub harmonic and super harmonic response, jump phenomena, hysteresis, internal resonance, parametric resonance Computational methods for nonlinear dynamics: Pseudo arc length continuation and shooting method Physical sources of nonlinearities Applications of nonlinear dynamics at the macroscopic and microscopic scale (MEMS/NEMS) Brief introduction to chaos	
Study Goals	In the nonlinear dynamics course, the students will learn to understand, interpret and analyse nonlinear dynamic response of conservative and dissipative mechanical systems in the presence of external excitations. More specifically the students will learn to: 1. Derive nonlinear equations of motion by identifying physical sources of nonlinear dynamics 2. Analyze a diverse variety of dynamic phenomena in nonlinear systems 3. Perform practical and quantitative analyses of nonlinear dynamics systems	
Education Method	Lecture	
Computer Use	The assignment will require using software like Matlab	
Literature and Study Materials	Lecture slides will be made available via Brightspaces	
Books	Nonlinear Dynamics and Chaos With Applications to Physics, Biology, Chemistry, and Engineering Steven H. Strogatz Second Edition July 29, 2014 513 pages https://westviewpress.com/books/nonlinear-dynamics-and-chaos/	
Assessment	Written exam+ assignment.	
Remarks	The course will be taught by Farbod Alijani, and Gerard Verbiest	
Elective	Yes	
Tags	Mathematics Mechanics Physics	
Department	3mE Department Precision & Microsystems Engineering	
Contact	f.alijani@tudelft.nl g.j.verbiest@tudelft.nl	

ME46095	Multiphysics Modelling using COMSOL	4
Responsible Instructor	Dr.ir. R.A.J. van Ostayen	
Contact Hours / Week x/x/x/x	2 hour lectures on a weekly basis	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	In applied mechanics one is often confronted with a multi-physics or coupled problem: A problem that requires the (simultaneous) solution of more than one type of physical process or phenomenon in order to accurately describe the problem. Examples of multiphysics problems are fluid-structure interaction, thermal-structure interaction and electro-thermal-structure interaction, possibly combined with a control problem. Particularly in the field of Mechatronic design and MEMS multiphysics problems are frequently encountered. COMSOL MultiPhysics is a finite element code, which can be used both as a MATLAB toolbox and as a standalone program, which is particularly suited for the simulation of multi-physics systems. In this course the student will learn to recognize different types of multi-physics coupling and methods for their efficient numerical solution using COMSOL. Short homework assignments are used to practise the use of COMSOL on different types of problems and in a final assignment the student is asked to study a multi-physics problem using COMSOL.	
Study Goals	The student must be able to: 1. recognize multiphysics coupling in complex problems 2. distinguish between different types of coupling one-directional vs. bi- or multi-directional interface vs. field strong vs. weak 3. describe numerical solution techniques applicable to coupled problems 4. use COMSOL MultiPhysics on coupled problems	
Education Method	(Online) Lectures (2 hours per week) / Self study	
Assessment	Report	
Remarks	Old course code: WB2454-07	
Department	3mE Department Precision & Microsystems Engineering	

ME46115-23	Compliant Mechanisms	4
Responsible Instructor	D. Farhadi Machekposhti	
Instructor	F.G.J. Broeren	
Instructor	Prof.dr.ir. J.L. Herder	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Prior knowledge or skills required in this course are: Statics (TU Delft B.Sc. course example: WB1630); Strength of Material (TU Delft B.Sc. course example: WB1631); Linear Algebra (TU Delft B.Sc. course example: WBMT1051); Rigid-Body Dynamics (TU Delft B.Sc. course example: WB2630)	
Course Contents	While engineering usually relies on complexity to achieve more functionality and better performance, compliant mechanisms can outperform classical solutions by the less is more principle. Compliant mechanisms transmit and transform force and motion by undergoing elastic deformation of their flexible segments, as opposed to rigid body parts and joints. While it may be easier to fabricate compliant mechanisms, it is challenging to understand their underlying systems as they have a monolithic structure and their design is complex because kinematics and kinetics cannot be treated separately. This course will enable students to synthesize, design, model, build and test a compliant mechanism and experience a blend of creativity, theory and hands-on. Based on a design assignment, students apply the knowledge from the lectures (learning objectives) to their project, create a 2D/3D design, produce this by laser cutting or additive manufacturing, and generate different theoretical models (Pseudo Rigid Body model/Beam theory and Finite Element model) that will be compared with experimental data of their design. The course will conclude with a mini-symposium where students will present their work in the format of poster, presentation and demo.	
Study Goals	After successful completion of the course, students will be able to : Describe the different classes of existing compliant mechanisms and rigid-body mechanisms, and the similarities and differences between them. Apply kinematic analysis and synthesis in rigid-body mechanism design. Apply pseudo-rigid-body modelling, beam theory, and FEM modeling to a compliant mechanism. Determine functional requirements, strategies, and specs in the design of compliant mechanisms for a given application. Design a compliant mechanism for a given application using kinematic synthesis, modeling methods, and experiments. Communicate the design assignment in the form of a report, presentation, poster, and pitch based on the given format.	
Education Method	Students will master the subject by applying theory and methods to their final design assignment. The course consists of lectures, invited lectures, in-class assignments, a design project in small groups of students, presentations, discussions, and a final report. In the case of the COVID-19 type of situations, the form of the course may be slightly changed in that the lectures may be partly online. As much as possible the hands-on nature of the course will be maintained.	
Books	Compliant Mechanisms 1st Edition, by Larry L. Howell (Author)	
Assessment	The grade is based on a final group design project (70% of the final grade) and a written exam (30% of the final grade). The final group design project consists of a design paper and presentation with a given format that addresses the learning objectives of the course. Timeline: Written exam: end of Q1. Final report submission: end of Q2.	
Remarks	For this course, you need to be interested in design and have a background in mechanical engineering, civil engineering, or aerospace engineering from your BSc. Knowing kinematics and FEM (in the format of software packages) is an advantage. You will learn most of the tools needed to do your design project during the course.	

ME46120	Predictive Modelling	4
Responsible Instructor	Dr.ir. R.A.J. van Ostayen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	- Mechatronic System Design - Engineering dynamics or courses with equivalent contents.	
Course Contents	Topics: - Control of high precision mechanical systems, in particular positioning stages: -- Determine required gain and bandwidth for set-point tracking -- Coarse PID setup -- Controller tuning using Bode and Nyquist -- Feedforward control -- (Non-)collocated control -- Disturbance sensitivity - Discrete and continuous modelling of mechanical systems - Modal analysis applied to mechanical (re-)design - Positioning stage design: -- Linear stage -- H-stage - Dynamic effects in high precision systems: -- Actuator flexibility -- Guidance flexibility -- Limited mass and stiffness of frame	
Study Goals	The student is able to understand, reproduce and develop mathematical models to describe the dynamic behaviour of high precision mechanical systems using standard computer software, and to use these models to modify or redesign said systems for an improved dynamic performance. More specifically, the student must be able to: - understand, reproduce and develop a computer model describing the dynamic behaviour of a high precision mechanical system, - and use this to analyse, evaluate and optimise the (modelled) dynamic behaviour of a high precision mechanical system; - understand and apply modal analysis to describe the internal dynamics of a high precision mechanical system, - and use this to understand, and improve the performance of collocated and non-collocated control principles applied to this system, - and use this to analyse, evaluate and redesign that high precision mechanical system with respect to its internal dynamics.	
Education Method	Lectures and computer room exercises	
Computer Use	During the lectures computer simulations are performed using a system modelling environment (Matlab + Simulink + Simscape Multibody). Students are expected to bring their laptops or work together.	
Literature and Study Materials	Robert Munnig Schmidt, Georg Schitter, Adrian Rankers, Jan van Eijk, "The Design of High Performance Mechatronics", 2nd Ed., 2014 Daniel J. Rixen, "Wb1418 lecture notes: Engineering dynamics", 2008 Further reader material is provided on BrightSpace	
Assessment	Assignment	
Percentage of Design	30%	
Design Content	System level design and modelling - Improving mechanical design	
Department	3mE Department Precision & Microsystems Engineering	

ME46125	Micro and Nanofabrication for Cell Biology and Tissue Engineering	3
Responsible Instructor	A. Accardo	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This course introduces students to applications of micro and nano-engineering in the field of life sciences focusing in particular on cell biology and tissue engineering aspects. The students will be exposed to selected state-of-art research in the field and learn about how to apply micro-fabrication strategies to efficiently interface engineered structures to cells and tissues. The topics addressed in the course are: 1. 2D & 3D microfabrication techniques (optical lithography, soft lithography, etching, stereolithography, bioprinting, two-photon polymerization) 2. Introduction to cell biology (mammalian cells, cytoskeleton, primary cells, stem cells, mechanobiology) 3. Engineered in vitro cellular microenvironments (disease models, drug-screening models, organ-on-chip) 4. Micro-structures for tissue engineering (tissue microscaffolds, spinal cord, stroke, bone tissue engineering) 5. Implantable microdevices (deep-brain-stimulators, retina implants, drug-delivery microsystems)	
Study Goals	After having followed this course, students will be able to: (1) Outline the fundamental microfabrication paradigms employed in the development of micro and nano-structures for cell biology applications; (2) Critically evaluate micro-manufacturing and characterization approaches of such engineered microenvironments; (3) Identify and describe modern research and industrial micro and nanoengineering challenges related to the fields of cell biology and tissue engineering; (4) Illustrate their findings both via a written report and an oral presentation.	
Education Method	Several educational methods will be employed during the course: lectures, guest lectures from external speakers (both from academia and industry), short recaps by the students preceding each lecture and post-classroom assignments.	
Literature and Study Materials	Handouts with lecture slides, selected literature provided on-line during course period.	
Assessment	Participation to the lectures is expected. Students will be assessed on a final group mini-project that will consist of a written report (60% of final grade) and an oral presentation/exam (40% of final grade).	
Enrolment / Application	For organisational reasons, the students who plan to take this course, must enroll on Brightspace within 1 week before the first lecture.	
Department	3mE Department Precision & Microsystems Engineering	

ME46310	Opto-Mechatronics	4
Responsible Instructor	Dr. N. Bhattacharya	
Instructor	Dr.ir. S. Iskander-Rizk	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	Optical sensors and actuators are being integrated at an accelerated rate into mechatronics systems making these systems more precise, autonomous and intelligent. In this course the principles of optics, optomechanics and optical detectors are taught and discussed from the perspective of integration in optomechatronics systems for increased performance and improvement/addition in functionality. Topics included in this course are: 1) The different models of optics 2) Optical measurement methods. 3) Tolerances and sensitivities of mechanical positioning which influence optical performance of a machine. 4) Total system design where optical requirements can be translated to mechanical and mechatronical requirements. 5) Sources of light and optical detectors which can influence system performance (SNR, reliability, accuracy).	
Study Goals	At the end of this course the student can LO 1: Understand the practical applications and limitations of the different models of optics (geometric optics, diffraction optics, electromagnetic waves). LO 2: Understand the basic principles of imaging, interferometry, and spectrometry. LO 3: Understand how aberrations are related to mechanical tolerances and imaging quality. LO 4: Evaluate the function and performance of an optical system including the electronic readout module LO 5: Design an optical system for given environmental and application requirements (stress, thermal, kinematic effects) LO 6: Justify the mechatronic system components choices which can improve the performance of a given optomechanical system LO 7: Understand the influence of material choices and types of mounting optics	
Education Method	Lectures and Quizzes	
Assessment	Written exam + Quiz grade + 2 Guest lecture summary.	
Elective	Yes	
Tags	Broad Challenging Industry Mechanics Mechatronics Physics Signals	
Department	3mE Department Precision & Microsystems Engineering	

MS43325	Application of Materials in High Tech Engineering	3
Responsible Instructor	Dr.ir. S.E. Offerman	
Instructor	V. Popovich	
Instructor	Dr. H.L. Castricum	
Instructor	Dr.ir. M.J.M. Hermans	
Instructor	Dr. E.M. Kelder	
Instructor	Dr. E. Mendes	
Contact Hours / Week x/x/x/x	0.4.0.0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basics of Materials Science and Engineering. Preferably, the BSc-course 'Materials Science and Engineering' at 3mE, TU Delft.	
Course Contents	The course aims at training students in performing material and process selection by using objective and quantitative criteria. A substantial part of the course consists of a compilation of actual cases from and presented by the Dutch High Tech industry (ASML, Ceratec, Euro-Techniek, Hittech, Ramlab, Thermo Fisher, VDL ETG, Tata Steel). TU lecturers subsequently provide further insight into the related materials science background and selection method. Cases encompass: additive manufacturing, steel and metal alloys, titanium, metallic glasses, coatings, ceramics, polymers. Themes: Materials in design - Engineering materials and their properties - High tech applications - Materials selection - Process selection - Case studies - Relation microstructure / technical properties / processing / design - Cost Considerations - Environmental Impact - Data sources.	
Study Goals	Understand the importance of and attain insight into material selection in the design and production of High Tech products, systems and components therein. Acquire the tools to carry out material and process selection based on objective and quantitative criteria. Acquire experience-based knowledge of applications and materials used in High Tech industry. Obtain a firmer knowledge base in the associated material science theory. Acquire adequate basic knowledge for one's future role as an academic engineer in industry: being able to make a good first selection of materials, as well as to understand and apply the advice of materials experts and processing experts. Being able to critically assess both industrial experience and advice from materials overviews and databases.	
Education Method	Case presentations; lectures	
Computer Use	Ansys Granta EduPack Software Package	
Literature and Study Materials	Michael F Ashby Materials Selection in Mechanical Design 5th edition, Butterworth-Heinemann eBook ISBN: 9780081006108 Paperback ISBN: 9780081005996 W.D. Callister & D.G. Rethwisch Materials Science & Engineering ISBN:978-1-118-31922-2 Reader background materials science for the case studies of the high-tech industry	
Assessment	Digital exam	
Department	3mE Department Materials Science & Engineering	

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Mechanical Engineering**

ME-HTE Common Elective Courses

AE4117	Fluid-Structure Interaction	4
Responsible Instructor	Dr.ir. A.H. van Zuijlen	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Fluid-Structure interaction is a topic that covers many important and complex problems in engineering where the interaction between the mechanical behaviour of a solid structure is significantly influenced by surrounding fluids (water, air, etc) and where, in turn, the aero/hydro-dynamic forces are modified by the deformation of the structure. Although it was pioneered by aeronautics engineers to study the static and dynamic deformation of wings under aerodynamic forces (aeroelasticity), fluid-structure interaction analysis involves also the description of interaction phenomenon in constructions (e.g. wind induced vibrations), vibro-acoutics, blood flow in elastic arteries or ink flow in an actuated printer head. In the past, many semi-analytical approaches were developed to describe fluid-structure interaction. Today, complex problems interaction problems are investigated using engineering codes that couple structural models to fluid models. In this course, we will recall the basics of fluid and solid mechanics and discuss some important numerical issues appearing when coupling fluid-structure models. In particular we will shortly introduce the Finite Volume Discretization of the fluid, discuss the expression of the fluid equation on moving meshes (Arbitrary Eulerian Lagragian formulation) and discuss time integration issues of the coupled problem. We will go in more details in discussion the issues of time-integration of the coupled problem. Also the issue on sharing forces/displacements across the interface between the fluid and the structural mesh will be handled in more details.	
Study Goals	In this course you will learn the modeling philosophies and computational techniques required to set up and simulate fluid-structure interaction (FSI) problems. The course focuses on the algorithmic aspects and provides the necessary background to choose (and motivate) the appropriate methodologies to simulate an FSI problem using a partitioned solution strategy in a robust and efficient manner. After the course you will be able to: 1. Derive the Navier-Stokes equations to describe fluid flow, 2. Explain the concepts of Eulerian, Lagrangian and Arbitrary Lagrangian-Eulerian reference frames and relate them to the governing equations, 3. Derive, apply and interpret the (Discrete) Geometric Conservation Law, 4. Formulate the FSI problem in terms of dynamic and kinematic interface (coupling) conditions, 5. Explain the different approaches to impose the coupling conditions when dealing with spatial and temporal discretization, 6. Give an overview of mesh deformation methods and their properties, 7. Explain basic concepts of time integration methods (implicit vs. explicit schemes and multi-step vs. multi-stage schemes), 8. Apply time integration on a coupled problem using a partitioned approach, 9. Motivate the choice for a specific (mesh deformation/interpolation, partitioning) method for a given application,	
Education Method	Lectures (2 hours per week), group assignments	
Computer Use	Use Matlab for assignment and illustration.	
Literature and Study Materials	Course material: Lecture notes (available through Brightspace) References from literature: Structural acoustics and vibration; mechanical models, variational formulations and discretization, R. Ohayon, C. Soize, Academic Press, 1998, ISBN 0-12-524945-4 A modern course in aeroelasticity, Earl H. Dowell, Kluwer Academic Pub.,1995, isbn 0-7923-2788-8 Fluid-Structure interaction: applied numerical methods, H. Morand. R. Ohayon, Wiley ed., 1995, isbn0-471-94459-9	
Assessment	Written report & oral examination	

AE4896	Space instrumentation	4
Responsible Instructor	Dr. J.J.D. Loicq	
Contact Hours / Week x/x/x/x	4/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	The aim of this course is to provide students with an understanding of instrumentation and its role in observational strategies for missions. The course will explore the main observables of the universe and how they are measured in space. For each of the observables, the course will provide: i) The physical context of the observables and how they are generated, ii) The physics of the measure, iii) A review and explanation of the existing and under development space instruments, iv) The missions related to these observables. The course will be divided into three main parts: - The Fundamentals of Instrumentation - Remote sensing space Instrumentation - In-situ space Instrumentation.	
Study Goals	This course aims to familiarize students with the wide range of space instrumentation and observables. By the end of the course, students will be able to: - Understand and describe the relationships between instruments, missions, and observation strategies - Identify and explain the physical observables that space instruments measure, including light, ions, magnetic and electric fields, etc and related phenomena - Distinguish between the different types of space instruments, and explain how they are designed and constructed - Derive instrument requirements from scientific questions - Perform preliminary analysis and use the key driver parameters to size space instruments. Overall, this course will equip students with a comprehensive understanding of space instrumentation, enabling them to dive into space observables diversity and the complex processes involved in their measurement.	
Education Method	Lectures	
Assessment	The exam will test all theories provided in the lectures and skills practised in the provided exercises. It will be a written exam (3h). The resist will have the same settings as the first exam.	

AE4ASM516	Material Selection in Mechanical Design	3
Responsible Instructor	Ir. J. Sinke	
Responsible Instructor	C.D. Rans	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The course aims at training the students in the field of material and process selection on the basis of objective and quantitative criteria. The concept of material selection on the basis of socalled Ashby diagrams will be presented in a number of lectures.	
Study Goals	Providing the student with the tools to make material and process selection on the basis of objective and quantitative criteria.	
Education Method	Lectures	
Literature and Study Materials	The book Materials Selection in Mechanical Design by M.F. Ashby. Butterworth Heinemann. 5th edition 2017	
Assessment	Written exam	
Set-up	The course will consist of a number of lectures in which the quantitative selection process involving an increasing number of constraints and objectives will be introduced. During the lectures students are invited to solve a number of simpeler cases.	

AE4ASM526	Spacecraft Thermal Design	3
Responsible Instructor	I. Uriol Balbin	
Instructor	N. van der Pas	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic knowledge of orbital mechanics Understanding of general design spacecraft systems	
Course Contents	General knowledge of space thermal control hardware, system design and analysis processes are presented. Using the principles of heat transfer, and more specific of radiation and conduction, the methods and tools for thermal design of a spacecraft are presented. Numerical analysis will also be introduced via the software ESATAN-TMS.	
Study Goals	The student is able to generate a thermal design of a satellite using heat transfer analysis and thermal engineering techniques. In more detail the student will be able to: - Understand the spacecraft thermal environment - Define the thermal subsystem and compare the different hardware options. - Understand the design cycle of the thermal subsystem. - Apply the concepts of heat transfer (radiation, conduction), heat balance and heat flows - Generate and use analytical thermal analysis models - Generate and use numerical thermal analysis models	
Education Method	Lectures + homework Lectures are given by an experienced thermal design engineer from Airbus NL. Students are given access to a discussion board provided with the course in Brightspace. The discussion board will be used as the main communication tool to ask questions on the material taught or in preparation of an assignment. Answers to the questions may be given by fellow students and/or the lecturer(s).	
Computer Use	Software licenses are available during the lecture period Software can be used in the computer room or in the personal computer (Windows/Linux)	
Literature and Study Materials	Lecture notes available on Brightspace.	
Assessment	The final grade of this course is based on 2 homework assignments (60%) and one final exam (40%). A minimum grade of 5 is required in each part.	
Enrolment / Application	This class has a limit of 80 students, due to the limit on the software licenses. Please enroll in Brightspace before 29th of March 2024 if you intend to take this class.	

AE4S12-23	Space systems engineering	4
Responsible Instructor	Prof.dr. E.K.A. Gill	
Contact Hours / Week x/x/x/x	4/x/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	General engineering knowledge at completed BSc level. AE1110, AE1111, AE2111, AE3211, or similar; AE3524 (for students, not having AE BSc)	
Parts	1. Introducing Space Systems Engineering 2. Designing Systems 3. Integrating Systems 4. Managing the Systems Engineering Process 5. Managing the technical effort 6. Managing interfaces and configuration; Conclusions	
Course Contents	The course covers Space System Engineering exemplified by the design of a spacecraft subsystem for a satellite mission of high societal impact. It introduces a process which allows the creation of successful systems, such as space applications, space missions and satellite subsystems. Methods and tools are presented and exercised which will improve the depth and breadth of Space Systems Engineering graduate level education at the TU Delft by emphasizing the need for the end-to-end approach and life cycle of space systems, including cost and risk handling and using state-of-the-art methods, such as elements of Model-based Systems Engineering. Upon completion, the student will have a firm understanding of advanced Systems Engineering and be able to apply adequate methods and tools which help to create successful systems.	
Study Goals	The course mission is to enable students to realize a successful space system in an end-to-end Systems Engineering approach. There are three high-level learning objectives: 1. Participants shall be able to explain the objectives and tasks of Systems Engineering for realizing successful systems together with their needs, potentials, benefits and limitations in a context which comprises Business Engineering and Management. 2. Participants shall be able to design an end-to-end Systems Engineering process demonstrating a smart balance of risks of cost, schedule, and performance. 3. Participants shall be able to use Systems Engineering methods and tools to design a high-level satellite subsystem, such as the Attitude and Orbit Control System (AOCS).	
Education Method	Lectures and tutorial Individual homeworks from lectures to practice material Group Assignment including peer-reviews to design a satellite sub-system, done by students in small groups The graded group report and graded individual homeworks comprise the formal course close-out.	
Literature and Study Materials	Required: Lecture Slides AE4-S12 Space Systems Engineering. Optional: Larson W. J.; Kirkpatrick D.; Sellers J. J.; Thomas L.D.; Verma D.: Applied Space Systems Engineering. McGraw Hill (2009). Optional: Space Mission Analysis and Design, 3rd edition (Space Technology Library, Vol. 8) Eds.: Wiley J. Larson, James R. Wertz (1999)	
Assessment	4 graded individual homeworks + 1 Group Assignment. Acceptance of 6 homeworks is required to pass the course. Passing both assessment elements is required to pass course.	
Enrolment / Application	Students have to enroll via Brightspace. Professionals are welcome to the course. Conditions for their participation and registration are available with the Management Assistants of the Department of Space Engineering.	
Set-up	The course comprises lectures, guest lectures, tutorial, group work, peer-review, and individual homeworks.	

AP3122	Advanced Optical Imaging	6
Responsible Instructor	Dr. J. Kalkman	
Instructor	Dr. K.S. Grußmayer	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Expected prior knowledge	BSc level courses on analysis, systems and signals, waves, and optics. Experience in programming with Python.	
Course Contents	We explain the basic properties of advanced optical imaging systems within the framework of Fourier optics and linear systems. We treat wave propagation, diffraction, phase transformation of lenses, modulation by apertures and objects. The spatial domain optical systems description will be transformed to the Fourier domain description of coherent and incoherent optical systems. Additionally, the course will treat various advanced imaging systems such as microscopes with different contrast mechanisms, optical coherence tomography, digital holography, and adaptive optics. We will show applications of microscopy in biomedical imaging. The course consists of lectures, pen and paper assignments, and (Python) programming assignments.	
Study Goals	The learning objectives of this course are: - Students can quantify the performance of optical imaging systems by applying Fourier optics methods - Students can theoretically describe the working of advanced optical imaging systems - Students can apply computational optics methods to model and analyze imaging concepts	
Education Method	Lectures, (pen and paper) homework, and computational optics assignments.	
Computer Use	Part of the course consists of making assignments in Python.	
Literature and Study Materials	Lecture notes and online material. The book: Introduction to Fourier Optics by J. Goodman is used as reference material.	
Assessment	Final written exam, computational optics assignments, and homework. The exam grade and the assignments grade both need a grade higher than 5.0. Homework is assessed by pass-fail. A maximum of one point bonus is provided on the end grade according to the ratio #passes/6. The final grade is computed with the following formula: Final grade = 0.6*(Exam grade) + 0.4*(Average assignment grade) + (Homework). The validity of result of the computational optics assignments and homework is restricted to one academic year.	
Permitted Materials during Tests	Handwritten notes page.	
Special Information	For students from Leiden University: registration as a guest student is required for Brightspace access and registration of grades! See: https://www.tudelft.nl/en/student/administration/enrolment/enrolling-as-a-minor-student-guest-student/	

AP3391	Geometrical Optics	6
Responsible Instructor	Dr. F. Bociort	
Instructor	Dr. B. Kruizinga	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge		
Course Contents	This course consists of two parts: the larger part is about the theoretical basis of geometrical optics, the second part is an introduction to optical system design. The topics are: - Fundamentals of geometrical optics (geometrical optics as a limiting case of wave optics, the eikonal function, rays and wave fronts, ray paths in inhomogeneous media) - Ray tracing (Snells law in vector form, formalism for reflection, refraction and transfer, ray failure, aspherical surfaces) - The paraxial approximation (paraxial and finite rays, matrix formalism, characteristics of ideal imaging, principal planes, telescopic systems, aperture and field stops, pupils, vignetting, marginal and chief rays, Lagrange invariant, F number, telecentric systems) - Aberrations (transverse ray aberration, wave front aberration and the relationship between them, power series expansions for optical systems with or without rotational symmetry, rotationally invariant combinations of ray parameters, defocusing, Seidel aberrations. Aberration balancing, caustic, Design aspects: situations when some aberrations are more important than others, aplanatic surfaces, ideal placement of aspheric surfaces.) - Optical design software, local and global optimization of optical systems.	
Study Goals	- Mastery of the concepts, theories and methods listed above at an advanced academic level. - Development of the ability to perform simple lens design tasks with state-of-the-art lens design software.	
Education Method	Oral lectures and independent work with lens design software. Because of the work with software, late enrollments in this course (more than 10 days after the 1st lecture) are not accepted.	
Literature and Study Materials	1. J. Braat, Diktaat Geometrische Optica , TU Delft 1991 (in Dutch; English-speaking students should use Born and Wolf (Ref 5) instead); 2. J. Braat, Paraxial Optics Handout (on Brightspace); 3. W.T. Welford, Aberrations of Optical Systems, Adam Hilger, 1986 (or the earlier version Aberrations of the Symmetrical Optical System, 1974); 4. F. Bociort, Optimization of optical systems (can be found on Brightspace). Supplementary reading (not required for the exam, just if you want extra depth on some subject): 5. M. Born and E. Wolf, Principles of Optics; 6. R.R. Shannon, The Art and Science of Optical Design, Cambridge University Press, 1997; 7. D. Sinclair, Optical Design Software, Handbook of Optics, Chapter 34.	
Assessment	The final mark will be for 70% the mark for the theoretical part (oral exam) and for 30% the mark of the design part (submitted solutions to exercises). To pass the exam, for both parts the minimal mark is 5.0.	

AP3401	Introduction to Charged Particle Optics	6
Responsible Instructor	Dr.ir. J.P. Hoogenboom	
Instructor	Dr. C.W. Hagen	
Instructor	Dr. A. Mohammadi Gheidari	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge		
Course Contents	Electron and ion lenses, aberrations, deflectors, multipoles, spectrometers, simulation programmes, transmission and scanning electron microscopes, lithography tools, electrical and magnetic fields in vacuum; Laplace equation, Fourier analysis, numerical methods, series expansion, flux lines, equipotential planes, making sketches of these; Geometrical optics: focal point, thick lens model, matrix description, phase space, Liouville, aberrations; Calculation of trajectories: paraxially in lenses, spherical and chromatic aberration constants, paraxially in multipoles, Lagrangians, manual calculations, analytically, numerically far from the axis, adiabatic, wave character; Partical optical elements: magnetic lenses, electrostatic lenses, electron sources, multipoles, analyzers; Partical optical systems: transmission electron microscope, scanning electron microscope (probe calculations), electron beam pattern generator, ion beam pattern generator.	
Study Goals	Understand electron and ion beam instruments and be able to design basic optical components (lenses, quadrupoles)	
Education Method	Explanation of principles, self study of material, assignments, discussion.	
Literature and Study Materials	course book and material on Brightspace.	
Reader	Reader digitally available, a hard copy can be obtained via the instructors (by payment of printing costs)	
Assessment	Assignments	
Elective	Yes	
Tags	Physics	
Studyload/Week	8 hours per week	

BM41155	3D Printing	4
Responsible Instructor	Dr. J. Zhou	
Responsible Instructor	Prof.dr. A.A. Zadpoor	
Instructor	Ing. M.A. Leeflang	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	in Track I (Musculoskeletal Biomechanics)and Track II (Medical Devices and Bioelectronics) of MSc Biomedical Engineering BME) and in the specialization of Bio-Inspired Technology (ME-BMD-BITE)	
Expected prior knowledge	BSc in mechanical engineering, materials engineering, industrial design engineering and aerospace engineering with a basic knowledge of materials (e.g. WB2330)	
Parts	- Overview of AM technologies, their principles and applications - Materials selection (polymers, ceramics, metals and composites) - Powder-bed fusion processes - Powdered materials and powder fabrication - Powder characteristics in relation to AM - Structural integration during melting and sintering - Extrusion-based systems - Design for functionality and 3D printability Software issues and AM limitations - Other powder-based manufacturing technologies as alternatives to AM - Post-processing of AM products - Mechanical testing of AM products - Case studies AM for engineering and medical applications - Assignment and presentations	
Course Contents	Additive manufacturing (AM), also known as 3D printing the process of joining materials to make objects from 3D model data, usually layer upon layer, as opposed to conventional subtractive manufacturing technologies, is set to revolutionize manufacturing across a wide range of industries. The course 3D printing - AM technologies is designed to equip students with the fundamental knowledge of AM and basic hands-on experience in AM. The topics include various AM processes, their principles, capabilities and limitations, materials used for AM processes, powdered material fabrication and characterization, CAD designs for AM, alternative powder-based manufacturing technologies, current and future AM applications, as well as the mechanical evaluation of AM parts. The primary focus is placed on powder-based AM for metallic parts. In addition to a series of lectures, project groups, each composed of 3-5 students, will create their own designs of the objects made of polymer PLA, formulate requirements in terms of geometry, dimensions, tolerances, mass, surface finish, functionality and mechanical properties of these objects, redesign them by making use of SolidWorks, print them, test and evaluate 3D printed products against the requirements, and document the projects in the form of reports and presentations. The new design is expected to reflect the distinctive features and freedom that AM has to offer (thin walls, lattice structures, form-free channels, integrated parts/functions, and customization), but within the capacity of the printing machines (bounding box size, minimum feature size or wall thickness). The report and presentation are expected to contain: reason for the selection of the object, weakness of the existing design, new design features, actual testing results and recommendations.	
Study Goals	Students will be familiar with the concept of AM, various AM processes, AM applications, materials for AM, product design for AM, post-processing techniques and evaluation methods. In addition, students will gain basic hands-on experience in product development via AM. Once successfully completing the course, the students will: - Have a fundamental knowledge of 3D printing/AM technologies for a wide range of applications - Be able to explain the benefits, advantages, challenges and limitations of individual AM technologies as well as alternative powder-based manufacturing technologies - Have an insight into selective laser sintering and melting (SLS/SLM) processes regarding process control in relation to the geometry and mechanical properties of the products - Become familiar with the common (metallic) materials for 3D printing and their characterization techniques - Understand printable geometry design rules and the common methods of digital file creation - Become familiar with the common techniques for post-processing - Become familiar with the common methods for the mechanical evaluation of 3D printed materials and functional parts.	
Education Method	Lectures and student projects Attending lectures and performing assignment are both compulsory	
Computer Use	- Object design with Software (SolidWorks Student Engineering Kit)	
Course Relations	Biomaterials, Tissue biomechanics and other engineering courses in MSc biomedical engineering, materials science and engineering, mechanical engineering, marine technology, aerospace engineering, design for interaction, integrated design and strategic design	
Literature and Study Materials	- Additive Manufacturing Technologies, Ian Gibson, David Rosen, Brent Stucker and Mahyar Khorasani, 3rd Edition, 2021, Springer (chapters: 1, 2, 3, 5, 6, 7, 10, 15, 16, 18, 19 and 21) - Lecture materials	
Books	- Additive Manufacturing Technologies, Ian Gibson, David Rosen, Brent Stucker and Mahyar Khorasani, 3rd Edition, 2021, Springer (chapters: 1, 2, 3, 5, 6, 7, 10, 15, 16, 18, 19 and 21)	
Reader	- Lecture materials	
Assessment	Written exam	
Department	3mE Department Biomechanical Engineering	

CH4011MS	Polymer Science	4
Responsible Instructor	Prof.dr. S.J. Picken	
Responsible Instructor	Dr. E. Mendes	
Instructor	Dr. G.A. Filonenko	
Contact Hours / Week x/x/x/x	The course description will be available soon.	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Course Contents In this course, you will be taught the basics of polymer science starting from definitions and various kinds of molecular structures of polymers until their properties in solution, melt and bulk material such as in elastomers, gels and polymer glasses. You will be taught how universal polymer chain models (ideal chain and excluded volume chain) can describe molecular behavior and chain conformations in bulk or solution for polymers with any chemical structure. Visco-elasticity of polymer solutions in dilute, semi dilute and melt will be considered from non-entangled to entangled regime in solutions and melts. Chain dynamics, reptation theory and tube models will be considered and you will be taught how to interpret experimental data in the light of those theories. You will learn how to extend visco-elasticity predictions to the glassy state and how free volume theories describe a polymer glass. Polymer miscibility and polymer blends, block copolymers self-assembly in solution and in melt and other ordered phenomena in polymers (crystallinity and liquid crystallinity) will also be taught. Finally, main aspects of polymer composites will be taught. We will give some examples of how polymers are applied and which one(s) to choose for the purpose.	
Study Goals	Study Goals Knowledge 1) be able to explain universal polymer chain models and chain statistics in relation to chemical structure 2) be able to explain what are the main quantities related to polymers such as, persistence length, radius of gyration, reptation time, glass transition temperature, etc. 3) be able to describe how various polymer characterisation methods work Understanding 4) be able to explain the role of entropy and of excluded volume interactions in polymer conformations and visco-elasticity 5) be able to formulate the concepts of miscibility, solvent quality and phase separation in polymer solutions and polymer blends 6) be able to formulate the concepts of entanglements and polymer tube Analysis (critical thinking) 7) be able to interpret quantitatively (characterization) experimental data on polymer materials 8) be able to calculate basic quantities related to structure and dynamics of polymer materials Application 9) be able to select or propose a polymer for a given application based on knowledge acquired in the course. 10) have a general overview of what polymers are used for, also discussing some examples.	
Education Method	Lectures	
Literature and Study Materials	Book part 1: Physics of Polymers M. Rubinstein & R. Colby Oxford Book part 2: From Polymers to Plastics A.K. van der Vegt VSSD 2002 Delft University Press	
Assessment	written exam with open slides	
Elective	Yes	

CIE5142	Computational Methods in Non-Linear Solid Mechanic	3
Responsible Instructor	Prof.dr.ir. L.J. Sluijs	
Contact Hours / Week x/x/x/x	4/0/0/0 - This course is discontinued. Final exam opportunities are still offered for this course in academic year 23/24. For details please read the information below under assessment and judgement. For more information on supporting materials and possible contact hours: please check the Brightspace page of this course.	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	In the lecture series computational techniques for the description of nonlinear behaviour of materials and structures will be treated. Topics of the course are: 1. Mathematical preliminaries; 2. Structure of nonlinear finite element programs; 3. Solution techniques for nonlinear static problems; 4. Solution techniques for nonlinear dynamic problems; 5. Plasticity models for metals and soils; 6. Fracture models; 7. Visco-elastic and viscoplastic models for time-dependent problems; 8. Computational analysis of failure and instabilities; 9. Geometrically nonlinear analysis.	
Study Goals	The course provides the student with the basic knowledge to adequately use nonlinear finite element packages.	
Education Method	Lectures	
Assessment	Oral examination on the basis of a set of exercises. Examination mark is final mark.	
Expected prior Knowledge	Basic knowledge on the finite element method Advise: CIE5123	
Academic Skills	--	
Literature & Study Materials	Lecture notes: "Computational methods in non-linear solid mechanics", R. de Borst and L.J. Sluys	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	--	
Collegerama	Yes	

ET4117	Electrical Machines and Drives	4
Responsible Instructor	G.R. Chandra Mouli	
Responsible Instructor	H. Vahedi	
Instructor	G.R. Chandra Mouli	
Responsible for assignments	H.J.M. Olsthoorn	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AC Machines (ET4121) Control of Electrical Drives (ET4291) Electric vehicles & charging technology (SET3100)	
Expected prior knowledge	Strongly advised: Bachelor program or fundamental courses in Electrical engineering that covers AC power, phasors, circuits, etc.	
Course Contents	The course gives an overview of different types of electrical machines and drives. Various types of mechanical loads are discussed. Maxwell's equations are applied to magnetic circuits including permanent magnets. DC machines, induction machines, synchronous machines, switched reluctance machines, brushless DC machines and single-phase machines are discussed with the power electronic converters used to drive them. Three lab practicals for DC machines, Induction machines and Synchronous machines, respectively.	
Study Goals	By the end of the course, students must be able to: - Apply Maxwell's equations to magnetic circuits including permanent magnets - Explain the construction and principle of operation of the DC machine, (single and three phase) induction and synchronous machine, permanent magnet AC machines, switched reluctance machine, stepper motor, hysteresis motor and universal motor - Derive equations describing the steady-state performance of DC machine, induction machine, synchronous machine and permanent magnet AC machines - Estimate the equivalent circuit parameters; torque-speed characteristics; iron, copper and mechanical losses; power factor; efficiency and operating mode of DC, induction, synchronous machine and permanent magnet AC machines - Verify the operation of DC, induction and synchronous machine using tests on an experimental setup - Choose the appropriate power electronic converter and method for speed control and starting of the machine	
Education Method	- Classroom lectures and problem solving workshops, - Demonstrations using machines and animation in the classroom, - Online exercises with self-evaluation, - Three lab practicals (half day each, done in groups) for DC machines, Induction machines and Synchronous machines.	
Literature and Study Materials	Primary reading: P.C. Sen, Principles of electric machines and power electronics, New York: John Wiley and Sons, 2013 (3rd edition), ISBN: 978-1-118-80434-6 Secondary reading: A.E. Fitzgerald, C. Kingsley, S.D. Umans, 'Electric Machinery', New York: McGraw-Hill, 2003 (6th edition) ISBN 0-07-112193-5	
Assessment	The final grade of the course consists of the following components: - Written exam (100% Weightage) - Closed book, Scientific (non-graphic) calculator can be used, No formula sheets. - Completion of the three lab practicals and approval of the lab reports is obligatory to complete the course. Repair possibilities: Written exam: Resit in next quarter In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	In order to complete the course and get a passing grade, students need to complete the three half day lab practicals and have their lab reports approved. If the practicals cannot be organized due to unforeseen circumstances, remote practicals are offered that are done individually or in smaller groups. If student does not complete the three half -day lab practicals, he/she will get a chance to resit that part only the year after.	

ET4257	Sensors and Actuators	4
Responsible Instructor	Prof.dr. P.J. French	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	P-study	
Course Contents	The course silicon sensors and Actuators gives an overview of the most important principles related to sensors and actuators fabricated in integrated silicon technology. The sensors are divided into those for optical, mechanical, thermal, magnetic and chemical signals. These domains will be dealt with from basic principles leading to the applications. The second part of the course will deal with actuators. The actuators lectures give the range from large machines down to silicon micromachined device in the micron range. The course is designed for students who will perform their thesis work in one of the laboratories within the faculty working on or using sensors	
Study Goals	The aim of this course is to learn about the physics and electronics of transducers. This brings together different disciplines develop these systems.	
Education Method	Lectures	
Literature and Study Materials	Lecture notes Sensors & Actuators	
Assessment	Written, essay or oral. Assessment material: at least 6 chapters from the book Sensors & Actuators. Students can choose which form of assessment they want to take. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4260	Microsystem Integration	4
Responsible Instructor	T. Manzaneque Garcia	
Instructor	K.M. Dowling	
Instructor	F. Arroyo Cardoso	
Contact Hours / Week x/x/x/x	0/0/0/3	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Electronic Instrumentation (EE4C08) Integrated circuits and MEMS technology (ET4289) Sensors & Actuators (ET4257)	
Course Contents	A microsystem is a complete instrument on a chip that can involve different domains of physics (electrical, mechanical, magnetic, etc.). The challenges associated with the combination of elements of different nature, including transducers and circuits into a single-chip integrated system, are more than compensated by the opportunities this concept offers in a wide range of applications. In this course, general concepts related to microsystems as sensors are first discussed. Secondly, approaches for multi-domain modeling are introduced and applied to the modelling electro-mechanical systems. Then, different transduction mechanisms, that allow for the conversion of signals between different physical domains, are introduced. Finally, all the modelling concepts are extended to the design of microsystem-based sensors. Throughout the course, the software package Comsol is introduced and used in several tutorials. The course includes theory (~8 lectures), Comsol tutorials, and tutorials for the realization of a final project, in which a microsystem is designed and modelled in groups.	
Study Goals	By the end of the course, you will be able to: - Identify parts of microsystems and their function. - Identify equations that govern the behaviour of different microsystem components, from different domains of physics. - Use electrical domain models for multidomain microsystems. - Use the finite element method (FEM) to model multidomain microsystems, using Comsol. - Analyze the dynamic behaviour of linear microsystems. - Design microsystems that meet given specifications, using both electrical domain modelling and FEM.	
Education Method	Lectures, tutorials and hands-on practical sessions with Comsol.	
Computer Use	The student requires access to computer systems for carrying out the Comsol tutorials and final assignment.	
Literature and Study Materials	Literature plus lecture notes.	
Assessment	The final grade of the course consists of the following components: - Final assignment report (50%) - Oral exam (50%) Final grade calculation: (0.5 * Final assignment report + 0.5 * Oral exam) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4277	Microelectronics Reliability	4
Responsible Instructor	Prof.dr.ir. W.D. van Driel	
Instructor	Prof.dr. G.Q. Zhang	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course aims to provide the students with a thorough understanding of the reliability issues involved in electronic components and systems. The following subjects will be treated. - Basic reliability definition and lifetime distributions. - Reliability prediction methods. - Physical failure mechanisms in electronic components - Package related failures - Reliability screening and reliability testing - Failure analysis methods - Reliability data handling - Design considerations - System reliability	
Study Goals	After the course the students: - should see reliability as a basic requirement that should receive attention throughout a products complete lifecycle: specification, design, production, exploitation and disposal. - should have a sound understanding of the physical background of failures and the reliability test and failure analysis methods available. - should be able to make a proper reliability prediction based on available reliability (test) data. - should be able to evaluate reliability and availability figures of complex systems based on component reliability data. - should be able to make the proper design and maintenance choices to optimize these figures.	
Education Method	Lecture	
Assessment	written exam or project (project only possible when homework assignments are passed with average grade 6 or more) In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4391	Advanced Microelectronics packaging	3
Responsible Instructor	Prof.dr. G.Q. Zhang	
Responsible Instructor	M. Mastrangeli	
Instructor	Dr.ir. R.H. Poelma	
Instructor	Prof.dr.ir. W.D. van Driel	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	This is a multi-diciplinary course, dealing with scientifically challenging and economically important technology issues. MSc students from but not limited to electrical engineering, mechanical engineering, material science and physics are encouraged to attend.	
Course Contents	1. Introduction of micro/nanoelectronics packaging - Basics and state of the art of advanced front-end and back-end technologies - International roadmaps and strategic research agenda - Major application trends 2. Interconnect technologies 3. MEMS & sensor packaging 4. 3D and wafer-level system integration 5. Application specific packaging -1 (IC packaging, etc.) 6. Application specific packaging -2 (LED, etc) 7. Application specific packaging -3 (RF, Power, automotive, healthcare, etc.)	
Course Contents Continuation	As the bridge between devices and various multi-funcational electronics systems, microelectronic system integration and packaging control more than 90% of the size, 60% of the cost, and largely the system performance and reliability. It is one of the most fascinating and rapidly developing technology and business fields in semiconductors, playing a dominant role in the development of future micro/nanoelectronics and systems.	
Study Goals	At the end of the course, the student should be able to: 1) Describe and assess basics and state-of-the-art in microelectronic packaging and reliability; 2) Describe and assess application-specific packaging and associated requirements, constraints and solutions for reliability; 3) Design, implement and characterize a standalone electronic and/or microelectromechanical system.	
Education Method	Lectures and field visit.	
Assessment	Assessment is based on mini-research projects, conducted preferably in teams. The projects are proposed by the instructors, and can also be proposed in agreement with the instructors by the students based on the content of the course or personal interests. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.ing. H.W. van Zeijl	
Co-Instructor	M. Mastrangeli	

ME41096	Bio Inspired Design	5
Responsible Instructor	Dr.ir. P. Breedveld	
Instructor	B. van Vliet	
Contact Hours / Week x/x/x/x	4.4.0.0	
Education Period	1 2	
Start Education	1	
Course Language	English	
Expected prior knowledge	Mechanics and basic design experience in hand sketching and CAD software, like SolidWorks.	
Course Contents	The course Bio-Inspired Design gives an overview of non-conventional mechanical approaches in nature and shows how this knowledge can lead to more creativity in mechanical design and to better (simpler, smaller, more robust) solutions than with conventional technology. The course discusses a large number of biological organisms with smart constructions and unusual internal mechanisms and presents a number of technical examples and designs of bio-inspired (surgical) instruments, devices, robots, vehicles and machines. Examples of topics: Life and its creation, engineering in biology, strength at low weight, stiffness with soft structures, robustness and redundancy, storing energy in compliant springs, energetically efficient muscle configurations, clamping with hands, claws, suction, glue, dry - and wet adhesion, biological vibration systems, biological walking, swimming, crawling and flying, locomotion of micro- and single-celled-organisms, biological sensors, soft robots, walking robots, flying robots, snake-like surgical instruments and devices. To teach students how to create smart and truly innovative designs, students are trained with the ACCREx creative design method that was developed at the 3mE department Biomechanical Engineering. Alternating intuitive brainstorming with logical, scientific abstracting and categorizing, ACRREx structures and guides a design process towards fundamentally new design solutions. Structure of the course: 1. Engineering & biology - Life and its creation - Finding biological information - Creating creativity 2. Bioconstruction (how are creatures constructed) - Mechanical stiffness & motion - Hydrostatic stiffness & motion 3. Biopropulsion (how do creatures move) - Flying at macro- & microscales - Floating, rolling & adhesive devices - Sliding, crawling & snake-like devices - Walking, compliant devices & compliant actuation 3. Bioclamping (how do creatures grasp) - Friction & biological grasping - Biological hands & prosthetic hands - Biological adhesion	
Study Goals	The student must be able to: 1. describe methods for creative design 2. identify mechanical working principles and phenomena of biological creatures - explain their construction, motion, and/or processing mechanisms - formalize the essence of these mechanisms in models - derive non-conventional design principles from these models 3. implement these design principles in innovative mechanical devices - summarize the transition process from the biological to the mechanical domain - present their design in drawings or preferably in working models	
Education Method	Interactive lectures Interludia with driven scientists in a talkshow-like setting Student presentations about the design process Reflective Q&A sessions	
Assessment	Students are subdivided into multidisciplinary student groups. Each student group receives a design assignment for which a novel, bio-inspired solution has to be found. During the design process, the student groups are coached by the teachers and have to give a number of presentations about their progress to the other student groups. The examination consists of three parts: 1. Each student group hands in a scientific paper covering a problem analysis, the biological background, the design process, the final solution and an evaluation. 2. Each student group gives a final presentation to the teachers and the other student groups. 3. Each student group fabricates a demonstration model that shows the essence of the final solution. The final mark of each individual student is based on a weighted average between these group results and his/her functioning within the student group.	
Department	3mE Department Biomechanical Engineering	

ME45050	Microfluidics: Applied Theory and Lab	3

Responsible Instructor	Dr. D.S.W. Tam
Instructor	H. Bazyar
Instructor	M.K. Ghatkesar
Instructor	Prof.dr. U. Staufer
Contact Hours / Week x/x/x/x	HC 0.0.2.0 + PT 0.0.0.4
Education Period	3 4
Start Education	3
Exam Period	3 4 5
Course Language	English
Expected prior knowledge	Bachelor degree level knowledge in fluid mechanics (WB2542). The measurement techniques course (ME46007) is recommended but not mandatory.
Course Contents	In this master level course, you will be introduced to the important theoretical concepts of fluid flow (low Reynolds number hydrodynamics, transport phenomena on the micron scale, electrical double layer, electroosmosis and electrophoresis) relevant to the general design and fabrication of microfluidic devices and circuits (e.g., channels, valves and flow controllers) . You will apply these general concepts in a group project, which includes the design, the fabrication, and the testing of a microfluidic device (e.g., droplet generator, liquid steering in channels with valves/pumps, flow rate measurement, visualization of laminar and parabolic flow, diffusional mixing and effect of flow rate, microchip gel electrophoresis for dye separation).
Study Goals	After completion of the course, you will be able to: 1. Apply the theoretical concepts of fluid flow to the design of a microfluidic device. 2. Be able to design, fabricate and motivate the chosen fabrication technique for a simple microfluidic device. 3. Set up and conduct experiments with microfluidic devices with pumps, flow controllers, digital imaging and data acquisition. 4. Analyze the acquired experimental data. 5. Produce well-structured technical figures, reports and presentations to communicate methods, results, and conclusions in a scientific manner.
Education Method	Q3 corresponds to 1 EC of work (indicative). It will include the lectures on theoretical concepts necessary to the design of the microfluidic device. You will work in a group of 4 students on the design aspect of the project. Q4 corresponds to 2 ECs of work (indicative). It will include lab sessions on microfabrication. You will work with the same group as in Q4 on the fabrication and the testing of your microfluidic device.
Books	Lecture material will include material from: 1. Henrik Bruus, Theoretical Microfluidics, Oxford University Press, 2007 2. Brian J. Kirby, Micro- and Nanoscale Fluid Mechanics: Transport in Microfluidic Devices, Cambridge University Press, 2010
Assessment	In Q3: Graded design group report (35%) In Q4: Graded final group report (50%) and Final presentation (15%)
Elective	Yes
Tags	Design Fluid Mechanics Project Transport phenomena
Contact	d.s.w.tam@tudelft.nl

ME46080	Enriched Finite Element Methods	4
Responsible Instructor	Dr. A.M. Aragon	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Students must take ME46050 prior to taking this course.	
Course Contents	The finite element method (FEM) is the de facto procedure for solving problems in solid mechanics. However, modeling of problems with complex/evolving geometries rapidly exposes the main pitfalls of FEM: creating matching meshes is a tedious and error-prone process that can take most of the time in a real-world simulation. In this advanced course on finite element analysis, we will delve into enriched finite element formulations. Students will be exposed to state-of-the-art methodologies for solving challenging linear boundary value problems. Particular emphasis will be placed in methodologies for solving problems with discontinuities, e.g., material interfaces, cracks, voids, etc. For these problem, we focus in decoupling the analysis from its underlying finite element discretization. By enriching the finite element space, it is possible to analyze problems with the same accuracy and rate of convergence as those of standard FEM on meshes that align to the problems geometry. This is a hands-on course where students develop on the finite element code used in ME46050 (Advanced Finite Element Methods).	
Course Contents Continuation	1. Introduction (overview of applications) 2. The eXtended/Generalized FEM (X/GFEM) Definitions The GFEM approximation and its reproducing condition Matrix notation, shifting, and scaling Linear dependency and Babusˇkas algorithm The 1-D elastostatics revisited GFEM for material interfaces GFEM for cracks Computer implementation details 3. Interface-enriched formulations Overview of applications The Interface-enriched GFEM (IGFEM) The Discontinuity-Enriched Finite Element Method (DE-FEM) Advantages over X/GFEM 4. Computer implementation FEM code basis (from discretization to solution) The assembly process and numerical integration Linear solvers (KU = F) Particular considerations for enriched formulations Pre-processing of geometry Assembly and matrix structure Dealing with singular systems Post-processing of results	
Study Goals	Through this course, students will: Discuss the different state-of-the-art enriched finite element formulations and identify problems where such formula- tions can be used; Extend a standard FEM package to include enrichments; Evaluate the different enriched methodologies and judge their performance on real problems; Use the software package to simulate complex problems that require an enriched formulation.	
Education Method	Online lectures supported by online material and coding.	
Computer Use	Python programming	
Assessment	Final project	
Department	3mE Department Precision & Microsystems Engineering	
Contact	a.m.aragon@tudelft.nl	

MS43100	Science of Failure	3
Responsible Instructor	V. Popovich	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	MS43025 - Mechanical Behaviour of Materials or equivalent knowledge	
Summary	Knowledge, tools and procedures to describe and assess fracture mechanical behaviour of materials.	
Course Contents	Roughly, the outline is as follows (# of lecture hours between brackets): Linear Elastic Fracture Mechanics: stress intensity factor, crack-tip plasticity, energy release rate (4) Elastic Plastic Fracture Mechanics: crack tip opening displacement, J integral (4) Failure: ductile and brittle fracture, ductile-to-brittle transition curve, failure assessment diagram (6) Fatigue: crack growth and crack initiation (4) Environment-assisted cracking: mechanisms and testing (2) Instructions: exercises in class (6) This totals 26 hours.	
Study Goals	The student is able to employ basic fracture mechanical quantities to describe crack growth in solids. Furthermore, the student is able to identify a number of common mechanical phenomena that cause material failure in terms of the mechanisms and the conditions for which such behaviour can be expected. The student understands how to experimentally determine relevant material properties and can make simple failure predictions. More specifically, the student is able to: 1. identify the principles, limitations and applications of the fracture mechanical concepts stress intensity, energy release rate, crack tip opening displacement and J integral; 2. explain experimental procedures to quantify fracture mechanics parameters; 3. identify the major microstructural aspects of brittle and ductile fracture in metals; 4. describe failure assessment in case of ductile and brittle fracture; 5. describe fatigue crack growth with delta K, explain the concept of crack closure, predict fatigue lifetime for constant amplitude loading and illustrate the effect of variable amplitude loading; 6. describe the mechanisms of fatigue crack initiation and relate this to surface roughness and microstructure; 7. illustrate the mechanisms for environment-assisted cracking in metals; 8. explain the principles for quantifying environmentally assisted cracking behaviour and predict lifetime under constant load.	
Education Method	Lectures and instructions	
Literature and Study Materials	Fracture Mechanics, M. Janssen, J. Zuidema and R.J.H. Wanhill, 2nd edition, DUP (2002) Collection of Exercises on Fracture Mechanics, available on Brightspace Slides (including notes) on Brightspace	
Assessment	Written examination during which the book Fracture Mechanics and the hand-outs provided may be consulted. Other forms of information, such as worked-out problems, are not allowed!	
Tags	Industry Mechanics Physics Research Methods Structural Mechanics Sustainability Technology	
Department	3mE Department Materials Science & Engineering	

MS43210	Advanced Characterisation	4
Responsible Instructor	Dr. M.W.E.M. Alfeld	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Required for	MS43800 Determination of Microstructures	
Expected prior knowledge	MS43005 Structure and Properties MS43010 Characterisation of Materials	
Course Contents	This course focusses on microstructure characterization of materials through interaction between x-ray photons, electrons, neutrons and ion beams and solids. This includes the emission of element characteristic x-ray radiation as well as Auger electrons and photoelectrons and phenomena such as: absorption, elastic and inelastic scattering, Sputtering with ions and depth profiling. How these effects are exploited for the characterization of materials will be discussed on the example of several common techniques, such as: electron microscopy, x-ray microanalysis, Auger electron spectroscopy, photoelectron spectroscopy, ion scattering spectroscopy, mass spectrometry and Rutherford backscattering spectroscopy. Further, state of the art synchrotron methods will be discussed. With these, the concepts of depth, spectral and lateral resolution and detection limits will be introduced. Further, common data reduction and evaluation approaches to the high-dimensional hyperspectral data sets acquired by aforementioned techniques will be discussed, as will be the use of tomography for the study of microstructures in 3D volumes. A special focus will be on Orientation Imaging Microscopy (OIM) based on Electron Back Scatter Diffraction (EBSD) and X-Ray Fluorescence (XRF) imaging. The principles of EBSD will be addressed and methods to determine grain size and shape, texture and misorientation between neighbouring grains. For XRF different sources, optics and detectors will be discussed in detail as well as applications of XRF in various field.	
Study Goals	The student is able to apply the physical principles of diffraction and X-ray, electron and ion spectroscopy to explain materials analysis techniques and select suitable techniques for typical problems. More specifically, the student is able to: Describe how the generation of electrons in the elastic and inelastic interaction between electrons and solids, including diffraction, is used to characterize the chemistry and microstructure of materials. Describe how the generation of X-rays in the elastic and inelastic interaction between X-rays and solids, including diffraction, is used to characterize the chemistry and microstructure of materials. Describe how the generation of ions in the elastic and inelastic interaction between ions and solids, is used to characterize the chemistry and microstructure of materials Describe the elastic and inelastic interaction between neutrons and solids is used to characterize the chemistry and microstructure of materials Understand how electron microscopes, diffractometers, and X-ray, electron and mass spectrometers are configurated and operated. Describe common data reduction and reconstruction techniques to hyperspectral imaging data sets, as acquired by the techniques discussed. Apply all the above to practical cases in which the chemical composition and microstructure of materials are determined.	
Education Method	Lectures and Assignment	
Literature and Study Materials	Lecture notes	
Books	Microstructural Characterization of Materials, David Brandon, Wayne D. Kaplan, 2008,John Wiley & Sons Ltd., ISBN 0470027851 Fundamentals of Nanoscale Film Analysis, Terry L. Alford, L.C. Feldman, James W. Mayer, Springer, 2007, ISBN 0387292608 Electron Backscatter Diffraction in Materials Science, Adam J. Schwartz, Mukul Kumar, Brent L. Adams, David P. Field, Springer, 2009, ISBN 9780387881355 An Introduction to Synchrotron Radiation, P. Willmott, Wiley, Chichester (UK), 2011, Print ISBN:9780470745793, Online ISBN:9781119970958, DOI:10.1002/9781119970958	
Assessment	The examination will be number of assignments including graded peer-review.	
Elective	Yes	
Tags	Chemistry Image processing Mathematics Physics Technology	
Department	3mE Department Materials Science & Engineering	

SC42030	Control for High Resolution Imaging	3
Responsible Instructor	Dr. O.A. Soloviev	
Responsible Instructor	C.S. Smith	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Required for	sc42065	
Expected prior knowledge	Basic knowledge of physics and control theory is expected. The first lectures recap in a condensed form the necessary knowledge in optics. Understanding of signal processing (in particular, the Fourier analysis) is expected for successful work on the homework assignments. The course involves some basic numeric simulations, so practical knowledge of Matlab/python/... is required.	
Course Contents	High-resolution imaging is crucial in scientific breakthroughs, such as discovering exoplanets in the habitable zone or understanding the origin and progress of diseases at a molecular level. For that purpose, special optical instruments like Extreme Large Telescopes or STED microscopy are developed. Optical aberrations are the key obstacle that hampers a clear vision and invites control engineers to step in. These are the disturbances induced by the medium, like turbulence in the case of astronomy, or by the specimen under investigation, like the change in refraction index due to inhomogeneities in the biological tissue. Adaptive Optics is a modern way to remove the effect of aberrations. This fascinating and expanding field in science is providing an excellent challenge to control engineers to improve image quality significantly by active control. This course will review the hardware necessary to control light waves in contemporary optical instruments, their modelling from a control engineering perspective, and discuss model-based control methodologies to do disturbance rejection.	
Study Goals	Understand the propagation of light, imaging and aberrations in the imaging process. Understand the operation principle of pupil-plane and focal-plane sensors to estimate the wavefront aberrations. Understand the design principles of opto-mechatronic wavefront corrector devices to correct wavefront aberrations. Develop spatial and temporal models of complete imaging systems and use these models in the design of model-based controllers for aberration correction.	
Education Method	Oral Presentations (seven lectures), homework assignments (4 in total), and self-study in the form of reading and presenting a scientific paper. Seven seminars (interactive discussions related to the homework assignments) are interleaved with the lectures, using the same time slots. Some homework assignments are allowed to be done in small groups; paper presentations are done in groups.	
Literature and Study Materials	Course Notes; selected chapters from published books and relevant scientific papers are provided as complementary reading material.	
Assessment	SC42030: (homework and paper presentation) 80% + oral examination 20%.	
Remarks	Old course code: SC4045	
Elective	Yes	
Tags	Broad Mathematics Optimalisation Physics Signals and Systems	
Department	3mE Department Delft Center for Systems and Control	

SC42061	Nonlinear Systems Theory	3
Responsible Instructor	Dr.ir. M. Jafarian	
Contact Hours / Week x/x/x/x	0.6.0.0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	- Linear state space models (e.g. SC42015 Control Theory) - Basic knowledge of (multivariable) calculus, differential equations and linear algebra	
Course Contents	- Fundamentals of nonlinear systems (basic properties, solvability, types of equilibrium points) - Analysis of planar systems (graphical representations, periodic orbits, elementary bifurcations) - Stability theory for autonomous nonlinear systems (Lyapunov's methods, invariance principle) - Advanced stability theory (time-varying systems, input-to-state stability, passivity) - Observers (local, global and high-gain, extended Kalman filter)	
Study Goals	Students can: - determine if a system is nonlinear, time-invariant, autonomous, etc. - determine time intervals over which a system is solvable - sketch and interpret graphical representations of 2d systems - check for the existence or absence of periodic orbits in 2d systems, compute bifurcation points, and classify elementary bifurcations - determine if equilibrium points of autonomous nonlinear systems are (asymptotically, exponentially, ...) stable - determine if equilibrium points of certain classes of non-autonomous systems are (asymptotically, exponentially, ...) stable - construct observers for certain classes of nonlinear systems	
Education Method	Lectures, problem sets and instruction sessions.	
Literature and Study Materials	- H. K. Khalil: "Nonlinear Control", Pearson, 1st Global Edition, 2015. ISBN 9781292060507. - Additional handouts	
Assessment	Written exam, depending on the COVID-19 situation either on campus or online.	
Department	3mE Department Delft Center for Systems and Control	

SC42065	Adaptive Optics Design Project	3
Responsible Instructor	Dr.ir. G.V. Vdovine	
Responsible Instructor	Dr. O.A. Soloviev	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Basic knowledge of signal/Fourier analysis, linear algebra, and optimisation is required. Basic skills in Python are required.	
Course Contents	The course consists of the realization of laboratory experiments to design and operate Adaptive Optics equipment to realize high-resolution imaging systems. Crucial in the design is the alignment of active optics systems and in the operation the development of algorithms for acquiring accurate information about the wavefront aberrations from intensity-based imaging components (like Shack-Hartmann sensors or CCD cameras) and using this wavefront information in the tuning of multivariable dynamic controllers to compensate in real-time the wavefront aberrations. The design is conducted under close supervision by world-leading experts in the field and is performed in groups of students. The size of the groups depends on the number of participants in this course. The course requires hands-on experiments, Python coding, and presenting the results in a report and/or a joint final presentation.	
Study Goals	Building insights about the key components in Adaptive Optics such as the wavefront reconstruction and the deformable mirror. As well as building the controller methodology to obtain a smart optics system for high-resolution imaging.	
Education Method	Project Based The work is subdivided into sessions. Each session results in a deliverable (e.g. some measurements or a piece of code). The work is performed in groups, which should be well balanced in, for instance, writing, coding, and theoretical skills.	
Computer Use	Lab computers are available, own laptops can be used	
Course Relations	Participation in SC42030 is required	
Practical Guide	A practical guide subdivided into sessions is provided.	
Assessment	Oral Presentation and evaluation of the written report in combination with the course sc42030	
Remarks	Old course code: SC4115	
Department	3mE Department Delft Center for Systems and Control	

SC42095	Control Engineering	3

Responsible Instructor	A. Dabiri
Contact Hours / Week x/x/x/x	0/4/0/0
Education Period	2
Start Education	2
Exam Period	2
Course Language	English
Expected prior knowledge	SC42015, SC42000 or similar. Knowledge of classical control techniques (systematic and realistic PID design, frequency domain approaches) as well as state space theory is required.
Course Contents	Computer control. Sampling of continuous-time signals. The sampling theorem. Aliasing. Discrete-time systems. State-space systems in discrete-time. The z-transform. Selection of sampling-rate. Analysis of discrete-time systems. Stability. Controllability, reachability and observability. Disturbance models. Reduction of effects of disturbances. Design methods. Approximations of continuous design. Digital PID-controller. State-space design methods. Observers. Pole-placement. Optimal design methods. Linear Quadratic control. Prediction. LQG-control. Implementation aspects of digital controllers.
Study Goals	The student must be able to: 1. describe the essential differences between continuous time and discrete-time control 2. transform a continuous time description of a system into a discrete-time description 3. calculate input-output responses for discrete-time systems 4. analyse the system characteristics of discrete-time systems 5. employ a pole-placement method on a discrete-time system 6. implement an observer to calculate the states of a discrete time system 7. apply optimal control on discrete-time systems 8. describe the functioning of the Kalman-filter as a dynamic observer
Education Method	Lectures and computer exercises
Computer Use	Matlab/Simulink is used to carry out the exercises of this course.
Literature and Study Materials	Course material: Lecture notes are made available on Brightspace Textbooks: K.J. Astrom, B. Wittenmark 'Computer-controlled Systems', Prentice Hall ,1997, 3rd edition. L. Keviczky et. al 'Control Engineering', Springer, 2019. L. Keviczky et. al 'Control Engineering: MATLAB Exercises', Springer, 2019. References from literature: B.C. Kuo 'Digital Control Systems', Tokyo, Holt-Saunders, 1980 G.F. Franklin, J.D. Powell 'Digital Control of Dynamic Systems', 1989, 2nd edition, Addison-Wesley
Assessment	Project assignment
Remarks	Old course code: WB2305 The project assignment can be completed only during the quarter when the course is offered (i.e. the project has no resit during other periods).
Department	3mE Department Delft Center for Systems and Control

SC42145	Robust Control	3
Responsible Instructor	Prof.dr.ir. J.W. van Wingerden	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	· Recap on background in linear systems theory and classical feedback control · Multivariable system control: Nyquist, interaction, decoupling · Directionality in multiloop control, gain and interaction measure · Stabilizing controllers and the concept of the generalized plant · Parametric uncertainty descriptions, approximations · The general framework of robust control · Robust stability analysis · Nominal and robust performance analysis · The H-infinity control problem · The structured singular value: Definition of mu · Mu synthesis, DK-iteration, role of uncertainty structure. · Design of robust controllers, choice of performance criterion and weights	
Study Goals	The student is able to reproduce theory and apply computational tools for robust controller analysis and synthesis. More specifically, the student must be able to: 1. substantiate relation between frequency-domain and state-space description of dynamical systems 2. define stability and performance for multivariable linear time-invariant systems 3. construct generalized plant for complex system interconnections 4. describe parametric and dynamic uncertainties 5. translate concrete controller synthesis problem into abstract framework of robust control 6. reproduce definition of the structured singular value 7. master application of structure singular value for robust stability and performance analysis 8. design robust controllers on the basis of the H-infinity control algorithm 9. apply controller-scalings iteration for robust controller synthesis	
Education Method	Lectures + Assisted lectures+Computer sessions	
Literature and Study Materials	S. Skogestad, I. Postlethwaite, Multivariable Feedback Control, 2nd Edition. John Wiley & Sons, 2005. References from literature: K. Zhou, J.C. Doyle, K. Glover, Robust and optimal control, Prentice Hall, 1996 D.-W. Gu, P.Hr. Petkov, M.M. Konstantinov, Robust Control Design with Matlab. Springer Verlag, London, UK, 2005	
Assessment	Controller design exercise (report and code)	
Remarks	Old course code: SC42010	
Department	3mE Department Delft Center for Systems and Control	

WI4014TU	Numerical Analysis	6
Responsible Instructor	Dr. N.V. Budko	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2	
Course Language	English	
Course Contents	Classification of partial differential equations and boundary and initial conditions. Maximum principle, uniqueness and non-negativity of the solution, elements of variational calculus. Finite- Difference, Finite-Volume and Finite-Element discretization methods. Solution methods for time-dependent and nonlinear problems. Direct and iterative linear solvers.	
Study Goals	Students will learn: the principles of finite-difference, finite-volume, and finite-element discretizations; the accuracy, consistency, and convergence of numerical solutions; the basic ideas of numerical linear algebra. They will be able to implement a simple PDE solver and to work with existing sophisticated numerical software.	
Education Method	Lectures (possibly, online), homework assignments, programming assignments, written report.	
Literature and Study Materials	[1] J.van Kan, A.Segal, and F.Vermolen, Numerical methods in scientific computing. VSSD, Delft, 2005, improved 2008, ISBN-13 978-90-71301-50-6 [2] Python programming: - Python Help: https://www.python.org/about/help/ - Numpy for Matlab users: https://docs.scipy.org/doc/numpy-dev/user/numpy-for-matlab-users.html [3] FENICS Project: https://fenicsproject.org/ - Installation: https://fenicsproject.org/download/ - Tutorial: https://fenicsproject.org/tutorial/	
Prerequisites	Calculus of functions of several variables, vector calculus, introductory course on numerical analysis (numerical integration, error analysis), introductory course on partial differential equations, some experience with Python or Matlab.	
Assessment	Theoretical and computational assignments, graded reports. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Tags	Mathematics Modelling Numeric Methods	
Co-Instructor	Prof.dr.ir. M.B. van Gijzen	

WI4019-SP	Non-linear Differential Equations	6
Responsible Instructor	Dr.ir. W.T. van Horssen	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Introductory course on Differential Equations e.g. Boyce and DiPrima Chapters 2-7, 9.	
Course Contents	Existence and uniqueness of solutions of initial value problems, Gronwall's lemma, Autonomous systems, Critical points, Periodic solutions, Stability theory, Linear systems and Floquet theory, Perturbation theory and asymptotic methods, Poincare-Lindstedt method, Averaging method, Multiple time-scales method, Elementary bifurcations.	
Study Goals	After following this course the student will be able to apply fundamental and advanced, analytical techniques and methods to study properties of solutions of nonlinear ordinary differential equations. The mathematical methods and techniques are mentioned under "course contents". After studying the lecture notes and after making the exercises successfully the student can start to work on research problems in the field of nonlinear differential equations.	
Education Method	Lectures and a MAXIMA practical work. In total 14 lectures of 2 hours each will be given. The MAXIMA practical work is about 4 hours work. Following and preparing the lectures and the MAXIMA practical work takes about 60 to 70 hours. Making the 30 take-home exercises takes about 60 hours. Preparing the 45 minutes oral exam takes about 30 hours.	
Literature and Study Materials	F. Verhulst, Nonlinear Differential Equations and Dynamical Systems, Springer-Verlag 2nd edition, ISBN 3-540-60934-2.	
Assessment	The final grade of the course consists of the following components: - Take home exam (50%) - oral exam (50%) Final grade calculation: (0.5 * take home exam) + (0.5 * oral exam) The take-home exam consists of 30 exercises. The oral exams will be held at the end of the 2nd quarter, and the student is only allowed to take part in this oral exam when at least 70% of the take-home exercises are made correctly and when the MAXIMA practical work is followed successfully. In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5 take home exam: resubmit deliverable oral exam: oral resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

WI4260TU	Scientific Programming for Engineers	3
Responsible Instructor	Dr.ir. H.X. Lin	
Practical Coordinator	Ir. C.W.J. Lemmens	
Contact Hours / Week x/x/x/x	0/0/2/0 + lab	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Required: experience with a programming language (e.g., Python or C), and linear algebra Strongly advised: experience with linux is a plus (but not a prerequisite).	
Course Contents	The course tries to bring students to a level where they are able to change algorithms from e.g., numerical analysis into efficient and robust programs. It comprises 1. Introduction to programming in general; 2. Floating point number rounding-off errors and numerical stability; 3. (Numerical) Software design; 4. Data Structures; 5. Testing, debugging, and profiling; 6. Efficiency issues in computing time and memory usage; 7. Optimization and dynamic memory allocation; 8. Scientific software sources and libraries. P.S. (i) This is not a general introductory course to learn how to write a computer program. Furthermore, the course concentrates mainly on sequential programming and only briefly introduces parallel programming (MPI and OpenMP). More advanced topics like threads or parallel (MPI/GPU) programming on supercomputers are not covered by this course (they are covered by other courses, e.g., Introduction to High-Performance Computing). (ii) This is a course aimed at master students, however, if you have followed the course Scientific Programming in the BSc CSE-minor, then you should not select this course as a Master elective course due to the large overlap.	
Study Goals	1. Is able to explain the problems related to the use of floating-point numbers in scientific computing; 2. Is able to implement, compile, and debug programs in a Linux programming environment, e.g., use of a makefile. 3. Is able to explain the impact of data structures on the performance of a program, e.g., data locality, cache and memory access. 4. 3. Is able to explain the impact of the program on the performance of a program, e.g., temporal locality, pipelining, and loop-carried dependency. 5. Is able to analyze the performance of a program, and can apply optimization, debugging, and profiling tools to improve of the program. 6. Is able to make the transition from a scientific model to a structured program;	
Education Method	Weekly there are 2-hour lectures and 3-hour lab sessions. Participation in a minimum of five lab sessions is required, with the optional homework, a bonus of max. 1 point can be obtained.	
Books	Textbook: Writing Scientific Software - A guide to good style, by Suely Oliveira and David Stewart, Cambridge University Press, ISBN-13 978-0-521-67595-6	
Assessment	1. Written exam: 3-hour exam comprising theory and practical (lab) questions. (90%) 2. Lab homework (10%) Final grade calculation: (0.9*Exam + 0.1*Lab homework) Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Written exam: resit in next quarter - Homework assignments: resubmit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see TER Art 29, sub 4).	
Tags	Numeric Methods Programming	

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Mechanical Engineering**

ME-HTE Second Year Project: Select 1 (15 ECTS)

ME56015P	ME-HTE Midterm Review	15
Responsible Instructor	Dr.ir. J.F.L. Goosen	
Course Coordinator	E. Matroos	
Contact Hours / Week x/x/x/x	x/x/x/x	
Course Contents	This project covers the initial steps in a comprehensive research project. The nominal period is 15 EC (2.5 - 3 months). This working environment can be our own department, another department within or outside 3mE/TU Delft, or within a company.	
Study Goals	The goal of this assignment is to take the initial steps in a larger research project. This period is concluded with an evaluation of the quality of the project and research performed, and a planning for the conclusion of the final MSc-project.	
Education Method	During this period the first steps in a large research project are covered. After the evaluation of the state-of-the-art in the preceding literature review and project proposal, in this period, the intended research activities are shaped further. The daily supervisor coaches the student during this period.	
Assessment	The midterm project is evaluated in two steps: First the student evaluates his own performance based on the relevant items in the rubric that will also be used in the evaluation of the final Msc-project. Then in a progress meeting, the supervisor evaluates the student and the self-evaluation. A pass or fail grade is determined at the end of this meeting.	
Remarks	Old course code: ME2400-15 The initial assignment is combined with the Literature Survey (ME56010) and Thesis Project (ME56035). For more information, please contact Eveline Matroos, e.matroos@tudelft.nl (room G-1-290).	
Contact	3mE Department Precision & Microsystems Engineering For more information, please contact Eveline Matroos, e.matroos@tudelft.nl (room G-1-290).	

ME56015S	ME-HTE Internship	15
Responsible Instructor	Dr.ir. J.F.L. Goosen	
Course Coordinator	E. Matroos	
Course Contents	Working as a starting engineer in a professional research, design or production environment. The nominal working period is 15 EC (2.5 - 3 months). This working environment can be our own department, another department within or outside 3mE / TUDelft, but it is more likely to be within a company	
Study Goals	The goal of this assignment is to gain experience in working as an engineer in a professional environment, such as a company. The student will search for this position, then contact the MSc-coordinator of the department for a suitable coach within the department. For an internship within a company, the mentor/coach has to give prior approval for the working period, on basis of a written assignment which has been communicated with the company.	
Education Method	Rules of the working period include: The assignment should contain problem solving activities (remember the working period is also part of your engineerings education). The assignment must be challenging and of sufficient level.(MSc) At the end you should be able to report on your work, which means that you must have performed a comprehensive task. Contributing to a variety of not-related things in a company, however interesting, is not a good working period. The field of work is the domain of the research of PME in its broadest sense as reflected in the research topics of the groups (see www.pme.tudelft.nl) For an Industrial traineeship abroad start searching at least three months before the intended due date. Always make sure you have prior consent of one of the senior PME staff members, before making any commitments.	
Assessment	The internship will be assessed with a pass or fail grade after the internship is competed and the report is handed in and graded by your TUD supervisor.	
Remarks	Old course code: ME2400-15	
Contact	3mE Department Precision & Microsystems Engineering For more information, please contact Eveline Matroos, e.matroos@tudelft.nl (room G-1-290).	

TUD4040	Joint Interdisciplinary Project	15
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Project Coordinator	Ir. B.J.E. de Bruin	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Summary	JIP consists of three sets of activities: 1. The project which takes place at a company. Students are responsible for the project work (collaboration, integration of different perspectives, the content quality of their work). They plan their activities via a scrum method, are challenged to realise field trips, consult with experts and realise the necessary research work. The team keeps a Scrum log, run a blog for interaction with the outside world and every team member keeps a personal development log. 2. Meeting Professionals lectures and workshops about specialised topics, by academic staff or senior professionals from the companies involved. 3. Plenary meetings and design reviews for all the teams, company coaches and academic staff. At the meetings the students present their intermediate resp. final outcomes and plans, and are provided with feedback.	
Course Contents	The aim of the Joint Interdisciplinary Project is to prepare students to contribute to solving impactful technological challenges. The projects not only demand good engineering working knowledge but also experience with interdisciplinary and systems theory, and both knowledge and mindsets of innovation and entrepreneurial behavior. The project brief is provided by renowned companies like Airbus, Arcadis, etc. Teams of interdisciplinary student teams guided by a company coach and offered academic and industry expertise, are invited to realise an innovative problem solution to a complex problem and contributing to the sustainable development goals.	
Study Goals	1. Cognitive abilities attributable to interdisciplinary learning Demonstrate the ability to engage in perspective-taking; Develop structural knowledge pertaining to the problem; Integrate knowledge and modes of thinking drawn from two or more disciplines; Produce an interdisciplinary understanding of complex problem or intellectual question. 2. Scientific and intellectual development Capable to analyze scientific and societal consequences (economic, social, cultural, environmental) of the innovation; 3. Research and design capabilities Demonstrate engineering skills: technical skills, interpreting results, creativity, usability for company/institute; Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; 4. Collaboration and communication in an interdisciplinary team Demonstrate behavioral competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to respect different disciplines and adapt to different cultures); Show ability to write a technical report: structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (30 pages); Present work performed in a structured way through an oral presentation to their peers and customer. 5. Self-adjustment and reflection capabilities Plan and control the project efficiently considering resources and methodology; Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career. Cognitive abilities attributable to interdisciplinary learning; The ability to integrate (scientific and practical technological)knowledge from different disciplines to solve complex problems Scientific and intellectual development The capacity to evaluate the ethical, scientific and societal consequences of the proposed innovation Research and design capabilities The ability to create reasonable and relevant research or design , according to the academic standards of the involved disciplines Collaboration and communication in an interdisciplinary team Demonstrate behavioural competences and skills relevant for teamwork and effective communication with different stakeholders Self-adjustment and reflection capabilities To carry out regular reflections on professional and personal development and being able to improve upon those reflections Understand contemporary and societal issues in their work.	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Assessment	The assessment criteria pertain to the process, product and presentation and are assessed at an individual and team level, during 3 review sessions at three levels of accomplishment (each review assesses at a different level): Interdisciplinary work Scientific reasoning and ethical mindset Innovation process Presentation and communication Academic staff, supported by the input from company coaches, grade the students during the Problem Statement, the Midterm and the Final Review. The final group mark for JIP is differentiated per person and is based on: The team presentations - 30% The Problem definition, progress report and final report 30% The individual contribution to the team 20% The final individual reflection report 5% The Blog 5%	
Remarks	Project Outcomes (product, model, system) are presented at: 1. Problem Statement Review 2. Midterm meeting 3. Final Review 4. Symposium presentation	

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Mechanical Engineering**

ME-HTE Graduation Project (45 ECTS)	
Introduction 1	The 2nd Master's year has a flexible structure, in which the Literature Research (10 ECTS) and the Master's thesis (35 ECTS) are compulsory for everyone. In a nominal situation this will start in the 2nd quarter of an academic year. In the first quarter, students can choose between: 1. 15 ECTS Joint Interdisciplinary Project; 2. 15 ECTS Internship or Research Assignment.

ME56010-22	ME-HTE Literature Research & Project Proposal	10
Responsible Instructor	Dr.ir. J.F.L. Goosen	
Course Coordinator	E. Matroos	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	None (Self Study)	
Course Language	English	
Course Contents	An individual assignment as part of 2nd year ME-HTE MSc Program The assignment is a literature study and initial phase of an MSc Thesis project, with the purpose to get acquainted with the scientific publications within the realm of the MSc Thesis Project, and to prepare for the specific topics to be investigated or focussed on a stand-alone topic. A project proposal for the final MSc Thesis project will be developed during this assignment.	
Course Contents Continuation	An individual assignment as part of 2nd year ME-HTE MSc Program The assignment is a literature study and initial phase of an MSc Thesis project, with the purpose to get acquainted with the scientific publications within the realm of the MSc Thesis Project, and to prepare for the specific topics to be investigated or focussed on a stand-alone topic. A project proposal for the final MSc Thesis project will be developed during this assignment.	
Study Goals	Demonstrate on an academic level to be able: -to perform a literature study and/or -to design in the physical or virtual world and/or -to set up and do experiments in hardware and/or software	
Education Method	Individual assignment A coach will be assigned	
Assessment	Two tests: T1: Literature Research & Project Proposal (written report) T2: Colloquium Attendance (pass/fail) student needs to attend a minimum of 10 colloquia to complete this part (colloquia points can be earned by visiting graduation talks of PME, lunchlectures organized by student association Taylor and by visiting come back meetings organized by the msc coordinators of PME)	
Department	3mE Department Precision & Microsystems Engineering	
Contact	Contact the Daily MSc coordinator of the PME department or your supervisor one month before your intended starting date	

ME56035	ME-HTE/OM MSc Thesis	35
Responsible Instructor	Dr.ir. J.F.L. Goosen	
Course Coordinator	E. Matroos	
Contact Hours / Week x/x/x/x	0/0/x/x	
Education Period	None (Self Study)	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	Students of the MSc track ME-HTE	
Expected prior knowledge	All MSc courses finished.	
Course Contents	The duration of the final (MSc) project is 35 EC. Students will preferably contribute to the on-going research of the PME Department. In some cases an assignment with a industrial partner can be accepted. Always check your coach or coordinator first. Main requirements: the assignment should offer possibilities to gain and sharpen MSc/ 'ingenieurs' (EN: Engineer) skills. This basically leads to the following requirements on an assignment and the environment in which it takes place: - The assignment must offer the possibility for a structured problem solving approach on academic level. This implies that the student has to address a 'grand problem' and not a series of small engineering problems, each of one standing mainly on its own. The problem must allow for a problem solving cycle which includes analysis, synthesis and where possible realisation and evaluation. On specific aspects, detailed analyses have to be made, for instance by means of (numerical) modelling and analysis, experimentation etc. The emphasis is of course different in each assignment. - The assignment includes a problem analysis phase. The typical starting point of an assignment is a first problem definition; the problem analysis phase should give clarity on what and where the real problems are and which problems are going to be addressed in subsequent phases - In the solution generation phase there should be some freedom as well. Assignments which have from the early start on a defined detailed solution direction from which the student may not deviate are usually not suitable for MSc projects. - The topics must be within the area of the research topics of PME and preferably follow the focus area chosen. For proper coaching and for being able to establish good links with the company involved, we are looking for subjects that match with the interest and coaching capabilities of our staff members. If a student undertakes the graduation project at a company, a graduation agreement should be signed. In this respect, the template of the TUD-3mE graduation agreement should be used, which is available at the following link: https://www.tudelft.nl/en/student/faculties/3me-student-portal/education/related/student-forms/msc-forms/.	
Study Goals	The overall objective of this assignment is to demonstrate a sufficient academic level in the field of Mechanical Engineering, focussing on research topics within the PME department(ME track HTE or OM) More specifically, the student is able to meet, to a sufficient level, the following qualifications: 1. Broad and profound knowledge of engineering sciences (applied physics and mathematics) and the capability to apply this knowledge at an advanced level in the variant-related discipline. 2. Broad and profound scientific and technical knowledge of the track related discipline and the skills to use this knowledge effectively. The discipline is mastered at different levels of abstraction, including a reflective understanding of its structure and relations to other fields, and reaching in part the forefront of scientific or industrial research and development. The knowledge is the basis for innovative contributions to the discipline in the form of new designs or development of new knowledge. 3. Thorough knowledge of paradigms, methods and tools as well as the skills to actively apply this knowledge for analysing, modelling, simulating, designing and performing research with respect to innovative variant-related systems, with an appreciation of different application areas. 4. Capability to independently solve technological problems in a systematic way involving problem analysis, formulating sub-problems and providing innovative technical solutions, also in new and unfamiliar situations. This includes a professional attitude towards identifying and acquiring lacking expertise, monitoring and critically evaluating existing knowledge, planning and executing research, adapting to changing circumstances, and integrating new knowledge with an appreciation of its ambiguity, incompleteness and limitations. 5. Capability to work both independently and in multidisciplinary teams, interacting effectively with specialists and taking initiatives where necessary. 6. Capability to effectively communicate (including presenting and reporting) about ones work such as solutions to problems, conclusions, knowledge and considerations, to both professionals and non-specialised public in the English language. 7. Capability to evaluate and assess the technological, ethical and societal impact of ones work, and to take responsibility with regard to sustainability, economy and social welfare. 8. Attitude to independently maintain professional competence through life-long learning.	
Education Method	Coaching on a regular basis.	
Assessment	The student will be assessed through: -> Final Report. Preferably no longer than 50 pages (excl. appendixes) Contact the secretariat of PME (room 34 G-1-270) about the administrative requirements (registration number, number of copies, etc) -> Oral Presentation (20 minutes) and discussion (10 minutes) as part of the graduation procedure. -> Defense before an exam committee as part of the graduation procedure. -> See student portal of 3mE, master forms for the MSc Thesis Graduation Rubric	
Exam Hours	Contact the secretariat of PME, 3ME-PME-Secretariaat@tudelft.nl (room G-1-270) at least one month (20 workdays) before your intended examination day to start all administrative procedures and schedule an examination date.	
Special Information	Students should not contact companies without prior consent of a senior staff member (intended coach)	
Remarks	Old course code: ME2490-35 The MSc thesis may be combined with the literature survey (ME56010) and midterm (ME56015P) or separate industrial traineeship (ME56015S)	

Department	3mE Department Precision & Microsystems Engineering
Contact	Contact the Daily MSc coordinator of the PME Department or your mentor/coach one month before the intended starting date.

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Mechanical Engineering**

ME-MME Track D. Multi-Machine Engineering

Contact for students	If you have questions regarding this track, contact the coordinator: Ir. Mark Duinkerken E-mail: m.b.duinkerken@tudelft.nl Tel: +31 (0)15 27 81790 (82889 secretary)
Program Coordinator	Ir. Mark Duinkerken T +31 15 278 1790 E M.B.Duinkerken@tudelft.nl
Introduction 1	Society faces tremendous challenges to meet demands on efficiency, sustainability, and safety of complex processes. In the logistics and production domain Multi-Machine Engineering address these challenges with an integrated perspective that combines core (mechanical systems) design with real-time operation and distributed machine-machine interactions. In the track MME you develop the skills necessary to design such integrated multi-machine systems, combining science-based methodologies, with state-of-the-art tools, and hands-on lab and industrial case experience.
Exit Qualifications	The exit qualifications for the master Mechanical Engineering can be found in the common section of this studyguide. For the track Multi-Machine Engineering, the specific exit qualifications are: 1. Competent in the scientific discipline Mechanical Engineering A graduate in Mechanical Engineering Multi-Machine Engineering is able to analyse and explain the characteristics and mechanical behaviour of material during transport and storage analyse and model different types of transport equipment and transport facilities analyse and model the logistics of complex transport systems and networks 3. Competent in designing A graduate in Mechanical Engineering Multi-Machine Engineering is able to model the dynamics of the interaction between equipment and materials design, control and automate transport equipment and facilities develop, monitor and control components for transport and logistic systems and networks 7. Considering the temporal and social context A graduate in Mechanical Engineering Multi-Machine Engineering is able to explain the importance of transport systems and logistics in society
Program Structure 1	First year: Obligatory courses for all ME masters: 17 ECTS Compulsory courses for the track MME : 29 ECTS Elective courses (including ethics): 14 ECTS Second year: Literature Assignment: 10 ECTS Research Assignment: 15 ECTS MSc master project: 35 ECTS

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Mechanical Engineering**

ME-MME Obligatory Courses (44 ECTS)

ME44101	Dynamics and Interaction of Material and Equipment	4
Responsible Instructor	Prof. Dr.ir. D.L. Schott	
Instructor	Dr.ir. Y. Pang	
Co-responsible for assignments	H. Shi	
Co-responsible for assignments	E.F.L. Stok	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	ME44115	
Course Contents	This course focuses on modelling and designing machine-cargo interactions and using those interactions to conceptually design equipment taking into account the transport demands and logistics processes. The course starts with the logistic context with emphasis on (bulk and container) terminal level, for which the functionality of equipment types will be addressed with regard to the cargo type (dry bulk, container, piece goods) and its characteristics. Specific equipment types will be discussed, analysed and designed, such as a belt conveying systems used for bulk materials (e.g. iron ore and coal) as well as for baggage and parcel handling. The interaction between cargo and equipment will be explicitly taken into account. Specifically for the bulk solid cargo type, an experimental assignment to determine the properties of a particular bulk solid material is part of the course. Bulk solid materials include granular materials such as coal, sand, limestone, iron ore, grain. These materials can be free flowing through bunkers and chutes as well as stored in silos, handled by stackers and reclaimers or transported by conveyors. With the experimentally determined properties conceptual design of a silo and belt conveyor will be undertaken. State-of-the-art particle based simulation with Discrete Element Method (DEM) will be introduced. The parameters, algorithms, and applications will be addressed, as well as the procedure for validation of DEM simulations to obtain realistic bulk material behaviour in a virtual environment. It will be shown how DEM has led to new breakthroughs and developments in the field of port handling equipment.	
Study Goals	The student will be able to: Differentiate among various types of cargo terminals based on the specific characteristics of cargo and their corresponding logistical requirements. Conduct experimental analysis to determine key properties of bulk solid materials, and utilize the findings to inform effective equipment design. Design equipment solutions that are tailored to specific logistical requirements, taking the performance of the equipment within the logistics context into account, supported by thorough calculations. Assess the effectiveness of equipment designs by considering the relationship between material characteristics, material behavior, and material equipment interaction parameters on equipment performance. Evaluate the role of Discrete Element Method (DEM) in the design of equipment, including understanding the theory and algorithm used for DEM calculations and analyzing the impact of input parameters (particle level, simulation level) on the output, specifically on bulk behavior and erratic behavior.	
Education Method	Lectures, including guest lectures from industry experts. Practical for testing and characterising bulk solid materials and it's material equipment interaction properties by experiments.	
Course Relations	ME44115 DEM simulations	
Literature and Study Materials	Schulze, D., Powders and Bulk Solids - Behavior, Characterization, Storage and Flow, Springer, 2008 ISBN 978-3-540-73767-4 (print), 978-3-540-73768-1 (online), DOI: 10.1007/978-3-540-73768-1 http://www.springerlink.com/content/l55416/?p=fbeb6748815f4e4c92f56519a15f8837&pi=0 Rhodes, M., Introduction to Particle Technology by Martin Rhodes, John Wiley & Sons, 2008. ISBN 978-0-470-01427-1 (print), 9780470727102 (online), DOI: 10.1002/9780470727102 http://www3.interscience.wiley.com/cgi-bin/bookhome/117932420?CRETRY=1&SRETRY=0 Slides and supplementary material provided on Brightspace and/or during lectures	
Assessment	Because of measures resulting from COVID-19, the prescribed assessment will be as follows: 1. practical assignment - remote (25% of the mark) 2. written exam - remote open book with several fraud prevention measures (75% of the mark) If the opportunity arises for examinations on campus, the method of assessment will be: 1. practical assignment (25% of the mark) 2. written exam - closed book (75% of the mark).	
Enrolment / Application	Please enroll yourself as soon as possible. It helps us to schedule and setup enrollment for practical (in pairs or threesomes)	
Remarks	MSc ME track Multi Machine Engineering MSc TIL Engineering specialisation	
Percentage of Design	20%	
Design Content	Conceptual design of material handling equipment considering cargo characteristics and interaction parameters between cargo and equipment Dynamics of Material and Equipment Interaction	
Department	3mE Department Maritime & Transport Technology	

ME44106	Structural Design with FEM	4
Responsible Instructor	Ir. W. van den Bos	
Instructor	X. Jiang	
Contact Hours / Week x/x/x/x	0.6.0.0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic mechanical engineering knowledge is expected. Experience with FEA programs is not required.	
Summary	Understand fundamental theory of finite element method and identify various failure mechanisms of metallic structure and its connection with relevant design standards. Design a ship-shore crane according to Dutch design Standards, check and evaluate the structural integrity applying a finite element model. For complete information see bright space!	
Course Contents	Understand fundamental theory of finite element method and identify various failure mechanisms of metallic structure and its connection with relevant design standards. Design a ship-shore crane according to the Dutch Standards (NEN 2017 to 2023) and control your design with the use of a finite element model (Femap and SDC Verifier) for detailed description and or examples see brightspace . This course is a continuation of the original course ME44105 and wb3416 Design with FEM Software used is: SDC Verifier (www.sdeverifier.com) and FEMAP (https://www.plm.automation.siemens.com/store/en-se/femap/index.html)	
Study Goals	The general objective of this course is to apply the finite element method in a proper way to design and evaluate the structural strength and durability of transport equipment as a crane. After the course the student is able to: 1. Understand the fundamental basis of FEM including the underlining linear and non-linear theory 2. Develop a reasonable FEM model and Judge the FEM results correctly 3. Apply the Finite Element method as a Design Tool 4. Design according to standards 5. Understand the structural failure mechanisms, buckling, plate buckling, crippling, fatigue and static overload (yielding) and how these failure modes influence the checks implemented in design standards	
Education Method	Weekly 2 hours theory lectures and 4 hours computer room time to work on the assignment.	
Assessment	The assessment consists of two parts- practical part and theoretical part. The practical part will be assessed through an oral exam of 1/2 hour on the final structural design report. The report contains the complete description of the structural design, the design specification and the translation of these specification into loads the design, and the final result of the FEA calculation. The FEA results should prove that the crane is structural save by complying with the design standard to be used. The theoretical part will be assessed based on a regular, closed book written exam. Detail information will be announced on Bright space in due time. The practical part and the theoretical part account for 75% and 25% of final grade respectively. Both parts are compulsory: a grade at least 5.0 is demanded per part in order to complete the course and obtain the final grade.	
Tags	Design Modelling Numeric Methods Structural Mechanics Transport & Logistics	

ME44110	Integrated Design Project for Multi-Machine Systems	5
Responsible Instructor	Ir. W. van den Bos	
Instructor	Dr. J. Jovanova	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Course Contents	The course Integration Project Multi-Machine Systems brings together the skills students develop throughout their studies. The students groups work together to develop designs of specific port equipment (ex: cranes; autonomous vehicles) and complex multi-machine systems (ex: floating terminals, bulk material terminals). Students are encouraged to use acquired knowledge in their previous studies and apply it to solve real engineering problems. They need to make multiple decisions in the design process and justify it based on their previous knowledge and experience. The students are expected to raise their technical confidence and learn how to combine the skills gained from other courses to design complex multi-machine systems. A graduate in Mechanical Engineering MME track is able to model the dynamics of the interaction between equipment and materials design, control and automate transport equipment and facilities develop, monitor and control components for transport and logistic systems and networks. This course Integration Project Multi-Machine Systems brings them a step closer to a real working scenario. Exit qualification Competent in designing. A graduate in Mechanical Engineering MME track is able to model the dynamics of the interaction between equipment and materials design, control and automate transport equipment and facilities develop, monitor and control components for transport and logistic systems and networks. The course Integration Project Multi-Machine Systems teaches to students of how to apply their knowledge in design of complex systems. The course is in Q3 and Q4 (the second semester) so the students are expected to already have high level of accumulated knowledge. As the course is project based, groups of students (6 -7) work on different tasks within a project. This allows for students from different entry levels to work together and support each other in the process. Another exit qualification is considering the temporal and social context. In this course the importance of transport systems and logistics in society is stressed as well as designs that are safer, more efficient and sustainable.	
Study Goals	The learning goals are: LO1: Review on design principles for multi-machines. LO2: Apply different design conceptual approaches (unit level, system level, integration level). LO3: Use standards for equipment design. LO4: Asses and recommend system integration (market availability, custom designs) LO5: Design a project for multi-machine system	
Education Method	Lectures, feedback sessions	
Assessment	Report	
Department	3mE Department Maritime & Transport Technology	

ME44200	Operations and Maintenance	3
Responsible Instructor	Dr.ir. Y. Pang	
Instructor	Dr. V. Reppa	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This course focuses on intelligent control to optimize the operational performance and to maintain the reliability of multi-machine systems in large-scale transport and logistics domain. Along with the understanding of diverse transport systems and logistic processes, including the dynamic interactions of the equipment and facilities involved, the control for multi-machine systems contains two folds, the integrated control for efficient operations and the intelligent decision-making for reliability control. To ensure efficient and reliable operations at both equipment level and system level, technologies and methodologies of automated data acquisition, remote communication, intelligent data analysis and integrated operational decision-making will be studied. Condition-based maintenance of systems and equipment will be analysed and discussed in combination with on-line techniques for monitoring the health of the sensors used for maintenance. During the course industrial case studies will be given to present the state-of-the-art development and foreseen challenges. The course is concluded by practical assignments where such intelligent control systems will be researched and conceptually designed.	
Study Goals	After successfully completing this course, you will be able to: - explain the concepts of intelligent control for operational support systems and advanced maintenance strategies; - identify the properties that determine efficient operation and reliable performance; - analyse the complexity of the operations and interactions of multi-machine systems; - derive models to represent the interactions and operational processes of multiple equipment and facilities; - design intelligent operational control systems in term of mathematical models and integrated control methods; - apply the methodologies and technologies to improve the efficiency and reliability for both individual machine and overall multi-machine system; - describe the effects of sensor faults in operations and maintenance of transport systems; - explain the design of model-based sensor fault detection methods; - explain the design of multiple sensor fault isolation methods; - design a sensor fault detection and isolation method for the operation and/or maintenance of a transport system.	
Education Method	Lectures (2 hours per week), case study, practical assignments	
Literature and Study Materials	Handouts and references to relevant literature and media. Additional teaching materials to be provided during lectures and available via Brightspace . Expected prior knowledge: ME44101 (not obligatory). Prior knowledge in system engineering (dynamical systems) is recommended.	
Assessment	Group assignment: Assignment Reports & Oral Exam Final grade is calculated by weighted average of Assignment Part I (75%) and Assignment Part II (25%). In case of a final grade less than 6, the group can have one opportunity to resubmit or repair the insufficient work via one of two modalities: 1. Repair: to improve the work of the same assignment towards a maximum retake grade of 6; 2. Resubmit: to choose a new assignment to be assessed by general assessment criteria towards a retake grade higher than 6. After successful delivery of the group assignment reports, the Oral Exam takes place in the exam week of the education period.	
Department	3mE Department Maritime & Transport Technology	

ME44206	Quantitative Methods for Logistics	5
Responsible Instructor	Dr. B. Atasoy	
Instructor	Ir. M.B. Duinkerken	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	The course contains of two parts; roughly 7 weeks each. In the first part of the course an introduction to basic operations research techniques will be presented: - linear programming (simplex, duality etc.) - integer programming (IP, BIP, MILP etc.) - basic probability and statistics; brief overview of models to deal with uncertainty This part of the course will be largely be based on the first chapters of the book by Hillier and Liebermann. During the first quarter students must complete an individual practical assignment where they develop a mathematical model and implement it using Python together with Gurobi as a solver. In the second part of the course several common quantitative methods for transportation and logistics are presented. - networks and routing: network flow problem, facility location problems, transportation problem, transshipment problem, vehicle routing problem and other standard problems; standard algorithms i.e. the branch and bound method. - queueing theory: overview of basic models and results. During the second quarter students must complete a practical assignment in groups on a case study with a transportation network again using Python.	
Study Goals	At the end of this course the students will be able to: - Explain the importance and use of operations research techniques to analyze and improve systems (i.e. queuing, simulation, forecasting, routing, scheduling) - Formulate mathematical models of real-life problems (e.g., resource allocation, production planning, transportation) with operations research techniques - Verify and validate mathematical models - Evaluate operation research techniques based on strengths and weaknesses (in terms of accuracy and efficiency) for improving real-life problems - Apply basic operation research techniques to small scale problems as well as realistic case studies	
Education Method	Flipped classroom approach: The students are expected to be prepared by studying the online material that is indicated every week. Lectures in two parts: First 45 min course content, Second 45 min assignment Course content includes a short intro and exercises related to the subject. Assignment part will be dedicated to introduction of the software (Python+Gurobi), introduction of the assignments and any further needed assistance and feedback. It is strongly recommended to bring a laptop to the lectures. There will be also additional reading material that will be provided during the course.	
Books	Hillier & Lieberman. Introduction to Operations Research. 10th Edition, McGraw-Hill, ISBN 9781259253188	
Assessment	Practical Q1 - 25% (individual: Mathematical Model, Python implementation and report) Practical Q2 - 25% (group: Mathematical Model, Python implementation and report) Exam Q2 - 50%. For all 3 parts a minimal grade of 5 is required. The exam will be given at the end of Q2, the resit for the exam in Q3 Both assignments are only given once a year; with a limited option to resubmit if the grade is insufficient. The highest grade for a resubmission is 6. In case of measures resulting from COVID-19, the prescribed assessment will be an online examination. If the opportunity arises for examinations on campus, the method of assessment will be on campus.	
Remarks	TIL-students: please be aware that there is an overlap in content between CIE4835 (elective T&P 3.1) and ME44206 (compulsory TIL). You are only allowed to choose one of the courses in your degree program	
Department	3mE Department Maritime & Transport Technology	

ME44210	Drive & Energy Systems	3
Responsible Instructor	Dr.ir. H. Polinder	
Contact Hours / Week x/x/x/x	3/0/0/0 (1st part Q1 2x2 hours, 2nd part Q1 1x2 hours)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	This course is designed for students without a BSc in Electrical Engineering. Proper knowledge of mathematics (differential equations, complex numbers) and mechanics is required.	
Course Contents	- Introduction to drive and energy systems - Electricity and electrical circuits - Magnetism and magnetic circuits - Power electronics - DC machines - AC voltages, currents, powers and phasors - Transformers - Synchronous machines (including PM machines) - Induction machines - Choices in electrical drive and energy systems	
Study Goals	After following this course, students should be able to - Calculate DC and AC (single phase and three-phase) voltages and currents in simple electrical circuits (such as equivalent circuits of electrical machines) with sources, resistive, capacitive and inductive loads using Kirchhoffs laws. - Sketch phasor diagrams with voltage and current phasors. - Calculate active, reactive and apparent power in AC circuits. - Calculate energy stored in capacitors and inductors. - Describe the most important power semiconductors (diodes, IGBTs, MOSFETs) and their switching behaviour. - Describe how power electronic converters use these switches to convert DC from one voltage level to another (choppers or phase legs using IGBTs or MOSFETs), AC into DC (passive rectifiers using diodes or active rectifiers using IGBTs or MOSFETs) or DC into AC (phase legs using IGBTs or MOSFETs). - Derive an expression for the output voltage of a single-phase or three-phase diode bridge rectifier as a function of input voltage in continuous conduction mode (neglecting commutation). - Derive an expression for the output voltage of a DC-DC converter as a function of duty cycle and input voltage in continuous conduction mode. - Calculate induced voltage using Faradays law. - Calculate electromagnetic forces and torques using Lorentz law or the power balance. - Derive voltage equations and equivalent circuits of transformers, DC machines, synchronous machines, induction machines and permanent magnet machines. - Derive performance characteristics based on these equivalent circuits (torque-speed characteristics). - Describe advantages and disadvantages of electrification of drives. - Select suitable machine and drives for different applications.	
Education Method	Lectures	
Books	T. Wildi, "Electrical Machines, Drives and Power Systems"	
Prerequisites	Mathematics, Mechanics	
Assessment	The method of assessment will be: written exam, closed book, partly multiple choice. An equations form will be handed out with the exam. If on campus assessment is not possible (for example, because of measures resulting from COVID-19) the assessment will be as follows: written exam, open book, online, partly multiple choice.	
Permitted Materials during Tests	An equations form will be handed out.	
Department	3mE Department Maritime & Transport Technology	
Contact	h.polinder@tudelft.nl	

ME44300	Multi-Machine Coordination for Logistics	3
Responsible Instructor	Dr. V. Reppa	
Instructor	Prof.dr. R.R. Negenborn	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	In this course students will get familiar with automatic control techniques and their benefits and applications for real-time coordination of actions of large-scale transport systems. Theoretical concepts mostly related to model predictive control are discussed, as well as their application to real-world transport systems. Illustrative examples of large-scale systems include among others intermodal / synchromodal (i.e. road, waterborne, rail) transport networks. The operation of these networks relies on the interoperability of various machines such as cargo ships, trains, road tracks, automated guided vehicles, quay/yard cranes and many more.	
Study Goals	Students are able to: LO1. apply system theory to describe large-scale machines and transport systems LO2. explain the design of an automatically controlled system and of the automatic coordination of several controlled systems LO3. explain the design of control architectures for large-scale interconnected machines and transport systems (centralized, distributed, single-agent, multi-agent, single-level, multi-level) LO4. evaluate the pros and cons of different control architectures and their impact on multi-machine coordination for transport LO5. design an optimal control strategy for a single machine used in the transport network (truck, ship, crane, etc) LO6. develop a multi-agent framework based on model predictive control for real-time logistics of large-scale transport systems	
Education Method	Study material will be provided to support the lectures during which new concepts will be discussed and explained. Small groups of students will be created to conduct a project that will help them to develop deeper understanding of the theoretical (optimal) control and coordination concepts applied to a specific machine that is part of a larger transport network. Throughout the course illustrative examples will be used for showing the application of new concepts. Feedback sessions will be organized to discuss questions about the project.	
Course Relations	The course is related to the exit qualifications 1.b, 1.c, 3.b, 3.c and 7.a of the track MME of the Master Programme Mechanical Engineering, and to the specialization Engineering of the Master Programme Transport, Infrastructure and Logistics. Prior knowledge in automation of transport systems, dynamical systems, optimization and simulation is recommended.	
Literature and Study Materials	We will discuss chapters from the book "Intelligent Infrastructures", Negenborn et al., Springer, Dordrecht, The Netherlands, 2010. ISBN: 978-90-481-3597-4. Additional handouts and references to relevant scientific literature and media will also be provided.	
Assessment	Assessment will be based on a written group project report (50%) and on a written individual (closed-book) exam (50%). In principle, the written exam will take place on campus. Resit options are offered for both the project and the exam. Note that the grade of the exam and the grade of the project are only valid for the academic year 2023-2024.	
Tags	Information & Communication Modelling Optimalisation Project Transport & Logistics	
Department	3mE Department Maritime & Transport Technology	
Contact	Dr. Vasso Reppa (v.reppa@tudelft.nl)	

ME44305	System Analysis and Simulation	5
Responsible Instructor	Ir. M.B. Duinkerken	
Instructor	F. Schulte	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Required for	Multi-Machine Engineering	
Expected prior knowledge	Basic programming	
Parts	Systems and Data Analysis Discrete-Event and Agent-Based Simulation	
Course Contents	The course is an introduction to "Problem Engineering"; how to formulate a well-defined problem in a complex situation. The course introduces a systems approach to define the elements and structure of a problem situation and the process approach of discrete simulation to understand and quantify the time-dependent relations (behavior)	
Study Goals	Learning objectives: - Describe all relevant elements and their relations for at least one complex industrial or transportation system (e.g., cyber-physical production systems) using a combination of at least two conceptual models: process models, system models, steady-state models; - Collect data for the described system and analyse the system using the collected data and introduced data analysis methods; - Design industrial and transportation systems and processes applying an established innovation model; - Transform the developed conceptual models into a process-oriented simulation model, using a suitable language (e.g., Delphi/Tomas/Lazarus or Python/Salabim/Simpy).	
Education Method	lectures + group assignments	
Computer Use	Windows laptop with Python	
Literature and Study Materials	"The Delft Systems Approach, Analysis and Design of Industrial Systems", 2008, H.P.M.Veeke, J.A.Ottjes,G. Lodewijks, Springer, ISBN 978-1-84800-176-3 Lecture notes	
Assessment	Report of group assignment	
Exam Hours	80 for assignment	
Tags	Abstract Broad Challenging Design Industry Logics Modelling Process Programming Programming concepts Prototyping Stochastics Transport & Logistics	
Percentage of Design	50%	
Design Content	Logistic systems	
Department	3mE Department Maritime & Transport Technology	

ME46000	Nonlinear Mechanics	4
Responsible Instructor	Dr. C. Ayas	
Instructor	Dr. A.M. Aragon	
Instructor	Dr. F. Alijani	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	

Course Contents

Introduction

Nonlinear mechanics (NM) builds upon the engineering mechanics (statics and dynamics) courses as embedded within the BSc curriculum. This MSc course will complete the education in engineering mechanics for a large group of students. Thus, this course should prepare them as good as possible for common problems faced by mechanical engineers without a specialisation in engineering mechanics. On the other hand, a group of students will continue their training in dynamics and/or statics. For this reason, this course serves as a first step into the more advanced engineering mechanics topics.

Given the broad background of the students participating in this course, abstract formulations will be avoided. Moreover, many topics will be presented as introductions, creating awareness among the students.

Detailed content:

Introduction to nonlinearities in statics and dynamics
Principle of virtual work
Principle of minimum potential
Green-Lagrange strain tensor
2nd Piola-Kirchhoff stress tensor and equilibrium equations
Alternative stress and strain tensors,
Geometrical nonlinearity (finite rotations, finite deformations, geometric stiffness effects)

Introduction to nonlinear material models
Nonlinear elastic models
Simple plasticity models (elastic-ideally plastic, isotropic and kinematic hardening)
Residual stresses
Application to simple truss and beam problems
3D plasticity models work (yield surface, flow rule)

Introduction to nonlinear dynamics
Studying simple discrete nonlinear systems using the phase plane
Quantitative analysis of weakly nonlinear single-degree-of freedom systems using general perturbation theory
Periodic behaviour of simple nonlinear oscillators, limit cycles and softening/hardening responses
Basics of bifurcation theory and stability for simple nonlinear dynamic systems

Introduction to the finite element method in elastostatics (strong, weak or variational, and Galerkin forms of the boundary value problem with linearized kinematics)
The finite element method for large deformation (total Lagrangian framework)
Computational tools needed to solve the discrete system of equations

Study Goals

Learning goals:

Regarding nonlinear mechanics in general (25%), students:
are aware of the range of nonlinearities that are encountered in practical problems, including both static as well as dynamic settings,
are capable of detecting when the nonlinear analysis might be required for practical problems,
have an awareness of the role of nonlinearities in both static as well as dynamic systems,
are capable of formulating and solving simple nonlinear engineering mechanics problems.
can distinguish when geometrical nonlinearity may be relevant (finite rotations, finite deformations, geometric stiffness effects),
can make analytical estimates for nonlinear problems, e.g., based on minimum of potential energy.

Regarding physical nonlinearity (25%), students:
have an awareness of different nonlinear material models,
are familiar with and know when to use nonlinear elastic models,
are able to describe the essential features of a nonlinear elastic model,
are aware of simple plasticity models (elastic-ideally plastic, isotropic and kinematic hardening) to solve simple problems,
are aware and can describe the practical relevance of yielding and the generation of residual stresses
are able to describe how 3D plasticity models work (yield surface, flow rule),

Regarding nonlinear dynamics (25%), students:
can perform a qualitative analysis of simple discrete nonlinear systems using the phase plane,
can carry out quantitative analysis of weakly nonlinear single-degree-of-freedom systems using general perturbation theory,
can discuss the periodic behaviour of simple nonlinear oscillators, limit cycles and softening/hardening responses,
are aware of the basics of bifurcation theory and stability for simple nonlinear dynamic systems.

Finally, regarding computational nonlinear mechanics and the finite element method (25%), students:
can perform relatively simple nonlinear simulations,
are aware of computational tools needed to solve nonlinear equations, including topics such as rate equations, physical stiffness, geometric stiffness, iterative solutions procedures, incremental procedures, incremental iterative procedures, path following, dynamic and static transient simulations, convergence, stability points,
can state the strong form of a boundary value problem of an elliptic equation, its variational form, and derive the discrete equations using linearized kinematics
understand the procedure to derive the discrete equations in finite deformation setting using a total Lagrangian formulation (students understand the linearization process)
can state the strong form of a boundary value problem of an elliptic equation, its variational form, and derive the discrete equations using linearized kinematics
understand the procedure to derive the discrete equations in finite deformation setting using a total Lagrangian formulation (students understand the linearization process)

Education Method	Contact hours: Lectures (4 hours per week) Weekly tutor session (1 hour per week) Study materials: lecture slides Suggested literature: Nonlinear Finite Element Analysis of Solids and Structures, 2nd Edition; Rene De Borst, Mike A. Crisfield, Joris J. C. Remmers, Clemens V. Verhoosel; ISBN: 978-0-470-66644-9; 540 pages August 2012 Nonlinear Finite Elements for Continua and Structures, 2nd Edition; Ted Belytschko, Wing Kam Liu, Brian Moran, Khalil Elkhodary; ISBN: 978-1-118-63270-3; 830 pages December 2013, ©2013 Nonlinear Solid Mechanics: A Continuum Approach for Engineering Gerhard A. Holzapfel; ISBN: 978-0-471-82319-3; 470 pages February 2000 Applied Mechanics of Solids. Allan F. Bower, CRC Press, ISBN:978-1-4398-0247-2 (available freely from www.solidmechanics.org) Lecture Notes on Nonlinear Vibrations, Richard H. Rand, freely available at http://audiophile.tam.cornell.edu/randdocs/nlvibe52.pdf. Nayfeh, A. H., & Mook, D. T. (2008). Nonlinear oscillations. John Wiley & Sons.
Assessment	Written (closed-book) exam.
Department	3mE Department Precision & Microsystems Engineering

ME46007	**Measurement Technology**	**3**
Responsible Instructor	Dr.ir. G.E. Elsinga	
Instructor	Dr.ir. M.J. Tummers	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Summary	This master level course covers fundamental aspects of measurements useful for a mechanical engineer. It covers measurement of various physical quantities, signal conditioning, error analysis and statistical analysis of the data. A bachelor degree in mechanical, electrical, aerospace, industrial engineering or applied physics is considered prerequisite.	
Course Contents	topics: - General introduction to measurements - Measurement system behavior - Hypothesis testing - Uncertainty analysis - Sensor probes: variety of experimental techniques - Signal conditioning - Data acquisition and processing - Imaging and image processing - 3D techniques	
Study Goals	The general aim of this course is to provide students with a background in the theory of engineering measurements. At the end of the course, the students should be able to (1) analyze measurement systems and devices using the laws of mechanics and general physics, (2) evaluate the accuracy, sensitivity, spatial and temporal resolution of a measurement, (3) analyze and interpret datasets that contain errors, (4) design a measurement scheme and signal conditioning in order to enhance the significance of the results.	
Education Method	The main learning methods include lectures, take-home assignments and a textbook. The lectures cover the different topics of the course. (In case of unforeseen circumstances or measures resulting from COVID-19, the lectures will be replaced by pre-recorded videos of the lectures and/or live online-sessions. Such changes will be communicated on Brightspace.) Lecture slides will be made available on Brightspace. The recommended textbook is given below, however, you are encouraged to refer other books as well. Additional reading material, if necessary, will be made available on Brightspace. Assignments will be provided to practice the relevant problems and the concepts discussed in the class. They will be made available on Brightspace. The solutions to the assignments will be discussed in the class. The course ends with a written exam.	
Literature and Study Materials	The prescribed books will be followed during the course. Lecture slides will be made available on Brightspace. Additional study materials will be provided on Brightspace if necessary.	
Books	Theory and design for mechanical measurements, 6th edition by Richard S. Figliola & Donald E. Beasley	
Assessment	The examination will be as follows: Closed book written exam. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be: Remote (online) open book with several fraud prevention measures.	
Department	3mE Department Process & Energy	

SC42001	Control System Design	5
Responsible Instructor	Dr.ir. A.J.J. van den Boom	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	State-space description of single-input, single-output linear dynamic systems, interconnections, block diagrams. Linearization, equilibria, stability, Lyapunov functions and the Lyapunov equation Dynamic response, relation to modes, the matrix exponential. Realization of transfer function models by state space descriptions, coordinate changes, canonical forms. Controllability, stabilizability, uncontrollable modes and pole-placement by state-feedback. Application of LQ regulator. Observability, detectability, unobservable modes, state-estimation observer design Output feedback synthesis and separation principle. Reference signal modeling, integral action for zero steady-state error; Analysis in robust stability and robust performance. Basics of model predictive control (MPC). Different model-structures. Prediction models in state-space setting. Standard predictive control scheme. Relation standard form with GPC, LQPC and other predictive control schemes. Solution of the standard predictive control problem. Stability of MPC.	
Study Goals	By taking this course, the student - will be able to master the introduced theoretical concepts in systems theory and feedback control design - will be able to practically apply these concepts to design projects and tasks - and will be able to relate the learned concepts and techniques to other more specialized ones, to potentially integrate them by taking adjacent courses. - will be able to translate a predictive control problem into a standards setting and solve the predictive control problem. More specifically, the student will be able to: - Translate differential equation models into state-space and transfer function descriptions - Rationalize differences between state-space and transfer function approaches - Linearize a system, determine its equilibrium points, analyze directly its local stability, leverage Lyapunov theory to study general stability properties - Describe the effect of eigenvalue/pole locations to the dynamic system response in time/frequency domain. Contrast step and impulse responses. Analyze transients and steady-state - Investigate model controllability. Formulate and apply the procedure of pole-placement by state-feedback, as well as LQ optimal state-feedback control - Derive observability properties. Formulate and apply the procedure of state estimation and build converging observers - Formulate the separation principle and employ it for the design of output feedback - Build reference models and achieve zero steady-state error using integral control. - set up a predictive control problem. - Solve the standard predictive control problem; - Master the main analytical details in stability proofs of predictive control schemes;	
Education Method	Lectures 4/0/0/0	
Literature and Study Materials	Textbook (its use is strongly recommended): K.J. Astrom, R.M. Murray, Feedback Systems: An Introduction for Scientists and Engineers; second edition; Princeton University Press, Princeton and Oxford, 2009 Available online for download: http://www.cds.caltech.edu/~murray/books/AM08/pdf/fbs-public_24Jul2020.pdf The link for downloading the lecture notes for model predictive control will be available in the first quarter.	
Assessment	Written examination. Due to COVID-19 the exam may be an online exam. Disclaimer: information may change depending on the developments around the coronavirus.	
Department	3mE Department Delft Center for Systems and Control	

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Mechanical Engineering**

ME-MME Elective Courses

ME44115	Discrete Element Method (DEM) simulation	4
Responsible Instructor	Prof. Dr.ir. D.L. Schott	
Instructor	R.N. Roeplal	
Instructor	A.H. Hadi	
Instructor	Ir. M.C. van Benten	
Instructor	H. Shi	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	ME44101	
Course Contents	This hands on course teaches the use of particle based simulations using Discrete Element Method (DEM). DEM can be applied to any system that handles or consists out of (groups of) particles. DEM models individual particles (microscopic level) with the aim to model realistic behaviour of a the system of particles (macroscopic level). DEM can be used as an engineering tool to provide insight in influencing factors (e.g. particle size, shape and density) in particle processes, such as segregation and mixing. It can help to analyse and solve bottlenecks in industrial systems, but then validation of the model is required. After validation DEM can be used as a virtual prototyping environment for design of equipment that handles particulate materials. The application range is wide: from powder handling (small particle sizes, cohesive materials in food or pharmaceutical industry) to large scale systems such as rock dumping (e.g. 300 kg stones, 500 mm in size), and anything in between including bulk handling. Next to the theoretical background of DEM, the major part of this hands on course is to perform an assignment using the software EDEM. In this assignment you will model and redesign or optimise a (part of a) handling system with a given particulate material. This might also require laboratory work to measure real material properties on a macroscopic level such as angle of repose and wall friction angle. Your DEM model has to represent realistic material behaviour and realistic interaction between material and equipment. With the properly working DEM model you will analyse influential parameters and where possible improve performance of equipment. Typical application areas for DEM modelling are: bulk handling of materials coal and iron ore, also including solid biomass such as wood pellets or biocoal mining offshore food industry, handling of powders pharmaceutical industry, powders and tablets agricultural industry, for planting and soil preparation, post harvesting, fruit handling and sorting machines pavement engineering, creating mixtures and more.	
Study Goals	During the course the student will learn to: 1. explain the Discrete Element Method (DEM) theory 2. define the performance criteria of the system to be modeled 3. choose an appropriate contact model for material-material and material-equipment interaction in a DEM simulation 4. choose and implement a simplified but appropriate representation of the system in the simulation environment 5. evaluate the stability of the model by investigating a range of timesteps 6. calibrate and verify the model with experiments, experimental data or analytical solutions 7. test and evaluate the influence of different simulation parameters on the performance of the system 8. analyse datasets obtained from simulations linked to but not limited to the defined performance criteria 9. define confidence intervals for simulation and experimental results due to numerical and measurement errors	
Education Method	Lectures and assignment (computer and possibly lab)	
Computer Use	60%	
Course Relations	ME44101	
Literature and Study Materials	Study materials will be provided throughout the course	
Assessment	The examination will be as follows: 1. presentation on model development (20%) 2. report and presentation on model and results (80%) In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be: 1. remote or recorded presentation on model development (20%) 2. report and remote or recorded presentation on model and results (80%).	
Enrolment / Application	As soon as possible via Brightspace	
Special Information	Design Lab Research Mechanics Modelling Practicals Prototyping Software Transport & Logistics Contact Models Offshore Mining Ballast Railway Scour protection Penetration Particle Interaction Equipment Machine Cargo Interaction Bulk material	

	Design of Experiments Experimental plan Bulk
Department	3mE Department Maritime & Transport Technology

ME44125	**Reliability and Maintenance of Transport Equipment**	**3**
Responsible Instructor	X. Jiang	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Technical academic BSc	
Course Contents	Transport equipment is in general a mechanical system, growing in size, capacity and complexity in order to accommodate continuous development of transport. It is essential to ensure sufficient safety, serviceability and durability of the equipment at the reduced cost in order to maintain the effectiveness of the transport system. This belongs to the research context of system reliability. Typically, a transport equipment consists of multiple subsystems or components with different function, and they would fail in different failure modes with different mechanism behind. Contributed by various non-destructive techniques, online sensor system and data analytics, potential failure of the equipment (fault, defect, damage) could be detected, predicted and mitigated in such a way that the occurrence of a catastrophic accident would be prevented, although with some extent of uncertainty caused by, i.e., the sensitivity of NDTs & sensors, environmental noise, etc. Upon the detection of a fault(or damage), a follow-up maintenance strategy shall be made in order to determine whether, when, where and how to repair/maintain the equipment, constrained by available budget, facilities and resources, etc. The overall objective of the course is to prepare engineering graduates with an in-depth knowledge of system reliability, maintainability and availability. A series of modules will be given covering the fundamental theory embedded in the reliability and maintenance of transport equipment. Application of the theory in practice will be exemplified and practiced through in-class exercises and discussion, weekly assignments and final written report. One or two guest lectures will be introduced to demonstrate the implication of system reliability in the operation and maintenance of the transport equipment and the infrastructure.	
Study Goals	After successfully completing the course, students will be able to: -Define system Reliability, Maintainability and Availability(RAM)and describe associated basic measures using statistic method. -Carry out system reliability analysis to identify critical component / subsystem using statistic models, like FMECA, FTA, ETA and Markov process among others. -Apply the acquired data to predict Remaining Useful Life (RUL) using statistic, data driven or hybrid models. -Analyze common maintenance strategies at both component and system levels based on statistic, data driven or hybrid models. -Develop and reflect a maintenance strategy for a simplified transport equipment/system based on statistic, data driven or hybrid models.	
Education Method	Lectures and supplementary material provided during lectures, in class exercises and discussions, weekly assignment, group work on final written report and presentation.	
Assessment	The final grade is based on 30% of the grade for weekly assignments plus 70% of the grade for the final report and associated presentation(50/50). Four to five students are anticipated to work as a group on the final report. Weekly assignments are compulsory and they can be done either individually or as a group work (the same group on the final report). The grade for assignments will be the average grade of all weekly assignments. To pass the course, the grade for each weekly assignment is not allowed to be lower than a 5.0. The final presentation will take place on campus. In case of unforeseen circumstances or measures resulting from i.e., COVID-19 etc., the final presentation will then be given online. Detail information will be announced on Brightspace in due time. The final grade must be a 5.75 or higher in order to pass the course.	
Remarks	Reference books: 1. Marvin Rausand, Arnljot Høyland, System Reliability Theory: Models, Statistical Methods, and Applications. 2. Kumar, U. D., Crocker, J., Knezevic, J., & El-Haram (2000), M. Reliability, Maintainability and Logistic Support - A Lifecycle Approach. 3. Charles E. Ebeling (2010), An Introduction to Reliability and Maintainability Engineering.	
Department	3mE Department Maritime & Transport Technology	

ME44311	Advanced Operations and Production Management	5
Responsible Instructor	A. Napoleone	
Instructor	Ir. M.B. Duinkerken	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge	To enroll in the course, students must have a BSc in Engineering. The course is especially for students following TIL, MME and SDPO. The course requires high engagement and commitment, and includes both self-study and group work. Students are expected to be willing to work in teams, to collaborate and use social and project management skills.	
Course Contents	The course revolves around different topics of operations and production management, including: system analysis and process mapping, modularity in product design and process design, smart manufacturing and Industry 4.0, reconfigurability and adaptability in operations management. With this course, the student is expected to develop a structured approach to address operational problems using the methods provided in the course and/or methods identified with individual study.	
Study Goals	After successfully completing this course, students will be able to: Describe the fundamental concepts of operations and production management Search for and use methods to solve operational problems in different applications Have a structured approach to manage familiar as well as unknown operational problems	
Education Method	Problem-based and interest-driven approach. PART ONE in Q3: (i) Plenary lectures / Cases by industrial speakers AND (ii) Individual study and deepening of one/more operational topics of interest. Attending lectures is required to acquire the fundamentals of production and operations management. Active participation during plenary lectures is strongly advised. At the end of Q3, students submit individual reports (one report per student). PART TWO in Q4: Group projects (up to 6 students per group) with case companies AND Group supervision (on demand). In Q4, students work in groups at different projects with case companies. The course leader can assist students in finding a company. Groups work autonomously and ask for supervision when needed. At the end of Q4, students submit group reports that include multiple analyses of operational problem at the case company and the solution to the problem. Q5 is available groups if preferable for time constraints.	
Assessment	The assessment is based on the evaluation of the individual report (Q3) and group report (Q4).	
Department	3mE Department Maritime & Transport Technology	
Judgement	The report is judged by Dr A. Napoleone The final grading considers the individual report (50%) and the group report (50%)The grading is based on a Rubric. Results can be discussed in feedback sessions (on demand)	
Contact	a.napoleone@tudelft.nl	

ME44312	Machine Learning for Transport and Multi-Machine Systems	3
Responsible Instructor	F. Schulte	
Instructor	Dr. B. Atasoy	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Basic programming	
Course Contents	The course gives an introduction to the basic concepts of machine learning (ML) (supervised, unsupervised, reinforcement learning) and puts them into relation to artificial intelligence and data analytics. Important fundamental machine learning algorithms including regression, classification, clustering, and neural networks are introduced and implemented in Python. The course also gives a brief overview of more advanced ML techniques that are used in the transportation and multi-machine domains such as reinforcement learning, graph-neural networks, and recommender systems. On this basis, machine learning methods are applied in group projects to transport and multi-machine systems using different types of data from different domains (e.g., AIS, GPS, New York Taxi, or machine failure data).	
Study Goals	-Understand the basic concepts of machine learning -Learn to implement widely-used machine learning algorithms -Apply machine learning methods to transport and multi-machine systems -Evaluate different machine learning methods for a problem setting -Conduct small ML projects	
Education Method	-Lectures -Group Work -Individual exercises for each machine learning algorithm	
Computer Use	Python on laptop	
Assessment	Group report	
Elective	Yes	
Tags	Abstract Algorithmics Artificial intelligence Broad Business Design Group work Industry Modelling Optimalisation Programming Research Methods Stochastics Sustainability Technology Transport & Logistics Transport phenomena	
Percentage of Design	40%	
Design Content	Designing the right machine learning methodologies for different problem settings in the context of transport and multi-machine systems	
Department	3mE Department Maritime & Transport Technology	
Contact	f.schulte@tudelft.nl b.atasoy@tudelft.nl	

MT44001	Mechatronics in MT	5
Responsible Instructor	A. Coraddu	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	BSc courses on control engineering (incl. Proportional-Integral-Derivative control), hydromechanics, marine engineering, and signal analysis and processing. Software Skills: Matlab/Simulink.	
Course Contents	Overview: Unlock the fascinating intersection of mechanics, electronics, and information technology in maritime settings through this cutting-edge course. Our main objective is to guide you in designing a Dynamic Positioning (DP) systema complex mechanism to stabilize a model boat against various natural disturbances like waves, wind, and currents. What You'll Learn: Integrated Systems Approach: Explore the synergy between mechanics, electronics, and IT in maritime projects. Dynamic Positioning (DP) System: Delve into designing and constructing a DP system to stabilize a model boat against environmental variables. Hands-on Experience: Use the Matlab/Simulink environment to create your DP model. Real-world Implementation: Transition from theoretical design to hands-on testing in our state-of-the-art lab. Critical Evaluation: Assess the strengths and weaknesses of your DP model in terms of performance and reliability. Core Components: Marine Engineering: Understand vessel design, propulsion, and other marine systems fundamentals. Control Engineering: Gain in-depth knowledge of control theory, including PID controls essential for dynamic systems. Digital Signal Processing: Master techniques for analyzing, interpreting, and manipulating signals critical for effective control systems. Why Take This Course? Interdisciplinary Learning: Equip yourself with a unique blend of skills from mechanics to digital signal processing, setting a robust foundation for your engineering career. Hands-on Application: Bridge the gap between theoretical knowledge and real-world applications through lab experiments. Career Relevance: Prepare for an ever-evolving marine, aerospace, and control systems engineering job market. Holistic Understanding: Gain a 360-degree perspective on maritime mechatronic systems, with the DP system serving as a quintessential example. Performance Analysis: Acquire the skills to critically evaluate and optimize complex engineering systems for real-world challenges. By the end of this course, you'll have a working DP model and a comprehensive understanding of how to design, analyze, and evaluate complex maritime mechatronic systems.	
Study Goals	1 - Develop an optimal DP solution for a given ship by understanding the theory and practice. 2 - Apply the basics of computer-based modeling, measuring, and control systems. 3 - Apply basics of digital signal processing relevant to mechatronics systems (sampling, filtering, signal resolution, A/D and D/A conversion). 4 - Describe the basic concepts of mechatronics and the working principles of sensors and actuators. 5 - Analyse and interpret the performance of mechatronics systems. 6 - Evaluate the differences between the theory and practice of control systems. 7 - Evaluate the strengths and weaknesses of mechatronic systems.	
Study Goals continuation	After successfully completing the course, students will be able to: - Describe the basic principles of mechatronics - Describe sensors and actuators in general and the ones specific to the project, - Apply basics of signal processing relevant to mechatronic systems (sampling, filtering, signal resolution, A/D and D/A conversion), - Use and tune computer-based measuring and control systems, - Analyse and interpret the performance of a miniature Dynamic Positioning system regarding theoretical and practical aspects, - Evaluate the strengths and weaknesses of a mechatronic system that the students set up.	
Education Method	Lectures + Assignments. In the lectures, course material from the course books will be elaborated upon using lecture sheets and interactive Q&A sessions. The assignments enable students to obtain experience with the required calculations during the course period. In this way, the assignments support students in developing the required insight into the course contents to submit the deliverables.	
Books	Bolton, W. (2003). Mechatronics: electronic control systems in mechanical and electrical engineering. Pearson Education. Karnopp, D. C., Margolis, D. L., & Rosenberg, R. C. (2012). System dynamics: modeling, simulation, and control of mechatronic systems. John Wiley & Sons. Cetinkunt, S. (2015). Mechatronics with experiments. John Wiley & Sons. Fossen, T. I. (1994). Guidance and control of ocean vehicles. John Wiley & Sons Inc. Wadoo, S., & Kachroo, P. (2016). Autonomous underwater vehicles: modeling, control design and simulation. CRC Press. Karris, S. T. (2006). Introduction to Simulink with engineering applications. Orchard Publications.	
Reader	Reader (available on Brightspace). Week assignments (available on Brightspace). Additional material supplied on Brightspace.	
Assessment	Students are evaluated based on their performance with respect to the following deliverables: D1: Demonstrate a selected DP solution using a working (online) control system computer model - 30%. D2: Demonstrate multiple DP solutions using a working computer model (offline) - 30%. D3: Document your solutions in a report, discuss the validity of the offline model as a representation of the online system, and reflect on the learning process - 40%.	
Remarks	The maximum capacity of MT44001 is 35 students per period. The reason is the availability of the flume tanks, boats, required charging time, etc.	
Department	3mE Department Maritime & Transport Technology	
Contact	Dr Andrea Coraddu - a.coraddu@tudelft.nl	

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Mechanical Engineering**

ME-MME Second year

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Mechanical Engineering**

ME-MME Second Year Project: Select 1 (15 ECTS)

ME54015	ME-MME Research Assignment	15
Responsible Instructor	Ir. M.B. Duinkerken	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	None (Self Study)	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	All ME-MME obligatory courses finished.	
Course Contents	The research assignment can be either a design, experimental or programming/simulation assignment on a wide range of topics often related to ongoing research. To get an idea of subjects for assignments, you can check titles and summaries of the assignments that were carried out by other students. To get an assignment contact our assignment coordinator Yusong Pang. You will be supervised by a staff member. Design assignment (15 EC) Your assignment may vary from (re)designing a part of a piece of equipment to the design of a transportation process with accompanying equipment. Experimental assignment (15 EC) You will set-up and carry out experiments in close cooperation with and under the authority of a staff member. These experiments are often concerned with ongoing research at our department. Programming/simulation assignment (15 EC) You will use software or develop software/algorithms/procedures for solving technical problems, controlling, measuring, designing, data analysis related to technical problems. Developing or choosing the appropriate model and implementing the model in a simulation language, testing a model in the laboratory can also be part of the assignment.	
Study Goals	Design The main objective for the assignment is to create a design, thereby following a methodological approach and using all the necessary tools and calculations to converge to a most suitable design. Experimental The main objective for the assignment is to acquire experience in designing, building a testing methodology and measuring the relevant parameters. In scientific research experiments have to be repeatable in order to be valid, this means that the experiments have to have the same results when performed by a third party. Computer/Programming The main objective for the assignment is to acquire experience in designing and building a representative environment for experiments. The test environment can be a computer or physical model and should be designed "to purpose" which means the performance indicators are relevant and the results accurate enough to answer the research question.	
Education Method	Self study with regular supervision of a staff member	
Assessment	Report and research process evaluation. More details: a list of assignments is available at the Brightspace organisation MME	
Remarks	Old course code: ME2130-15	
Department	3mE Department Maritime & Transport Technology	

TUD4040	Joint Interdisciplinary Project	15
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Project Coordinator	Ir. B.J.E. de Bruin	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Summary	JIP consists of three sets of activities: 1. The project which takes place at a company. Students are responsible for the project work (collaboration, integration of different perspectives, the content quality of their work). They plan their activities via a scrum method, are challenged to realise field trips, consult with experts and realise the necessary research work. The team keeps a Scrum log, run a blog for interaction with the outside world and every team member keeps a personal development log. 2. Meeting Professionals lectures and workshops about specialised topics, by academic staff or senior professionals from the companies involved. 3. Plenary meetings and design reviews for all the teams, company coaches and academic staff. At the meetings the students present their intermediate resp. final outcomes and plans, and are provided with feedback.	
Course Contents	The aim of the Joint Interdisciplinary Project is to prepare students to contribute to solving impactful technological challenges. The projects not only demand good engineering working knowledge but also experience with interdisciplinary and systems theory, and both knowledge and mindsets of innovation and entrepreneurial behavior. The project brief is provided by renowned companies like Airbus, Arcadis, etc. Teams of interdisciplinary student teams guided by a company coach and offered academic and industry expertise, are invited to realise an innovative problem solution to a complex problem and contributing to the sustainable development goals.	
Study Goals	1. Cognitive abilities attributable to interdisciplinary learning Demonstrate the ability to engage in perspective-taking; Develop structural knowledge pertaining to the problem; Integrate knowledge and modes of thinking drawn from two or more disciplines; Produce an interdisciplinary understanding of complex problem or intellectual question. 2. Scientific and intellectual development Capable to analyze scientific and societal consequences (economic, social, cultural, environmental) of the innovation; 3. Research and design capabilities Demonstrate engineering skills: technical skills, interpreting results, creativity, usability for company/institute; Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; 4. Collaboration and communication in an interdisciplinary team Demonstrate behavioral competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to respect different disciplines and adapt to different cultures); Show ability to write a technical report: structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (30 pages); Present work performed in a structured way through an oral presentation to their peers and customer. 5. Self-adjustment and reflection capabilities Plan and control the project efficiently considering resources and methodology; Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career. Cognitive abilities attributable to interdisciplinary learning; The ability to integrate (scientific and practical technological)knowledge from different disciplines to solve complex problems Scientific and intellectual development The capacity to evaluate the ethical, scientific and societal consequences of the proposed innovation Research and design capabilities The ability to create reasonable and relevant research or design , according to the academic standards of the involved disciplines Collaboration and communication in an interdisciplinary team Demonstrate behavioural competences and skills relevant for teamwork and effective communication with different stakeholders Self-adjustment and reflection capabilities To carry out regular reflections on professional and personal development and being able to improve upon those reflections Understand contemporary and societal issues in their work.	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Assessment	The assessment criteria pertain to the process, product and presentation and are assessed at an individual and team level, during 3 review sessions at three levels of accomplishment (each review assesses at a different level): Interdisciplinary work Scientific reasoning and ethical mindset Innovation process Presentation and communication Academic staff, supported by the input from company coaches, grade the students during the Problem Statement, the Midterm and the Final Review. The final group mark for JIP is differentiated per person and is based on: The team presentations - 30% The Problem definition, progress report and final report 30% The individual contribution to the team 20% The final individual reflection report 5% The Blog 5%	
Remarks	Project Outcomes (product, model, system) are presented at: 1. Problem Statement Review 2. Midterm meeting 3. Final Review 4. Symposium presentation	

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Mechanical Engineering**

ME-MME Graduation Project (45 ECTS)

ME54010-20	ME-MME Literature Research	10
Responsible Instructor	Ir. M.B. Duinkerken	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	None (Self Study)	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Literature Assignment	
Course Contents Continuation	The course assesment consists of three parts: T1 is an individual assignment, studying literature on a specific subject and resulting in a written report. The report will be graded by the assignment supervisor. T2 is giving a presentation at one of the sections theme colloquia. Because currently these colloquia are not organized this part will be graded with a pass. T3 is assessing your attendance at graduation presentation. You must attend at least 5 presentations of MSc-students ME-TEL, ME-MME or other relevant tracks in order to prepare for your own 2nd year assignments. A short evaluation describing the link between these previous projects and your assignments is sufficient to obtain a pass.	
Study Goals	1. Get acquainted with the major information sources of a subject 2. Consolidate a large amount of information into a systematic, well-organized and logical argument 3. Judge often mutual contradictory literature and to arrive at a critical attitude 4. Make unordered literature accessible	
Education Method	Self study	
Assessment	Three tests: T1 Literature Research (written report) T2 Presentation (pass/fail) T3 Colloquium Attendance (pass/fail)	
Remarks	Old course code: ME2110-10	
Department	3mE Department Maritime & Transport Technology	

ME54035	ME-MME MSc Thesis	35
Responsible Instructor	Ir. M.B. Duinkerken	
Contact Hours / Week x/x/x/x	0/0/x/x	
Education Period	None (Self Study)	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	all obligatory and elective courses on the Individual Study Program ME54010 ME54015	
Course Contents	You will set up and carry out your own research project in the field of Multi-Machine Engineering. The subject of research can be provided by the staff or a company, however it is also possible to propose your own project. The final project can be carried out in company or within the framework of ongoing research at the university, e.g. in our laboratory. If a student undertakes the graduation project at a company, a graduation agreement should be signed. In this respect, the template of the TUD-3mE graduation agreement should be used, which is available at the following link: https://www.tudelft.nl/en/student/faculties/3me-student-portal/education/related/student-forms/msc-forms/. If a student identifies a potential master assignment within a company or institute then he/she can only commit to that assignment after written confirmation of approval by the professor. If any case, it is not allowed to take on a master assignment that is part of a critical path in a project within a company. Your supervisors will be a staff member and a professor, and if applicable a supervisor from the company. More information is available from the MME brightspace organisation and the assignment coordinator Xiaoli Jiang.	
Study Goals	The overall objective of this assignment is to demonstrate a sufficient academic level in the field of Multi-Machine Engineering	
Education Method	Self study with regular supervision from staff members and professor	
Assessment	Written report, oral presentation, oral defense, research process evaluation	
Remarks	Old course code: ME2190-35	
Department	3mE Department Maritime & Transport Technology	
Contact	Xiaoli Jiang, x.jiang@tudelft.nl	

Year	**2023/2024**
Organization	**Mechanical, Maritime and Materials Engineering**
Education	**Master Mechanical Engineering**

ME-OM Track Opto-Mechatronics	
Contact for students	E. Matroos T +31 15 27 86581 E e.matroos@tudelft.nl
Program Title	ME-OM Track Opto-Mechatronics
Program Coordinator	Dr.ir. J.F.L. (Hans) Goosen T +31 15 278 6500 E J.F.L.Goosen@tudelft.nl
Introduction 1	The track OM is currently offered as a focus area within the Track HTE. This has no consequence on the content or formalities but gives students more flexibility in choosing electives and liaising with a larger student population. For more information on the contents of this track, please read the OM-track information on the page of HTE.

Dr.ir. D.A. Abbink

Unit	Mech, Maritime & Materials Eng
Department	Human-Robot Interaction
Telephone	+31 15 27 82077
Room	34.F-1-470

A. Accardo

Department	Micro and Nano Engineering
Telephone	+31 15 27 81610

Dr. A.P. Afghari

Department	Safety and Security Science
Telephone	+31 15 27 83026
Room	31.C1.020

A.C. Akyildiz

Department	Medical Instruments & Bio-Inspired Technology

Dr. M.W.E.M. Alfeld

Unit	Mech, Maritime & Materials Eng
Department	Team Matthias Alfeld
Telephone	+31 15 27 88253

Dr. F. Alijani

Unit	Mech, Maritime & Materials Eng
Department	Dynamics of Micro and Nano Systems
Telephone	+31 15 27 86739

Dr. J. Alonso Mora

Unit	Mech, Maritime & Materials Eng
Department	Learning & Autonomous Control
Telephone	+31 15 27 85489
Room	34.F-2-100

Dr. A.M. Aragon

Unit	Mech, Maritime & Materials Eng
Department	Computational Design and Mechanics
Telephone	+31 15 27 82267

F. Arroyo Cardoso

Department	Electronic Instrumentation
Telephone	+31 15 27 87740

Dr. B. Atasoy

Unit	Mech, Maritime & Materials Eng
Department	Transport Engineering and Logistics
Telephone	+31 15 27 82706
Room	34.B-4-290

Dr. C. Ayas

Unit	Mech, Maritime & Materials Eng
Department	Computational Design and Mechanics
Telephone	+31 15 27 81550

H. Bazyar

Department	Engineering Thermodynamics
Telephone	+31 15 27 82760
Room	34b.K-1-240

Ir. M.C. van Benten

Department	Marine and Transport Technology
Room	34.B-3-210

Dr. N. Bhattacharya

Unit	Technische Natuurwetenschappen
Department	Optische Technologie
Telephone	+31 15 27 85785

Dr. F. Bociort

Unit	Technische Natuurwetenschappen
Department	ImPhys/Bociort group
Telephone	+31 15 27 81457

Prof.dr.ir. B.J. Boersma

Unit	Mech, Maritime & Materials Eng
Department	Marine and Transport Technology
Telephone	+31 15 27 87979
Room	34.B-1-320

Dr.ir. A.J.J. van den Boom

Unit	Mech, Maritime & Materials Eng
Department	Team Ton van den Boom
Telephone	+31 15 27 84052
Room	34.C-3-180

Ir. W. van den Bos

Unit	Mech, Maritime & Materials Eng
Department	Transport Engineering and Logistics
Telephone	+31 15 27 86766
Room	34.B-3-310

L. Botto

Department	Complexe Vloeistof Processen

Dr.ir. P. Breedveld

Unit	Mech, Maritime & Materials Eng
Department	Medical Instruments & Bio-Inspired Technology
Telephone	+31 15 27 85232
Room	34.E-3-340

Dr.ir. W.P. Breugem

Unit	Mech, Maritime & Materials Eng
Department	Multi Phase Systems
Telephone	+31 15 27 88663
Room	34.B-K-170

F.G.J. Broeren

Department	Precision and Microsystems Engineering
Telephone	+31 15 27 85905
Room	34.G-1-465

Ir. B.J.E. de Bruin

Department	Cognitive Robotics

Department	Cognitive Robotics

Dr. A.J.L.L. Buchner

Department	Fluid Mechanics
Telephone	+31 15 27 85748

Dr. N.V. Budko

Unit	Elektrotechn., Wisk. & Inform.
Department	Numerical Analysis
Telephone	+31 15 27 86050
Room	36.HB 03.050

Dr. J.G. Buijnsters

Unit	Mech, Maritime & Materials Eng
Department	Micro and Nano Engineering
Telephone	+31 15 27 85396
Room	34.G-1-440

Dr. S. Caneva

Department	QN/van der Zant Lab
Telephone	+31 15 27 81673
Room	34.G-1-420

Department	Dynamics of Micro and Nano Systems
Telephone	+31 15 27 81673
Room	34.G-1-420

Dr. H.L. Castricum

Department	Materials Science and Engineering
Telephone	+31 15 27 84398

Department	Team Joris Dik
Telephone	+31 15 27 84398

G.R. Chandra Mouli

Unit	Elektrotechn., Wisk. & Inform.
Department	DC systems, Energy conversion & Storage
Telephone	+31 15 27 81898
Room	36.LB 03.670

T.N. Coggins

Department	Ethiek & Filosofie van Techniek
Telephone	+31 15 27 89870

Unit	Techniek, Bestuur & Management
Department	Ethiek & Filosofie van Techniek
Telephone	+31 15 27 89870

K. Cools

Department	Numerical Analysis
Telephone	+31 15 27 83442
Room	36.HB 03.280

Dr. S.M. Copeland

Unit	Techniek, Bestuur & Management
Department	Ethiek & Filosofie van Techniek
Telephone	+31 15 27 87723

A. Coraddu

Department	Ship Design, Production and Operations
Telephone	+31 15 27 87809
Room	34.B-4-180

Dr. L.L. Cutz IJchajchal

Unit	Mech, Maritime & Materials Eng
Department	Large Scale Energy Storage
Telephone	+31 15 27 85797
Room	34b.K-1-140

A. Dabiri

Department Team Azita Dabiri
Telephone +31 15 27 82259

Prof.dr.ir. B.H.K. De Schutter

Unit Mech, Maritime & Materials Eng
Department Delft Center for Systems and Control
Telephone +31 15 27 85113

Dr. R. Delfos

Unit Mech, Maritime & Materials Eng
Department Energy Technology
Telephone +31 15 27 82963
Room 34b.K-0-270

F. Delgado Medina

Department Delft Centre for Entrepreneurship

Dr. C. Della Santina

Department Learning & Autonomous Control
Telephone +31 15 27 81630
Room 34.F-2-320

Prof.dr. J.J. van den Dobbelsteen

Unit Mech, Maritime & Materials Eng
Department Medical Instruments & Bio-Inspired Technology
Telephone +31 15 27 89515
Room 34.E-2-230

Prof.dr.mr.ir. N. Doorn

Unit Techniek, Bestuur & Management
Department Ethiek & Filosofie van Techniek
Telephone +31 15 27 88059
Room 31.B4.050

K.M. Dowling

Department Electronic Instrumentation
Telephone +31 15 27 86550

Prof.dr.ir. W.D. van Driel

Department Electronic Components, Technology and Materials
Telephone +31 15 27 89528

Department Electronic Components, Technology and Materials
Telephone +31 15 27 89528

Unit Elektrotechn., Wisk. & Inform.
Department Electronic Components, Technology and Materials
Telephone +31 15 27 89528

Ir. M.B. Duinkerken

Unit Mech, Maritime & Materials Eng
Department Transport Engineering and Logistics
Telephone +31 15 27 81790
Room 34.B-3-320

Dr. J.M. Duran

Unit Techniek, Bestuur & Management
Department Ethiek & Filosofie van Techniek
Telephone +31 15 27 87879

Dr.ir. G.E. Elsinga

Unit	Mech, Maritime & Materials Eng
Department	Fluid Mechanics
Telephone	+31 15 27 88179
Room	34b.K-2-270

R.M. Ensing

Unit	Mech, Maritime & Materials Eng
Department	Intelligent Vehicles
Telephone	+31 15 27 89767
Room	34.E-0-200

Dr. H.B. Eral

Unit	Mech, Maritime & Materials Eng
Department	Complexe Vloeistof Processen
Telephone	+31 15 27 86715
Room	34b.K-0-250

D. Farhadi Machekposhti

Department	Mechatronic Systems Design
Telephone	+31 15 27 82745

Dr. G.A. Filonenko

Department	Team Georgy Filonenko

Department	ChemE/Inorganic Systems Engineering

Prof.dr. P.J. French

Unit	Elektrotechn., Wisk. & Inform.
Department	Bio-Electronics
Telephone	+31 15 27 84729
Room	36.HB 16.310

A.R. Gammon

Department	Ethiek & Filosofie van Techniek
Telephone	+31 15 27 81387

Prof.dr. D. Gavrila

Unit	Mech, Maritime & Materials Eng
Department	Intelligent Vehicles
Telephone	+31 15 27 88504
Room	34.E-0-330

Prof.dr.ir. P.H.A.J.M. van Gelder

Unit	Techniek, Bestuur & Management
Department	Safety and Security Science
Telephone	+31 15 27 86544
Room	31.c1.170

M.K. Ghatkesar

Unit	Mech, Maritime & Materials Eng
Department	Micro and Nano Engineering
Telephone	+31 15 27 82299
Room	34.G-1-400

Dr.ir. F.J.H. Gijsen

Department	Medical Instruments & Bio-Inspired Technology

Department	Medical Instruments & Bio-Inspired Technology

Prof.dr.ir. M.B. van Gijzen

Unit Elektrotechn., Wisk. & Inform.
Department Numerical Analysis

Telephone +31 15 27 82519
Room 36.HB 03.060

Prof.dr. E.K.A. Gill

Unit Luchtvaart- & Ruimtevaarttechn
Department Space Systems Egineering

Telephone +31 15 27 87458
Room 62.8.12

Dr.ir. J.F.L. Goosen

Unit Mech, Maritime & Materials Eng
Department Computational Design and Mechanics

Telephone +31 15 27 86500

Ir. J.D.C. Groenhuis

Department Biomechanical Engineering

Department HR Administratie Centre

Dr. J.B. van Grunsven

Unit Techniek, Bestuur & Management
Department Ethiek & Filosofie van Techniek

Telephone +31 15 27 84715

Dr. K.S. Grußmayer

Department BN/Kristin Grussmayer Lab

Telephone +31 15 27 84245
Room 58.F1.350

A.H. Hadi

Department Transport Engineering and Logistics

Dr. C.W. Hagen

Unit Technische Natuurwetenschappen
Department ImPhys/Hagen group

Telephone +31 15 27 86073
Room 22.F 064

Dr. R.M. Hartkamp

Unit Mech, Maritime & Materials Eng
Department Complexe Vloeistof Processen

Telephone +31 15 27 86674

Dr. L. Hartmann

Unit Techniek, Bestuur & Management
Department Delft Centre for Entrepreneurship

Telephone +31 15 27 88073
Room 31.C2.140

J.W. Haverkort

Unit Mech, Maritime & Materials Eng
Department Energy Technology

Telephone +31 15 27 86651
Room 34b.K-0-260

Prof.dr.ir. J. Hellendoorn

Department College van Bestuur

Telephone +31 15 27 89007
Room 34.F-2-440

Unit	Mech, Maritime & Materials Eng
Department	Cognitive Robotics
Telephone	+31 15 27 89007
Room	34.F-2-440

Prof.dr. F.C.T. van der Helm

Unit	Mech, Maritime & Materials Eng
Department	Biomechatronics & Human-Machine Control
Telephone	+31 15 27 85616
Room	34.E-1-340

Prof.dr.ir. R.A.W.M. Henkes

Unit	Mech, Maritime & Materials Eng
Department	Fluid Mechanics
Telephone	+31 15 27 81323

Prof.dr.ir. J.L. Herder

Unit	Mech, Maritime & Materials Eng
Department	Precision and Microsystems Engineering
Telephone	+31 15 27 84713
Room	34.G-1-270

Dr.ir. M.J.M. Hermans

Unit	Mech, Maritime & Materials Eng
Department	Team Marcel Hermans
Telephone	+31 15 27 82286
Room	34.H-1-300

Dr. E.B.A. Hinderink

Department	Multi Phase Systems

Ir. J.W. Hoftijzer

Unit	Industrieel Ontwerpen
Department	Human Information Communication Design
Telephone	+31 15 27 89325
Room	32.C-1-570

Dr.ir. J.P. Hoogenboom

Unit	Technische Natuurwetenschappen
Department	ImPhys/Hoogenboom group
Telephone	+31 15 27 85920
Room	22.F 088

L. Hoogendam

Department	Human-Robot Interaction

Prof.dr. K. Hooman

Department	Process and Energy

G.A. van der Hoorn

Unit	Mech, Maritime & Materials Eng
Department	Robot Dynamics
Telephone	+31 15 27 88643
Room	34.E-K-290

Dr.ir. T. Horeman

Unit	Mech, Maritime & Materials Eng
Department	Medical Instruments & Bio-Inspired Technology
Telephone	+31 15 27 85097
Room	34.F-1-360

Dr.ir. W.T. van Horssen

Unit	Elektrotechn., Wisk. & Inform.

Department	Mathematical Physics
Telephone	+31 15 27 83524
Room	36.HB 05.080

Dr. S.H. Hossein Nia Kani

Unit	Mech, Maritime & Materials Eng
Department	Mechatronic Systems Design
Telephone	+31 15 27 84248

Dr. A. Hunt

Unit	Mech, Maritime & Materials Eng
Department	Micro and Nano Engineering
Telephone	+31 15 27 86657

Dr.ir. S. Iskander-Rizk

Department	Optische Technologie
Telephone	+31 15 27 81324

Dr. L.C.M. Itard

Unit	Bouwkunde
Department	Environmental & Climate Design
Telephone	+31 15 27 86341
Room	08.01WEST540

Dr.ir. M. Jafarian

Department	Team Matin Jafarian
Telephone	+31 15 27 82422

X. Jiang

Unit	Mech, Maritime & Materials Eng
Department	Transport Engineering and Logistics
Telephone	+31 15 27 88511
Room	34.B-3-320

Prof.dr.ir. W. de Jong

Unit	Mech, Maritime & Materials Eng
Department	Large Scale Energy Storage
Telephone	+31 15 27 89476

Ir. B.R. Joseph

Unit	Techniek, Bestuur & Management
Department	Delft Centre for Entrepreneurship
Telephone	+31 15 27 89831

A.I. Joseph Thaddeus

Department	Environmental & Climate Design
Telephone	+31 15 27 81533

Dr. J. Jovanova

Department	Transport Engineering and Logistics
Telephone	+31 15 27 83376
Room	34.B-4-220

Dr. J. Kalkman

Unit	Technische Natuurwetenschappen
Department	ImPhys/Kalkman group
Telephone	+31 15 27 83727

Drs. A.E. Kam

Unit	Techniek, Bestuur & Management

Department	Instituut voor Talen en Academische Vaardigheden
Telephone	+31 15 27 89728
Room	31.c0.100

Dr. E.M. Kelder

Unit	Technische Natuurwetenschappen
Department	RST/Storage of Electrochemical Energy
Telephone	+31 15 27 83262

Prof.dr.ir. A. van Keulen

Unit	Mech, Maritime & Materials Eng
Department	Werktuigbouwkunde, Maritieme Techniek en Technische Materiaalwetenschappen
Telephone	+31 15 27 86515
Room	34.G-1-430

H. Khodaei

Unit	Techniek, Bestuur & Management
Department	Delft Centre for Entrepreneurship
Telephone	+31 15 27 82759
Room	31.C2.130

Prof.dr.ir. S.A. Klein

Department	Energy Technology
Telephone	+31 15 27 85840

D. de Klerk

Department	Dynamics of Micro and Nano Systems
Room	34.G-1-200

A.P. Konijnenberg

J.F.P. Kooij

Unit	Mech, Maritime & Materials Eng
Department	Intelligent Vehicles
Telephone	+31 15 27 89271
Room	34.E-0-260

Dr. R. Kortlever

Unit	Mech, Maritime & Materials Eng
Department	Large Scale Energy Storage
Telephone	+31 15 27 86640
Room	34b.K-1-150

Dr. B. Kruizinga

O. Kudina

Department	Ethiek & Filosofie van Techniek
Telephone	+31 15 27 83956

Dr.ir. M. Langelaar

Unit	Mech, Maritime & Materials Eng
Department	Computational Design and Mechanics
Telephone	+31 15 27 86506
Room	34.G-1-300

Dr. A. Laskari

Department	Multi Phase Systems
Telephone	+31 15 27 85662
Room	34b.K-2-310

Ing. M.A. Leeflang

Unit	Mech, Maritime & Materials Eng

Department	Biomaterials & Tissue Biomechanics
Telephone	+31 15 27 82242
Room	34.E-3-230

Ir. C.W.J. Lemmens

Department	Support Delft Institute of Applied Mathematics
Telephone	+31 15 27 87224
Room	36.HB 04.080

Unit	Elektrotechn., Wisk. & Inform.
Department	Support Delft Institute of Applied Mathematics
Telephone	+31 15 27 87224
Room	36.HB 04.080

M.J. Ley

Department	Ethiek & Filosofie van Techniek
Telephone	+31 15 27 87795

Dr.ir. H.X. Lin

Unit	Elektrotechn., Wisk. & Inform.
Department	Mathematical Physics
Telephone	+31 15 27 87229
Room	36.HB 05.060

Dr. J.J.D. Loicq

Department	Spaceborne Instrumentation
Telephone	+31 15 27 83259
Room	62.10.19

K. Maheshwari

Department	Ethiek & Filosofie van Techniek
Telephone	+31 15 27 87930

T. Manzaneque Garcia

Unit	Mech, Maritime & Materials Eng
Department	Electronic Instrumentation
Telephone	+31 15 27 85981
Room	36.HB 15.300

Dr.ing. L. Marchal Crespo

Department	Human-Robot Interaction
Room	34.F-1-470

M. Mastrangeli

Unit	Elektrotechn., Wisk. & Inform.
Department	Electronic Components, Technology and Materials
Telephone	+31 15 27 83835
Room	36.LB 01.600

E. Matroos

Unit	Mech, Maritime & Materials Eng
Department	Precision and Microsystems Engineering
Telephone	+31 15 27 86581
Room	34.G-1-290

A. Melnyk

Department	Ethiek & Filosofie van Techniek
Telephone	+31 15 27 87925

Dr. E. Mendes

Unit	Technische Natuurwetenschappen
Department	ChemE/Advanced Soft Matter

Telephone	+31 15 27 82623
Room	58.D2.320

Dr. A. Mohammadi Gheidari

Department	ImPhys/Hoogenboom group

J.K. Moore

Department	Biomechatronics & Human-Machine Control
Telephone	+31 15 27 83556

Dr. O. Moultos

Unit	Mech, Maritime & Materials Eng
Department	Engineering Thermodynamics
Telephone	+31 15 27 81307
Room	34b.K-0-250

Dr.ir. W. Mugge

Unit	Mech, Maritime & Materials Eng
Department	Biomechatronics & Human-Machine Control
Telephone	+31 15 27 88461
Room	34.F-1-320

Dr. M. Möller

Unit	Elektrotechn., Wisk. & Inform.
Department	Numerical Analysis
Telephone	+31 15 27 89755
Room	36.HB 03.040

A. Napoleone

Department	Transport Engineering and Logistics

Prof.dr. R.R. Negenborn

Unit	Mech, Maritime & Materials Eng
Department	Transport Engineering and Logistics
Telephone	+31 15 27 86718
Room	34.B-3-300

Dr.ir. L.F.P. Noel

Department	Computational Design and Mechanics
Telephone	+31 15 27 87467

R.A. Norte

Department	QN/Groeblacher Lab
Telephone	+31 15 27 86070

Unit	Mech, Maritime & Materials Eng
Department	Dynamics of Micro and Nano Systems
Telephone	+31 15 27 86070

Dr.ir. S.E. Offerman

Unit	Mech, Maritime & Materials Eng
Department	Team Erik Offerman
Telephone	+31 15 27 82198

H.J.M. Olsthoorn

Unit	Elektrotechn., Wisk. & Inform.
Department	ESP LAB
Telephone	+31 15 27 86163
Room	36.LB 02.620

Dr.ir. R.A.J. van Ostayen

Unit Mech, Maritime & Materials Eng
Department Mechatronic Systems Design
Telephone +31 15 27 81647
Room 34.G-1-460

Dr.ir. Y. Pang

Unit Mech, Maritime & Materials Eng
Department Transport Engineering and Logistics
Telephone +31 15 27 88685
Room 34.B-3-310

Dr. E. Papadimitriou

Unit Techniek, Bestuur & Management
Department Safety and Security Science
Telephone +31 15 27 84913

N. van der Pas

Department Aerospace Structures & Computational Mechanics

Department Space Systems Egineering

Prof.dr. R. Pecnik

Department Energy Technology
Telephone +31 15 27 89153
Room 34b.K-1-220

Dr.ir. J.W.R. Peeters

Unit Mech, Maritime & Materials Eng
Department Energy Technology
Telephone +31 15 27 86628
Room 34b.K-1-150

Dr.ir. M.A. Peeters-Reus

Department Multi Phase Systems

Department Multi Phase Systems

Dr.ir. M.P. Peirlinck

Department Medical Instruments & Bio-Inspired Technology

Dr.ir. S.F. Pereira

Unit Technische Natuurwetenschappen
Department ImPhys/Pereira group
Telephone +31 15 27 82474

L. Peternel

Unit Mech, Maritime & Materials Eng
Department Human-Robot Interaction
Telephone +31 15 27 85695
Room 34.F-1-420

Prof.dr. S.J. Picken

Unit Technische Natuurwetenschappen
Department ChemE/Advanced Soft Matter
Telephone +31 15 27 86946
Room 58.D2.380

Dr.ir. R.H. Poelma

Unit Elektrotechn., Wisk. & Inform.
Department Electronic Components, Technology and Materials

Room	36.LB 01.660

Dr.ir. H. Polinder

Unit	Mech, Maritime & Materials Eng
Department	Transport Engineering and Logistics
Telephone	+31 15 27 81844
Room	34.B-3-290

V. Popovich

Unit	Mech, Maritime & Materials Eng
Department	Team Vera Popovich
Telephone	+31 15 27 89568
Room	34.H-2-200

Dr.ir. M.J.B.M. Pourquie

Unit	Mech, Maritime & Materials Eng
Department	Fluid Mechanics
Telephone	+31 15 27 82997
Room	34.F-1-600

M. Ramdin

Unit	Mech, Maritime & Materials Eng
Department	Engineering Thermodynamics
Room	34b.K-1-290

C.D. Rans

Unit	Luchtvaart- & Ruimtevaarttechn
Department	Structural Integrity & Composites
Telephone	+31 15 27 81680

A. Rasooli

Department	Environmental & Climate Design
Telephone	+31 15 27 87383

Prof.dr.ir. G.L.L.M.E. Reniers

Unit	Techniek, Bestuur & Management
Department	Safety and Security Science
Telephone	+31 15 27 83749
Room	31.c1.100

Dr. V. Reppa

Unit	Mech, Maritime & Materials Eng
Department	Transport Engineering and Logistics
Telephone	+31 15 27 88545

R.N. Roeplal

Department	Transport Engineering and Logistics
Telephone	+31 15 27 82039

Prof.dr. S. Roeser

Unit	Techniek, Bestuur & Management
Department	Values Technology and Innovation
Telephone	+31 15 27 88779
Room	31.b4.110

M.L. van de Ruit

Department	Biomechatronics & Human-Machine Control
Telephone	+31 15 27 83149
Room	34.F-1-320

M. Sand

Unit	Techniek, Bestuur & Management
Department	Ethiek & Filosofie van Techniek
Telephone	+31 15 27 87885
Room	31.B4.250

Dr. F. Santoni De Sio

Unit	Techniek, Bestuur & Management
Department	Ethiek & Filosofie van Techniek
Telephone	+31 15 27 85141

Dr.ing. V.E. Scholten

Unit	Techniek, Bestuur & Management
Department	Delft Centre for Entrepreneurship
Telephone	+31 15 27 89596
Room	31.C2.070

Prof. Dr.ir. D.L. Schott

Unit	Mech, Maritime & Materials Eng
Department	Transport Engineering and Logistics
Telephone	+31 15 27 83130
Room	34.B-4-300

Prof.dr.ir. A.C. Schouten

Unit	Mech, Maritime & Materials Eng
Department	Biomechanical Engineering
Telephone	+31 15 27 85247
Room	34.F-1-240

Dr.ir. F.F.J. Schrijer

Unit	Luchtvaart- & Ruimtevaarttechn
Department	Aerodynamics
Telephone	+31 15 27 86386
Room	64.HSL 0.39

F. Schulte

Unit	Mech, Maritime & Materials Eng
Department	Transport Engineering and Logistics
Telephone	+31 15 27 84414

Dr. M.V. van der Seijs

Department	Dynamics of Micro and Nano Systems
Room	34.G-1-420

A. Seth

Department	Biomechatronics & Human-Machine Control
Telephone	+31 15 27 86985
Room	34.E-1-230

H. Shi

Department	Transport Engineering and Logistics
Room	34.B-3-340

Dr. P. Simões Costa

Department	Energy Technology

Ir. J. Sinke

Unit	Luchtvaart- & Ruimtevaarttechn
Department	Aerospace Manufacturing Technologies
Telephone	+31 15 27 85137
Room	62.NB 0.54

Prof.dr.ir. L.J. Sluijs

Unit Civiele Techniek & Geowetensch
Department Materials- Mechanics- Management & Design

Telephone +31 15 27 82728
Room 23.HG 6.42

Dr.ir. G. Smit

Unit Mech, Maritime & Materials Eng
Department Medical Instruments & Bio-Inspired Technology

Telephone +31 15 27 81688

C.S. Smith

Department BN/Nynke Dekker Lab

Telephone +31 15 27 82411

Department ImPhys/Rieger group

Telephone +31 15 27 82411

Unit Mech, Maritime & Materials Eng
Department Team Carlas Smith

Telephone +31 15 27 82411

Dr. O.A. Soloviev

Department Team Michel Verhaegen

Telephone +31 15 27 85493
Room 34.C-1-250

Department Team Michel Verhaegen

Telephone +31 15 27 85493
Room 34.C-1-250

Unit Mech, Maritime & Materials Eng
Department Team Michel Verhaegen

Telephone +31 15 27 85493
Room 34.C-1-250

Dr.ir. W.H. van der Spoel

Unit Bouwkunde
Department Environmental & Climate Design

Telephone +31 15 27 83386

Prof.dr. U. Staufer

Unit Mech, Maritime & Materials Eng
Department Micro and Nano Engineering

Telephone +31 15 27 86804
Room 34.G-1-350

Prof.dr. P.G. Steeneken

Department QN/Steeneken Lab

Telephone +31 15 27 89545
Room 34.G-1-390

Unit Mech, Maritime & Materials Eng
Department Dynamics of Micro and Nano Systems

Telephone +31 15 27 89545
Room 34.G-1-390

Unit Technische Natuurwetenschappen
Department QN/Steeneken Lab

Telephone +31 15 27 89545
Room 34.G-1-390

E.F.L. Stok

Unit Mech, Maritime & Materials Eng
Department Support Marine and Transport Techology

Telephone +31 15 27 86839
Room 34.B-2-240

Dr. B.J. van Straten

Department Medical Instruments & Bio-Inspired Technology

Room 34.F-1-360

Prof.dr.ir. B. Taebi

Unit Techniek, Bestuur & Management
Department Ethiek & Filosofie van Techniek

Telephone +31 15 27 87511

Dr. D.S.W. Tam

Unit Mech, Maritime & Materials Eng
Department Fluid Mechanics

Telephone +31 15 27 82991

Dr.ir. M. Tichem

Unit Mech, Maritime & Materials Eng
Department Micro and Nano Engineering

Telephone +31 15 27 81603
Room 34.G-1-310

Dr. B.P. Tighe

Unit Mech, Maritime & Materials Eng
Department Multi Phase Systems

Telephone +31 15 27 81103
Room 34b.K-1-250

R.H. van Till

Unit Mech, Maritime & Materials Eng
Department Ship Design, Production and Operations

Telephone +31 15 27 86564
Room 34.B-4-200

Dr.ir. M.J. Tummers

Unit Mech, Maritime & Materials Eng
Department Fluid Mechanics

Telephone +31 15 27 82477
Room 34b.K-2-270

I. Uriol Balbin

Department Aerospace Structures & Computational Mechanics

Telephone +31 15 27 82355

H. Vahedi

Department DC systems, Energy conversion & Storage

Telephone +31 15 27 89086

Dr. Y. Vardar

Department Human-Robot Interaction

Telephone +31 15 27 84964
Room 34.F-1-450

Dr.ir. G.V. Vdovine

Department Team Michel Verhaegen

Telephone +31 15 27 85610
Room 34.C-1-330

Unit Mech, Maritime & Materials Eng
Department Team Mulders

Telephone +31 15 27 85610

Room	34.C-1-330

Prof.dr. H.E.J. Veeger

Unit	Mech, Maritime & Materials Eng
Department	Biomechatronics & Human-Machine Control
Telephone	+31 15 27 89530
Room	34.E-1-250

Dr. G.J. Verbiest

Unit	Mech, Maritime & Materials Eng
Department	Dynamics of Micro and Nano Systems
Telephone	+31 15 27 86504

B. van Vliet

Unit	Mech, Maritime & Materials Eng
Department	Support Biomechanical Engineering
Telephone	+31 15 27 83436
Room	34.E-3-330

Prof.dr.ir. T.J.H. Vlugt

Unit	Mech, Maritime & Materials Eng
Department	Engineering Thermodynamics
Telephone	+31 15 27 87551
Room	34b.K-0-150

Prof.dr.ir. W. van de Water

Department	Fluid Mechanics
Telephone	+31 15 27 85531

Department	Fluid Mechanics
Telephone	+31 15 27 85531

Dr.ir. W.J. Westerveld

Department	Dynamics of Micro and Nano Systems
Telephone	+31 15 27 88128
Room	34.G-1-300

Prof.dr.ir. J. Westerweel

Unit	Mech, Maritime & Materials Eng
Department	Fluid Mechanics
Telephone	+31 15 27 86887

Dr. M. Wiertlewski

Unit	Mech, Maritime & Materials Eng
Department	Human-Robot Interaction
Telephone	+31 15 27 82736
Room	34.F-1-420

Prof.dr.ir. J.W. van Wingerden

Unit	Mech, Maritime & Materials Eng
Department	Team Jan-Willem van Wingerden
Telephone	+31 15 27 81720
Room	34.C-2-340

Prof.dr.ir. J.C.F. de Winter

Unit	Mech, Maritime & Materials Eng
Department	Human-Robot Interaction
Telephone	+31 15 27 81078

W. Wolfslag

Department	Biomechatronics & Human-Machine Control

Dr. M. Yang

Department	Safety and Security Science
Telephone	+31 15 27 82891
Room	31.C1.130

Prof.dr. A.A. Zadpoor

Unit	Mech, Maritime & Materials Eng
Department	Biomaterials & Tissue Biomechanics
Telephone	+31 15 27 81021
Room	34.E-3-290

Dr.ing. H.W. van Zeijl

Unit	Elektrotechn., Wisk. & Inform.
Department	Electronic Components, Technology and Materials
Telephone	+31 15 27 81092
Room	36.EKL 01.060

Prof.dr. G.Q. Zhang

Unit	Elektrotechn., Wisk. & Inform.
Department	Electronic Components, Technology and Materials
Telephone	+31 15 27 89495
Room	36.DI 00.030

Dr. J. Zhou

Unit	Mech, Maritime & Materials Eng
Department	Biomaterials & Tissue Biomechanics
Telephone	+31 15 27 85357
Room	34.E-3-310

Dr.ir. A.H. van Zuijlen

Unit	Luchtvaart- & Ruimtevaarttechn
Department	Aerodynamics
Telephone	+31 15 27 82046
Room	64.HSL 0.39

Ir. S.J. Zwart